

Transaction Processing Manual

BASE24-atm[®]



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What's New

The following tables highlight the major changes that have been made to the **BASE24-atm Transaction Processing Manual** since its publication for BASE24 release 6.0 version 10. The first column of each table lists the sections and appendixes in which major changes have been made. The second column of each table describes the major changes for the section or appendix

January 2012

Section/ Appendix	Major Changes
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A	Updates Standard Internal Message field descriptions for ORIG-PRO-NAME and HLD-ORIG-PRO-NAME.
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December 2009

Section/ Appendix	Major Changes
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3	Updates the FAST CASH ACCOUNT TYPE description and balance inquiry usage.
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Updates Surcharging transactions.

A	Updates the Standard Internal Message deposit type.
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B	Updates Authorization Response Codes 57 TRANSACTION NOT SUPPORTED and 72 DESTINATION NOT AVAILABLE.
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Preface

The ***BASE24-atm Transaction Processing Manual*** describes how BASE24-atm processes transactions. It includes basic transaction processing concepts and definitions, as well as specifics on how transactions are identified and processed by BASE24-atm. It also describes how to control and configure various aspects of BASE24-atm transaction processing.

Audience

This manual is intended for the BASE24 managers and operational staff who are responsible for configuring the BASE24-atm database for transaction processing and understanding how the transactions flow through the BASE24-atm system.

Prerequisites

Persons using this manual should be familiar with the BASE24-atm product, its basic functions, its terminology, and the BASE24 files it uses.

Additional Documentation

The BASE24 documentation set is arranged so that each BASE24 manual presents a topic or group of related topics in detail. When one BASE24 manual presents a topic that has already been covered in detail in another BASE24 manual, the topic is summarized and the reader is directed to the other manual for additional information. Information has been arranged in this manner to be more efficient for readers who do not need the additional detail and at the same time provide the source for readers who require the additional information.

This manual contains references to the following BASE24 publications:

- The ***BASE24 Logical Network Configuration File Manual*** describes, in detail, each assign and param used in configuring BASE24-atm Device Handler, Authorization, and Host Interface processes.
- The ***BASE24 Base Files Maintenance Manual*** and ***BASE24-atm Files Maintenance Manual*** provide detailed descriptions and valid values for the configuration fields discussed in this manual.
- The ***BASE24 Files and Record Structures Manual*** provides general information about the structure of BASE24 files and records, along with instructions for using the ACI DDLDOC program.
- The ***BASE24 Tokens Manual*** presents detailed field descriptions and configuration information for the tokens used by BASE24-atm.
- The ***BASE24 ISO Host Interface Manual***, ***BASE24 ANSI Host Interface Manual***, ***BASE24 BIC ISO Interface Manual***, and ***BASE24 BIC ANSI Interface Manual*** provide detailed processing information for each of the specified interfaces.
- The ***BASE24 BIC ISO Standards Manual*** contains information about the standards used with the BIC ISO Interface.
- The ***BASE24 External Message Manual*** presents detailed field descriptions and configuration information for the ISO-based external message used between BASE24 and a host. (The ANSI-based external message is documented in the ***BASE24 ANSI Host Interface Manual***.)
- The ***BASE24-atm Settlement and Reporting Manual*** describes, in detail, the settlement and reporting functions supported by BASE24-atm and how to configure them.
- The ***BASE24 Transaction Security Manual*** describes BASE24 support of PIN verification and encryption, card verification, message authentication, and dynamic key management, along with the database settings required to implement this support.
- The ***BASE24-atm NCR 5XXX Device Support Manual*** describes BASE24-atm support of NCR 5XXX-series ATMs.
- The ***NET24-XPNET Network Control Command Reference Manual*** provides detailed information on configuring a BASE24 network of lines, stations, and processes.
- The ***BASE24-atm Diebold 10XX/478X Device Support Manual*** describes BASE24-atm support of Diebold 10XX/478X ATMs.

- The ***BASE24-atm Multiple Currency Support Manual*** describes the BASE24-atm Multiple Currency add-on product.
- The ***BASE24-atm EMV Manual*** describes the BASE24-atm EMV add-on product.

Software

This manual documents standard processing as of its publication date. Software that is not current and custom software modifications (CSMs) may result in processing that differs from the material presented in this manual. The customer is responsible for identifying and noting these changes.

Manual Summary

The following is a summary of the contents of this manual.

“Conventions Used in this Manual” follows this preface and describes notation and documentation conventions necessary to understand the information in the manual.

Section 1, “Introduction,” introduces the reader to the basics of BASE24-atm transaction processing. It includes definitions and discussions of transaction originators and authorizers, issuers and acquirers, hosts and DPCs, cards and accounts, supported transactions, transaction messages, and transaction logging.

Section 2, “BASE24-atm Transaction Path,” discusses the BASE24-atm processes involved in transaction processing and how they are linked together to form various transaction paths. It also describes the timers they use to control processing.

Section 3, “BASE24-atm Transaction Processing Controls,” describes the various transaction processing control options that enable a logical network to be tailored to the needs of its institutions. It includes discussions of such topics as authorization levels, authorization methods, sharing, and cardholder limits and accumulation.

Section 4, “BASE24-atm Routing and Authorization,” describes—and provides examples of—how BASE24-atm routes transactions and selects authorization parameters for processing transactions.

Section 5, “BASE24-atm Message Flows,” presents a series of diagrams illustrating the paths taken by BASE24-atm transactions under various types of processing situations.

Section 6, “BASE24-atm Transaction Impacts on the Database,” describes the impacts of each type of BASE24-atm transaction on the Cardholder Authorization File (CAF), Positive Balance File (PBF), BASE24-atm Terminal Data files, and Usage Accumulation File (UAF).

Appendix A, “BASE24-atm Internal Message,” describes the fields in the BASE24-atm Standard Internal Message (STM) and identifies the BASE24-atm tokens that can be carried in the internal message.

Appendix B, “BASE24-atm Card Disposition and Authorization Response Codes,” describes the response codes that can be carried in the STM to identify the results of processing a transaction.

Appendix C, “BASE24-atm Device Handler Processing,” summarizes the transaction processing performed by a BASE24-atm Device Handler process.

Appendix D, “Pregen Utility,” describes the Pregen utility and the procedures used to build and start using a new or revised SPROUTE File.

Appendix E, “Base Extended Memory Table Build Utility,” provides detailed instructions for using the Base Extended Memory Table Build utility. This utility is used to build the extended memory tables for the Acquirer Processing Code File (APCF), Institution Definition File (IDF), Issuer Processing Code File (IPCF), Receipt File (RCPT), and Terminal Receipt File (TRF).

Publication Identification

Three entries appearing at the bottom of each page uniquely identify this BASE24 publication. The publication number (for example, AT-AE000-07 for the ***BASE24-atm Transaction Processing Manual***) appears on every page to assist readers in identifying the manual from which a page of information was printed. The publication date (for example, Dec-2009 for December, 2009) indicates the issue of the manual. The software release information (for example, R6.0v10 for release 6.0, version 10) specifies the software that the manual describes. This information matches the document information on the copyright page of the manual.

Conventions Used in This Manual

This section explains how syntax and required blank characters are documented in this manual. It also explains BASE24-atm Terminal Data files acronyms.

Syntax Notation

Syntax descriptions in this manual use several symbols and conventions to denote how commands must be entered. These conventions are described in the table below.

Notation	Meaning
UPPERCASE LETTERS	Indicate key words and literals that you must enter exactly as shown.
<i>lowercase italics</i>	Identify variables that you must provide.
Brackets []	Enclose optional syntax items that you can enter in the command.
Braces { }	Enclose a group of items from which you are required to choose one item. The items are separated within the braces by a vertical bar ().
Vertical bar	Separates alternatives in a list of items.
Ellipsis ...	Indicates the immediately preceding syntactical item can occur one or more times.
Punctuation	Must be entered as shown. This includes parentheses, commas, semicolons, and other symbols not previously described.

Notation	Meaning
Spaces	Are required as delimiters wherever it would be otherwise impossible to discriminate between adjacent words or symbols. You can insert additional spaces between any adjacent words or symbols, but not within a word or multiple character symbol.
Indented lines	Indicates a continuation of the preceding line of syntax. If the syntax of a command is too long to fit on a single line, each continuation line is indented.

Required Blank Spaces

Throughout this manual when discussing the impact of required blanks or spaces in entered field data, the symbol *b* is used to denote a required blank character or space.

BASE24-atm Terminal Data Files Acronyms

The BASE24-atm Terminal Data files can be abbreviated using a number of acronyms, depending on the Device Handler process used in the BASE24-atm system, whether you are accessing the file through the BASE24 user access system, or whether the data is static or dynamic. The table below summarizes BASE24-atm Terminal Data files acronyms by Device Handler process:

Device Handler Process	BASE24-atm Terminal Data Files Acronym			
	Screen	File	Static	Dynamic
Diebold 10XX/478X	ATD	ATD	ATDS1	ATDD x
NCR 5XXX	ATD	ATD	ATDS1	ATDD x
Other	ATD	TDF	Not applicable	Not applicable

In the above table, the acronym ATDD x , where x is the value 1 or 2, can describe the dynamic data stored in the BASE24-atm Terminal Data files. The data in ATDD1 is general dynamic data. The data in the ATDD2 is token data for the last transaction processed by BASE24 and acquired at this terminal.

To eliminate confusion when referring to the file, the name BASE24-atm Terminal Data files is used instead of a file acronym.

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1: Introduction

This section provides an introduction to BASE24-atm transaction processing. It provides readers with some basic information on how BASE24-atm processes transactions.

BASE24-atm Transaction Processing

BASE24-atm provides the means by which a transaction originator can request and receive authorization for an action on a customer card or account from a transaction authorizer. The role of BASE24-atm in transaction processing includes the following:

- Routing transactions from the transaction originator to the appropriate authorizer.
- Participating in or carrying out authorization on behalf of the institutions for which it processes.
- Logging all transactions for later processing and settlement.

BASE24-atm can accept transactions from a variety of sources. Likewise, it can involve a number of different authorizers in its processing. The following describes the various transaction originators and authorizers that can exist in a BASE24-atm system.

Transaction Originators

BASE24-atm can accept transactions for authorization from several different types of originators. These originators are described below.

ATMs

Automated teller machines (ATMs) can be directly attached to BASE24-atm, in which case transactions are originated by customers or service personnel at the ATM. Communications between the ATM and BASE24-atm are controlled by BASE24-atm Device Handler processes.

Hosts

Host computers can be set up to control ATM networks and forward transactions to BASE24-atm. In this case, communications between these acquirer hosts and BASE24-atm are controlled by BASE24-atm Host Interface processes.

Interchanges

Interchanges can act as switches to forward transactions to BASE24-atm for authorization. In this case, communications between the interchange acquiring the transaction and BASE24-atm are controlled by BASE24-atm Interchange Interface processes.

Co-Networks

Co-networks can be linked to a BASE24-atm system through a BASE24 Interchange (BIC) Interface process. In this case, transactions can be originated in any of the ways noted above and forwarded through BASE24 into another BASE24 system as if through an interchange. Communications between the BASE24 networks is controlled by BASE24 BIC Interface processes on each BASE24 system.

Transaction Authorizers

BASE24-atm can authorize transactions or route them to several types of authorizers. The possible transaction authorizers are described below.

BASE24-atm

BASE24-atm can be set up to authorize transactions in those situations where cardholder files are maintained on the BASE24-atm system.

Hosts

Hosts can be set up to authorize transactions in situations where cardholder files are maintained on the host computer. In this case, BASE24-atm can screen the transactions before sending them to the host and can also be set up to stand in and authorize transactions for a host in situations where communications between BASE24-atm and the host system are down.

Interchanges

Interchanges can be used to forward transactions to other networks for authorization.

Co-Networks

BASE24-atm can be linked to co-networks through a BIC Interface process, and transactions can be forwarded through BASE24 to another BASE24 system or co-network for authorization. BASE24-atm can also be set up to stand in and authorize transactions for another co-network in situations where communications between BASE24-atm and the co-network are down.

Customer-Specific Application Processes

The BASE24-atm Diebold 10XX/478X and NCR 5XXX Device Handler processes and certain Interchange Interface processes can route transactions directly to a process running outside of the BASE24-atm environment (using a customer-specific XPNET process) for authorization. For example, an ATM bill payment transaction acquired by an enhanced endpoint could be sent directly to a bill payment service. The authorization response returned from this application process can then be optionally logged to the Transaction Log File (TLF).

The following Interchange Interface processes support this functionality:

- Banknet Interchange Interface process
- BIC ISO Interface process
- MDS/MDSM Interchange Interface process
- PLUS ISO Interchange Interface process
- VisaNet Interchange Interface process

Issuers and Acquirers

Transaction originators and authorizers are generally defined by the EFT industry terms acquirer and issuer. An acquirer, or transaction acquirer, is the party that originates a transaction. An issuer, or card issuer, is an institution that issues cards or owns accounts that can be used in EFT transactions.

Whether an entity is defined as an issuer or an acquirer depends on the perspective from which the transaction is viewed. Intermediary networks and processors that receive transactions from a transaction originator and authorize them or forward them elsewhere for authorization can be acting on behalf of both the acquirer and the issuer.

From the acquirer point of view, intermediary networks are acting on behalf of the issuer. For example, a cardholder initiates a transaction at an ATM belonging to institution A. In this case, institution A is the transaction acquirer. Institution A sends the transaction to an intermediary network that, in turn, sends the transaction to institution B for authorization. Institution B owns the cardholder's account and issued the cardholder the card; therefore, institution B is the true issuer. However, from the point of view of institution A, the intermediary network is performing as an issuer on behalf of institution B since institution A had to send the transaction to the intermediary network before it could reach its authorizing destination.

From the issuer point of view, intermediary networks are acting on behalf of the acquirer. As in the above example, institution B views the intermediary network as an acquirer since the intermediary network is the entity that sent it a transaction to be authorized.

BASE24 can process transactions on behalf of an issuer or an acquirer, depending on where a transaction originates and who is to authorize the transaction. For example, when BASE24 sends a transaction to a back-end host for authorization, BASE24 represents the acquirer in the message exchange and the back-end host represents the issuer. On the other hand, when a transaction is sent to BASE24 for authorization, the sending host represents the acquirer and BASE24 represents the issuer.

Hosts and DPCs

Hosts can play a prominent role in the transaction processing performed by BASE24. A host is an external mainframe computer system with which BASE24-atm interacts—either online or by batched tape—to authorize transactions and/or update cardholder records.

Although the term host generally refers to an actual computer or system of computers, the term also refers to the institution or organization responsible for the host computer system and its processing. For example, in the phrase, “the host can opt to receive advice messages,” the term host actually refers to the BASE24-defined institution in control of the host computer, rather than the computer itself.

BASE24 also refers to hosts as data processing centers (DPCs). Generally, the terms DPC and host are synonymous. However, the term host implies either a host computer or the organization responsible for a host computer; the term DPC always characterizes the host computer or computer network itself.

A DPC can be one mainframe computer communicating with BASE24, or a DPC can represent several mainframe computers networked together at a host site. Regardless of the machine configuration in relation to BASE24, a DPC represents a single host from the BASE24 point-of-view. A host site can establish multiple DPCs; however, BASE24 considers each of these DPCs to be a separate host.

DPCs are assigned identification numbers and are defined to BASE24 in the Host Configuration File (HCF). Each DPC must have a record in the HCF to be recognized by BASE24. The HCF contains options at the DPC-level that allow hosts to specify individual processing parameters for each DPC.

Cards and Accounts

Transactions in the BASE24-atm system can be defined as specific actions—usually financial—carried out on accounts of some type using a card or a card number as a basis for initiating the action.

BASE24 makes a distinction between cards and accounts. Transaction limits and accumulators can be maintained at the card or account level. Balances, if maintained, are maintained at the account level. The following discusses cards and accounts as they are defined and used in BASE24.

Cards

Plastic cards serve as evidence of an account and as a mechanism for accessing the account from electronic funds transfer (EFT) devices such as automated teller machines (ATMs) and point-of-sale (POS) terminals. Cards are given card numbers, which may or may not match the account numbers of the accounts that they are used to access. There can be a one-to-one relationship between cards and accounts, one card can access multiple accounts, or one account can be accessed by multiple cards, depending on card issuance procedures and BASE24 parameters.

Credit Cards and Debit Cards

In BASE24-atm, credit cards and debit cards can access both noncredit and credit accounts. BASE24-atm does not distinguish between credit and debit cards for the purpose of transaction processing.

Card Type Codes

BASE24 uses codes to identify the type of card being used to initiate transactions. Card type codes are either reserved by BASE24 or user-defined.

Reserved BASE24 Card Types

BASE24 reserves specific card types across all product lines. These reserved codes are to be used only as defined. Refer to the ***BASE24 Base Files Maintenance Manual*** for a list of reserved card types.

User-Defined Card Types

Users can define and use any one- or two-character code, not reserved by BASE24, according to the following guidelines:

- The first character must be alphabetic: A, B, D through O, and Q through Z.
- The second character can be A through Z, 0 through 9, or a blank.

A valid COBNAMES table entry is recommended for each user-defined code.

Note: Codes starting with C and P can also be user-defined, but these are subject to restrictions imposed by BASE24 (refer to the reserved BASE24 card types list above). The BASE24 standard COBNAMES table already contains entries for these codes; however, the COBNAMES entries can be modified as needed by the user. Refer to the ***BASE24 Base Files Maintenance Manual*** for more information about card types.

Accounts

BASE24 classifies all accounts as one of two types: credit or noncredit. This classification determines the account-level limits and accumulators that apply to a particular transaction.

Credit and Noncredit Accounts

Credit accounts involve funds advanced to an accountholder, by a financial institution or retailer, based on a credit agreement with the accountholder.

Noncredit accounts involve accountholder funds on deposit with a financial institution (for example, checking or savings).

BASE24 allows processors to set up various transaction limits and accumulators. Throughout BASE24, the account classification determines which limits and accumulators apply to a particular transaction.

Account Type Codes

BASE24-atm identifies accounts by account type as described in the table below. Institutions can set up other account types in the account type ranges, but BASE24-atm treats these account types as if they are the first in the number range. For example, an institution may want to distinguish between different types of checking accounts for their own record keeping and reporting purposes. They can assign type 02 to checking account type A, 03 to checking account type B, and 04 to checking account type C. However, during processing, BASE24-atm treats account types 01, 02, 03, and 04 the same.

Type	Account Description	Credit	Noncredit
01	Checking (DDA)		✓
02–09	Checking (DDA) as set up by the institution. These types are processed the same as type 01.		✓
11	Savings		✓
12–19	Savings as set up by the institution. These types are processed the same as type 11.		✓
31	Credit	✓	
32–39	Credit account as set up by the institution. These types are processed the same as type 31.	✓	
60	Other. Accounts of this type are used with multiple account selection by account qualifier.	✓	✓

Account Type Names

Screens for certain BASE24 files use account type names defined in the Account Type Table File (ATT) instead of the account type codes listed above. Account type names are used on Acquirer Processing Code File (APCF), Issuer Processing Code File (IPCF), and Terminal Receipt File (TRF) screens, which BASE24-atm uses for defining a set of transactions allowed and the accounts on which the transactions can be performed based on the processing code and transaction profile.

ACI provides a default ATT that contains account type names for a number of account type codes. The ATT uses account type codes based on the International Organization for Standardization (ISO), rather than the account type codes used internally by BASE24-atm. The following table shows how ISO and BASE24 account type codes relate to account type names provided in the default ATT. The first two columns of the table list the ISO and BASE24-atm account type codes, respectively. The third column lists the corresponding account type name in the ATT, and the last column describes the account.

Account Type		Account Type Name	Description
ISO	BASE24		
00	Not applicable	NONE	
10	11	SAV	Savings accounts
1A	14	SAV1	
1B	15	SAV2	
1C	16	SAV3	
1D	17	SAV4	
1E	18	SAV5	
1F	19	SAV6	
20	01	DDA	Demand deposit (checking) accounts
2A	02	DDA1	
2B	03	DDA2	
2C	04	DDA3	
2D	05	DDA4	
2E	06	DDA5	
2F	07	DDA6	
2G	08	DDA7	
2H	09	DDA8	

Account Type		Account Type Name	Description
ISO	BASE24		
30	31	CR	Credit accounts
3A	33	CR1	
3B	34	CR2	
3C	35	CR3	
3D	36	CR4	
3E	37	CR5	
3F	38	CR6	
3G	39	CR7	
9M	60	OTHER	Other accounts used with multiple account selection by account qualifier processing

BASE24-atm Transactions

BASE24-atm supports the transactions described on the following pages. Unless noted to the contrary, BASE24-atm supports these transactions in all respects. However, not all ATMs and interchanges support all of the listed transactions. Refer to the appropriate BASE24-atm device support manual for a complete list of transactions supported by a particular device. Refer to the appropriate Interchange Interface documentation for a complete list of transactions supported by a specific Interchange Interface process.

Transaction Identifiers

Transactions are identified in BASE24-atm by a series of three values known as the processing code. The processing code is carried in the TRAN-CDE, FROM-ACCT-TYP, and TO-ACCT-TYP fields of the BASE24-atm Standard Internal Message (STM), and consists of a transaction code, *from* account type, and *to* account type. For more information about these STM fields, refer to appendix A.

Note: Instead of a *from* account and a *to* account, the transaction code for a split deposit transaction contains a first *to* account, and the second *to* account. This transaction is only available through NCR and Diebold self-service banking (SSB).

The following pages provide the six-digit transaction code and a brief description for each of the BASE24-atm transactions, with similar transactions grouped together.

In addition, some transactions have subtypes associated with them. Transaction subtypes enable BASE24-atm customers to distinguish among the types of transactions that have identical processing codes.

Withdrawal Transactions

BASE24-atm supports the following withdrawal transactions.

Note: Subtype APT0 is used to identify a preferred transaction.

Tran Code	Transaction Type and Description
10 00 00	Fast Cash Withdrawal — The withdrawal of a predetermined amount of money or purchase of a noncurrency item from a predefined customer account. This transaction reduces the cardholder's selection process to one or two simple key entries at the ATM, decreasing the overall transaction time. Note: The <i>from</i> account type for fast cash withdrawals cannot be set to 00 if the IDF routing parameters differ by account type.
10 01 00	Withdrawal from Checking Account — A cash withdrawal from a cardholder's checking account. This transaction type includes the purchase of a noncurrency item from the ATM.
10 11 00	Withdrawal from Savings Account — A cash withdrawal from a cardholder's savings account. This transaction type includes the purchase of a noncurrency item from the ATM.
10 31 00	Withdrawal from Credit Account — The advance of funds from a cardholder's credit account. This transaction type includes the purchase of a noncurrency item from the ATM.
10 60 00	Withdrawal from Other Account — A cash withdrawal or advance of funds from an account identified by an account qualifier.

Deposit Transactions

BASE24-atm supports the following deposit transactions.

Notes: BASE24-atm self service banking (SSB) allows a check to be deposited either in an envelope or directly into the ATM. The transaction code is the same for either method.

In addition, subtype APT0 is used to identify a preferred transaction.

Tran Code	Transaction Type and Description
20 00 01	Deposit to Checking — An unverified deposit of funds to a cardholder's checking account.
20 00 11	Deposit to Savings — An unverified deposit of funds to a cardholder's savings account.
20 00 60	Deposit to Other — An unverified deposit of funds to an account identified by an account qualifier.
24 00 01	Deposit to Checking with Cash Back — An unverified deposit of funds to a cardholder's checking account, with part or all of the deposit amount back in cash.
24 00 11	Deposit to Savings with Cash Back — An unverified deposit of funds to a cardholder's savings account, with part or all of the deposit amount back in cash.
24 00 60	Deposit to Other with Cash Back — An unverified deposit of funds to a cardholder's account identified by an account qualifier, with part or all of the deposit amount back in cash.
20 01 01	Split Deposit—Two Checking Accounts — An unverified deposit of funds into two separate checking accounts. This transaction is available only with BASE24-atm self-service banking (SSB).
20 01 60	Split Deposit—Checking and Other — An unverified deposit of funds into a checking account and an account identified by an account qualifier. This transaction is available only with BASE24-atm self-service banking (SSB).

Tran Code	Transaction Type and Description
20 01 11	Split Deposit—Checking and Savings — An unverified deposit of funds into a checking account and a savings account. This transaction is available only with BASE24-atm self-service banking (SSB).
20 11 01	Split Deposit—Savings and Checking — An unverified deposit of funds into a checking account and a savings account. This transaction is available only with BASE24-atm self-service banking (SSB).
20 11 11	Split Deposit—Two Savings Accounts — An unverified deposit of funds into two separate savings accounts. This transaction is available only with BASE24-atm self-service banking (SSB).
20 11 60	Split Deposit—Savings and Other — An unverified deposit of funds into a savings account and an account identified by an account qualifier. This transaction is available only with BASE24-atm self-service banking (SSB).
20 60 01	Split Deposit—Other and Checking — An unverified deposit of funds into an account identified by an account qualifier and a checking account. This transaction is available only with BASE24-atm self-service banking (SSB).
20 60 11	Split Deposit—Other and Savings — An unverified deposit of funds into an account identified by an account qualifier and a savings account. This transaction is available only with BASE24-atm self-service banking (SSB).
20 60 60	Split Deposit—Two Other Accounts — An unverified deposit of funds into two different accounts identified by account qualifiers. This transaction is available only with BASE24-atm self-service banking (SSB).

Mobile Top-Up Transactions

The following mobile top-up transactions are defined for the definitions-based model. The BASE24-atm Transaction Context Manager (TCM) will log this transaction code representing the secondary service authorization.

Tran Code	Transaction Type and Description
25 00 00	Mobile Top-Up — A mobile top-up transaction requests a secondary provider (e.g., telephone company) to load additional funds or minutes to a mobile telephone account. Note: The Authorization process handles the funds authorization transaction as a withdrawal (transaction code 10).
26 00 00	Mobile Top-Up with Funds — A mobile top-up using funds transaction request is logged to the Transaction Log File (TLF) by the Transaction Context Manager when the TCM is unable to process transactions. Note: The Authorization process handles the funds authorization transaction as a withdrawal (transaction code 10).

Balance Inquiry Transactions

BASE24-atm supports the following balance inquiry transactions.

Note: Subtype APT0 is used to identify a preferred transaction.

Tran Code	Transaction Type and Description
30 00 00	Balance Inquiry from Default — An inquiry on the amount where the authorizer determines the account type.
30 01 00	Balance Inquiry from Checking — An inquiry on the amount in a cardholder's checking account.
30 01 11	Balance Inquiry from Checking and/or Savings — An inquiry on the amount in a cardholder's checking account and/or savings account.
30 11 00	Balance Inquiry from Savings — An inquiry on the amount in a cardholder's savings account.

Tran Code	Transaction Type and Description
30 11 01	Balance Inquiry from Savings and Checking — An inquiry on the amount in a cardholder's savings account and checking account.
30 31 00	Balance Inquiry from Credit Account — An inquiry on the amount in a cardholder's credit account.
30 60 00	Balance Inquiry from Other — An inquiry on the amount in a cardholder's account that is identified by an account qualifier.

Transfer Transactions

BASE24-atm supports the following transfer transactions.

Note: Subtype APT0 is used to identify a preferred transaction. .

Tran Code	Transaction Type and Description
40 01 01	Transfer from Checking to Checking — A transfer of funds from one of a cardholder's checking accounts to another.
40 01 11	Transfer from Checking to Savings — A transfer of funds from a cardholder's checking account to the cardholder's savings account.
40 01 60	Transfer from Checking to Other — A transfer of funds from a cardholder's checking account to another account identified by an account qualifier.
40 11 01	Transfer from Savings to Checking — A transfer of funds from a cardholder's savings account to the cardholder's checking account.
40 11 11	Transfer from Savings to Savings — A transfer of funds from one of a cardholder's savings accounts to another.
40 11 60	Transfer from Savings to Other — A transfer of funds from a cardholder's savings account to another account identified by an account qualifier.

Tran Code	Transaction Type and Description
40 31 01	Transfer from Credit Account to Checking — A transfer of funds from a cardholder's credit account to the cardholder's checking account.
40 31 11	Transfer from Credit Account to Savings — A transfer of funds from a cardholder's credit account to the cardholder's savings account.
40 31 60	Transfer from Credit Account to Other — A transfer of funds from a cardholder's credit account to the cardholder's account identified by an account qualifier.
40 60 01	Transfer from Other to Checking — A transfer of funds from a cardholder's account identified by an account qualifier to the cardholder's checking account.
40 60 11	Transfer from Other to Savings — A transfer of funds from a cardholder's account identified by an account qualifier to the cardholder's savings account.
40 60 60	Transfer from Other to Other — A transfer of funds from a cardholder's account identified by an account qualifier to another account identified by an account qualifier.

Payment Transactions

BASE24-atm supports the following payment transactions.

Tran Code	Transaction Type and Description
50 01 31	Payment from Checking to Credit Account — A transfer of funds, as a payment, from a cardholder's checking account to the cardholder's credit account.
50 11 31	Payment from Savings to Credit Account — A transfer of funds, as a payment, from a cardholder's savings account to the cardholder's credit account.

Tran Code	Transaction Type and Description
50 31 31	Payment from Credit Account to Credit Account — A transfer of funds, as a payment, from a cardholder's credit account to another credit account belonging to the same cardholder.
50 60 31	Payment from Other to Credit Account — A transfer of funds, as a payment, from a cardholder's account identified by an account qualifier to the cardholder's credit account.
51 00 00	Payment Enclosed — A payment toward outstanding debts, such as loans, made by deposit.

Check-Related Transactions

BASE24-atm supports the following check-related transactions.

Note: BASE24-atm self-service banking (SSB) allows a check to be deposited either in an envelope or directly into the ATM. The transaction code is the same for either method.

Tran Code	Transaction Type and Description
03 01 00	Check Guarantee — A cardholder can verify that the amount of the check is available in the checking account and place the amount of the check on hold. Note: Only the BASE24-atm Authorization process supports this transaction. This transaction is not supported by standard Device Handler, Interchange Interface, or Host Interface processes.
04 01 00	Check Verification — A cardholder can verify that the amount of the check is available in the checking account. Note: Only the BASE24-atm Authorization process supports this transaction. This transaction is not supported by standard Device Handler, Interchange Interface, or Host Interface processes.
11 00 00	Cash Check — A cardholder can deposit checks and receive that cash back immediately using this transaction.

Log-Only Transactions

BASE24-atm supports the following log-only transactions.

Tran Code	Transaction Type and Description
60 00 00	Message to Financial Institution — Written communication from the cardholder to the financial institution. Subtype AER0 is used for exchange rate notification processing. Subtype APS0 is used for preferred transaction setup processing.
61 01 00 61 02 00 61 03 00 61 04 00	Log-Only Transaction — Messages displayed as menu selections on the ATM screen. Cardholders can select specific messages to the institution with key selections at the ATM. Possible log-only transactions include “REORDER CHECKS” and “SEND MORTGAGE LOAN INFO.”
62 00 00	Card Review — A verification of accounts, status, and card limits associated with a card. Only the Olivetti Device Handler and the BASE24-atm Authorization processes support this transaction; therefore, this transaction is available only with Olivetti terminals. Subtype API0 is used for preferred transaction inquiry processing.

Statement Print and Passbook Update Transactions

BASE24-atm supports the following statement print and passbook update transactions.

Note: Subtype APU0 is used for passbook update processing.

Tran Code	Transaction Type and Description
70 01 00	Statement Print Transaction for Checking — A request from a cardholder for a list of statement items posted to the cardholder’s checking account.

Tran Code	Transaction Type and Description
70 11 00	Statement Print Transaction for Savings Account — A request from a cardholder for a list of statement items posted to the cardholder's savings account.
70 31 00	Statement Print Transaction for Credit Account — A request from a cardholder for a list of statement items posted to the cardholder's credit account.
70 60 00	Statement Print Transaction for Other — A request from a cardholder for a list of statement items posted to an account identified by an account qualifier.

PIN Change Transaction

BASE24-atm supports the following PIN change transaction.

Tran Code	Transaction Type and Description
81 00 00	PIN Change Request — A request from a cardholder to change the Personal Identification Number (PIN) associated with his or her card.
81 01 00	PIN Change Request from Checking — A request to change the Personal Identification Number (PIN) associated with a cardholder's checking account.
81 11 00	PIN Change Request from Savings — A request to change the Personal Identification Number (PIN) associated with a cardholder's savings account.
81 31 00	PIN Change Request from Credit — A request to change the Personal Identification Number (PIN) associated with a cardholder's credit account.

EMV PIN Unblock Transaction

BASE24-atm supports the following PIN unblock transaction:

Tran Code	Transaction Type and Description
82 00 00	EMV PIN Unblock Request — A request from a cardholder to remove the block from the Personal Identification Number (PIN) associated with his or her card.
82 01 00	EMV PIN Unblock Request from Checking — A request from a cardholder to remove the block from the Personal Identification Number (PIN) associated with his or her checking account.
82 11 00	EMV PIN Unblock Request from Savings — A request from a cardholder to remove the block from the Personal Identification Number (PIN) associated with his or her savings account.
82 31 00	EMV PIN Unblock Request from Credit — A request from a cardholder to remove the block from the Personal Identification Number (PIN) associated with his or her credit account.

Administrative Transactions

BASE24-atm supports the following administrative transactions entered from ATMs directly connected to the system. Administrative transactions that are sent to the BASE24 Authorization process must be performed using a card number that BASE24 has been configured to recognize as belonging to an administrative card. This may be a physical card that is used at the front of the terminal or a number that is specified in the ATM's configuration data for rear administrative requests.

When the Authorization process determines that a transaction is an administrative request, it examines the first two digits of the processing code for a value of 98, which indicates that the request was initiated at the rear of the ATM.



If the ATM-REAR-ADMN-TRAN-PIN-ALWD param is present in the LCONF and has the value N, the Authorization process will deny any rear administrative transaction that contains a PIN. Payment Card Industry (PCI) standards strongly discourage entering PINs at the rear of an ATM because of the security risk. If the ATM-REAR-ADMN-TRAN-PIN-ALWD param is not present in the LCONF or does not have the value N, or if the transaction does not contain a PIN, the Authorization process will handle the rear administrative request identically to a

front administrative request. Except for the special transaction code of 98, the processing code in an administrative request is specific to the Device Handler and is not used by the Authorization process.

When the Authorization process receives an administrative request that was initiated with an administrative card, the Authorization process identifies the card type as administrative, verifies whether the transaction is an on-us transaction, and determines the appropriate response code depending on the balance flag and balancing window.

Only certain device handlers, such as the Diebold 10XX/478X and NCR 5XXX Device Handlers, send administrative requests to the Authorization process for approval. Administrative requests from the Device Control Terminal (DCT) and from other Device Handlers are not sent to the Authorization process for approval.

Tran Code	Transaction Type and Description
-- -- --	Terminal Balancing — A request from an ATM service person to clear the ATM totals and close out the processing day.
-- -- --	Mid-Point Cash Adjustment — A request from an ATM service person to change the ATM hopper totals to reflect an addition or removal of cash from a hopper.
-- -- --	Terminal Sub-Total — A request from an ATM service person for a listing of the ATM totals.

BASE24-atm Message Types

All transactions are carried out by means of messages exchanged between the originator and authorizer of the transaction. However, mobile top-up transactions can be routed to multiple secondary routing destinations based on the transaction subtype used in the Split Transaction Routing File (STRF). These messages are assigned numbers to identify their message type and function.

Message Functions

Messages can be grouped into the following functional categories.

Request — A request from a transaction originator to a transaction authorizer to authorize a financial transaction.

Response — A response from a transaction authorizer to a transaction originator approving or declining a financial transaction request.

Advice/Force Post — A message from a transaction originator to a transaction authorizer so the authorizer can update its files. Advices are also known as force post transactions because the authorizer must accept the transactions.

Reversal — A message from the transaction originator to the transaction authorizer indicating that any database impacts from an earlier transaction should be reversed because the transaction either could not be completed or was completed incorrectly.

Adjustment — A message from a transaction originator to a transaction authorizer requesting that the amount of a previously approved transaction be adjusted or corrected.

Notification/Selection — A message from a transaction authorizer to a transaction originator when actions are required by the customer to either acknowledge a surcharge/tax or to make a selection in order for the transaction to continue.

Message Types

Message types are used inside and outside of the BASE24-atm system. The following table identifies the message types used inside BASE24-atm along with the message type as they would be sent to, or received from, a host or a BIC co-network. The internal message types are defined in the table column identified by INT. External message types are provided for both ISO and ANSI communications with hosts and BIC co-networks. These external message types are identified in the table in the ISO and ANSI columns.

INT	ISO	ANSI	Description
0200	0200	0200	Financial Transaction Request — A request to authorize a transaction.
0205	0205	0205	Statement Print Transaction Request — A request to print statement information. Note that the BIC ISO Interface does not support this transaction. Passbook update transactions enter BASE24-atm as statement print transactions with the subtype APU0.
0210	0210	0210	Financial Transaction Request Response — An approval or denial response to an authorization request.
0215	0215	0215	Statement Print Transaction Response — A response carrying the statement information for printing. Note that the BIC ISO Interface does not support this transaction. Passbook update transactions enter BASE24-atm as statement print transactions with the subtype APU0.
0220	0220	0220	Financial Transaction Advice — A completion message generated as notification to a host or co-network that BASE24-atm has approved a transaction. When BASE24-atm receives a 0220 message, it considers it a “force post” transaction.
Not used	0221	0221	Financial Transaction Advice Repeat — A reissue of the completion message (0220) generated to notify a host or co-network that BASE24-atm has approved a transaction.

INT	ISO	ANSI	Description
Not used	0230	0230	Financial Transaction Advice Response — An acknowledgment of receipt generated by a host or co-network in response to a financial transaction advice (0220). This message type can also be sent by an Interchange Interface process.
Not used	0231	0231	Financial Transaction Advice Response Repeat — A reissue of the acknowledgment of receipt generated by a host or co-network in response to a financial transaction advice (0220).
0420	0420	0400	Reversal — A message indicating that a transaction was not completed as authorized.
Not used	0421	0401	Reversal Repeat — A reissue of the reversal message (0420) indicating that a transaction was not completed as authorized.
Not used	0430	0410	Reversal Response — An acknowledgment of receipt generated by a host or co-network in response to a reversal (0420). This type can also be sent by an Interchange Interface process.
Not used	0800	0800	Network Management Message Request — A request to perform echo-test, logon, and logoff.
Not used	0810	0810	Network Management Message Response — An acknowledgment of receipt generated in response to a network management message request (0800).
5400	5400	5400	Financial Transaction Correction — An adjustment to a previously completed transaction. An adjustment (5400) can be sent by an Interchange Interface process.
Not used	Not used	9220	Store-and-Forward Financial Transaction Advice — A financial transaction advice or reversal that is being sent to the host from the BASE24 Store-and-Forward File (SAF). When an advice cannot be sent to a host because communications to the host are down, the advice is stored in the SAF for sending when host communications are restored.

INT	ISO	ANSI	Description
Not used	Not used	9221	Store-and-Forward Financial Transaction Advice Repeat — A reissue of a store-and-forward financial transaction advice or reversal (9220). A store-and-forward financial transaction advice repeat is identical to the original advice it duplicates, except that it denotes to the receiver that it is a possible duplicate message. A store-and-forward financial transaction advice repeat is used when an acknowledgment was not received for a store-and-forward advice.
Not used	Not used	9230	Store-and-Forward Financial Transaction Advice Response — An acknowledgment of receipt generated by a host in response to a store-and-forward financial transaction advice (9220).
Not used	Not used	9231	Store-and-Forward Financial Transaction Advice Response Repeat — A reissue of an acknowledgment of receipt (9230) generated by a host in response to a store-and-forward financial transaction advice.
9901	Not used	Not used	Surcharge Notification — A message indicating an acquirer fee (surcharge) is involved in the transaction. Acknowledgment of the surcharge is required by the cardholder.
9906	Not used	Not used	Account Data Selection — A message containing account data for selection by a cardholder. BASE24 uses 9906 messages when selections are required for transactions involving multiple accounts.
9907	Not used	Not used	Card Review Account Selection — A response message containing all accounts associated with a card. This message is used by the Olivetti Device Handler process only.
9909	Not used	Not used	Mobile Top-Up Notification/Selection — A message containing mobile top-up data for acknowledgment/selection by a cardholder. BASE24 uses 9909 messages when an acknowledgment/selection is required for transactions involving mobile top-up activity.

INT	ISO	ANSI	Description
9980	Not used	Not used	Informational (Money in Drawer) — An informational message advising the BASE24-atm system that the cardholder left money in the drawer. This message is used by the Diebold, Fujitsu, and NCR Device Handler processes only.
9991	Not used	Not used	Log-Only Transaction — A message that must be logged to the Transaction Log File (TLF).

Transaction Logging

The BASE24-atm Authorization process logs all of the transactions it processes to the Transaction Log File (TLF). The Interchange Interface processes and BIC Interface processes log all the transactions they process to the appropriate Interchange Log Files (ILFs) if configured to do so.

Message Types Logged

The Authorization and the Interchange Interface processes convert messages from the internal message into the TLF or the ILF format before logging them to the TLF or the ILF. The following BASE24-atm message types are logged to the TLF or the ILF:

Type	Description
0210	Authorization response
0215	Statement print response—first only Passbook update transactions with the subtype APU0 are logged to the TLF.
0220	Advice—incoming only
0420	Reversal (refer to note below)
5400	Transaction adjustment (refer to note below)
9980	Informational (money in drawer)
9991	Log-only transaction Note: If the TLF file is unavailable, log the ATM settlement and balancing messages to the EMS log.

Note: Neither the 0420 or 5400 message is logged to the ILF. Both messages are used to update existing 0210 and 0220 records that have already been written to the ILF.

The following BASE24-atm message types are not logged to the TLF or ILF:

Type	Description
0200	Authorization request
0205	Statement print transaction request
0220	Advice—created locally (that is, completion)
9901	Surcharge notification
9906	Account data selection
9907	Card review account selection
9909	Mobile top-up notification/selection

Tokens Logged

The Token File (TKN) controls which tokens in the internal message are logged to the TLF or ILF. All tokens that can be logged to the TLF or ILF are displayed on TKN screen 2. As each message is logged to the TLF or ILF, tokens identified to be logged are arranged in the TLF or ILF record in the order in which they were added to the internal message.

2: BASE24-atm Transaction Path

A number of processes and processing entities form the BASE24-atm transaction path—that is, the series of BASE24-atm processes and processing entities through which a specific transaction passes in order to be authorized.

The actual path each transaction takes varies based on a number of factors, including the type of transaction being processed, the transaction originator, the transaction authorizer, and the processing set up by the BASE24 processor.

This section discusses the following aspects of the BASE24-atm transaction path:

- BASE24-atm processes that can be part of the transaction path.
- Process linkages (a description of how the various BASE24-atm processes and processing entities are linked together for the purpose of forwarding and receiving transaction messages).
- Service-based routing (a description of service-based routing within BASE24-atm).
- Timers (a description of the various timers used by BASE24-atm processes to ensure the timely completion of transactions).

BASE24-atm Processes

Along the transaction path, processing is carried out in stages by several BASE24-atm processes, each of which has its own responsibilities. The BASE24-atm processes that can be involved in transaction processing are listed below and described on the following pages:

- Device Handler processes
- Authorization processes
- Host Interface processes
- Interchange Interface processes
- BIC Interface processes
- Transaction Context Manager process

BASE24-atm processes all communicate with one another through the XPNET process.

Device Handler Processes

BASE24-atm Device Handler processes are responsible for communicating with ATMs (that is, devices) directly attached to BASE24-atm. Device Handler processes are designed specifically for each type of device; however, all Device Handler processes perform essentially the same functions which include the following:

- Recognizing native-mode messages sent from the ATM and translating them into the BASE24 internal message format.

Each type of ATM sends and receives messages in a device-specific format called native mode. BASE24-atm is not capable of using the various native-mode message formats internally. Therefore, the Device Handler process translates the messages between the ATM and BASE24-atm.

- Determining whether the transaction needs to be authorized by the BASE24-atm Authorization process or sent to a different authorization destination. For example, a bill payment transaction might be authorized by a bill payment service running under the same XPNET system.

Note: The APCF routing functionality is supported for the Diebold 10XX/478X and NCR 5XXX Device Handler processes only.

In addition, mobile top-up transactions are sent to the Transaction Context Manager process for split routing, unless enhanced mobile top-up routing support is being used.

- Setting and stopping transaction response timers as transactions pass in and out of the Device Handler process.

These response timers control the maximum length of time the Device Handler process waits for a response from the Authorization process before cancelling the transaction. For example, when the Device Handler process receives a transaction from a terminal, it sets a response timer and then passes the transaction on for processing. When processing completes the Device Handler process deletes the timer. If the Device Handler does not receive a response before the timer expires, the transaction is declined.

Also, if the transaction is declined because the Device Handler timer expired, and if the Device Handler process receives a late response from the Authorization process, the Device Handler process generates an appropriate reversal.

- Updating the accumulator fields and status flags in the BASE24-atm Terminal Data files record for each ATM.

When transactions pass through the Device Handler process, the Device Handler process updates the BASE24-atm Terminal Data files accumulation fields and other BASE24-atm Terminal Data files processing fields and flags depending on the type of transaction and processing required.

- Cutting over terminals for daily settlement using either a terminal balancing transaction or a Device Handler force cutover option.

If the Device Handler process does not cut over the terminal, the Settlement Initiator process cuts the terminal over. Refer to the ***BASE24-atm Settlement and Reporting Manual*** for more information about terminal cutover.

- Facilitating PIN and message authentication support between the terminal and BASE24-atm. For information on the PIN and message authentication support provided by BASE24-atm Device Handler processes, refer to the ***BASE24 Transaction Security Manual***.
- Downloading terminal configuration information to ATMs.

The information is downloaded from terminal-specific configuration and load files.

Authorization Processes

BASE24-atm Authorization processes are responsible for many transaction processing tasks. These tasks include the following:

- Authorizing or routing transactions to the appropriate interface process, which sends the transaction to the entity that is to authorize the transaction.
- Logging transactions to the Transaction Log File (TLF).
- Generating completion messages to notify hosts and co-networks regarding transactions approved by BASE24-atm.
- Calculating surcharge fees when applicable.

Host Interface Processes

BASE24-atm Host Interface processes are responsible for communicating with host systems. They become involved in processing when the host is the appropriate authorizer or is to receive completion messages for transactions authorized by BASE24-atm. They also become involved in processing when BASE24 is the authorizer, and the host is the transaction acquirer.

There are two types of Host Interface processes: one that communicates with hosts using an ANSI-based message format and one that communicates with hosts using an ISO-based message format. For information on the ANSI Host Interface process, refer to the ***BASE24 ANSI Host Interface Manual***. For information on the ISO Host Interface process, refer to the ***BASE24 ISO Host Interface Manual*** and ***BASE24 External Message Manual***.

The Host Interface process performs the following functions:

- Translating messages from the BASE24-atm internal message format to the ISO or ANSI external format recognizable to a host.
- Providing a store-and-forward function, whereby if a host is unavailable, certain types of messages bound for the host can be stored and sent to the authorizing entity at a later time.
- Setting and stopping transaction timers that monitor the time it takes for messages to return to the Host Interface process from another entity.
- Providing for PIN, MAC, and key management between BASE24-atm and the host. For information on the transaction security functions performed by Host Interface processes, refer to the ***BASE24 Transaction Security Manual***.

Interchange Interface Processes

BASE24-atm Interchange Interface processes are responsible for communicating with interchanges. They become involved in processing when the interchange is the appropriate authorizer or when BASE24 is the authorizer, and the interchange is the transaction acquirer.

Each interchange has its own Interchange Interface process because each interchange has its own message format and message handling requirements. Interchange Interface processes are typically responsible for the following:

- Translating messages between the BASE24-atm internal message format and the external message format recognizable to the interchange.
- Determining whether outbound transactions are allowed to be sent from BASE24 to the interchange.
- Determining whether inbound transactions are allowed from the interchange, and if so, whether they need to be authorized by BASE24-atm or sent to a different authorization destination. For example a bill payment transaction might need to be authorized by a bill payment service.

Note: Inbound transaction allowed checking and APCF routing functionality is supported for the following Interchange Interface processes only:

- ❑ Banknet Interchange Interface process
 - ❑ BIC ISO Interface process
 - ❑ MDS/MDSM Interchange Interface process
 - ❑ PLUS ISO Interchange Interface process
 - ❑ VisaNet Interchange interface process
- Logging transactions to the Interchange Log File (ILF).
- Providing a store-and-forward function, whereby, if the interchange is unavailable, certain types of messages bound for the interchange can be stored and sent to the interchange at a later time.
- Setting and stopping transaction timers that monitor the time it takes for messages to return to the Interchange Interface process from another entity.
- Providing for PIN, MAC, and key management between BASE24-atm and the interchange. For information on the transaction security functions performed by Interchange Interface processes, refer to the ***BASE24 Transaction Security Manual***.

Transactions entering BASE24-atm from an interchange must first pass through the Interchange Interface process. The Interchange Interface process translates the message from the interchange external message to the ATM Standard Internal Message (STM) before routing the message to the other BASE24 processes.

Transactions bound for an interchange that originated at a host or at an ATM connected to BASE24-atm must first pass through the appropriate Interchange Interface process. The Interchange Interface process translates the message into the interchange external message format before sending it to the interchange.

BIC Interface Process

BASE24 Interchange (BIC) Interface processes are responsible for communicating with other BASE24 systems and co-networks. They become involved in processing when a transaction must be forwarded to another BASE24 system or co-network for authorization or a transaction is being sent from another BASE24 system or co-network for authorization.

There are two types of BIC Interface processes: one that communicates with co-networks using an ANSI-based message format and one that communicates with co-networks using an ISO-based message format. For information on the BIC ANSI Interface process, refer to the ***BASE24 BIC ANSI Interface Manual***. For information on the BIC ISO Interface process, refer to the ***BASE24 BIC ISO Interface Manual*** and ***BASE24 BIC ISO Standards Manual***.

The BIC Interface process is responsible for the following:

- Translating messages between the BASE24-atm internal message format and the external message format recognizable to the BIC co-network.
- Determining whether outbound transactions are allowed to be sent from BASE24 to the interchange.
- Determining whether inbound transactions are allowed from the interchange, and if so, whether they need to be authorized by BASE24-atm or sent to a different authorization destination. For example a bill payment transaction might need to be authorized by a bill payment service.

Note: Inbound transaction allowed checking and APCF routing functionality is supported for the BIC ISO Interface process only.

- Logging transactions to the Interchange Log File (ILF).

- Providing a store-and-forward function, whereby, if the BIC co-network is unavailable, certain types of messages bound for the co-network can be stored and sent to the co-network at a later time.
- Setting and stopping transaction timers that monitor the time it takes for messages to return to the BIC Interface process from another entity.
- Providing for PIN, MAC, and key management between BASE24-atm and a co-network. For information on the transaction security functions performed by BIC Interface processes, refer to the ***BASE24 Transaction Security Manual***.

Transactions entering BASE24-atm from a BIC co-network must first pass through the BIC Interface process. The BIC Interface process translates the message from the co-network external message to the BASE24-atm Standard Internal Message (STM) before routing the message to the other BASE24 processes.

Transactions bound for a BIC co-network that originated at a host or at an ATM connected to BASE24-atm must first pass through the appropriate BIC Interface process. The BIC Interface process translates the message into the co-network external message format before sending it to the co-network.

Transaction Context Manager Process

The BASE24-atm Transaction Context Manager processes are responsible for splitting the transaction routing to different destinations. They become involved in processing for mobile top-up transactions. However, in an enhanced mobile top-up routing environment, where the interchange network splits and routes the mobile top-up transactions, the Transaction Context Manager (TCM) process must not be utilized.

Transaction Context Manager processes sequentially route mobile top-up transactions to a top-up authorizer and a funds authorizer for authorization. The Transaction Context Manager process is responsible for the following:

- Determining the routing hierarchy for the mobile top-up transaction.
- Sending and receiving messages to/from the primary and secondary authorizers.
- Logging the mobile top-up transaction response to the Transaction Log File (TLF).
- Sending the final mobile top-up transaction response to the acquiring Device Handler process.

Process Linkages

Since there are multiple processes involved in transaction processing, transactions must be sent between them for processing. The links between these processes are configurable. Each link in the transaction path is defined below.

Inbound Messages from ATMs, Hosts, and Interchanges

All transactions inbound to BASE24-atm are sent by stations. All ATMs, host links, and interchange links are defined as stations. A station is a physical point in the BASE24 system that is designated to send or receive messages.

The XPNET process controls the interaction of BASE24 with its stations. The XPNET process receives inbound transactions and routes them to the appropriate processes. The XPNET process determines the process to which it is to send a message using the destination table.

Each station definition includes a DESTINATION attribute, which identifies the process to which all messages from the station are to be sent.

- In the case of ATMs, the DESTINATION attribute should specify a Device Handler process.
- In the case of host stations, the DESTINATION attribute should specify a Host Interface process.
- In the case of interchange stations, the DESTINATION attribute should specify an Interchange Interface process.

Using the DESTINATION attribute, the XPNET process forwards messages to the designated process.

Once the designated process receives the message from the station, that process verifies that it should be receiving messages from the station. This verification is discussed below.

Inbound Messages to a Device Handler Process

When a Device Handler process receives a message from an ATM, it verifies that it is supposed to receive messages from that particular ATM. It does this based on the value in the DH PROCESS NAME field on screen 1 of the BASE24-atm Terminal Data files (ATD) record for that ATM. If its name is not in the DH PROCESS NAME field, the Device Handler process logs an event message indicating the situation and drops the incoming message.

Inbound Messages to a Host Interface Process

When a Host Interface process receives a message from a host station, it verifies that it is supposed to receive messages from that particular station. It does this based on an internal table of stations that it builds at initialization from the STATION fields on screen 2 of its records in the Host Configuration File (HCF). If a message is received from a station that is not in the table, the Host Interface process logs a message indicating the situation and writes the incoming message to the log.

Inbound Messages to an Interchange Interface Process

When an Interchange Interface process receives a message from an interchange station, it verifies that it is supposed to receive messages from that particular station. It does this based on an internal table of stations that it builds at initialization from the STATION fields on screen 1 of its record in the Interchange Configuration File (ICF) or Enhanced Interchange Configuration File (ICFE). If a message is received from a station that is not in the table, the Interchange Interface process logs a message indicating the situation and writes the incoming message to the log (this action can vary by Interchange Interface process).

Device Handler Process to Authorization Process

The Device Handler process determines the Authorization process to which it is to send a non mobile top-up transaction request using the value in the AUTH PROCESS field on screen 1 of the BASE24-atm Terminal Data files (ATD). This field can indicate either a specific Authorization process or a service. Service-based routing is described later in this section.

Device Handler Process to the Transaction Context Manager Process

For mobile top-up transactions, the Device Handler process forwards the transaction to the Transaction Context Manager process specified in the SPLIT-TXN-RTG-DEST param, which can route the transaction to multiple authorizing entities. If the param is not present, the transaction is declined. In an enhanced mobile top-up routing environment, the SPLIT-TXN-RTG-DEST param will be sent to the BASE24-atm Authorization process.

After the Device Handler process retrieves the destination name from the SPLIT-TXN-RTG-DEST param in the Logical Network Configuration File (LCONF), it adds the Split Transaction Routing token with the name of the Authorization process in the FUNDS-AUTH-DEST field.

Device Handler Process to Customer-Specific Application Process

The Diebold 10XX/478X and NCR 5XXX Device Handler processes can route a transaction directly to a customer-specific application process for authorization instead of to the Authorization process based on settings in the Acquirer Processing Code File (APCF). A value in the AUTHORIZATION DESTINATION field on APCF screen 2 specifies whether a transaction is sent to a application process. If this field is blank, the Device Handler process routes the transaction to the Authorization process specified in the AUTH PROCESS field on BASE24-atm Terminal Data files ATD screen 1.

Transaction Context Manager Process to Authorization Destinations

The Transaction Context Manager process updates the Split Transaction Routing token using information from the Split Transaction Routing File (STRF) and sends mobile top-up transactions to the primary authorizer based on settings in the STRF. The Transaction Context Manager process then forwards any subsequent request messages to the secondary authorizer.

Customer-Specific Application Process to Device Handler Process

When a customer-specific application process returns a transaction response to the Device Handler process based on APCF routing, the Diebold 10XX/478X and NCR 5XXX Device Handler processes determine whether the response is logged to the Transaction Log File (TLF). If the LOG AUTH DEST RESPONSE field on APCF screen 2 is set to a value of Y, the Device Handler process sends the transaction response as a log only transaction to the Authorization process specified in the AUTH PROCESS field on ATD screen 1 for logging to the TLF.

Host Interface Process to Authorization Process

The Host Interface process, handling requests from acquirer networks, determines the Authorization process to which it is to send inbound transaction requests from the host using the value in the AUTH PROCESS field on screen 5 of the Host Configuration File (HCF).

Authorization Process to Host Interface Process

The Authorization process determines whether a transaction must be sent to a host using IDF routing. This routing method is described in section 4 under the topic “Routing Transactions with Recognized Prefixes.” If a transaction is to be sent to a host, the IDF routing steps determine the Host Interface process to which the transaction must be sent for forwarding to the host.

Interchange Interface Process to Authorization Process

The Interchange Interface process sends incoming transactions to the process named in the AUTH PROCESS field specified on screen 8 of the Interchange Configuration File (ICF) or Enhanced Interchange Configuration File (ICFE).

Interchange Interface Process to Customer-Specific Application Process

When one of the following Interchange Interface processes receives a transaction, it determines whether the transaction is allowed and whether the transaction is to be authorized by the Authorization process or by a customer-specific application process:

- Banknet Interchange Interface process
- BIC ISO Interface process
- MDS/MDSM Interchange Interface process
- PLUS ISO Interchange Interface process
- VisaNet Interchange Interface process

If the transaction is allowed and a value is present in the AUTHORIZATION DESTINATION field on Acquirer Processing Code File (APCF) screen 2, the transaction is sent to a customer-specific application process instead of to the Authorization process specified in the ICFE.

Customer-Specific Application Process to Interchange Interface Process

When a customer-specific application process returns a transaction response to an Interchange Interface process, the process determines whether the response is logged to the Transaction Log File (TLF). If the LOG AUTH DEST RESPONSE field on APCF screen 2 is set to a value of Y for the transaction, the Interchange Interface process sends the transaction response as a log-only transaction to the Authorization process specified in the AUTH PROCESS field on ICFE screen 8 for logging to the TLF.

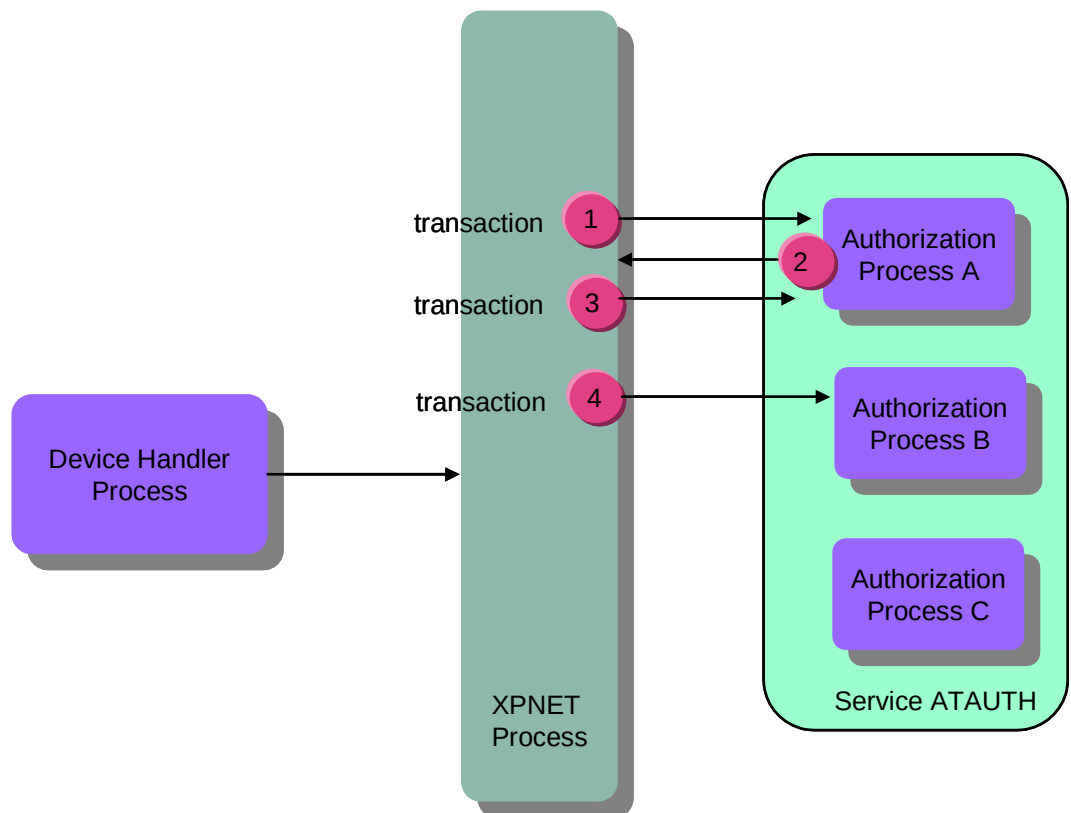
Authorization Process to Interchange Interface Process

The Authorization process determines whether a transaction must be sent to an interchange using SPROUTE routing. This routing method is described in section 4 under the topic “Routing Transactions with Unrecognized Prefixes.” If a transaction is to be sent to an interchange, the SPROUTE routing steps determine the Interchange Interface process to which the transaction must be sent for forwarding to the interchange.

Service-Based Routing

When multiple Authorization processes are in use, BASE24-atm allows Device Handler, Host Interface, and Interchange Interface processes to send transaction requests to a service instead of directly to a specific Authorization process. This option is designed to avoid the processing inefficiencies that could develop when a specific process is assigned to receive transaction requests. Without service-based routing, the requesting processes assigned to one server process must wait for that server process to complete its transactions even if other server processes are available. With service-based routing, transactions can be distributed efficiently among multiple server processes as they become available.

The following illustration shows how service-based routing functions with the Device Handler process and Authorization processes.



In this example, the XPNET process can send transactions to three Authorization processes that have been associated to the service ATAUTH in the XPNET configuration. The Device Handler process sends all BASE24-atm messages it

receives to the ATAUTH destination. The XPNET process sends each transaction it receives to the first available Authorization process for the ATAUTH service, as follows:

1. The XPNET process sends the first transaction to Authorization process A, the first provider available for the ATAUTH service.
2. Authorization process A completes its processing of the first transaction and passes the transaction through the XPNET process to another process for further processing.
3. The XPNET process sends a second transaction to Authorization process A. Because the first transaction processing is complete, process A is still the first available provider for the ATAUTH service.
4. The XPNET process sends a third transaction to Authorization process B because Authorization process A is still processing the second transaction when the XPNET process receives the third transaction for the ATAUTH service.

The XPNET process uses a service-based routing algorithm to determine which provider receives each message sent to a service. Messages sent to a service are distributed on a rotary basis. A provider is considered to be available if it is started and is not currently busy. Providers configured with a startup logic of DEMAND that are stopped are started only if they are needed to handle the processing load. Once started, they remain in rotation. For a detailed description of the service-based algorithm used by the XPNET process, refer to the ***NET24-XPNET Network Control Operations Guide***.

Processes are configured to a service using the SERVICE attribute. For detailed information on configuring the SERVICE attribute, refer to the ***NET24-XPNET Network Control Command Reference Manual***.

Transaction Timers

Timers enable processes to take action in situations where responses have not been received as expected. BASE24-atm uses timers for logical network processing, device handler processing, host processing, and interchange processing. These various timers are set in the Logical Network Configuration File (LCONF), the Host Configuration File (HCF), the Interchange Configuration File (ICF), and the Enhanced Interchange Configuration File (ICFE).

General Timers

The following general timer is used by multiple BASE24 processes.

Security Device Timer Limit — The SECURE-DEV-TIM-LMT param in the LCONF specifies the maximum amount of time a BASE24 process (Device Handler, Authorization, Host Interface, or Interchange Interface process) that uses a security module waits for a response from the security module before the message is considered timed out.

Device Handler Process Timers

The following timers are used in transaction processing by the BASE24-atm Device Handler processes.

Device Handler Timer Limit — The ATM-DH-AUTH-TIMER param in the LCONF specifies the maximum amount of time that the BASE24-atm Device Handler process waits for an authorization response from the BASE24-atm Authorization process before declining the request.

Device Handler Queue Timer Limit — The ATM-DH-STALE-RQST-TIMEOUT param in the LCONF specifies the number of seconds an incoming transaction request can remain in the Device Handler process queue before being discarded by the Device Handler process.

Host Interface Process Timers

The following timers are used by the Host Interface processes. Not all are directly related to transaction processing; some are related to network management.

Network Management Timer Limit — Specifies the amount of time a BASE24 Host Interface process waits for a response after sending a network management message (0800 series) to a data processing center (DPC) station (that is, a host). If a response message (0810) is not received within the time specified, the Host Interface process marks the station down. This timer limit is set in the NETWORK MANAGEMENT field on HCF screen 1.

Extended Network Timer Limit — Specifies the amount of time a Host Interface process waits, once a station has been marked down, before sending an echo-test message. This timer limit is set in the EXTENDED NETWORK field on HCF screen 1.

Store-and-Forward Timer Limit — Specifies the amount of time a Host Interface process waits for an acknowledgment after sending a store-and-forward message to the DPC. This timer limit is set in the STORE AND FORWARD field on HCF screen 1.

Wait-for-Traffic Timer Limit — Specifies the amount of time a Host Interface process waits for traffic on the line to a station after sending a network management message. Once this amount of time has passed and no traffic has occurred on the line, the Host Interface process sends another network management message. This timer limit is set in the WAIT FOR TRAFFIC field on HCF screen 1.

Outbound Timer Limit — Specifies the amount of time a Host Interface process waits for a response message after sending a BASE24-atm transaction message to the host. This timer limit is set in the OUTBOUND LIMIT field on HCF screen 5. The EXTRN-RQST-TIMEOUT param in the LCONF determines the actions taken when a transaction request times out.

If this param is set to a value of 0 and a request times out, the Host Interface process fails the transaction and returns it to the Authorization process for alternate or stand-in routing. The Host Interface process generates a reversal and sends it to the host if it receives a late response.

If the EXTRN-RQST-TIMEOUT param is set to a value of 1 and a request times out, the Host Interface process fails the transaction and returns it to the Authorization process for alternate or stand-in routing and also generates a reversal that it sends to the host. If the Host Interface process then receives a late response from the host, it drops the message.

Inbound Timer Limit — Specifies the amount of time a Host Interface process waits for a response message after sending a financial transaction message or a balance inquiry request to a BASE24-atm Authorization process. This timer limit is set in the INBOUND LIMIT field on HCF screen 5.

Completion Acknowledgment Message Timer Limit — Specifies the amount of time a Host Interface process waits for an acknowledgment message after sending a BASE24-atm advice, reversal, or adjustment message to the DPC. This timer limit is set in the COMPLETION ACK field on HCF screen 5.

Queue Subtract Timer Limit — This timer limit is set in the QUEUE SUBTRACT field on HCF screen 5. It specifies the amount of time the XPNET process is to subtract from the value in the OUTBOUND TIMER LIMIT field to compute the maximum time the XPNET process can wait before sending the BASE24-atm message to the host. If the XPNET process cannot send the message to the host within this time, it returns the message to the Host Interface process as stale. This option circumvents processing when it becomes apparent that sending a message will not give a host time to respond before the message times out.

Interchange Interface Process Timers

The following timers are used by the Interchange Interface processes. Not all are directly related to transaction processing; some are related to network management.

Network Management Timer Limit — Specifies the amount of time an Interchange Interface process waits for a response after sending a network management message (sign on, sign off, and so on) to an interchange. This timer limit is set in the NETWORK MANAGEMENT field on ICF or ICFE screen 3.

Extended Network Timer Limit — Specifies the amount of time an Interchange Interface process waits before attempting to retry a network management message during periods when communications to the interchange are down. This timer limit is set in the EXTENDED NETWORK field on screen 3 of the ICF or ICFE.

Wait-for-Traffic Timer Limit — Specifies the amount of time an Interchange Interface process waits for traffic on the line from the interchange after sending a network management message. Once this amount of time has passed and no traffic has occurred on the line, the Interchange Interface sends another network management message. This timer limit is set in the WAIT FOR TRAFFIC field on screen 3 of the ICF or ICFE.

Performance Period Timer Limit — Specifies the length of the time that the performance of the interchange is monitored. This timer is set in the PERFORMANCE PERIOD field on screen 3 of the ICF or ICFE.

Store-and-Forward Timer Limit — Specifies the amount of time an Interchange Interface process waits for an acknowledgment after sending a BASE24-atm store-and-forward message to the interchange. This timer limit is set in the STORE AND FORWARD field on screen 8 of the ICF or ICFE.

Outbound Timer Limit — Specifies the amount of time an Interchange Interface process waits for a response message after sending a BASE24-atm financial transaction message to the interchange. This timer limit is set in the OUTBOUND field on screen 8 of the ICF or ICFE.

Inbound Timer Limit — Specifies the amount of time an Interchange Interface process waits for a response message after sending a financial transaction message from the interchange to the Authorization process. This timer limit is set in the INBOUND field on screen 8 of the ICF or ICFE.

Completion Timer Limit — Specifies the amount of time an Interchange Interface process waits for a completion message from either the interchange or BASE24 after passing a response message from one entity to another. This timer is set in the COMPLETION field on screen 8 of the ICF or ICFE.

Completion Acknowledgment Message Timer Limit — Specifies the amount of time an Interchange Interface process waits for an acknowledgment message after sending an advice, reversal, or adjustment message to the interchange. This timer limit is set in the COMPLETION ACK field on screen 8 of the ICF or ICFE.

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3: BASE24-atm Transaction Processing Controls

This section describes the basic BASE24-atm transaction processing controls that are available at the terminal, institution, host, and interchange levels. It provides information on the following BASE24-atm topics and features:

- Authorization levels and methods
- Transaction screening
- Sharing
- Transaction security
- Cardholder limits and usage
- Cardholder usage accumulation
- Completion messages to a host
- Adjustment processing
- Store-and-forward processing
- Second card processing
- Expiration date checking
- Multiple account selection
- Statement print processing
- Passbook update processing
- Optional receipts processing
- Surcharging
- Token handling
- Deposit hold notification processing
- Exchange rate notification processing

- Add-on module processing—including discussions of self-service banking, Non-Currency Dispense, Mobile Top-Up, Multiple Currency, EMV, Automated Key Distribution System, Dynamic Key Management, and Dynamic Currency Conversion support

Note: This section should be used in conjunction with section 4, which provides additional information on BASE24-atm transaction routing and authorization. For information on additional configuration issues, along with screen layouts and valid values for file settings presented in this section, refer to the following manuals: ***BASE24 Base Files Maintenance Manual, BASE24-atm Files Maintenance Manual, BASE24 ANSI Host Interface Manual, BASE24 ISO Host Interface Manual, and BASE24 Logical Network Configuration File Manual.***

Authorization Levels and Methods

The Authorization process authorizes transactions based on the authorization levels and methods specified by each institution. These levels and methods are described below, along with other authorization functions that the Authorization process performs.

Authorization Levels

The authorization level defines the level of participation that the institution has in authorizing transactions. BASE24 can perform all of the authorizations for an institution, or BASE24 can be set up to forward transactions to a back-end host processor for full or final approval. BASE24 can also authorize transactions in place of the host in cases where communications with the host are down.

The authorization level is specified in the AUTH LVL fields on IDF screen 9. The possible authorization levels are online authorization (level 1), offline authorization (level 2), and online/offline authorization (level 3).

A single institution can process transactions using multiple authorization levels, depending on the number of hosts it uses and the types of transactions it processes. An institution can control processing with multiple authorization levels, depending on the number of hosts it uses and the types of transactions it processes. An institution can control processing with multiple authorization levels using the IDF routing table on screen 9 of the IDF. For more information on the IDF routing table, refer to section 4.

Online Authorization

Online authorization is also referred to as level 1 authorization. At this level, the host performs all transaction authorizations. If the host is offline and, therefore, unable to authorize transactions, the Authorization process declines all transaction requests. Typically, BASE24 does not maintain cardholder account files at this level of authorization. Online authorization can be used only with the online host processing authorization method.

Note: Although the Authorization process cannot authorize transactions when an institution is using the online authorization level, BASE24-atm can optionally screen, or prescreen, transactions before sending them to the host. If a transaction does not pass the prescreening checks, the Authorization process can decline the transaction. Prescreening is described later in this section.

Offline Authorization

Offline authorization is also referred to as level 2 authorization. At this level, all authorizations are performed by BASE24; no authorization requests are forwarded to the host.

With the offline authorization level, BASE24 must maintain cardholder accounts in its database. These account files, as well as transaction log files, are then periodically extracted to tape or disk and used to update host files. These accounts are also refreshed periodically using information from host files.

Online/Offline Authorization

Online/offline authorization is also referred to as level 3 authorization. At this level, BASE24 optionally screens transactions based on prescreening settings in the IDF and forwards those transactions that pass the screening checks to the host for final approval. The Authorization process declines transactions that do not pass the screening checks.

The Authorization process also stands in for the host and authorizes transactions when communications to the host are down. The transactions BASE24 authorizes are then stored and forwarded to the host when communications with the host are restored.

With the online/offline authorization level, BASE24 must maintain cardholder account files in its database. These account files, as well as transaction log files, are periodically extracted to tape or disk and used to update host files. These accounts are also refreshed periodically using information from host files.

Authorization Methods

The Authorization method chosen by an institution establishes the file configuration and authorization checks that the BASE24-atm Authorization process performs during the authorization process.

Institutions specify the authorization method or methods in the AUTH TYPE field on IDF screen 9. Possible authorization methods are online host processing, the Negative Authorization with Usage Accumulation method, the Positive Authorization method, the Positive Balance Authorization method, and the Negative Authorization without Usage Accumulation method.

A single institution can be set up to process using multiple authorization methods, depending on the files it wishes BASE24 to maintain and the types of checks it wishes BASE24 to perform. This is controlled by the routing table on IDF screen 9. An explanation of the IDF routing table is provided in section 4.

Online Host Processing

Online host processing must be used in conjunction with the online authorization level (level 1). Online authorization does not support the use of BASE24 authorization files; therefore, the host must perform all authorizations online. BASE24-atm can, however, screen transactions for an institution using the online host processing authorization method. This screening involves the Institution Definition File (IDF) and the Card Prefix File (CPF).

Negative Authorization With Usage Accumulation Method

The Negative Authorization with Usage Accumulation method uses the Negative Card File (NEG) and the Usage Accumulation File (UAF). The Authorization process authorizes transactions using negative identification. If a record is found in the NEG, the Authorization process takes the action specified in the record.

If no record is found in the NEG, the Authorization process assumes the account is good and, if the transaction is a cash disbursement, checks the usage accumulation totals in the UAF against the institution transaction limits in the CPF. Based on those totals and limits, the Authorization process approves or declines the transaction request and updates the UAF totals appropriately.

Positive Authorization Method

The Positive Authorization method uses the Cardholder Authorization File (CAF). The Authorization process authorizes transactions using positive identification.

For the transaction to be valid, a CAF record must exist for the card that initiated the transaction and the status of the account in the CAF must be good. Also, if the transaction is a cash disbursement, the amount previously disbursed plus any holds plus the new transaction amount cannot exceed the cardholder's transaction limits. Transaction limits and usage limits can be set for the card prefix in the CPF or for the individual cardholder in the CAF.

Based on these totals and limits, the Authorization process approves or declines the transaction request and updates CAF totals appropriately.

Positive Balance Authorization Method

The Positive Balance Authorization method uses the Cardholder Authorization File (CAF) and the Positive Balance File (PBF). The Authorization process authorizes transactions using positive identification.

For the transaction to be valid, a CAF record must exist for the card that initiated the transaction and the status of the account in the CAF must be good. Also, if the transaction is a cash disbursement, the amount previously disbursed plus any holds plus the new transaction amount cannot exceed the cardholder's transaction limits. A PBF record must also exist for the account, and the amount of the transaction cannot exceed the balance in the account.

Transaction limits can be set for the card prefix in the CPF or for the individual cardholder in the CAF.

Based on these totals and limits, the Authorization process approves or declines the transaction request and updates CAF and PBF totals appropriately.

Negative Authorization Without Usage Accumulation Method

The Negative Authorization without Usage Accumulation method uses the Negative Card File (NEG). BASE24 does not maintain card usage totals for this method. The Authorization process authorizes transactions using negative identification. If a record is found in the NEG, the Authorization process takes the action specified in the NEG record.

If no record is found in the NEG, the Authorization process assumes the account is good and, if the transaction is a cash disbursement, checks the card usage limits and checks the amount of the transaction against the institution transaction limits in the CPF. Based on these limits, the Authorization process approves or declines the transaction request.

Transaction Screening

Transaction screening, also referred to as prescreening or preauthorization, is a term used frequently in situations where BASE24 shares authorization responsibilities with the host. In such situations, the Authorization process can make preliminary standard checks on the transaction and decline those that do not meet the preliminary requirements. This prevents unnecessary processing by the host or co-network.

BASE24-atm can perform prescreening only when configured to do so by the institution and when the institution is processing transactions using the online or online/offline authorization level. When the institution is using the offline authorization level, these checks are included automatically as a part of the authorization processing that BASE24-atm performs.

Transaction screening checks include those described below.

Card Status

The card status indicates whether the card that initiated the transaction is valid, has some type of restriction on it, or is a special-use card. Depending on the status, BASE24-atm denies the transaction or sends it on to the host for authorization. Institutions indicate whether they want BASE24-atm to prescreen the card status in the CARD STATUS field on IDF screen 2.

The Authorization process uses card status information set in the Cardholder Authorization File (CAF) when the institution is using the Positive Authorization method or the Positive Balance Authorization method. A CAF record can have one or two cards associated with it. Refer to “Second Card Processing” later in this section for more information about configuring card status checks.

When the institution is using the Negative Authorization with Usage Accumulation method or the Negative Authorization without Usage Accumulation method, the Authorization process uses the value set in the REASON ON FILE field on Negative Card File (NEG) screen 1 to determine the card status.

Note: The card status is not checked when the institution is processing transactions using online host processing. The CAF and NEG do not exist for the institution when using the online authorization level.

Expiration Date

A series of checks can be performed to determine whether the card has expired. In the EXP DATE field on IDF screen 2, institutions indicate whether the BASE24-atm product should prescreen the card's expiration date. If the EXP DATE field indicates that the expiration date should be checked during prescreening, the Authorization process then uses values set in the CPF to determine if the expiration date should be checked for this card prefix. Several configuration options are available for expiration date checking. Refer to "Second Card Processing" and "Expiration Date Checking" later in this section for a description of these options.

PIN Check

Institutions can configure BASE24-atm to determine whether a cardholder entered the correct PIN when initiating a transaction. This is configured in the PIN field of IDF screen 2 and the CHECK IF HOST ONLINE PIN field of CPF screen 2.

The PIN field in the IDF establishes PIN prescreening at the institution level, meaning that all transactions coming into that host should or should not be prescreened for valid PINs. However, an institution may want to prescreen all PINs except for those belonging to cards with specific prefixes or prescreen only those cards with certain prefixes. In that case, the CHECK IF HOST ONLINE PIN field in the CPF overrides the PIN field in the IDF.

Withdrawal Limits

Institutions can configure BASE24-atm to determine during prescreening whether the requested transaction will cause the cardholder to exceed the withdrawal limits the institution established for either the card prefix or the cardholder. Withdrawal limits are prescreened only for withdrawal, cash check, and deposit with cash back transactions. Institutions can configure this prescreening option in the LIMITS field on screen 2 of the IDF.

If the LIMITS field indicates that BASE24-atm should prescreen the withdrawal limits and the transaction is a withdrawal, cash check, or deposit with cash back transaction, the Authorization process checks the ACTIVITY LIMITS fields on CPF screen 6 or CAF screen 10. The values in the ACTIVITY LIMITS fields on CAF screen 10 override those on CPF screen 6. This enables institutions to establish limits for entire card prefixes while allowing the flexibility to establish

different limits for specific cardholders. For example, if a cardholder is consistently delinquent, the institution can establish lower limits than those established for the prefix of the card the cardholder carries.

PIN Verification and Card Verification

The Authorization process performs PIN and card verification when the BASE24-atm ISO Host Interface process sets the RQST.MULT-ACCT field in the STM to a value of 3 in host-acquired 0200 request messages. A value of 3 in the RQST.MULT-ACCT field in the STM forces the Authorization process to prescreen the transaction even though accounts are already present in the request message. (Normally, the Authorization process does not perform transaction prescreening when accounts are already present in the message.)

Other Prescreening Checks

In addition to the prescreening checks described above, the Authorization process also performs the following:

- Verifies the card if the CHECK IF HOST ONLINE CV field on CPF screen 2 is set to a value of Y for the card prefix.
- Determines whether the transaction is allowed for the institution.
- Verifies the account types for transfers and split deposits.
- Fills in the account type for fast cash transactions with the value from the FAST CASH ACCOUNT TYPE field on IDF screen 13.
- Fills in the account type for balance inquiry transactions with the value from the FAST CASH ACCOUNT TYPE field on IDF screen 13 when no account type has been specified. The default account type is 00.
- For two-sided transactions, the process validates the *to* account if the account is maintained in the BASE24 database.
- Determines whether the transaction is a log-only transaction. If so, the process performs as indicated by the HOST LOG-ONLY OPTION field on IDF screen 13.
- For PIN change and EMV PIN unblock transactions, the process checks the HOST PIN CHANGE OPTION field on IDF screen 13. If the value of this field is 0, the process attempts to process the transaction. If the PIN offset is not maintained in the BASE24 database, the transaction is denied.

Sharing

The relationship between the financial institution that owns a cardholder's account and the financial institution handling the cardholder's transaction impacts the way BASE24-atm processes the transaction.

Sharing is an option that allows cardholders belonging to a BASE24-atm institution to perform transactions at ATMs that belong to other BASE24-atm institutions or institutions serviced by interchanges. Sharing also allows an interchange cardholder to perform transactions at a terminal that belongs to a BASE24-atm institution.

BASE24-atm supports sharing controls that enable an institution to determine the extent to which it participates in a sharing arrangement with other institutions and interchanges. Institutions can define the extent of sharing according to transaction type at the terminal, cardholder, and interchange level. Institutions can also define sharing criteria according to county, state, or country locations and according to online and offline modes.

On-us and Not-on-us Transactions

A transaction performed at an ATM owned by the institution that issued the ATM card is an on-us transaction. A transaction performed at an ATM owned by an institution that did not issue the card is a not-on-us transaction.

Sharing Groups

A card-issuing institution, a terminal-owning institution, and an interchange can belong to up to twenty-four sharing groups. When card-issuing and terminal-owning institutions and interchanges are part of a common sharing group, it indicates they have established an agreement to accept and/or process transactions for each other. BASE24-atm identifies the sharing arrangements that exist by checking the value in the SHARING GROUP field on IDF screen 2; by the GROUP field on ATD screen 1; and by the SHARING GROUPS field on Interchange Configuration File (ICF) or Enhanced Interchange Configuration File (ICFE) screen 8.

The Authorization process searches the sharing groups fields whenever the terminal owner and card issuer are not the same to determine whether they belong to any of the same sharing groups. If a match is found, the terminal owner and the card issuer have a sharing arrangement.

Allowed Transactions

Institutions can define which transactions they allow at their terminals and at the interchange. The ability to restrict the transactions allowed at the terminal, cardholder account and interchange levels enables the institution to better control its transaction processing.

Transactions Allowed at the Terminal, Diebold 10XX/478X and NCR 5XXX Devices

An institution with the following Device Handler processes uses enhanced transactions allowed processing with information from the Acquirer Processing Code File (APCF) extended memory table to define the transactions allowed at its ATMs.

- Diebold 10XX/478X Device Handler process
- NCR 5XXX Device Handler process

BASE24 terminal owners have the option to define the ISO processing codes allowed by the individual terminal or by groups of terminals. A 16-character value called the *acquirer transaction profile* can be assigned to individual terminals or used to group terminals together for defining allowed transactions. The acquirer transaction profile is defined on ATM BASE24-atm Terminal Data files (ATD) screen 2 and Institution Definition File (IDF) screen 9. Acquirer transaction profiles configured at the ATD level override those set at the IDF level. Refer to the **BASE24 Base Files Maintenance Manual** for a description of the APCF and IDF screens. Refer to the **BASE24-atm Files Maintenance Manual** for a description of the ATD screens.

The Device Handler process uses the acquirer transaction profile specified in the ATD to read the APCF extended memory table. If no acquirer transaction profile is specified in the ATD, the Device Handler process uses the profile specified in the IDF to read the APCF extended memory table. Finally, if an acquirer transaction profile is not defined in the ATD or IDF, a value of all wildcard characters (asterisks) is used to read the APCF extended memory table.

Once the acquirer transaction profile value used to read the APCF extended memory table has been determined, the Device Handler process attempts to read the APCF extended memory table using the acquirer transaction profile value, the message category code from the transaction, and the ISO processing code for the

transaction. The Device Handler process calls a utility to map the internal BASE24 transaction code for the transaction to the corresponding ISO processing code before reading the APCF extended memory table.

Using the acquirer transaction profile, message category, and ISO processing code, the Device Handler process steps through the records in the APCF extended memory table looking for a match. If an exact match is not found, certain other alternatives are allowed. Below is the hierarchy used for selection:

Preference	Acquirer Transaction Profile	Message Category	Processing Code
1	Exact Match	Exact Match	Exact Match
2	Exact Match	*	Exact Match
2	*****	Exact Match	Exact Match
4	*****	*	Exact Match

If none of the searches return a match, the transaction is declined.

If a match is found, the Device Handler process fills in the RQST.TERM-TRAN-ALLOWED field of the STM with the value from the TRANSACTION ALLOWED field on APCF screen 2 indicating if and how the transaction is allowed. The Authorization process checks the value in this field if the transaction request is from a not-on-us cardholder only. It does not check this field if the transaction request is from an on-us cardholder.

Note: The transactions allowed at the terminal can be performed only if the terminal is capable of performing them. Some terminals have limited capabilities, and, although the institution might allow a transaction and BASE24-atm can process that transaction, the terminal itself may not have the capability of handling that particular transaction.

Transactions Allowed at the Terminal, Other Devices

A terminal-owning institution can specify which transaction types are allowed for not-on-us cardholders at its ATMs using the TRANSACTION CLASS and SPECIAL TRANSACTION CLASS fields on ATD screens 2 and 3 (for terminals other than the Diebold 10XX/478X and NCR 5XXX). Refer to the **BASE24-atm Files Maintenance Manual** for a description of the screen and a list of possible allowed transactions.

Note: The transactions allowed at the terminal can be performed only if the terminal is capable of performing them. Some terminals have limited capabilities and, although the institution may allow a transaction and BASE24-atm can process that transaction, the terminal itself may not have the capability of handling that particular transaction.

Transactions Allowed by the Institution

A card-issuing institution can specify which transaction types are allowed for its cardholders at its ATMs and at other institutions' ATMs using information from the Issuer Processing Code File (IPCF).

The institution can define by cardholder, prefix, institution, groups of prefixes, or groups of institutions which transactions are allowed. A 16-character value called the *issuer transaction profile* can be used to group prefixes or institutions together for transactions allowed processing purposes. This profile, plus the ISO processing code (the ISO transaction code, *from* account type and *to* account type) and message category (the second byte of the message type), are used to access information configured in the IPCF extended memory table. The issuer transaction profile can be specified in the ISSUER TXN PROFILE field on Cardholder Authorization File (CAF) screen 8, Card Prefix File (CPF) screen 4, or Institution Definition File (IDF) screen 9 for transactions authorized by the BASE24-atm Authorization process. Issuer transaction profiles configured at the CAF level override those set at the CPF and IDF levels. Issuer transaction profiles configured at the CPF level override those set at the IDF level. Refer to the **BASE24 Base Files Maintenance Manual** for descriptions of the CAF, CPF, IDF, and IPCF screens.

The Authorization process searches the IPCF extended memory table using the issuer transaction profile to determine whether the transaction is allowed. If a profile is specified in the CAF, it is used in the search. If a profile is not specified in the CAF, or the Authorization method does not use the CAF, the profile from the

CPF is used in the search. If a profile is not specified in the CPF, the Authorization process uses the profile specified in the IDF. After determining the profile to use, the Authorization process searches the table using the following hierarchy:

Preference	Issuer Transaction Profile	Message Category	Processing Code
1	Exact Match	Exact Match	Exact Match
2	Exact Match	*	Exact Match
3	*****	Exact Match	Exact Match
4	*****	*	Exact Match

If none of the searches return a match, the Authorization process declines the transaction.

If a match is found, the Authorization process uses the value from the RQST. TERM-TRAN-ALLOWED field of the STM to determine whether the transaction is allowed at the terminal. If the transaction is not allowed at the terminal, the Authorization process declines the transaction. If the transaction is allowed at the terminal, the Authorization process checks the value of the ON-US OR SWITCH OUTBOUND (ATM/POS) or NOT-ON-US (ATM) field on IPCF screen 2 to determine whether the transaction is allowed.

The Authorization process generates a completion message and sends it to the host, depending on the value in the COMPLETION REQUIRED TO HOST field on IPCF screen 2. If completions are not required, this field is set to 0. Otherwise, this field is set to a nonzero value. Refer to Appendix A for more information about the values used for the COMPL-REQ field.

Transactions Allowed Between BASE24 and an Interchange

An institution can define which transactions it sends to a particular interchange for authorization and—for some interchanges—which transactions it will receive from an interchange for authorization.

Outbound Transactions

Institutions define the transactions that can be sent to a particular interchange on screen 7 of the Interchange Configuration File (ICF) or by specifying an issuer transaction profile value in the ISSUER TXN PROFILE field on screen 8 of the Enhanced Interchange Configuration File (ICFE).

The following Interchange Interface processes must use the ICFE if they are running in the current release (i.e., 6.0):

- Banknet Interchange Interface process
- BIC ISO Interface process
- MDS/MDSM Interchange Interface process
- PLUS ISO Interchange Interface process
- VisaNet Interchange Interface process

Other Interface Interchange processes or any of the above process running in a previous release use the ICF.

For interchanges using the ICFE, the value of the issuer transaction profile is used to read the Issuer Processing Code File (IPCF) extended memory table, which defines all the transactions allowed for the profile. If an issuer transaction profile is not specified for the interchange, the Interchange Interface process uses a value of all asterisks (wildcard characters) to read the IPCF extended memory table.

If BASE24 attempts to send a transaction to the interchange, and the transaction is not allowed, the interchange interface process denies the transaction and the transaction is never sent to the interchange.

Issuer Transaction Profile Processing. Once the issuer transaction profile value used to read the IPCF extended memory table has been determined, the Interchange Interface process attempts to read the IPCF extended memory table using the issuer transaction profile value, the message category code from the transaction, and the ISO processing code for the transaction. The Interchange Interface process calls a utility to map the internal BASE24 transaction code for the transaction to the corresponding ISO processing code before reading the IPCF extended memory table.

Using the issuer transaction profile, message category, and ISO processing code, the Interchange Interface process steps through the records in the IPCF extended memory table looking for a match. If an exact match is not found, certain other alternatives are allowed. Below is the hierarchy used for selection.

Preference	Issuer Transaction Profile	Message Category	ISO Processing Code
1	Exact match	Exact match	Exact match
2	Exact match	*	Exact match
3	*****	Exact match	Exact match
4	*****	*	Exact match

Once a match is found, the record is read to determine whether the transaction is allowed. If the ON-US OR SWITCH OUTBOUND (ATM/POS) field on IPCF screen 2 is set to a nonzero value, the transaction is allowed.

ICF Processing. The NOT ON US field on ICF screen 7 is used to determine whether the BASE24 cardholder is allowed to perform a particular transaction through an interchange and under what circumstances. The TRANSACTION ALLOWED TO SWITCH field on ICF screen 7 is used to determine whether BASE24 allows the transactions to be sent to an interchange. Refer to the ***BASE24 Base Files Maintenance Manual*** for a description of the screen and a list of possible allowed transactions.

Inbound Transactions

For Interchange Interface processes configured in the ICFE, the value in the ACQUIRER TXN PROFILE field on ICFE screen 8 defines the transactions allowed from the interchange. The value of the acquirer transaction profile is used to read the Acquirer Processing Code File (APCF) extended memory table, which defines all the transactions allowed for the profile and the accounts on which the transactions can be performed. If an acquirer transaction profile is not specified for the interchange the Interchange Interface process uses a value of all asterisks (wildcard characters) to read the APCF extended memory table. If BASE24 receives a transaction from an interchange that has not been defined as allowed, the Interchange Interface process denies the transaction.

Acquirer Transaction Profile Processing. Once the acquirer transaction profile value used to read the APCF extended memory table has been determined, the Interchange Interface process attempts to read the APCF extended memory table using the acquirer transaction profile value, the message category code from the transaction, and the ISO processing code for the transaction. The Interchange Interface process calls a utility to map the internal BASE24 transaction code for the transaction to the corresponding ISO processing code before reading the APCF extended memory table.

Using the acquirer transaction profile, message category, and ISO processing code, the Interchange Interface process steps through the records in the APCF extended memory table looking for a match. If an exact match is not found, certain other alternatives are allowed. Below is the hierarchy used for selection.

Preference	Issuer Transaction Profile	Message Category	ISO Processing Code
1	Exact match	Exact match	Exact match
2	Exact match	*	Exact match
3	*****	Exact match	Exact match
4	*****	*	Exact match

Once a match is found, the record is read to determine whether the transaction is allowed.

If none of the searches return a match, the transaction is declined.

If a match is found, the Interchange Interface process reads the APCF record to determine whether the transaction is allowed and fills in the RQST.TERM-TRAN-ALLOWED field of the STM with a value indicating if and how the transaction is allowed. The Authorization process checks the value in this field if the transaction request is from a not-on-us cardholder.

Transaction Security

BASE24-atm offers several forms of transaction security, including PIN and card verification, PIN encryption, and message authentication. The information presented here summarizes the transaction security features. Refer to the **BASE24 Transaction Security Manual** for detailed information. Transaction security parameters are specified for the institution on Institution Definition File (IDF) screen 2, for the card prefix on Card Prefix File (CPF) screen 2, and for the ATM on BASE24-atm Terminal Data files (ATD) screen 9. Transaction security parameters for each financial institution and DPC are also specified in the Key Authorization File (KEYA); transaction security parameters for each Host Interface process and Interchange Interface process are specified in the Key File (KEYF) or Key 6 File (KEY6).

PIN Verification

PIN verification can be performed for transactions authorized by BASE24-atm or as a prescreening step for transactions authorized by a host. PIN verification can be performed locally at the ATM, by the BASE24-atm Authorization process, or by an authorizing host.

When the PIN is to be verified locally at the ATM, the PIN offset must be carried on Track 1, Track 2 or Track 3 of the card. This local PIN verification method requires specific field settings in the IDF, CPF, CAF, and BASE24-atm Terminal Data files.

When the PIN is to be verified by the BASE24-atm Authorization process or by the authorizing host, PIN verification is controlled at the institution or the card prefix level. PIN verification controlled at the institution level requires a record in the Key Authorization File (KEYA) for each institution performing PIN verification. PIN verification controlled at the card prefix level requires a record in the KEYA that corresponds with the KEYA group identified in the PIN VERIFICATION KEYA GROUP field in the CPF.

PIN verification information is maintained on screens 2 through 5 of the KEYA in addition to the IDF, CPF, and ATD screens mentioned earlier.

BASE24 can use software or security devices to perform PIN verification. Hardware security options include interfaces to various models of Atalla and Thales e-Security (Racal) security hardware. Software security options include several methods of PIN encryption and PIN verification.

Card Verification

While BASE24 can perform PIN verification based on parameters established at the institution or card prefix level, card verification parameters must be established at the card prefix level. The Track data on the card must be captured by a magnetic stripe reader. BASE24 uses security devices to perform card verification. Card verification information is maintained on screen 6 of the KEYA in addition to the CPF screen mentioned earlier.

PIN Encryption

PINs can be sent between BASE24 and a device, host, or interchange encrypted or in the clear. PINs are encrypted and decrypted within BASE24 by the BASE24 processes that send transactions to and receive transactions from processing entities outside the system. These processes include Host Interface processes, Interchange Interface processes, and Device Handler processes.

When BASE24 receives a transaction, the PIN may need to be encrypted, decrypted, or translated from one encrypted form to another, depending on how the BASE24 system and the external processor are set up to handle their PINs. The same is true for transactions that BASE24 is sending to another entity.

PIN encryption data for messages sent between the ATM and a Device Handler process is configured on ATD screen 9. PINs can also be sent from the Host Interface process or the Interchange Interface process encrypted or in the clear. PIN encryption data for messages sent between the Host Interface process and a host or between the Interchange Interface process and an interchange is configured on screens 1 through 3 of the KEYF or screens 1 through 4 of the KEY6.

Message Authentication

Message authentication is a method used to monitor the integrity of data flowing between EFT devices, such as POS devices and ATMs, and the transaction authorizer. A message can change as the result of tampering or because of communications difficulties. Message authentication identifies whether or not the message has changed. BASE24-atm supports the generation and verification of message authentication code (MAC) values in Thales e-Security and Atalla hardware and in software.

BASE24-atm supports message authentication for messages exchanged between BASE24 and the ISO Host Interface process or the BIC ISO Interface process as well as between BASE24 and the ATM.

Message authentication is supported by field settings on ATD screen 9, on Key File (KEYF) screens 1 through 3 or Key 6 (KEY6) screens 1 through 4, and External Message File (EMF) screen 2.

Note: Only certain devices support the generation and verification of MAC values. These devices are the Diebold 910, Diebold 912, Diebold 10XX/478X, NCR 5XXX, and IBM 4730 devices. Also, not all interchanges and their respective Interchange Interface processes support message authentication.

Dynamic Key Management

BASE24 processes use communication keys (also known as transaction keys or working keys) to protect the PINs and message authentication codes (MACs) moving through the BASE24 network. Communication keys should be changed regularly to maintain proper security. Monitoring and changing these keys is known as key management, and can be performed manually or automatically. Automated key management is known as dynamic key management.

Communication Between BASE24 and the Host

Dynamic key management support is provided between BASE24-atm and the ISO Host Interface process and the Interchange Interface processes. The value of the ATM-DH-CONS-MAC-ERR-LMT param in the Logical Network Configuration File (LCONF) controls the number of consecutive MAC errors that can occur before the Device Handler process sends a shutdown message to the ATM. Key management information is maintained on KEYF screens 1 through 3 or KEY6 screens 1 through 4.

Dynamic key management requires the use of security modules to perform PIN encryption, PIN verification, and message authentication. It is not supported when these functions are performed in software.

Note: Not all interchanges and their respective Interchange Interface processes support dynamic key management.

Communication Between BASE24 and ATMs

The NCR 5XXX Device Handler process or the Diebold 10XX/478X Device Handler process use the ATM-DH-CONS-MAC-ERR-LMT param to monitor the number of consecutive MAC errors that can occur before the Device Handler process sends a shutdown message to the ATM. Operators can then manually download new transaction keys to the ATM and reopen the ATM using the BASE24 Device Control Terminal (DCT).

Institutions using the NCR 5XXX, Diebold 10XX/478X, or Triton Device Handler processes can purchase the Dynamic Key Management add-on feature to automatically load the keys and reopen the ATM. Using this feature, the ATM shuts down when a key must be changed. Keys are changed when any of the following events occurs:

- The number of PIN verify transactions exceeds a predetermined limit
- The number of MAC verify and MAC generate transactions exceeds a predetermined limit
- The number of PIN verify failures exceeds a predetermined limit
- The number of MAC verify and MAC generate failures exceeds a predetermined limit
- The number of MAC consecutive failures exceeds a predetermined limit
- A specified time interval has passed since the last MAC or PIN change

After the ATM shuts down, the Device Handler process loads the new transaction key and MAC key, and restarts the ATM. Refer to the device-specific device support manual for more information about the Dynamic Key Management add-on feature.

Cardholder Limits and Usage

Institutions can establish variable amount and usage limits for cards, individual cardholders, and specific transactions. These limits are configured in the Card Prefix File (CPF), the Cardholder Authorization File (CAF), and the Positive Balance File (PBF). BASE24 and BASE24-atm use these limits when authorizing transaction requests. Transactions that violate the limits as set by the institution are declined.

In addition, accumulator fields in the CAF and the Usage Accumulation File (UAF) are used to track transaction activity performed by BASE24-atm during a single usage accumulation period. A usage accumulation period is a set amount of time during which transaction amounts and counts are accumulated. At the end of the usage accumulation period, the amounts and counts are cleared. The accumulator fields in the CAF and UAF are compared to corresponding limits established in the CAF or CPF to determine if a transaction should be approved or declined.

The Negative Authorization without Usage Accumulation method and host-only authorization do not use accumulators. CPF limits are checked on a per-transaction basis, so they should be set accordingly.

Card and Cardholder Limits

Institutions can establish card and cardholder limits as described in the following paragraphs.

Card Prefix File

The Card Prefix File (CPF) defines each card prefix that can be processed within the BASE24 logical network. One record exists in the CPF for each card prefix to be processed. Each CPF record contains processing parameters and authorization limits for that specific card prefix.

Screen 1 — Using the CARD PROCESSING INFORMATION fields on CPF screen 1, institutions can limit the transaction activity allowed by BASE24 for cards with a specific prefix during a single usage accumulation period. These card usage limits apply to both BASE24-atm and BASE24-pos transactions combined.

When the institution is using the Negative Authorization without Usage Accumulation method, the authorization level is online, and the LIMITS field on IDF screen 2 indicates to do so, the values in these fields are checked with every transaction. When the institution is using the host-only authorization method and the LIMITS field on IDF screen 2 indicates to do so, the values in these fields are checked with every transaction.

When the institution is using the Positive Authorization method or the Positive Balance Authorization method, the limits established on CPF screen 1 are used if the value in the TOTAL AGGR field on Cardholder Authorization File (CAF) screen 1 is set to zero; otherwise, the limits set in the CAF are used.

The transactions controlled by these limits include cash disbursements against credit and noncredit accounts.

Screen 4 — Using the ACTIVITY LIMITS fields on CPF screen 4, institutions can limit the transaction activity allowed by BASE24-atm for cards with a specific prefix during a single usage accumulation period. These card usage limits apply only to BASE24-atm cash disbursement transactions against credit and noncredit accounts.

When the institution is using the Negative Authorization without Usage Accumulation method, the authorization level is online, and the LIMITS field on IDF screen 2 indicates to do so, the values in these fields are checked with every transaction. When the institution is using the host-only authorization method and the LIMITS field on IDF screen 2 indicates to do so, the values in these fields are checked with every transaction.

Using the TIMES USED PER PERIOD LIMIT field on CPF screen 4, institutions can specify the maximum number of times a card with the specified prefix can be used for cash disbursements by means of BASE24-atm during a single usage accumulation period.

The limits established on CPF screen 4 are used if the value in the TOTAL AGGR field on CAF screen 1 is set to zero; otherwise, the limits set in the CAF are used. If the BASE24-atm limits set on CPF screen 4 are all set to zero and the TOTAL AGGR field on CAF screen 1 is set to zero, then BASE24-atm uses only the base limits established on CPF screen 1.

If you have the BASE24-atm self service banking (SSB) Enhanced Check Processing Application, use the fields on CPF screen 14 to set limits and percentages for checks and registered checks.

Screen 14 — Using the DEPOSIT CREDIT PERCENT—CHECK and MAXIMUM CREDIT PER CHECK DEPOSIT—CHECK to represent the non-registered check deposits and the DEPOSIT CREDIT PERCENT—CSF CHECK and MAXIMUM CREDIT PER CHECK DEPOSIT—CSF CHECK fields to represent the registered check deposits, institutions can set deposit credits and percentages on CPF screen 14. These credits and percentages apply only to checks and registered checks deposited.

If any of these fields on CPF screen 14 is set to a value of 0, the deposit credit percentages and amounts from CPF screen 4 are used.

Refer to the **BASE24 Base Files Maintenance Manual** for more information about the fields on screens 1 and 4 of the CPF. Refer to the device-specific SSB manuals for more information about CPF screen 14.

Cardholder Authorization File

The Cardholder Authorization File (CAF) contains one record for each cardholder whose card-issuing institution uses the Positive Authorization method or Positive Balance Authorization method. Each CAF record contains authorization parameters and usage accumulation information for a specific cardholder.

Screen 1 — Using the ACTIVITY LIMITS fields on CAF screen 1, institutions can establish usage accumulation limits for the cardholder. The limits defined in these fields are for both BASE24-atm and BASE24-pos transactions combined. The BASE24-atm transactions affected by these usage limits are cash disbursements made against credit and noncredit accounts.

Screen 8 — Using the ACTIVITY LIMITS fields on CAF screen 8, institutions can establish usage accumulation limits for the cardholder. The limits defined in these fields are for BASE24-atm transactions only.

Using the TIMES USED PER PERIOD LIMIT field on CAF screen 8, institutions can specify the maximum number of times a card can be used for cash disbursements by means of BASE24-atm during a single usage accumulation period.

If the BASE24-atm limits on CAF screen 8 are all set to zero, BASE24-atm uses only the base limits established on CAF screen 1.

Refer to the **BASE24 Base Files Maintenance Manual** for more information about the fields on screens 1 and 8 of the CAF.

Deposit Credit Limits

Institutions using the Positive Balance Authorization method can establish deposit credit limits by card prefix or by cardholder for a single usage accumulation period. These limits are configured in the Card Prefix File (CPF) or the Cardholder Authorization File (CAF).

In addition, accumulator fields in the CAF are used to track the number and amount of deposit credits a cardholder has accumulated during the current usage accumulation period.

Institutions can establish deposit credit limits as described in the following paragraphs.

Card Prefix File

The Card Prefix File (CPF) defines each card prefix that can be processed within the BASE24 logical network. One record exists in the CPF for each card prefix to be processed. Each CPF record contains processing parameters and authorization limits for that specific card prefix.

Screen 4 — Using the DEPOSIT CREDIT INFORMATION fields on CPF screen 4, institutions using the Positive Balance Authorization method can specify limits for deposit credits allowed by BASE24 for cards with a specific prefix during a single accumulation period.

The DEPOSIT CREDIT PERCENT field specifies the percentage of each deposit that will be added to a cardholder's available funds. The maximum number and amount of deposit credits and the maximum credit per deposit are controlled by the MAXIMUM NUMBER OF DEPOSIT CREDITS, the MAXIMUM DEPOSIT CREDIT AMOUNT, and the MAXIMUM CREDIT PER DEPOSIT fields on CPF screen 4. The amount established in the MAXIMUM DEPOSIT CREDIT AMOUNT can be overridden by the MAXIMUM DEPOSIT CREDIT AMOUNT field on CAF screen 8.

If the BASE24-atm limits on CPF screen 4 are all set to zero, BASE24-atm uses the base limits set on CPF screen 1.

Cardholder Authorization File

The Cardholder Authorization File (CAF) contains one record for each cardholder whose card-issuing institution uses the Positive Authorization method or Positive Balance Authorization method. Each CAF record contains authorization parameters and usage accumulation information for a specific cardholder.

Screen 8 — Using the MAXIMUM DEPOSIT CREDIT AMT field on CAF screen 8, institutions can establish deposit credit limits for the cardholder. A nonzero value in this field overrides the amount in the MAXIMUM DEPOSIT CREDIT AMOUNT field on CPF screen 4.

CAF screen 8 also displays the number and amount of deposit credits a cardholder has performed during the current usage accumulation period for which deposit credits have been granted.

If the BASE24-atm limits on CAF screen 8 are all set to zero, BASE24-atm uses the base limits set on CAF screen 1.

Overdraft Limits

Institutions using the Positive Balance Authorization method can specify an overdraft limit for a particular account on Positive Balance File (PBF) screen 1.

Positive Balance File

Institutions can establish an overdraft limit for a particular account using the OVERDRAFT LIMT field on screen 1 of the PBF. This amount is added to the amount in the AVAILABLE BALANCE/AVAILABLE CREDIT field in the PBF to obtain the maximum transaction amount available to an accountholder. This field makes it possible for an accountholder to perform transactions even when the transaction amount exceeds the balance in the AVAILABLE BALANCE/AVAILABLE CREDIT field.

Cardholder Usage Accumulation

BASE24 keeps card usage and amount totals in the Cardholder Authorization File (CAF) and Usage Accumulation File (UAF) if the Positive Balance Authorization method or the Negative Authorization with Usage Accumulation method is used. Institutions can specify whether they clear usage accumulation totals in the Cardholder Authorization File (CAF) and the Usage Accumulation File (UAF) every day of the week, every day but weekends, every day but weekends and holidays, or after a specified number of days.

Note: Card usage totals are updated only for transactions involving a cash disbursement. This includes Non-Currency Dispense transactions and load value transactions.

Institutions establish their usage accumulation periods and how they clear their usage accumulation totals on Institution Definition File (IDF) screen 4.

Establishing Usage Accumulation Periods

A usage accumulation period is a set amount of time during which each cardholder's transaction statistics, such as the number of times a card is used, are tracked and compared to established transaction limits.

For example, an institution can set a usage accumulation period of one week and restrict the number of times a card can be used for cash disbursements during that period to 10. Each time a cardholder uses his or her card to obtain cash, a counter is increased by one. If the counter reaches 10 within the usage accumulation period, the transaction may be declined. At the end of the usage accumulation period, BASE24 clears the counter and all other usage accumulation totals.

An institution defines the length of its usage accumulation period in the PERIOD LENGTH field and the first day of the usage accumulation period in the BEGINNING DATE field on IDF screen 4. The setting in the PERIOD LENGTH field is impacted by the setting in the WORK DAY CODE field on the same screen. Refer to the ***BASE24 Base Files Maintenance Manual*** for a detailed description of how to set these fields.

Clearing Usage Accumulation Totals

Institutions can choose to clear their usage accumulation totals every day of the week, every day but weekends and holidays, every day but Saturdays and holidays, every day but Sundays and holidays, or after a specified number of days regardless of weekends and holidays. An institution defines how often usage accumulation totals are cleared by setting the WORK DAY CODE field on IDF screen 4. If totals are not cleared on holidays, the institution must also specify the holiday dates in the HOLIDAYS field on the same screen.

Refer to the ***BASE24 Base Files Maintenance Manual*** for more information about these fields.

Generating Completions to a Host

Completions (0220 messages) notify the host of authorizations made by BASE24-atm. The institution controls completion generation at the institution level. The Authorization process generates completions based on the COMPLETION REQUIRED TO HOST field on Issuer Processing Code File (IPCF) screen 2. This field enables BASE24-atm customers to configure completions required at the processing code level.

The Authorization process generates a completion message and sends it to the host, depending on the value in the COMPLETION REQUIRED TO HOST field on IPCF screen 2. Refer to Appendix A for more information about the values used for the COMPL-REQ field.

If the COMPLETION REQUIRED TO HOST field indicates that completions are not required (value of 0), the Authorization process generates completions only under the following two circumstances:

- When the authorization level is online/offline, the host is unavailable for authorizing a 0200 message, and the Authorization process stands in to approve the transaction, the Authorization process generates a 0220 message to notify the host of the approval, regardless of the setting in the COMPLETION REQUIRED TO HOST field.
- When a two-sided transaction is split between different DPCs (that is, one DPC handles the *from*-side of the transaction and another DPC handles the *to*-side of the transaction), the Authorization process uses 0220 messages as notification to the *to*-side DPC that a transaction has been approved. As in the first case, completions are generated only if a transaction is approved and regardless of the setting in the COMPLETION REQUIRED TO HOST field.

Since completions are controlled at the institution level, all DPCs processing for an institution either receive completions or do not. Completions cannot be enabled or disabled by specific DPC.

Adjustment Processing

Adjustment are transactions which can be sent from certain interchanges to adjust previously-posted transactions. Institutions must specify how BASE24 is to handle these transactions.

Adjustment transactions, received as 5400 messages by the Authorization process, can be processed manually by the institution, in which case BASE24 denies any adjustment transactions it receives. Or, an institution can specify that adjustments be processed only by BASE24 and not be sent to a host. Or, an institution can specify that adjustments be processed by BASE24 and sent to a back-end host.

An institution controls adjustment processing with the HOST ADJ. PROCESSING field on Institution Definition File (IDF) screen 3.

Store-and-Forward Processing

When communications to a host system or an interchange are down for maintenance or other processing needs, BASE24-atm can continue to process transactions, storing them in a data file to be transmitted when communications are reestablished. This function is known as store-and-forward processing. The data file used in this processing is the Store-and-Forward File (SAF).

Interchange Store-and-Forward Processing

When communications are reestablished with an interchange, BASE24-atm can forward the stored transactions to the interchange, interspersing the stored transactions with live transactions. Some interchanges also provide the option of forwarding the stored transactions to the interchange consecutively. This option, however, is specific to particular interchanges.

The maximum number of times the Interchange Interface process should attempt to transmit a transaction from an interchange SAF to the interchange before the transaction is logged to the error log and deleted from the SAF is specified in the MAX SAF RETRY field on Interchange Configuration File (ICF) or Enhanced Interchange Configuration File (ICFE) screen 3.

For more information on store-and-forward processing to the interchange, refer to the appropriate interchange documentation or the ***BASE24 Base Files Maintenance Manual***.

Host Store-and-Forward Processing

When communications are reestablished with a host, BASE24-atm can forward the stored transactions to the host by interspersing the stored transactions with live transactions or by sending the stored transactions consecutively. The method used to send stored transactions to the host is specified in the STORE AND FORWARD PROCESSING FLAGS field on Host Configuration File (HCF) screen 1.

HCF screen 1 also contains other fields used for store-and-forward processing with the host. The MAXIMUM OUTSTANDING SAFS field is used to specify the maximum number of store-and-forward messages that can be outstanding to a multithreaded host at one time. The STORE AND FORWARD field is used to specify the number of seconds the Host Interface process waits for an acknowledgment after transmitting a stored message to the DPC. The MAX SAF RETRY field is used to specify the number of times the BASE24-atm Host

Interface process should attempt to send the same store-and-forward message before logging it to the error log and deleting it from the Store-and-Forward File (SAF).

For detailed information on store-and-forward processing to the host system, refer to the ***BASE24 ISO Host Interface Manual*** or the ***BASE24 ANSI Host Interface Manual***.

Second Card Processing

Institutions can issue two cards with the same primary account number (PAN) and member number. The only way to differentiate between the two cards is by using the expiration date information. An example is when an institution reissues a card shortly before the existing card expires.

The BASE24-atm product maintains card status and expiration date information in the Cardholder Authorization File (CAF), which means the institution must be using the Positive Authorization method or the Positive Balance Authorization method to perform second card processing. The PAN and the member number form the primary key to the CAF, so a single CAF record has both cards associated with it. Therefore, a single CAF record must be able to hold two card expiration dates. Similarly, as the card status could be different for the two cards, e.g. one could be *stolen*, the other could be *open*, a single CAF record must be able to hold two card status values.

When the CAF record has one card, the Authorization process uses the card status set in the CARD STATUS field on CAF screen 1 and the expiration date set in the EXPIRATION DATE field on CAF screen 2. When the CAF record has two cards, the Authorization process uses the information on CAF screens 1 and 2 for the first card and information set in the CARD STATUS and EXPIRATION DATE fields on CAF screen 7 for the second card.

If a CAF record has two cards associated with it, the Authorization process first checks the expiration date received with the transaction. If the expiration date matches the value in the EXPIRATION DATE field on CAF screen 2, the Authorization process uses that expiration date and the value set in the CARD STATUS field on CAF screen 1. If the expiration date matches the value in the EXPIRATION DATE field on CAF screen 7, the Authorization process uses the card status set in the CARD STATUS field on CAF screen 7. If no expiration date is received in the transaction the Authorization process uses the EXPIRATION DATE field on CAF screen 2 and the value set in the CARD STATUS field on CAF screen 1.

Expiration Date Checking

If the expiration date is checked, either as a prescreening check or during transaction authorization, the following fields control the processing performed:

- The EXP CHECK DISP field on IDF screen 2 controls the response code (return card or retain card) the Authorization process returns to the merchant when a card fails the expiration date check.
- The EXP CHECK TYPE field on CPF screen 1 controls whether to check the expiration date and tells where the expiration date is located. Options are do not check, use Track 1 or Track 2, and use the CAF.

The Authorization process can use the CAF option only if the institution is using the Positive Authorization method or the Positive Balance Authorization method.

- The TRACK 1 SETTINGS EXP DATE and TRACK 2 SETTINGS EXP DATE fields on CPF screen 2 provide the location of the expiration date on card tracks.
- The EXPIRATION DATE PROCESSING FLAG field on CPF screen 3 indicates whether an expiration date is required in transaction requests.
- The EXPIRATION DATE COMPARISON field on CPF screen 3 indicates whether the expiration date on Track 1 or Track 2 of the card is compared against the expiration date in the CAF.

The EXP CHECK TYPE field on CPF screen 1 must be set to Track 1 or Track 2 if expiration date comparison processing is performed.

The CAF can contain card expiration dates in the EXPIRATION DATE fields on screens 2 and 7. Expiration date comparison processing must be performed for the Authorization process to consider the second card expiration date.

- The CARD VALIDITY PERIOD field on CPF screen 3 indicates the maximum number of years into the future a card expiration date is valid.

Multiple Account Selection

Plastic cards serve as a mechanism for accessing funds from an ATM. A card can be associated with a single account or multiple accounts. When a single card is associated with multiple accounts, the BASE24-atm system can use multiple account selection processing to enable the cardholder to select the correct account or accounts for a financial transaction. This is configured at the terminal level using the MULTIPLE ACCT SELECT field on BASE24-atm Terminal Data Files (ATD) screen 1. If multiple account selection is not configured at a terminal (the MULTIPLE ACCT SELECT field is set to a value of N) and an account qualifier is not specified, the primary account for the card is used.

BASE24-atm customers can use multiple account selection by account number (also known as the open account relationship) or multiple account selection by account qualifier.

Multiple Account Selection by Account Number

Multiple account selection by account number enables the cardholder to select accounts from a list of account numbers presented on an ATM screen.

The Authorization process provides a list accounts of a given type to the Device Handler process in a 9906 account selection message. The Diebold 10XX/478X, Fujitsu 7000, and NCR 5XXX Device Handler processes can process up to 16 different accounts per card with no maximum per account type. Other Device Handler processes can process up to five accounts per account type.

The Device Handler process formats and sends an account selection menu in a native-mode message to the ATM. The message contains the account numbers of the first five accounts for the account type. The cardholder can choose one of the accounts shown or, if applicable, request to view more accounts. If the cardholder wants to view more accounts, the Device Handler process sets the MORE field of the Multiple Account Selection token, adds the token to the 0200 message and sends it to the Authorization process. The Authorization process responds with more accounts of the given account type.

You can configure multiple account selection by account number (open account relationship) by setting the MULTIPLE ACCT SELECT field on ATD screen 1 to a value of Y.

Multiple Account Selection by Account Qualifier

Multiple account selection by account qualifier enables the cardholder to select an account or accounts for a transaction using a predefined set of account qualifiers. The selection of accounts by qualifier takes place using a series of menus before the transaction is sent to the BASE24-atm system for authorization. This processing is supported by the following Device Handler processes:

- Diebold 910
- Diebold 10XX/478X
- Fujitsu 7000
- IBM 3624
- NCR 5XXX

An account qualifier is a predefined code designating an account. Account qualifiers are assigned as follows:

Account	Account Qualifier			
	Checking	Savings	Credit	Other
Primary	101	201	301	401
Secondary	102	202	302	402
Tertiary	103	203	303	403

Account qualifiers 401, 402, and 403 are used when a cardholder has more than three accounts of one type.

In a transaction that uses multiple account selection by account qualifier, the cardholder selects an account by its qualifier. The ATM sends the qualifier to the Device Handler process in a native-mode message. The Device Handler process maps the qualifier to the Account Qualifier token. The account qualifier token is recognized by the Authorization, ISO Host Interface, BIC ISO Interchange Interface, MAC ISO Interchange Interface and NYCE ISO Interchange Interface processes. The account type in the message is set to a value of 60 for *other*.

If the BASE24-atm Authorization process authorizes the transaction, it performs the Cardholder Authorization File (CAF) lookup using the account type from the STM and the third digit of the account qualifier passed in the Account Qualifier token. If the Authorization process does not authorize the transaction, it passes the token to the Host Interface or Interchange Interface process.

The ISO Host Interface and BIC ISO Interchange Interface processes map the Account Qualifier token data to the BASE24-atm Additional Data data element (S-126).

You can configure multiple account selection by account qualifier by setting the MULTIPLE ACCT SELECT field on ATD screen 1 to a value of N. You also must download the multiple account selection load group (group 200) from the configuration file for the terminal.

Statement Print Processing

Statement print transaction messages that route through the BASE24-atm system to or from the host do so in a series of transaction messages. These messages consist of request messages to the host and response messages from the host, or vice versa. Because of the amount of information carried in the statement print transaction message, several request and response messages can be sent between BASE24-atm and the host.

A statement print transaction can be sent to the host even if the ATM routing table on IDF screen 9 indicates that the transaction should be authorized offline. This option is set in the STATEMENT PRINT ONLINE field on IDF screen 13. Refer to the ***BASE24 Base Files Maintenance Manual*** for more information about this field.

BASE24-atm Statement Print Message Types

The transaction types BASE24-atm uses for statement print messages are shown below:

Type	Description
0200	Financial Transaction Request — A request to authorize a transaction.
0205	Statement Print Transaction Request — A request for additional statement information for a transaction in progress.
0215	Statement Print Transaction Response — A response carrying the statement information for printing.

BASE24-atm Statement Print Transaction Processing

The following steps describe the processing for statement print transactions being sent from BASE24-atm to the host.

1. BASE24-atm receives a request for statement information. This request is then sent to the host as a 0200 request message (a 0205 message with the ANSI Host Interface process).
2. The host processes the 0200 message and returns a 0215 response message to BASE24-atm. The 0215 message carries the statement information.
3. If additional pages of statement information are required, BASE24-atm sends a 0205 request message to the host requesting additional information.
4. The host processes the 0205 message and returns a 0215 message carrying additional statement information.

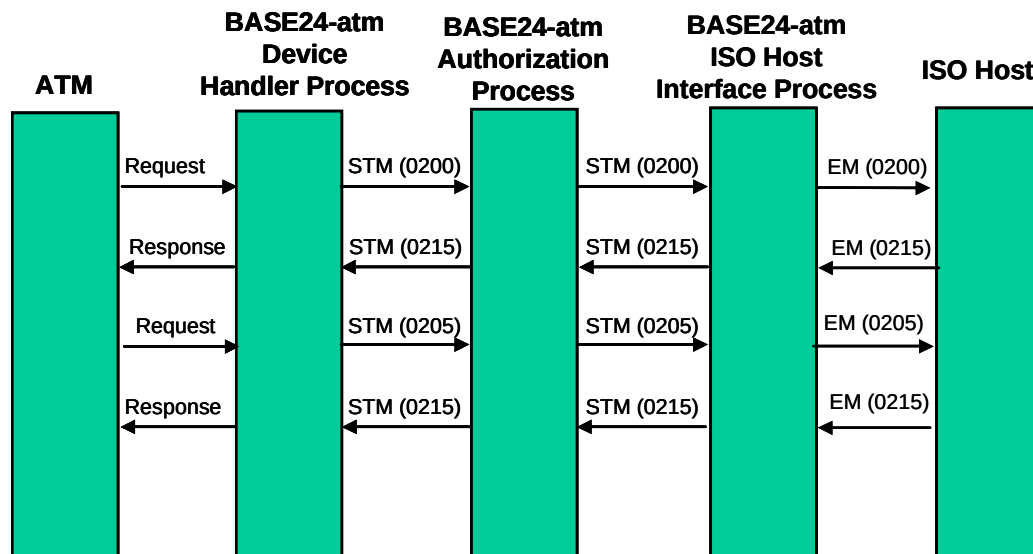
Steps 3 and 4 are repeated until all of the statement information has been sent.

ISO Statement Print Messages

The ISO-based external message is based on the standard external message developed by the International Organization for Standardization (ISO) and is used to route messages between a host and BASE24-atm. The BASE24-atm Standard Internal Message (STM) is used to route messages between BASE24-atm processes. The BASE24-atm ISO Host Interface process provides the message interface between BASE24-atm and the host.

For detailed information on the ISO-based external message (identified in the following diagram by the code EM), refer to the **BASE24 External Message Manual**. For information on the BASE24-atm Standard Internal Message (STM) and the BASE24-atm ISO Host Interface process, refer to appendix A and the **BASE24 ISO Host Interface Manual**, respectively.

The following diagram illustrates the message flow when BASE24-atm is sending a statement print transaction to an ISO host. Although this diagram shows only two request/response cycles, each statement print transaction is different and requires a different number of cycles to complete.

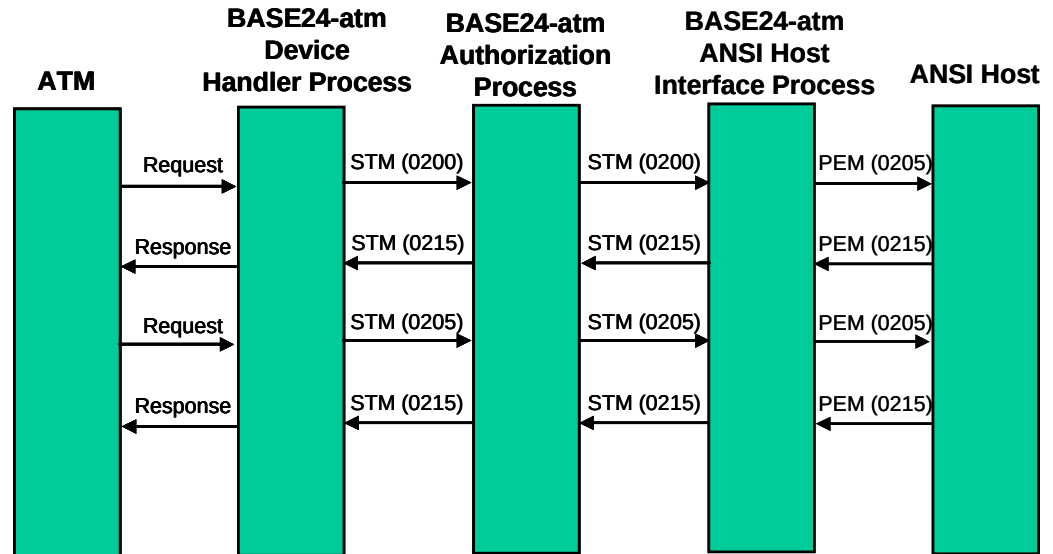


ANSI Statement Print Messages

The Print External Message (PEM) is used to exchange messages with hosts that have yet to upgrade their communications with BASE24-atm to the ISO-based external message. The BASE24-atm Standard Internal Message (STM) is used to route messages between BASE24-atm processes. The BASE24-atm ANSI Host Interface process provides the message interface between BASE24-atm and the host.

The PEM external message format is used to route statement print data messages (0215 and 0205) between a host and BASE24-atm. For detailed information on the ANSI-based external message, and the BASE24-atm ANSI Host Interface process refer to the ***BASE24 ANSI Host Interface Process Manual***.

The following diagram illustrates the message flow when BASE24-atm is sending a statement print transaction to an ANSI host. Although this diagram shows only two request/response cycles, each statement print transaction is different and required a different number of cycles to complete.



BASE24-atm Statement Print Data Compression

BASE24-atm Statement Print Data Compression establishes data compression standards that allow compressed data to be carried in a statement print transaction message.

The amount of information contained in a statement print transaction requires that a series of messages be sent back and forth between BASE24-atm and the host. These messages consist of request messages from BASE24-atm to the host and response messages from the host to BASE24-atm. As a result, compared to other types of transactions, statement print transactions take longer to send, make greater use of communication lines, and place a heavier burden on transaction processing efficiencies.

Data compression reduces the amount of data passed between BASE24-atm and the host during statement print transaction processing. This feature is currently available for use with statement print transactions initiated at NCR 5XXX ATMs that are supported by the BASE24-atm NCR 5XXX Device Handler process. The statement print data compression feature is also available with the Diebold 10XX/478X Device Handler process.

Refer to the ***BASE24-atm Diebold 10XX/478X Device Support Manual*** or the ***BASE24-atm NCR 5XXX Device Support Manual*** for more information about statement print data compression.

Passbook Update Processing

Passbooks are issued by a bank or savings institution to record deposits, withdrawals, and interest earned in a savings account.

Passbook update transaction messages that route through the BASE24-atm system to or from the host do so in a series of transaction messages. These messages consist of request messages to the host and response messages from the host, or vice versa. Because of the amount of information carried in the passbook update transaction message, several request and response messages can be sent between BASE24-atm and the host with the subtype APU0.

BASE24-atm Passbook Update Message Types

BASE24-atm uses the 0200, 0205, and 0215 transaction codes for passbook update messages. Passbook update transactions enter BASE24-atm as statement print transactions with the subtype APU0.

BASE24-atm Passbook Update Transaction Processing

Passbook update transactions are processed in the same manner as statement print transactions, except that only the first 0215 response message is logged to the Transaction Log File (TLF). The initial 0200 request and any additional 0205 requests are not logged to the TLF.

The Passbook Processing token, AK, contains the passbook update information.

Optional Receipts Processing

BASE24-atm optional receipts processing allows an institution to offer the cardholder the option to complete a transaction without a receipt. This option is available with every type of transaction offered by the ATM except statement print transactions. (The statement print transaction requires that a receipt be printed.)

The institution can offer optional receipts in two scenarios: every time the ATM is used or on a receipt printer failure only.

The cardholder can be asked if a receipt is necessary at the beginning of the first transaction of the session or the beginning of each transaction in the session.

Optional receipts processing is configured using the device-specific screens of the BASE24-atm Terminal Data files (ATD). Refer to the ***BASE24-atm Files Maintenance Manual*** for more information about the ATD screens.

For more information about optional receipts processing, refer to the device-specific device support manual.

Note: The IBM 3624 Device Handler process only supports offering optional receipts in the case of a receipt printer failure only, and the cardholder is only asked once at the first transaction of the session.

Surcharging

An institution can configure the BASE24-atm system to assess acquirer fees (surcharges) on transactions initiated at selected ATMs. The following transactions can have a surcharge added by the BASE4-atm system:

- Withdrawal from checking account
- Withdrawal from savings account
- Fast cash withdrawal
- Cash advance from credit account
- Noncurrency dispense
- Deposit
- Deposit with cash back
- Split deposit

The withdrawal transactions listed above also include noncurrency dispense transactions.

BASE4-atm supports a surcharge on the following transactions when received from an acquirer system or an interchange:

- Balance inquiry
- PIN change
- PIN unblock

In addition to transaction-based criteria, surcharges can be based on attributes such as terminal type (for example, the location of the ATM), card prefix or transaction routing, card type, and minimum transaction amount. In addition, surcharges can be based on a flat fee per transaction or on a percentage of the transaction amount.

A surcharge can be positive (as in an acquirer fee) or negative (as in a rebate or incentive).

Surcharge fees can be assessed by transaction code or transaction subtype. Transaction subtypes enable BASE24-atm customers to distinguish among the types of transactions that have identical transaction codes. Using the transaction subtype a customer can configure multiple rate structures for a single transaction code.

Customers also have the option to log transactions in which the customer declines the surcharge.

Surcharging is configured in the Surcharge File (SURF). Fields in the Card Prefix File (CPF) and the BASE24-atm Terminal Data files link the SURF with a card prefix or terminal group.

Some customer contracts with telecommunications services providers prohibit surcharging on mobile top-up transactions at an ATM. If you use the surcharging feature at your ATMs and want to offer mobile top-up transactions without a surcharge, you must configure the SURF to prevent surcharges.

Using the transaction subtype defined in the Transaction Code/Subtype Relationship File (TSRF), you can specify transactions by subtype to not have a surcharge. This is done by setting the MIN TXN, FLAT FEE, and PERCENT FEE fields to the value 0.00 and the USE MIN/MAX field to a blank.

Refer to the **BASE24 Base Files Maintenance Manual** for more information about configuring the SURF, CPF, and TSRF for surcharging. Refer to the **BASE24-atm Files Maintenance Manual** for more information about configuring the BASE24-atm Terminal Data files.

Token Handling

Tokens provide BASE24-atm with flexibility in the information carried in its internal messages; logged to its log files; extracted by the Super Extract process; and sent to a host, interchange, or co-network by way of the ISO Host Interface or ISO BASE24 Interchange (BIC) Interface processes.

A token is a collection of related data, which may or may not be needed to process a particular transaction. A token can be made up of one field or a series of fields. If the information is needed to process a transaction, it can be added to the internal message, logged, and extracted. If the information is not needed, the token need not be involved in the transaction at all—that is, carried in the message, logged, or extracted.

The tokens used by BASE24-atm are referred to briefly in appendix A. They are described in detail in the ***BASE24 Tokens Manual***.

BASE24-atm processes use the Token File (TKN) to determine which tokens are logged, extracted, or sent by way of the ISO Host Interface and ISO BASE24 Interchange (BIC) Interface processes to other entities.

Logging Tokens

TKN screen 2 is used to specify which tokens contained in the internal message are logged to the Transaction Log File (TLF) or any Interchange Log File (ILF). These tokens are arranged in the log record in the order they were added to the internal message by BASE24-atm processes. The option of logging or not logging exists for all BASE24-atm tokens; however, to process reversals correctly, tokens always need to be logged to the ILF.

Extracting Tokens

TKN screen 3 is used to specify which tokens in the TLF or ILF records are extracted by the Super Extract process. This screen is also used to specify the order in which these tokens are arranged within each TLF or ILF extract record. The option of extracting or not extracting exists for all BASE24-atm tokens; however, a token must be logged to the TLF or ILF in order to be extracted.

Retrieving Tokens

An institution has the option to have the Diebold 10XX/478X or NCR 5XXX Device Handler process include tokens in reversal messages and, if included, from where the token data is retrieved. Token retrieval for reversal transactions is configurable at the terminal or institution level. Settings at the terminal level can override settings for the institution.

The three token processing options are as follows:

- Adding no tokens to the reversal message.
- Writing tokens to and retrieving them from the BASE24-atm Terminal Data files—scratch pad (ATDD2) and appending them to the STM.
- Retrieving tokens from the TLF and appending them to the STM.
- Retrieving tokens from the TLF and appending them to the STM. This is the default setting. In addition, this option impacts transaction processing as follows:
 - The request rate to the TLF is doubled, even if the transaction is not a reversal.
 - If the TLF is unavailable for any reason, such as the primary TLF being unavailable during backup processing, the Device Handler process forwards the reversal transaction to the Authorization process without any token data. The Authorization process logs the reversal as an exception item that must be reconciled manually.

At the terminal level the token retrieval option is configured using the TOKEN RETRIEVAL OPTION field on ATD screen 2. The option is configured at the institution level using the TOKEN RETRIEVAL OPTION field on IDF screen 13. The option set at the terminal level overrides the option set at the institution level.

The reversal processing option is carried in the TLF token. The Device Handler process adds the TLF token to the request message and sets a field indicating the token reversal processing option.

The Authorization process updates the TLF token on the response message (if it indicates TLF retrieval) with the information needed for the Device Handler process to retrieve the TLF record during reversal processing.

Refer to the ***BASE24 Tokens Manual*** for more information about the TLF token. Refer to the device-specific device support manuals for more information about how the Device Handler process handles reversals.

Sending Tokens to a Host or Co-Network

TKN screen 4 is used to specify which tokens contained in the internal message are sent to a host by way of the ISO Host Interface process or to a co-network by way of the BIC ISO Interface process. This screen is also used to specify the order in which the tokens are carried in element S-126 (BASE24-atm Additional Data) of the BASE24 external message. If tokens are to be sent in the external message, data element S-126 must be set to conditional or mandatory. Refer to the ***BASE24 External Message Manual*** for detailed descriptions of the BASE24 external message.

Carrying Tokens in the BASE24-atm Internal Message

Information in the TKN does not affect the tokens carried in the internal message. Tokens are added to the internal message if the token data is present and the token is required for processing. Tokens are then used in the internal message throughout the transaction path.

Deposit Hold Notification Processing

BASE24-atm uses deposit hold notification processing to accept deposit hold information from the host and return this information to the cardholder. The cardholder then has the option to either accept the hold or cancel the transaction before they deposit any funds into the ATM. This is accomplished through a set of transaction messages. These messages consist of request messages to the host and response messages from the host.

BASE24-atm Deposit Hold Notification Processing

If BASE24 is configured to stand-in for deposits, BASE24 will authorize the deposit. Hold information that is not available in the Cardholder Authentication File (CAF) will not be returned to the ATM; nor will it be used as criteria in the authorization process. If no hold is to be applied to the deposit, the transaction is processed as a standard deposit.

1. BASE24-atm receives a deposit request, which is then sent to the host.
2. The host sets the ISO response code to 19 (re-enter transaction) and, if a hold is to be applied to the deposit, sends the Hold token (AL) in the response containing the date the deposit will be released and the hold amount.
3. The BASE24-atm HISO process translates this ISO response code in the STM response code to 68 according to standard deposit processing.
4. BASE24 will log the declined deposit and the AL token (if configured in the TKN file) to the TLF, and then send the response to the ATM.
5. If the cardholder chooses to continue with the transaction, the ATM will send another deposit request message.
 - a. The BASE24-es IFX Device Handler will attach the AL token received in the response to the request.
 - b. The HLD-ACCEPTED flag in the AL token will be set to Y.
6. The request is sent to the host. The host will post the transaction to the cardholder's account according to standard deposit processing and return a response to BASE24-atm.

7. The response is logged to the TLF.
8. If a problem occurs during the deposit or if the customer cancels the deposit transaction, the ATM will reverse the deposit and then standard reversal processing will apply.

Exchange Rate Notification Processing

Exchange rate notification processing is used to notify the cardholder that they may accept or decline the exchange rate offered by the financial institution (via transaction code 60, subtype AER0). The BASE24-atm Authorization process will return the exchange rate from the Exchange Rate File (ERF) to the device handler. At this point, the cardholder can choose to accept the exchange rate, or decline it and no further processing will occur.

Add-on Module Processing

The following add-on modules can be purchased for the BASE24-atm product. These modules provide institutions with additional processing capabilities.

Self-Service Banking

The BASE24-atm self-service banking (SSB) add-on product consists of three separate software applications which can be purchased and installed individually or in combination. These software applications are the SSB Device Handler Extension, the SSB Base Application, and the SSB Enhanced Check Processing Application.

The features of the BASE24-atm SSB add-on product allow the financial institution to expand its automated financial services to include a variety of banking functions previously available only at a branch location through a human teller or a personal banker. The SSB product is specific to particular terminals.

With the SSB product, an institution can expand the available consumer transactions, provide more card and check services, enhance terminal capabilities, and optimize their processing capabilities. The features provided by each SSB software application are listed below. Refer to the appropriate self-service banking (SSB) manual for more information about the SSB features supported by the device.

SSB Device Handler Extension

The SSB Device Handler Extension can be purchased as a stand-alone application or in conjunction with the SSB Base Application and the SSB Enhanced Check Processing Application. The SSB Device Handler Extension includes the following:

- Check cashing transactions
- Envelope-free check deposit transactions
- Envelope-free deposit with cash back transactions
- Envelope-free cash deposit transactions
- Check proof listing
- Flexible deposit sorting

- Coin dispensing
- Check imaging support
- MICR support

SSB Base Application

The SSB Base Application provides a number of configurable features and is a prerequisite to the SSB Enhanced Check Processing Application. The institution must install the SSB Device Handler Extension before implementing the SSB Base Application. The SSB Base Application includes the following:

- Split deposit transactions
- Transaction time windows
- Configurable bill dispensing
- Configurable deposit printing
- Super Teller support
- Statement print host communications

SSB Enhanced Check Processing Application

The SSB Enhanced Check Processing Application expands the check cashing and processing features of the SSB Base Application to provide the institution with the full capabilities of the SSB features. The SSB Device Handler Extension and the SSB Base Application are both prerequisites for the SSB Enhanced Check Processing Application. The SSB Enhanced Check Processing Application includes the following:

- Check-cashing cards
- Registered check cashing
- Preapproved and predeclined check cashing
- Stop payment verification
- Enhanced MICR data processing and modification
- Enhanced check authorization
- Enhanced deposit sorting

BASE24-atm Non–Currency Dispense

The BASE24-atm Non–Currency Dispense add-on product enables an ATM to dispense cash value items (such as traveler’s checks or postage stamps) and nonvalue items (such as coupons or advertisements).

The institution can configure up to four noncurrency item per ATM. This corresponds to the number of note hoppers available in NCR 5XXX and Diebold 10XX/478X ATMs.

Refer to the appropriate device support manual for more information about the Non–Currency Dispense add-on product.

BASE24-atm Mobile Top-Up Device Handler Extension

The BASE24-atm Mobile Top-Up Device Handler Extension add-on product enables cardholders to replenish, or *top up*, their mobile telephone accounts at an ATM. The replenishment transaction is a noncurrency dispense transaction. The institution must purchase both the Non–Currency Dispense add-on product and the Mobile Top-Up Device Handler extension to offer mobile top-up services.

A telecommunications services provider or an individual mobile operator, rather than the institution that owns the ATM or issues cards, is responsible for authorizing the minutes purchase, maintaining the consumer mobile telephone account, and calculating any taxes applied to the transaction. The BASE24-atm product authorizes the funds portion of the transaction.

Refer to the appropriate device support manual for more information about the Non–Currency Dispense and the Mobile Top-Up add-on products.

BASE24-atm Multiple Currency

The BASE24-atm Multiple Currency add-on product allows institutions authorize transactions in multiple currencies or dispense multiple currencies from an NCR 5XXX or Diebold 10XX/478X ATM.

The authorization portion of the Multiple Currency add-on product enables institutions to offer accounts in different currencies, with the ability to define each account with its own currency. It also allows cardholders to perform transactions in different currencies at different ATMs.

The BASE24-atm Multiple Currency Dispense add-on feature enables an ATM to dispense cash in multiple currencies. The authorization portion of the Multiple Currency add-on product can work with the cash dispensing feature but is not required.

Refer to the ***BASE24-atm Multiple Currency Support Manual*** for more information about the BASE24-atm Multiple Currency add-on product. Refer to the ***BASE24-atm Diebold 10XX/478X Device Support Manual*** or the ***BASE24-atm NCR 5XXX Device Support Manual*** for more information about the Multiple Currency Dispense add-on feature.

BASE24-atm EMV

Europay, MasterCard, and Visa (EMV) have developed integrated circuit card (ICC) credit and debit specifications to facilitate interoperability across competing payment systems. The BASE24-atm EMV add-on product supports processing for ICC and magnetic stripe transactions from a single terminal.

Refer to the ***BASE24-atm EMV Manual*** for more information about the BASE24-atm EMV add-on product.

BASE24-atm Automated Key Distribution System

The BASE24-atm Automated Key Distribution System add-on product enables an institution to remotely load a unique master key to an ATM. In the past, master keys, which are used to encrypt all other keys, were loaded manually at ATMs. Using the BASE24-atm Automated Key Distribution System add-on product, you can load the master key to the ATM from BASE24. In addition, the key is randomly generated, ensuring a unique key for all ATMs in the network. This add-on product is available for the NCR 5XXX Device Handler process.

Refer to the ***BASE24-atm Diebold 10XX/478X Device Support Manual*** and ***BASE24-atm NCR 5XXX Device Support Manual*** for more information about the BASE24-atm Automated Key Distribution System add-on product.

BASE24-atm Dynamic Key Management

The BASE24-atm Dynamic Key Management add-on product enables a BASE24 system to determine when PIN and MAC keys need to be changed and start key change procedures. The BASE24-atm Dynamic Key Management add-on product

shuts down the ATM and initiates the command to change the keys. This add-on product is available for the NCR 5XXX, Diebold 10XX/478X, and Triton Device Handler processes.

Refer to the device-specific device support manuals for more information about the Dynamic Key Management add-on product.

BASE24-atm Dynamic Currency Conversion

The Dynamic Currency Conversion (DCC) add-on product enables BASE24-atm to support DCC on withdrawal transactions that originate from an ATM that is driven by the Diebold 10XX/478X or NCR 5XXX Device Handler. This functionality gives the consumer the option of having the currency conversion performed on the acquirer side by locking in a rate displayed at the ATM before the transaction is sent to a network for authorization. Typically, transactions would have been sent to the interchange in the currency of the acquiring network, making the issuing network responsible for performing the necessary currency conversion.

DCC processing applies only to withdrawals that originate at a BASE24 ATM and are routed (using the SPROUTE file) to a separate authorizer via an interchange that allows DCC. It does not apply to non-withdrawal transactions or to withdrawals authorized locally by BASE24 or a host (i.e., CPF-routed transactions).

The Authorization DCC module uses two files to perform DCC processing: the Exchange Rate File (ERF) and the Dynamic Currency Conversion Data file (DCCD).

The ERF is used by the EMT Build process to create the ERFEMT, which the DCC module shares with the Multiple Currency and Multiple Currency Dispense modules. This file provides a single exchange rate and effective date for each currency supported by the BASE24 network.

The DCCD is used by the EMT Build process to create the DCCDEMT, which is used only by the DCC module. This file provides DCC processing parameters specific to the terminal, issuer, and currency, including the mark-ups that are used in conjunction with the rates from the ERF to calculate the rates used in DCC conversion.

Refer to the **BASE24 Base Files Maintenance Manual** for more information about these files.

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4: BASE24-atm Routing and Authorization

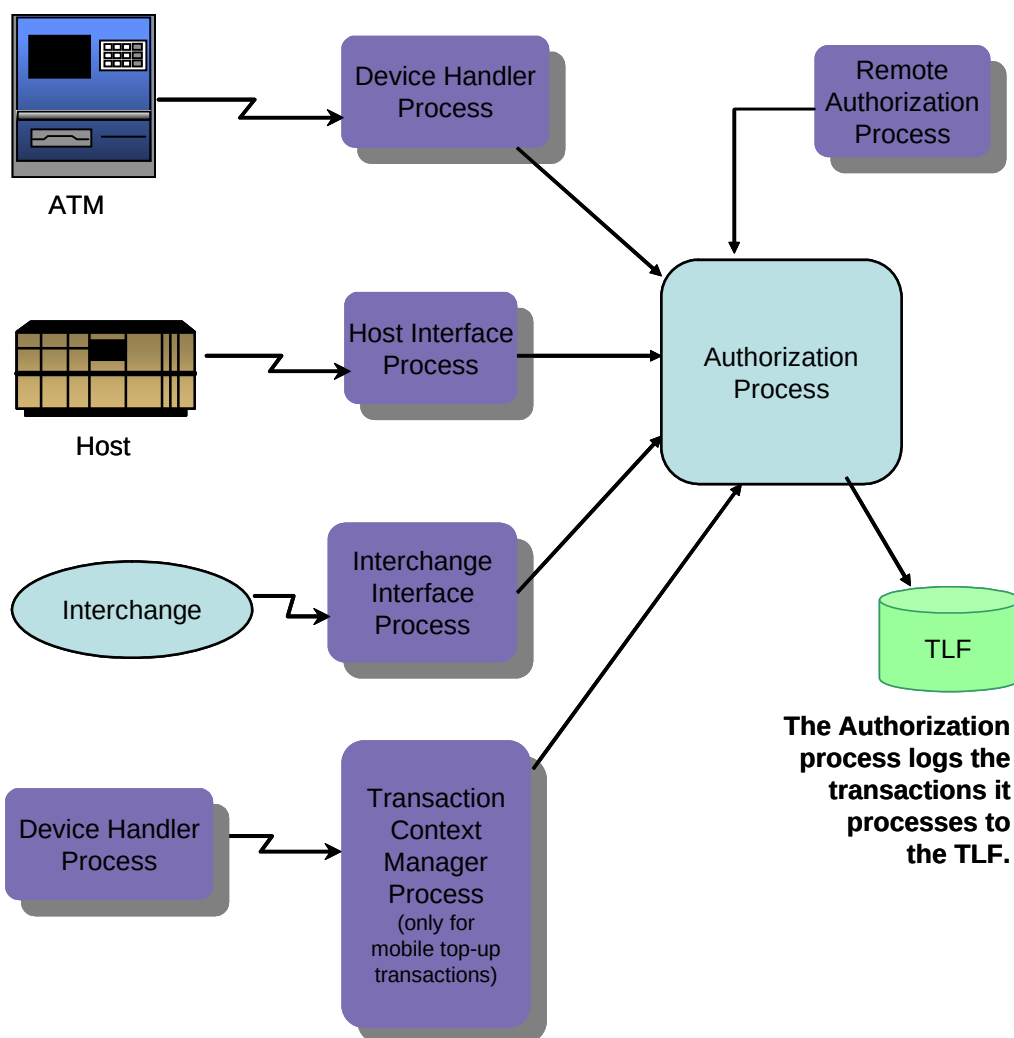
This section summarizes the processing BASE24-atm performs during transaction routing and authorization. The information should be used with section 3, which describes the controls available for all of the transaction processing performed by BASE24-atm. This section includes information about the following topics:

- Authorization processing. A brief overview of the role of the Authorization process in routing and authorizing transactions.
- Routing transactions with unrecognized prefixes. A description of routing using the SPROUTE File (SPROUTE).
- Processing transactions with recognized prefixes. A description of routing and authorization using the Institution Definition File (IDF).
- Bypassing the BASE24-atm Authorization process. A description of routing using the Acquirer Processing Code File (APCF).
- Split transaction routing of mobile top-up transactions. A description of routing using the Split Transaction Routing File (STRF).

Authorization Processing

The Authorization process authorizes transactions according to the authorization parameters for an institution and controls and returns responses to the source of the transaction request. The authorization controls are described in section 3. The Authorization process also logs transactions to the Transaction Log File (TLF), which can then be extracted for affecting settlement immediately without waiting for transaction receipts.

As shown below, the Authorization process receives transactions from a number of sources, including Device Handler processes, Host Interface processes, Interchange Interface processes, Transaction Context Manager process, and Authorization processes in other logical networks. Each of these processes can send or forward requests for authorization.



The Authorization process handles requests according to the card-issuing institutions' prescribed authorization criteria, logs the transactions to the TLF, and returns responses to the requesting sources.

The following pages summarize the transaction handling performed by the Authorization process.

Routing Transactions with Unrecognized Prefixes

To process a transaction, the Authorization process must be able to locate the prefix of the card used to initiate the transaction in the Card Prefix File (CPF). If a CPF record that corresponds with the prefix of the card used in a transaction is located, the Authorization process can locate a record in the Institution Definition File (IDF) for the card-issuing institution. If the prefix cannot be found in the CPF for a transaction, the transaction must be sent elsewhere for processing.

For example, when an interchange cardholder uses a local terminal, the Authorization process does not have access to the cardholder record and the transaction must be routed elsewhere—to the interchange—for processing. Another example would be a cardholder associated with an institution in another BASE24 logical network. Here again, the Authorization process does not have access to the cardholder record (because it would be in another logical network) and, therefore, could not find the prefix in the CPF.

The Authorization process routes transactions for which it does not recognize the prefix using SPROUTE, which is a BASE24 configuration file built when the BASE24 system is configured. From SPROUTE, the Authorization process can derive primary and alternate destinations for transactions where it does not recognize the prefix.

The SPROUTE File

The SPROUTE File (SPROUTE) is built during BASE24 configuration by the Pregen utility. It contains information from the PREMAP and SPREFIX Files, two other BASE24 configuration files built by the Pregen utility. The PREMAP File (PREMAP) contains the actual routing logic of the BASE24 system and the SPREFIX File (SPREFIX) contains all of the prefixes from the prefix files in the system (i.e., all the CPFs in the BASE24 system, if you have more than one logical network, plus any interchange prefix files in the system). If you have only one logical network in the BASE24 system, the CPF is not included in SPREFIX. SPREFIX is a temporary file that the Pregen utility creates to build SPROUTE. Once SPROUTE is built, the utility purges the file. The Authorization process uses SPROUTE to look up prefixes which are recognized as valid by the BASE24 system.

SPROUTE is a CPU memory-resident file that contains all of the information needed and available to determine proper destinations for transactions whose prefixes are not recognized by a BASE24-atm Authorization process. It is maintained in a common CPU memory area and is available to all processes

running in a particular CPU. SPROUTE is paged in and out of memory for each CPU as needed. If a process running on another CPU needs to read SPROUTE, SPROUTE is paged into memory for that CPU. The Authorization process uses SKEL library procedures to access SPROUTE and perform prefix routing. SKEL is an XPNET library that is used by BASE24 application processes to route transactions.

When a BASE24-atm Authorization process cannot find a CPF record for the card originating the transaction, the process calls the SPROUTE^LOOKUP routing procedure in the SKEL library. The SKEL library procedures access SPROUTE to determine a primary or alternate authorizer for the transaction. This information is returned to the application process, and the transaction is routed accordingly.

The SPROUTE created by the Pregon utility resides on the XPNET system data subvolume. The Pregon utility also copies SPROUTE to any remote HP NonStop systems in the XPNET system, if required.

The SPREFIX File

SPREFIX serves as a temporary point of reference for all prefixes used in prefix routing. The Pregon utility builds SPREFIX on the XPNET system data subvolume from all of the Card Prefix Files (CPFs) and Interchange Prefix Files (IPFs) specified in the Pregon Command File (PRECMD). Once the file is built, the utility uses it to build the SPREFIX table in SPROUTE. After SPROUTE is built, the utility purges SPREFIX.

The Basics of Routing (PREMAP)

PREMAP contains the routing logic for the BASE24 system as a whole. It is a multiple entry table, built by the Pregon utility from commands entered in the Pregon Command File (PRECMD). Refer to appendix D for detailed procedures on setting up the PRECMD and using the Pregon utility. Once PREMAP is created, the utility reads it to build the PREMAP table in SPROUTE. Each entry in the PREMAP table contains the following fields:

```
source-ln, sym-source, routing-group, routing-order,  
method, card-type, dest, alt-dest, file-name, type, sprefix-key  
method, card-type, dest, alt-dest, file-name, type, sprefix-key  
method, card-type, dest, alt-dest, file-name, type, sprefix-key  
...
```

PREMAP entries can be divided into two parts: key fields and action fields. The key fields allow processes to select the appropriate entry from the PREMAP data. The action fields prescribe the actions to take once a PREMAP entry has been selected.

PREMAP contains one record for each transaction source, symbolic source, routing group, and destination combination in the system. A transaction source is any logical network or interchange from which transactions can be routed.

A description of each of the fields in a PREMAP entry is included below. These descriptions summarize the fields for the purpose of understanding the routing performed by the BASE24-atm Authorization process.

Key Fields

The *source-ln*, *sym-source*, *routing-group*, and *routing-order* fields are the key fields of the table entry. The Authorization process compares transactions to these four fields to determine whether the subsequent routing actions should be taken.

source-ln — The logical network or interchange from which the transaction originated (e.g., PRO1, PRO2, TES1, VISA, BNET). This field must always contain the name of a logical network or interchange.

sym-source — The symbolic name of the Authorization process that received the transaction. This field can contain an actual process name or an asterisk (*) if the source process need not be specific.

routing-group — The routing group associated with the transaction originator. This field can contain an actual routing group identifier, or, if the routing group need not be specific, this field can contain an asterisk (*).

An asterisk can be entered in this field if routing based on the routing group is not required. The asterisk is used with the last two entries in the selection hierarchy when the Authorization process searches SPROUTE for a PREMAP entry.

You can enter zeros in this field if routing based on the routing group and source process is not required. When the routing group is zeros and the source process contains an asterisk, the Authorization process matches on the source logical network or interchange only as the second entry in the selection hierarchy. The selection hierarchy is described later in this section.

routing-order — A field generated by the Pregen utility that specifies the order in which routing methods are to be applied for this record. This order is determined by the order in which you enter routing methods in the PRECMD.

Action Fields

The *method*, *card-type*, *dest*, *file-name*, *type*, and *alt-dest* fields specify the routing actions to take if a transaction matches on the key fields of the table entry. Multiple actions can be set to allow for several routing alternatives. The Authorization process checks multiple actions in the order they appear in the PREMAP table in SPROUTE.

method — The method to use for taking action on this table entry. The methods available are LGNT, AMEX, BASE or VISA, DSCV, INAS or BNET, and DFLT. Each is discussed later in this section.

card-type — The card type to use in determining how to route this transaction. The *card-type* field is used as part of the key into the Surcharge File (SURF).

dest*, *alt-dest — The symbolic names of the primary and alternate destinations to use for routing the transaction if the account number meets the criteria set forth for the selected method.

Note: Alternate destinations are intended for use by BASE24 Interchange Interface processes to use if their respective interchange links are down. They are not intended to allow for alternate routing by the Authorization process. While an Authorization process will place an alternate destination in the internal message if one is found in SPROUTE, it does not route to an alternate destination if a message is found to be undeliverable to the primary destination.

file-name*, *type — The fully qualified name of the prefix file, and its type, for use with the LGNT method. Valid values for *type* are as follows:

CP = Card Prefix File
IP = Interchange Prefix File

sprefix-key — A field generated by the Pregen utility that contains the key used to access SPREFIX for this routing method.

Selection Hierarchy

To select a PREMAP table entry, the Authorization process provides the following information:

- The logical network or interchange from which the transaction originated.
- Its own symbolic Authorization process name.
- The routing group associated with the terminal or interchange originating the transaction (to provide this information, the Authorization process uses the value from the RQST.RTE-GRP field of the STM).

The routing group listing above indicates the group to which the terminal belongs for routing and processing of transactions initiated by cardholders that are outside the logical network.

For transactions originating at BASE24-atm terminals, the Device Handler process sets this field in the STM using the alternate routing group field of the BASE24-atm Terminal Data files.

For transactions originating through an interchange, the Interchange Interface process sets this field in the STM using the default routing group field in the BASE24-atm segment of the Interchange Configuration File (ICF) or Enhanced Interchange Configuration File (ICFE).

Using the logical network, symbolic name, and routing group, the Authorization process steps through the PREMAP entries in SPROUTE looking for a match. If an exact match is not found, certain other alternatives are allowed. Below is the hierarchy used for selection.

Preference	Logical Network	Process Name	Routing Group
1	match	match	match
2	match	*	zeros
3	match	*	match
4	match	match	*
5	match	*	*

If an entry is found that meets the above selection criteria, the routing actions specified in the entry are applied to derive the primary and alternate destinations. If the Authorization process cannot find an entry based on the above hierarchy, the transaction cannot be processed.

Routing Methods

Once the PREMAP entry is selected, the Authorization process uses the action fields for the entry to determine the primary and alternate destinations.

To evaluate a transaction against the PREMAP action fields, the Authorization process must supply the account number used in the transaction (taken from the RQST.TRACK2 field in the STM) and the account number length.

The methods of checking the account number are described below.

LGNT (Logical Network)

In a logical network search, the Authorization process searches the designated SPREFIX data in SPROUTE—specified by the file name (and type)—for a card prefix that matches the account number. If the account number prefix is located in the SPREFIX data, the primary and alternate destinations are taken from that set of action fields.

AMEX (American Express Algorithm)

The Authorization process checks the account number against an AMEX algorithm. In this case, if the account number starts with 34 or 37, the primary and alternate destinations are taken from that set of action fields.

BASE or VISA (VISA Algorithm)

The Authorization process checks the account number against a VISA algorithm. In this case, if the account number is 13 or 16 positions in length and the first position is a 4, the primary and alternate destinations are taken from that set of action fields.

DSCV (Discover Algorithm)

The Authorization process checks the account number against a Discover algorithm. In this case, the primary and alternate destinations are taken from that set of action fields only when the leading digits fall into one of the following ranges.

60110000 - 60119999
62212600 - 62292599
64400000 - 64499999
650000 - 659999

INAS or BNET (MasterCard Algorithm)

The Authorization process checks the account number against a MasterCard algorithm. In this case, if the leading digits are 51, 52, 53, 54, or 55 for any card number length, the primary and alternate destinations are taken from that set of action fields.

DFLT (Default Routing)

No checking is done. The Authorization process takes the primary and alternate destinations from that set of action fields.

Setting Up Routing Tables

The order in which multiple entries are made in the PREMAP routing table should be based on business and performance considerations. Some of the factors to take into account include:

- Does the destination accept invalid prefixes? DFLT routing has the lowest overhead because no prefix checks are preformed.
- Can an algorithm check be performed? AMEX, BASE, DSCV, and INAS routing methods require less overhead than the prefix table lookup performed with LGNT.
- Which set of action fields will handle the most traffic? The earlier a routing method can be selected, the less time and overhead are required.

Routing Example

The following pages provide an example of how the Authorization process derives primary and alternate destination from SPROUTE.

PREMAP Data

The example uses the PREMAP information shown below. This PREMAP information is for illustrative purposes only. It is not intended to represent a complete or recommended version of PREMAP. For information on how to use the Pregen utility to build the PREMAP data for an actual system, refer to appendix D of this manual.

PREMAP Table

```

PRO1, P1A^AUTH1, 00000000000,
      INAS, card-type, pri-dest,           , alt-dest,
      BASE, card-type, pri-dest,           , alt-dest,
      LGNT, card-type, pri-dest, $DATA.PRO2DATA.CPF, CP, alt-dest,
      DFLT, card-type, pri-dest,           , alt-dest;
PRO1, P1A^AUTH1, 00000000001,
      INAS, card-type, pri-dest,           , alt-dest,
      BASE, card-type, pri-dest,           , alt-dest,
      AMEX, card-type, pri-dest,           , alt-dest,
      LGNT, card-type, pri-dest, $DATA.PRO2DATA.CPF, CP, alt-dest,
      DFLT, card-type, pri-dest,           , alt-dest;
PRO1, P1A^AUTH1,      *      ,
      INAS, card-type, pri-dest,           , alt-dest,
      BASE, card-type, pri-dest,           , alt-dest,
      LGNT, card-type, pri-dest, $DATA.PRO2DATA.CPF, CP, alt-dest,
      DFLT, card-type, pri-dest,           , alt-dest;
PRO1, P1B^AUTH1, 00000000000,
      INAS, card-type, pri-dest,           , alt-dest,
      BASE, card-type, pri-dest,           , alt-dest,
      LGNT, card-type, pri-dest, $DATA.PRO2DATA.CPF, CP, alt-dest,
      DFLT, card-type, pri-dest,           , alt-dest;
PRO1, P1B^AUTH1, 00000000001,
      INAS, card-type, pri-dest,           , alt-dest,
      BASE, card-type, pri-dest,           , alt-dest,
      AMEX, card-type, pri-dest,           , alt-dest,
      LGNT, card-type, pri-dest, $DATA.PRO2DATA.CPF, CP, alt-dest,
      DFLT, card-type, pri-dest,           , alt-dest;
PRO1, P1B^AUTH1,      *      ,
      INAS, card-type, pri-dest,           , alt-dest,
      BASE, card-type, pri-dest,           , alt-dest,
      LGNT, card-type, pri-dest, $DATA.PRO2DATA.CPF, CP, alt-dest,
      DFLT, card-type, pri-dest,           , alt-dest;

```

```

PRO1,      *      , 000000000001,
      INAS, card-type, pri-dest,      ,      , alt-dest,
      BASE, card-type, pri-dest,      ,      , alt-dest,
      AMEX, card-type, pri-dest,      ,      , alt-dest,
      LGNT, card-type, pri-dest, $DATA.PRO2DATA.CPF, CP, alt-dest,
      DFLT, card-type, pri-dest,      ,      , alt-dest;
PRO1,      *      ,      *      ,
      INAS, card-type, pri-dest,      ,      , alt-dest,
      BASE, card-type, pri-dest,      ,      , alt-dest,
      AMEX, card-type, pri-dest,      ,      , alt-dest,
      LGNT, card-type, pri-dest, $DATA.PRO2DATA.CPF, CP, alt-dest,
      DFLT, card-type, pri-dest,      ,      , alt-dest;

```

Processing Steps

The following steps describe the processing that the Authorization process goes through in order to derive primary and alternate destinations from the example SPROUTE above.

In this example, an Authorization process, P1C^AUTH1, receives a transaction from a cardholder who is part of logical network PRO2. The transaction originated from a terminal in routing group 00000000002.

1. After not finding the cardholder's prefix in the CPF for logical network PRO1, the Authorization process looks for an entry in SPROUTE. It first looks for an exact match on the following combination of values:

PRO1, P1C^AUTH1, 00000000002

The Authorization process steps through the table sequentially as follows:

Step 1	PRO1, P1A^AUTH1, 00000000000, ...	No match
Step 2	PRO1, P1A^AUTH1, 00000000001, ...	No match
Step 3	PRO1, P1A^AUTH1, * , ...	No match
Step 4	PRO1, P1B^AUTH1, 00000000000, ...	No match
Step 5	PRO1, P1B^AUTH1, 00000000001, ...	No match
Step 6	PRO1, P1B^AUTH1, * , ...	No match
Step 7	PRO1, * , 000000000001, ...	No match
Step 8	PRO1, * , * , ...	No match

2. Since no match was found, the Authorization process looks for its second preference:

PRO1, *, , 00000000000

The Authorization process steps through the table sequentially as follows:

Step 1	PRO1, P1A^AUTH1, 00000000000, ...	No match
Step 2	PRO1, P1A^AUTH1, 000000000001, ...	No match
Step 3	PRO1, P1A^AUTH1, *, , ...	No match
Step 4	PRO1, P1B^AUTH1, 00000000000, ...	No match
Step 5	PRO1, P1B^AUTH1, 000000000001, ...	No match
Step 6	PRO1, P1B^AUTH1, *, , ...	No match
Step 7	PRO1, *, , 000000000001, ...	No match
Step 8	PRO1, *, , *, , ...	No match

3. Since no match was found, the Authorization process looks for its third preference:

PRO1, *, , 000000000002

The Authorization process steps through the table sequentially as follows:

Step 1	PRO1, P1A^AUTH1, 00000000000, ...	No match
Step 2	PRO1, P1A^AUTH1, 000000000001, ...	No match
Step 3	PRO1, P1A^AUTH1, *, , ...	No match
Step 4	PRO1, P1B^AUTH1, 00000000000, ...	No match
Step 5	PRO1, P1B^AUTH1, 000000000001, ...	No match
Step 6	PRO1, P1B^AUTH1, *, , ...	No match
Step 7	PRO1, *, , 000000000001, ...	No match
Step 8	PRO1, *, , *, , ...	No match

4. Since no match was found, the Authorization process looks for its fourth preference:

PRO1, P1C^AUTH1, *

The Authorization process steps through the table sequentially as follows:

Step 1	PRO1, P1A^AUTH1, 00000000000, ...	No match
Step 2	PRO1, P1A^AUTH1, 000000000001, ...	No match
Step 3	PRO1, P1A^AUTH1, *, ...	No match
Step 4	PRO1, P1B^AUTH1, 00000000000, ...	No match
Step 5	PRO1, P1B^AUTH1, 000000000001, ...	No match
Step 6	PRO1, P1B^AUTH1, *, ...	No match
Step 7	PRO1, *, 000000000001, ...	No match
Step 8	PRO1, *, *, ...	No match

5. Since no match was found, the Authorization process looks for its fifth preference:

PRO1, *, *

The Authorization process steps through the table sequentially as follows:

Step 1	PRO1, P1A^AUTH1, 00000000000, ...	No match
Step 2	PRO1, P1A^AUTH1, 000000000001, ...	No match
Step 3	PRO1, P1A^AUTH1, *, ...	No match
Step 4	PRO1, P1B^AUTH1, 00000000000, ...	No match
Step 5	PRO1, P1B^AUTH1, 000000000001, ...	No match
Step 6	PRO1, P1B^AUTH1, *, ...	No match
Step 7	PRO1, *, 000000000001, ...	No match
Step 8	PRO1, *, *, ...	Match found

6. Once an appropriate table entry is located, the Authorization process checks the card number involved in the transaction, shown below, against the actions specified in the selected entry.

CARD NUMBER: 8 1 0 0 6 4 3 2 5 4 9 8 7 6 1 1

9. The card number does not meet the AMEX criteria, so the Authorization process tries a logical network search using the card number. In this case, the Authorization process checks the \$DATA.PRO2DATA.CPF portion of the SPREFIX information in SPROUTE:

PRO1, *	, *	,	
INAS, card-type, pri-dest,		, alt-dest	
BASE, card-type, pri-dest,		, alt-dest	
AMEX, card-type, pri-dest,		, alt-dest	
LGNT, card-type, pri-dest, \$DATA.PRO2DATA.CPF, CP, alt-dest			
DFLT, card-type, pri-dest,		, alt-dest	

10. Because the cardholder is a part of the PRO2 logical network, the Authorization process locates the cardholder prefix in the designated SPREFIX information. Since a prefix can be located, the Authorization process uses the primary and alternate destinations from that entry:

LGNT, card-type, pri-dest, \$DATA.PRO2DATA.CPF, CP, alt-dest

Processing Transactions with Recognized Prefixes

To process a transaction, the Authorization process must be able to locate the prefix of the card used to initiate the transaction in the Card Prefix File (CPF). If a CPF record that corresponds with the prefix of the card used in a transaction is located, the Authorization process can locate a record in the Institution Definition File (IDF) for the card-issuing institution. From that point, the Authorization process can determine how to process the transaction using the institution IDF routing table.

The IDF routing table enables institutions to specify multiple DPCs for processing authorizations. This five-entry table allows institutions to segregate transactions by account type and prefix group for processing by separate DPCs and under separate authorization parameters.

Note: The fast cash account type code cannot not be set to 00 if the IDF routing parameters differ by account type.

Institutions can set up a table entry on screen 9 of the IDF for each DPC involved in authorizing transactions and then establish the authorization parameters individually for each DPC. For example, one DPC could authorize transactions for a specific group of card prefixes, while another handles authorizations on credit accounts.

Using the routing table, an institution can distribute its processing according to its database. An institution whose account database is distributed among several DPCs can also tailor the authorization process to involve all of the DPCs according to the types of accounts and card prefixes for which they have files.

IDF Routing Table Entries

The routing table in the IDF can contain up to five entries. Each entry contains the fields listed and briefly described below. For more information about these fields and the valid values that can be entered in these fields, refer to the ***BASE24 Base Files Maintenance Manual***.

PRIMARY DPC — The DPC number of the host computer to which transactions included under the processing parameters for this table entry are to be routed.

SYMBOLIC NAME — The symbolic name of the Host Interface process used for sending this group of transactions to a back-end authorizing host.

ACCT TYPE — The transaction account type to be processed according to the table entry parameters. For transactions involving two account types, such as transfers, the *from* account and *to* account sides may be processed differently according to different table entries. The withdrawal side of the transaction may be sent to one host computer while the deposit side is sent to another.

Whenever the routing table contains multiple entries and includes an entry with a value of AL (all account types) in this field and a value of A (all card prefixes) in the PRFX RTG field, the entry with a value of AL in this field and a value of A in the PRFX RTG field must follow all of the other entries.

Note: Any DPC that is to receive PIN change or EMV PIN unblock transactions must have at least one IDF entry that has an ACCT TYPE value of AL.

In addition, the fast cash account type code cannot not be set to 00 if the IDF routing parameters differ by account type.

PRFX RTG — The prefix group to be processed according to the table entry parameters. Card prefixes can be grouped in the CPF (using the PREFIX ROUTING field on CPF screen 1) for different types of authorization processing. This field ties the card prefix groups to this table entry.

In an enhanced mobile top-up routing environment, the Authorization process sends the mobile top-up transaction to the host process configured in the Institution Definition File (IDF) based on the MTU PREFIX ROUTING field on CPF screen 4. This process, configured in the IDF, will be sent to a network interface. This network interface will send the transaction to the network interchange. The network interchange is then responsible for splitting the transaction into the funds side and the top-up side, and finally, sending it to the appropriate entities for authorization. The funds side will be sent to a funds authorizer; in this case the BASE24-atm Authorization process for authorization.

At this time, the transaction will no longer use the MTU PREFIX ROUTING field on CPF screen 4, but will use either the EMV PREFIX ROUTING field on CPF screen 11 or the default PREFIX ROUTING field on CPF screen 1 for authorization. The transaction will be authorized and returned to the interchange network. The interchange network will coordinate the merging of the funds response and the top-up response. It will return a single response to BASE24-atm.

Whenever the routing table contains multiple entries and includes an entry with a value of A (all card prefixes) in this field and a value of AL (all account types) in the ACCT TYPE field, the entry with a value of A in this field and a value of AL in the ACCT TYPE field must follow all of the other entries.

AUTH TYPE — The authorization method to be used for processing transactions that use this table entry.

AUTH LVL — The authorization level to be used for processing transactions that use this table entry.

Using the IDF Routing Table

The BASE24-atm routing table on IDF screen 9 was designed for one entry per DPC, whereby authorization parameters could be controlled individually for each DPC. Institutions can, however, set up multiple entries for the same DPC in order to modify the authorization level or method based on the account type or card prefix being processed.

For example, a DPC might maintain all of the files for a certain card prefix and be set up as authorization level 1 (online) for that prefix. At the same time, it might share authorization responsibilities with BASE24 in all other cases, such as authorization level 3 (online/offline). In this case, one DPC would have two table entries.

An institution might also set up their credit accounts for Negative Authorization with or without Usage Accumulation while they set up their checking and savings accounts for Positive Balance Authorization using the same DPC as a back-end host. Again, one DPC would have two entries.

The routing table allows for virtually any combination of authorization levels and methods, with the exception that an AUTH TYPE field value of 0 (online host processing) always requires a corresponding AUTH LVL field value of 1 (online) and vice versa.

The Authorization process checks the routing table for each financial message it processes and derives its authorization parameters according to the account types and card prefix being processed in the transaction.

Note: The fast cash account type code cannot not be set to 00 if the IDF routing parameters differ by account type.

IDF Routing Table Examples

The following pages provide several examples of routing tables with descriptions of how transactions would be processed under each.

Example 1: Single-Entry Routing Table (One DPC)

If only one host DPC is being used, the IDF routing table can contain a single entry. An example is shown below:

#	PRIMARY DPC	SYMBOLIC NAME	ACCT TYPE	PRFX RTG	AUTH TYPE	AUTH LVL
1	1	P1A^HISF1	AL (all)	A (all)	3 (CAF/PBF)	3 (on/off)

In this example, the Authorization process processes all transactions using authorization level 3 (online/offline) with the Positive Balance Authorization method. It sends messages to P1A^HISF1 (the Host Interface process) for sending to DPC 1. If DPC 1 is unavailable, the Authorization process authorizes or declines the transaction using the Positive Balance Authorization method.

Example 2: Single-Entry Routing Table (No Host)

If a host is not involved in authorizing transactions, the IDF routing table can contain a single entry. An example is shown below.

#	PRIMARY DPC	SYMBOLIC NAME	ACCT TYPE	PRFX RTG	AUTH TYPE	AUTH LVL
1	0	NO HOST ROUTING	AL (all)	A (all)	3 (CAF/PBF)	2 (off)

In this example, the Authorization process processes all transactions using the Positive Balance Authorization method. No messages are sent to a host for authorization.

Example 3: Multiple-Entry Routing Table

To process credit accounts differently from other account types using the same DPC, a routing table must be set up with multiple entries. An example is shown below.

Note: One entry in this table contains default values in the ACCT TYPE and PRFX RTG fields. To avoid the possibility of misrouting transactions, the entry containing two default values (i.e., the ACCT TYPE field is set to AL and the PRFX RTG field is set to A) must be placed last in the table.

#	PRIMARY DPC	SYMBOLIC NAME	ACCT TYPE	PRFX RTG	AUTH TYPE	AUTH LVL
1	1	P1A^HISF1	31 (ccd)	A (all)	1 (NEG/UAF)	2 (off)
2	1	P1A^HISF1	AL (all)	A (all)	0 (no files)	1 (on)

In this example, transactions on credit accounts (entry 1) are authorized offline using the Negative Authorization with Usage Accumulation method. Completions can be sent to the host. Transactions on all non-credit accounts (entry 2) are sent to DPC 1 for online authorization.

Example 4: Multiple-Entry Routing Table

This example contains a number of DPCs with varying parameters.

Note: One entry in this table contains default values in the ACCT TYPE and PRFX RTG fields. To avoid the possibility of misrouted transactions, the entry containing two default values (i.e., the ACCT TYPE field is set to AL and the PRFX RTG field is set to A) must be placed last in the table. The order of other table entries does not matter.

#	PRIMARY DPC	SYMBOLIC NAME	ACCT TYPE	PRFX RTG	AUTH TYPE	AUTH LVL
1	1	P1A^HISF1	01 (chk)	1	3 (CAF/PBF)	3 (on/off)

#	PRIMARY DPC	SYMBOLIC NAME	ACCT TYPE	PRFX RTG	AUTH TYPE	AUTH LVL
2	1	P1A^HISF1	11 (sav)	1	3 (CAF/PBF)	2 (off)
3	2	P1A^HISF1	31 (ccd)	1	0 (no files)	1 (on)
4	3	P1A^HISF1	AL (all)	2	0 (no files)	1 (on)
5	4	P1A^HISF1	AL (all)	3	0 (no files)	1 (on)

In this example, transactions on checking accounts (entry 1) initiated by cards in prefix group 1 are sent to DPC 1 if it is available. If DPC 1 is not available, transactions are authorized offline using the Positive Balance Authorization method.

Transactions on savings accounts (entry 2) initiated by cards in prefix group 1 are authorized offline using the Positive Balance Authorization method. Transactions on credit accounts (entry 3) initiated by cards in prefix group 1 are sent to DPC 2 for online authorization.

All transactions initiated by cards in prefix group 2 (entry 4) are sent to DPC 3 for online authorization. All transactions initiated by cards in prefix group 3 (entry 5) are sent to DPC 4 for online authorization.

Deriving Parameters from the IDF Routing Table

The following pages describe how the Authorization process uses the BASE24-atm IDF routing table to derive authorization and routing parameters for the transactions it processes.

Each time the Authorization process receives a transaction where it maintains an IDF record for the card issuer, it uses the table to derive the authorization and routing parameters it needs for processing the transaction. To do this, it compares the account type and prefix for the transaction to the routing table.

Note: The fast cash account type code cannot not be set to 00 if the IDF routing parameters differ by account type.

For one-sided transactions, such as withdrawals or deposits, the Authorization process derives a set of primary authorization and routing parameters.

For two-sided transactions, such as transfers, the Authorization process derives a set of primary and secondary authorization and routing parameters. In this case, the primary parameters are used to process the *from* side of the transaction and the secondary parameters are used to process the *to* side of the transaction.

Selection Hierarchy

The routing table in the IDF specifies the authorization parameters according to the account type involved in the transaction and the prefix group to which the card belongs.

To derive the routing and authorization parameters required for processing a transaction, the Authorization process compares the account type and prefix for the transaction to the routing table and chooses an entry based on the following hierarchy:

Preference	Account Type	Prefix Group
1	match	match
2	match	default
3	default	match
4	default	default

When the Authorization process finds an entry containing account type and prefix values that satisfy the requirements of the transaction, the Authorization process uses the parameters for that entry. If the Authorization process cannot find an entry based on account type and prefix values, the transaction cannot be processed.

IDF Routing Table Selection Examples

The following pages contain transaction examples to illustrate how the Authorization process selects the routing table entries it uses for processing.

The routing table used for these transaction examples is the routing table described in example 4 on the previous pages. This table is shown again below:

#	PRIMARY DPC	SYMBOLIC NAME	ACCT TYPE	PRFX RTG	AUTH TYPE	AUTH LVL
1	1	P1A^HISF1	01 (chk)	1	3 (CAF/PBF)	3 (on/off)
2	1	P1A^HISF1	11 (sav)	1	3 (CAF/PBF)	2 (off)
3	2	P1A^HISF1	31 (ccd)	1	0 (no files)	1 (on)
4	3	P1A^HISF1	AL (all)	2	0 (no files)	1 (on)
5	4	P1A^HISF1	AL (all)	3	0 (no files)	1 (on)

Note: This routing table is set up specifically for 3 prefix groups. If the Authorization process received a transaction where the prefix group was other than 1, 2, or 3, the transaction would be declined because no applicable entry would be found in the table.

Example 1: Credit Account Withdrawal — Prefix Group 1

The following illustrates how the Authorization process would derive authorization and routing parameters for a credit account withdrawal initiated by a card in prefix group 1.

1. The Authorization process starts by searching for its first preference.

Preference	Account Type	Prefix Group
1	Exact Match	Exact Match
2	Exact Match	Default
3	Default	Exact Match
4	Default	Default

The Authorization process searches for an exact match on Account Type (31) and Prefix Group (1). Entry 3 matches exactly on Account Type and Prefix Group.

#	PRIMARY DPC	SYMBOLIC NAME	ACCT TYPE	PRFX RTG	AUTH TYP	AUTH LVL
1	1	P1A^HISO1	01	1	3	3
2	1	P1A^HISO1	11	1	3	2
3	2	P1A^HISO1	31	1	0	1
4	3	P1A^HISO1	AL	2	0	1
5	4	P1A^HISO1	AL	3	0	1

- The Authorization process would select entry 3 for its primary authorization and routing parameters.

3	2	P1A^HISO1	31	1	0	1
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Example 2: Savings Deposit—Prefix Group 2

The following illustrates how the Authorization process would derive authorization and routing parameters for a savings deposit initiated by a card in prefix group 2.

- The Authorization process starts by searching for its first preference.

Preference	Account Type	Prefix Group
1	Exact Match	Exact Match
2	Exact Match	Default
3	Default	Exact Match
4	Default	Default

The Authorization process searches for, but cannot find, an exact match on Account Type (11) and Prefix Group (2).

#	PRIMARY DPC	SYMBOLIC NAME	ACCT TYPE	PRFX RTG	AUTH TYP	AUTH LVL
1	1	P1A^HISO1	01	1	3	3

2	1	P1A^HISO1	11	1	3	2
3	2	P1A^HISO1	31	1	0	1
4	3	P1A^HISO1	AL	2	0	1
5	4	P1A^HISO1	AL	3	0	1

2. Since its first preference could not be found, the Authorization process searches for its second preference.

Preference	Account Type	Prefix Group
1	Exact Match	Exact Match
2	Exact Match	Default
3	Default	Exact Match
4	Default	Default

The Authorization process searches for, but cannot find, an exact match on Account Type (11) and Prefix Group default (A).

#	PRIMARY DPC	SYMBOLIC NAME	ACCT TYPE	PRFX RTG	AUTH TYP	AUTH LVL
1	1	P1A^HISO1	01	1	3	3
2	1	P1A^HISO1	11	1	3	2
3	2	P1A^HISO1	31	1	0	1
4	3	P1A^HISO1	AL	2	0	1
5	4	P1A^HISO1	AL	3	0	1

3. Since its second preference could not be found, the Authorization process searches for its third preference.

Preference	Account Type	Prefix Group
1	Exact Match	Exact Match
2	Exact Match	Default
3	Default	Exact Match
4	Default	Default

The Authorization process searches for an Account Type default (AL) and an exact match on Prefix Group (2). Entry 4 matches the default on Account Type and exactly on Prefix Group.

#	PRIMARY DPC	SYMBOLIC NAME	ACCT TYPE	PRFX RTG	AUTH TYP	AUTH LVL
1	1	P1A^HISO1	01	1	3	3
2	1	P1A^HISO1	11	1	3	2
3	2	P1A^HISO1	31	1	0	1
4	3	P1A^HISO1	AL	2	0	1
5	4	P1A^HISO1	AL	3	0	1

4. The Authorization process would select entry 4 for its primary authorization and routing parameters.

4	3	P1A^HISO1	AL	2	0	1
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Example 3: Payment From Checking to Credit Account — Prefix Group 1

The following steps illustrate how the Authorization process would derive authorization and routing parameters for a payment from a checking account to a credit account initiated by a card in prefix group 1.

1. The Authorization process derives its *from*-side parameters first. To do this, it starts by searching for its first preference.

Preference	Account Type	Prefix Group
1	Exact Match	Exact Match
2	Exact Match	Default
3	Default	Exact Match
4	Default	Default

The Authorization process searches for an exact match on Account Type (01) and Prefix Group (1). Entry 1 matches exactly on Account Type and Prefix Group.

#	PRIMARY DPC	SYMBOLIC NAME	ACCT TYPE	PRFX RTG	AUTH TYP	AUTH LVL
1	1	P1A^HISO1	01	1	3	3
2	1	P1A^HISO1	11	1	3	2
3	2	P1A^HISO1	31	1	0	1
4	3	P1A^HISO1	AL	2	0	1
5	4	P1A^HISO1	AL	3	0	1

2. The Authorization process would select entry 1 for its primary (*from-side*) authorization and routing parameters.

1	1	P1A^HISO1	01	1	3	3
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3. The Authorization process then derives its *to-side* parameters. To do this, it starts by searching for its first preference.

Preference	Account Type	Prefix Group
1	Exact Match	Exact Match
2	Exact Match	Default
3	Default	Exact Match
4	Default	Default

The Authorization process searches for an exact match on Account Type (31) and Prefix Group (1). Entry 3 matches exactly on Account Type and Prefix Group.

#	PRIMARY DPC	SYMBOLIC NAME	ACCT TYPE	PRFX RTG	AUTH TYP	AUTH LVL
1	1	P1A^HISO1	01	1	3	3
2	1	P1A^HISO1	11	1	3	2
3	2	P1A^HISO1	31	1	0	1
4	3	P1A^HISO1	AL	2	0	1

5	4	P1A^HISO1	AL	3	0	1
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4. The Authorization process would select entry 3 for its secondary (*to-side*) authorization and routing parameters.

3	2	P1A^HISO1	31	1	0	1
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Network Routing and Surcharging

The BASE24-atm Authorization process uses a method similar to the one it uses for routing to search the Surcharge File (SURF) for the appropriate surcharge fee.

The Authorization process uses information from the BASE24-atm Terminal Data files and CPF or SPROUTE File (SPROUTE) to find a surcharge profile in the SURF. The Authorization process looks for exact matches to the key to the SURF. The key to SURF records is a combination of the TERM PROFILE field and either the CARD PROFILE or the CARD FIID field. The value in the CARD PROFILE field is used when the card has an unrecognized prefix; the value in the CARD FIID field is used when the card has a recognized prefix. Refer to the **BASE24 Base Files Maintenance Manual** for more information about the SURF.

For transactions routed using SPROUTE (that is, the card has an unrecognized prefix), the Authorization process searches the SURF for a record with key fields that match the SURCHARGE PROFILE field on ATD screen 1 and the primary destination and card type from SPROUTE. For transactions routed using the CPF (that is, the card has a recognized prefix), the Authorization process searches the SURF for a record with key fields that match the SURCHARGE PROFILE field on ATD screen 1 and the FIID and CARD PROFILE fields on CPF screen 1.

The Authorization process first searches the SURF for an exact match of all three key elements. If no match is found, the Authorization process searches on different variations of the key using wild card values. The search combinations are as follows:

Step	Element		
	ATD SURCHARGE PROFILE	SPROUTE Primary Routing Destination/ CPF FIID Field	SPROUTE Card Type/CPF CARD PROFILE Field
1	Exact	Exact	Exact
2	Exact	Exact	Wild card
3	Exact	Wild card	Exact
4	Exact	Wild card	Wild card
5	Wild card	Exact	Exact
6	Wild card	Wild card	Exact

Step	Element		
	ATD SURCHARGE PROFILE	SPROUTE Primary Routing Destination/ CPF FIID Field	SPROUTE Card Type/CPF CARD PROFILE Field
7	Wild card	Exact	Wild card
8	Wild card	Wild card	Wild card

The wild card value in each case is all asterisks (*). Wild card surcharging profiles can be configured when setting up the SURF. If the Authorization process cannot find a match in the SURF, no surcharge fee is assessed.

Note: A blank BASE24-atm Terminal Data files surcharge profile or CPF card profile indicates that no surcharge fee is assessed for the ATM or card, respectively. However, a blank card type in SPROUTE is considered a generic card type, and the Authorization process looks for a match in the SURF.

APCF Routing

Device Handler processes and Interchange Interface processes using enhanced transactions allowed processing can use the Acquirer Processing Code File (APCF) extended memory table to specify an authorization destination other than the Authorization process specified in the BASE24-atm Terminal Data files (ATD) or Enhanced Interchange Configuration File (ICFE). This mechanism allows you to route transactions received from an ATM or interchange to a customer-specific application process in the XPNET system. The customer-specific authorization destination is specified in the AUTHORIZATION DESTINATION field on APCF screen 2.

When the Device Handler or Interchange Interface process receives a request from the ATM or interchange, respectively, it reads the AUTH-DEST field in the APCF extended memory table. If the field contains blanks, the transaction is sent to the Authorization process indicated in the ATD or ICFE. If the field contains a value, the Device Handler or Interchange Interface process sends the request to the process specified in the field.

The following Device Handler processes and Interchange Interface processes can use APCF routing:

- Diebold 10XX/478X Device Handler process
- NCR 5XXX Device Handler process
- Banknet Interchange Interface process
- BIC ISO Interface process
- MDS/MDSM Interchange Interface process
- PLUS ISO Interchange Interface process
- VisaNet Interchange Interface process

Logging Transactions to the TLF

When you specify APCF routing for a transaction, you also can specify whether the transaction response should be logged to the Transaction Log File (TLF). This is set using the LOG AUTH DEST RESPONSE field on APCF screen 2. When the Device Handler or Interchange Interface process receives a response from a customer-specific process, it uses the value in LOG AUTH DEST RESPONSE field to determine whether to send a copy of the response message to the Authorization process specified in the AUTH PROCESS field on ATD screen 1 or

ICFE screen 8 for logging to the TLF. If the value of the LOG AUTH DEST RESPONSE field is Y, the response message is logged to the TLF. If the field contains a value of N, the response message is not logged to the TLF.

Refer to section 3 for more information on how Device Handler processes or Interchange Interface processes access the APCF extended memory table. Refer to section 5 for transactions flows for transactions using APCF routing.

Split Transaction Routing

The Diebold 10XX and NCR 5XXX Device Handler processes can send mobile top-up transactions to the Transaction Context Manager process to split route the transactions between multiple authorizers. In an enhanced mobile top-up routing environment, the Device Handler process sends the transaction to the BASE24-atm Authorization process, which will send the transaction to a network interchange to split route the transaction.

The Transaction Context Manager process splits the transaction routing to different destinations. In the case of standard mobile top-up processing, mobile top-up with funds transaction routing is split between the top-up authorizer and the funds authorizer. The Transaction Context Manager process uses the routing hierarchy specified in the Split Transaction Routing File (STRF) to determine the order to send the transaction message to the authorizing entities. The Transaction Context Manager process uses the Split Transaction Routing token to keep track of the transaction routing. The Transaction Context Manager process updates the TXN-RESP-IND, FUNDS-AUTH-STAT, and SCND-SVC-STAT fields in the Split Transaction Routing token each time it passes the transaction to either the funds authorizer or the top-up authorizer.

5: BASE24-atm Message Flows

Transactions that route to or through the BASE24-atm system for authorization do so in a series of transaction messages. This section provides examples of transaction message flows through the BASE24-atm system under a variety of processing situations. The transaction message flow diagrams are organized as follows:

- BASE24-atm Transactions: Authorization Level 1
- BASE24-atm Transactions: Authorization Level 2
- BASE24-atm Transactions: Authorization Level 3
- Interchange Transactions: Requests Sent to an Interchange
- Interchange Transactions: Adjustments from an Interchange
- Multinodal Transactions
- Administrative Transaction
- APCF-Routed Transactions
- Mobile Top-Up Transactions

Note: This section does not include all possible transaction message flows, only a representative sampling. The examples included are intended to familiarize you with typical message flows and give you a sense of how messages move through the BASE24-atm system under different types of processing situations.

Diagram Information

The following provides some general information on the transaction message flow diagrams in this section.

Message Types

The message types in the following table are used in the transaction message flow diagrams in this section.

Type	Description
0200	Financial transaction request
0205	Statement print transaction request
0210	Financial transaction request response
0215	Statement print transaction response
0220	Financial transaction advice
0230	Financial transaction advice response
0420	Reversal
0430	Financial transaction reversal advice response
5400	Financial transaction correction
9901	Surcharge notification
9906	Account data selection
9907	Card review account selection
9909	Mobile top-up notification/selection
9980	Informational (money in drawer)
9991	Log-only transaction

Special Diagram Notations

In addition to message types, the following special notations are used in the transaction message flow diagrams in this section.

Notation	Description
RQST	Authorization request initiated at an ATM connected to BASE24-atm. The Device Handler process reformats the native-mode language of the ATM into STM request messages (0200-series).
RESP	Authorization response sent to an ATM connected to BASE24-atm. The Device Handler process reformats the internal response message into the native-mode language of the ATM.
COMP	Transaction completion message generated by an ATM connected to BASE24-atm. The Device Handler process uses this message, in the native-mode language of the ATM, to determine whether a reversal (0420) transaction must be generated.
STM	The BASE24-atm Standard Internal Message (STM) format used to communicate within BASE24-atm.
AMEN	Selection menu sent to the ATM to allow cardholders to select from multiple accounts during a transaction. The Device Handler process reformats the data in an Account Data message (9906) to the native-mode language of the ATM.
ASEL	Message sent by the ATM containing accounts selected by the cardholder for a multiple account transaction. The Device Handler process uses this message to create an STM request message (0200) containing the selected account numbers.
FAI	A message failure, the result of a no-response situation with an external entity, such as a device host interchange.
IRQST	Authorization request sent to an interchange or a request from an interchange sent to BASE24-atm. If the message is being sent to the interchange, the Interchange Interface process reformats STM request messages (0200-series) into requests that are understandable to the interchange. If the request is coming from an interchange, the Interchange Interface process translates the message into a 0200-series message.

Notation	Description
IRESP	Authorization response sent to BASE24-atm from an interchange or a response sent to an interchange from BASE24-atm. If the response is being sent to an interchange, the Interchange Interface process reformats these responses into responses that are understandable to the interchange. If the response is being sent to BASE24-atm, the Interchange Interface process reformats these responses into STM 0210 response messages.
IADJ	Transaction adjustment sent by an interchange to BASE24-atm. The Interchange Interface process reformats these adjustments into adjustments understandable to BASE24-atm.
IADV	Transaction advice sent to an interchange. The Interchange Interface process generates transaction advices to interchanges.
IRVSL	Transaction reversal sent to an interchange. The Interchange Interface process reformats STM messages (0420-series) into messages understandable to the interchange.
TMEN	Selection menu sent to the ATM to allow cardholders to select from telco information during a transaction. The Device Handler process reformats the data in a Mobile top-up notification/selection message (9909) to the native-mode language of the ATM.
TSEL	Message sent by the ATM containing information selected by the cardholder for a message-based Mobile top-up transaction. The Device Handler process uses this message to create an STM request message (0200) containing the selected information.
TRQST	Telco Interface information-only request.
TRESP	Telco Interface information-only response.

In the illustrations, the term “Authorization process” has been shortened to “Auth process” to conserve space.

Processing Assumptions

Several processing assumptions have been made for the transaction message flow diagrams in this section. These assumptions are described below.

Dial-Up ATM

Unless specified, transaction message flow diagrams in this section assume that dial-up ATMs are not in use.

ISO Host Interface

All of the transaction message flow diagrams in this section assume the use of an ISO Host Interface process.

Host Multithreading/Text-Level Acknowledgments

Unless specifically stated to the contrary, all of the transaction message flow diagrams in this section assume that hosts are multithreaded and are not configured to provide or require text-level acknowledgments for the messages they receive and send.

Note: Single-threaded hosts and multithreaded hosts can be configured in the Host Configuration File (HCF) to respond with text-level acknowledgments or responses to each message they receive (for example, 0230 to a 0220, or 0430 to a 0420).

Completions Required

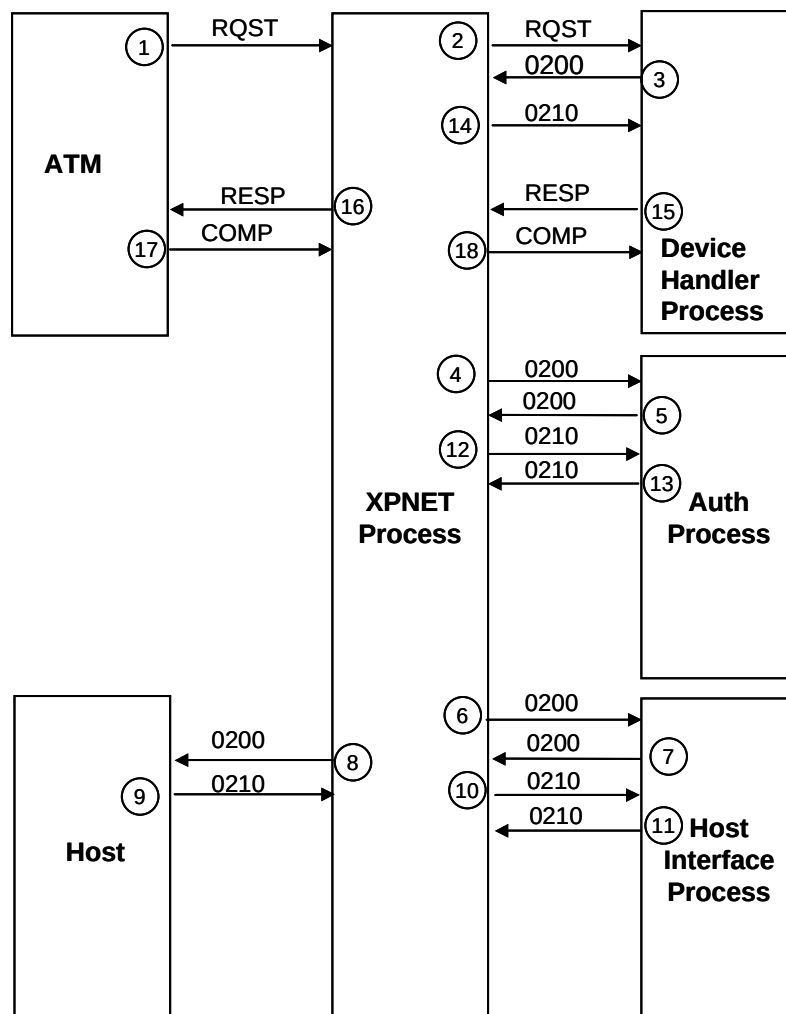
Most of the transaction examples in this section involving a back-end, issuer host indicate whether or not completions to the host are required. The COMPLETION REQUIRED TO HOST field on IPCF screen 2 controls the BASE24-atm Authorization process generation of 0220 messages to a back-end issuer host. If completions are not required, this field is set to 0. Otherwise, this field is set to a nonzero value. Refer to the message flows throughout this section and Appendix A for more information about the values used for the COMPL-REQ field.

BASE24-atm Transactions: Authorization Level 1

The following pages contain message flows for transactions being processed by BASE24-atm using level 1 (online) authorization processing. This authorization level requires that a host authorize all transactions. BASE24-atm performs several transaction screening checks, but only a host can authorize transactions. If a host is unavailable, transactions are declined by BASE24-atm.

Approved by Host — Completions Not Required

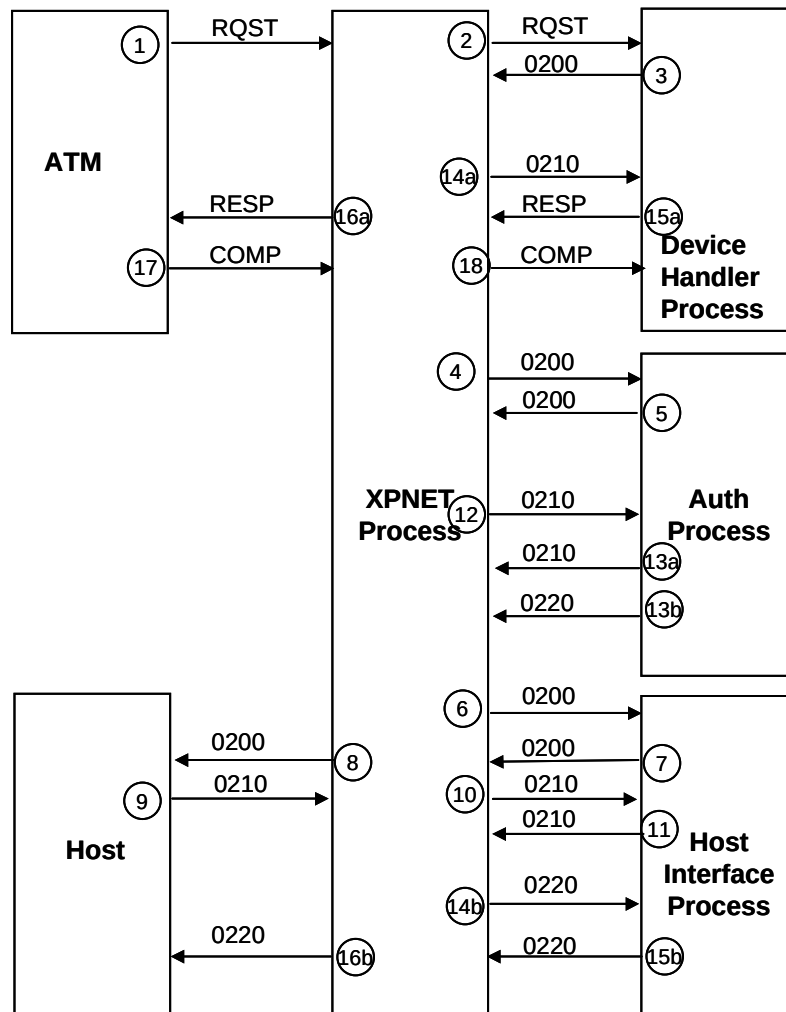
The diagram below illustrates the flow of a transaction that originates at an ATM and is approved by a host. The host does not require completions.



Steps	Processing
1, 2	The ATM sends a native-mode request message to the Device Handler process.
3, 4	The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Authorization process.
5, 6	From the Card Prefix File (CPF), the Authorization process locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 1. The Authorization process sends the 0200 request message to the Host Interface process.
7, 8	The Host Interface process reformats the internal message into an external message, sets an outbound request timer, and sends the 0200 request message to the host.
9, 10	The host approves the request and returns a 0210 response message to the Host Interface process.
11, 12	The Host Interface process deletes its outbound request timer, reformats the external message into an internal message, and sends the 0210 response message to the Authorization process.
13, 14	The Authorization process logs the response to the Transaction Log File (TLF) and sends the 0210 response message to the Device Handler process.
15, 16	The Device Handler process deletes its authorization timer and reformats the 0210 response message into a native-mode response that it sends to the ATM.
17, 18	The ATM sends a native-mode message to the Device Handler process to indicate that the transaction completed as authorized.

Approved by Host — Completions Required

The diagram below illustrates the flow of a transaction that originates at an ATM and is approved by a host. The host requires completions.



Steps

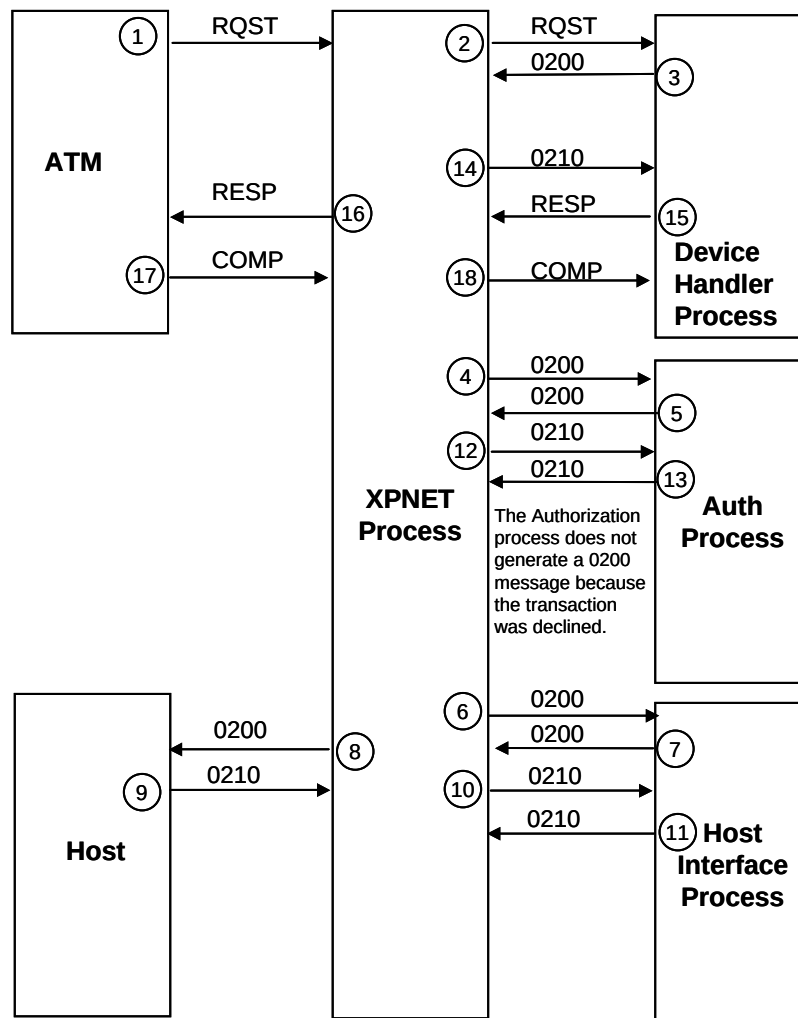
Processing

- 1, 2 The ATM sends a native-mode request message to the Device Handler process.
- 3, 4 The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Authorization process.

Steps	Processing
5, 6	From the Card Prefix File (CPF), the Authorization process locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 1. The Authorization process sends the 0200 request message to the Host Interface process.
7, 8	The Host Interface process reformats the internal message into an external message, sets an outbound request timer, and sends the 0200 request message to the host.
9, 10	The host approves the request and returns a 0210 response message to the Host Interface process.
11, 12	The Host Interface process reformats the external message into an internal message, deletes its outbound request timer, and sends the 0210 response message to the Authorization process.
13, 14	The Authorization process performs the following steps: a. Logs the transaction to the Transaction Log File (TLF) and sends the 0210 response message to the Device Handler process. Processing continues with step 15a. b. Sends a 0220 completion message to the Host Interface process when the COMPLETION REQUIRED TO HOST field on the Issuer Processing Code File (IPCF) screen is set to 1 or 4. Processing continues with step 15b.
15a, 16a	The Device Handler process deletes its authorization timer and reformats the 0210 response message into a native-mode response that it sends to the ATM. Processing continues with step 17.
15b, 16b	The Host Interface process reformats the internal message into an external message and sends the 0220 completion message to the host when the COMPLETION REQUIRED TO HOST field on the IPCF screen is set to 1 or 4. This leg of processing is now complete.
17, 18	The ATM sends a native-mode message to the Device Handler process to indicate that the transaction completed as authorized.

Declined by Host — Completions Required

The diagram below illustrates the flow of a transaction that originates at an ATM and is declined by a host. The host requires completions.



Steps

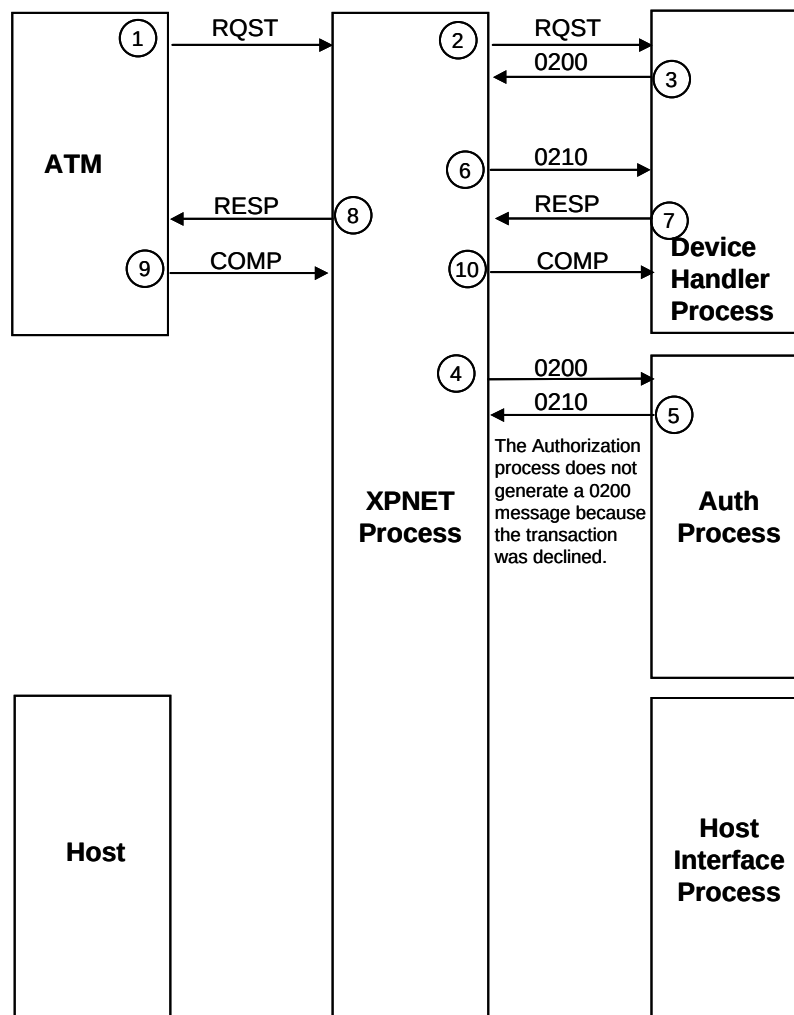
Processing

- | | |
|------|---|
| 1, 2 | The ATM sends a native-mode request message to the Device Handler process. |
| 3, 4 | The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Authorization process. |

Steps	Processing
5, 6	From the Card Prefix File (CPF), the Authorization process locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 1. The Authorization process sends the 0200 request message to the Host Interface process.
7, 8	The Host Interface process reformats the internal message into an external message, sets an outbound request timer, and sends the 0200 request message to the host.
9, 10	The host declines the request and returns a 0210 response message to the Host Interface process.
11, 12	The Host Interface process reformats the external message into an internal message, deletes its outbound request timer, and sends the 0210 response message to the Authorization process.
13, 14	The Authorization process logs the response to the Transaction Log File (TLF) and sends the 0210 response message to the Device Handler process. The Authorization process generates a 0220 completion message when the COMPLETION REQUIRED TO HOST field on the IPCF screen is set to 4.
15, 16	The Device Handler process deletes its authorization timer and reformats the 0210 response message into a native-mode response that it sends to the ATM.
17, 18	The ATM sends a native-mode message to the Device Handler process to indicate that the declined transaction was completed.

Declined by BASE24-atm —Transaction Fails Prescreening Checks — Completions Required

The diagram below illustrates the flow of a transaction that originates at an ATM and fails the prescreening checks performed by the Authorization process. The host requires completions.



Steps

Processing

- | | |
|------|---|
| 1, 2 | The ATM sends a native-mode request message to the Device Handler process. |
| 3, 4 | The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Authorization process. |

Steps**Processing**

5, 6

From the Card Prefix File (CPF), the Authorization process locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 1 with prescreening checks by BASE24-atm. The request fails the prescreening checks performed by the Authorization process. The Authorization process logs the response to the Transaction Log File (TLF) and sends a 0210 response message to the Device Handler process.

The Authorization process generates a 0220 completion message when the COMPLETION REQUIRED TO HOST field on the IPCF screen is set to 4 or 6.

7, 8

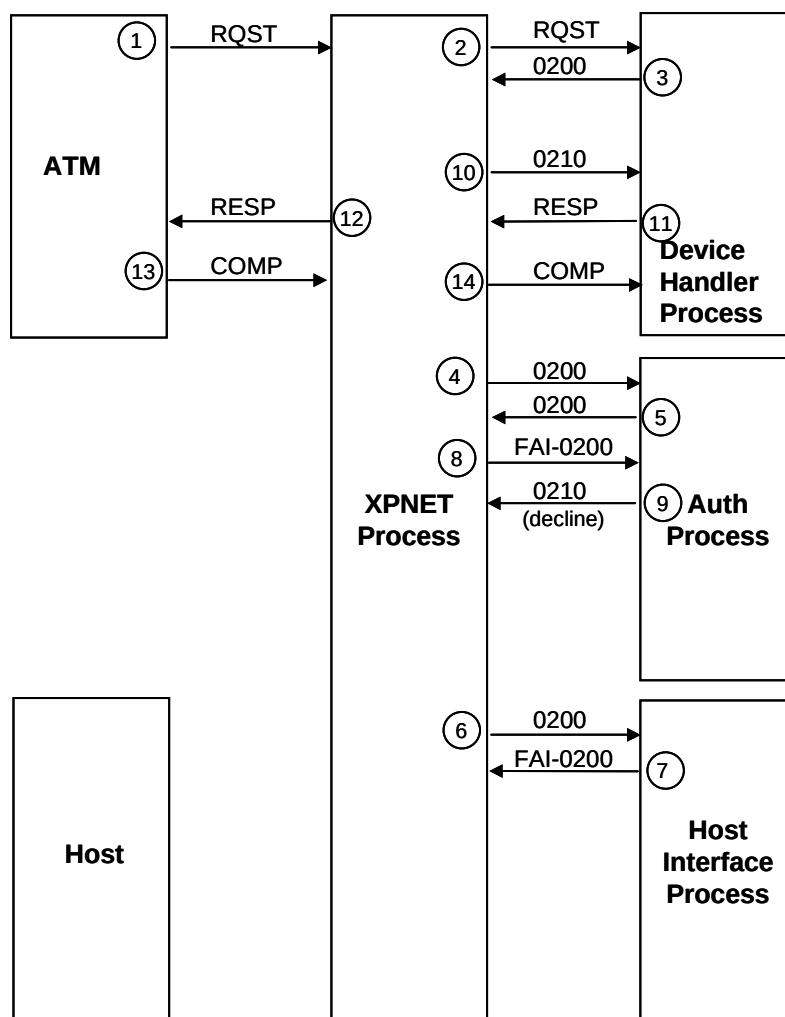
The Device Handler process deletes its authorization timer and reformats the 0210 response message into a native-mode response that it sends to the ATM.

9, 10

The ATM sends a native-mode message to the Device Handler process to indicate that the declined transaction was completed.

Declined by BASE24-atm — Host Not Available — Completions Required

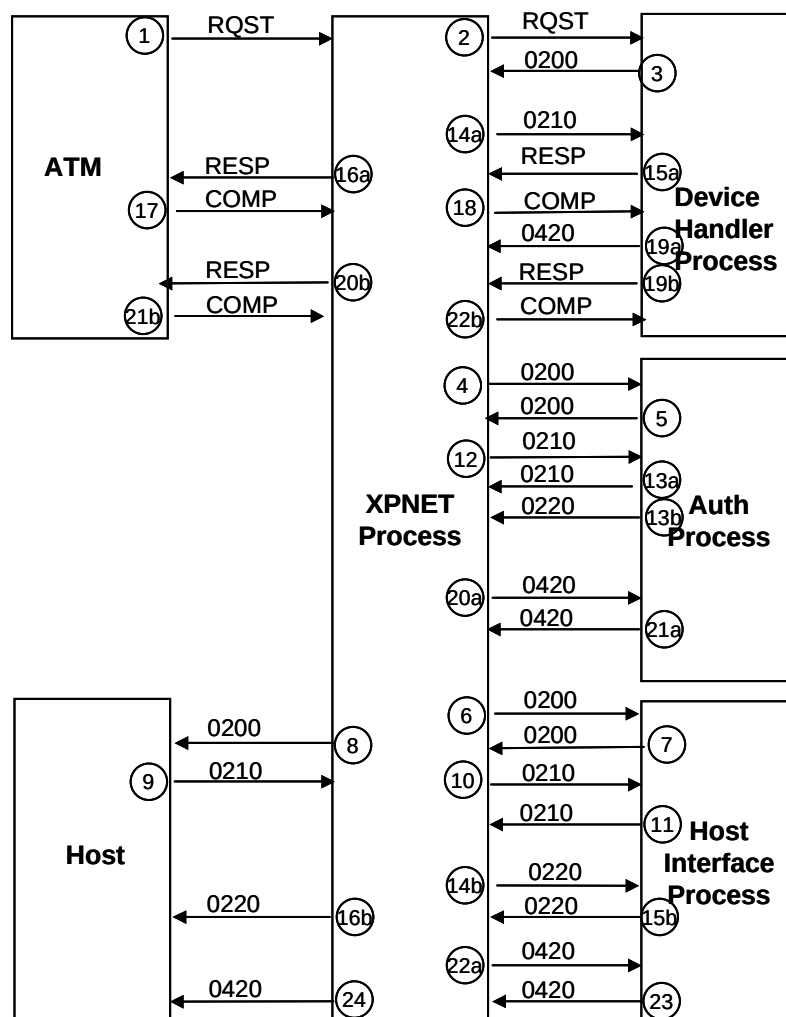
The diagram below illustrates the flow of a transaction that originates at an ATM and is declined by BASE24-atm because the host is not available. The host requires completions. Prior to this transaction, the BASE24 system became aware that the host was down and marked the host as down.



Steps	Processing
1, 2	The ATM sends a native-mode request message to the Device Handler process.
3, 4	The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Authorization process.
5, 6	From the Card Prefix File (CPF), the Authorization process locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 1. The Authorization process sends the 0200 request message to the Host Interface process.
7, 8	Because the host is unavailable, the Host Interface process changes the 0200 to a failed 0200 request message and returns it to the Authorization process.
9, 10	The Authorization process changes the failed 0200 to a 0210 response message that declines the request, logs the response to the Transaction Log File (TLF), and sends it to the Device Handler process. The Authorization process generates a 0220 completion message when the COMPLETION REQUIRED TO HOST field on the IPCF screen is set to 4 or 6.
11, 12	The Device Handler process deletes its authorization timer and reformats the 0210 response message into a native-mode response that it sends to the ATM.
13, 14	The ATM sends a native-mode message to the Device Handler process to indicate that the declined transaction was completed.

Approved by Host — Transaction Reversed — Completions Required

The diagram below illustrates the flow of a transaction that originates at an ATM, is approved by a host, and is then reversed. The host requires completions.



Steps

Processing

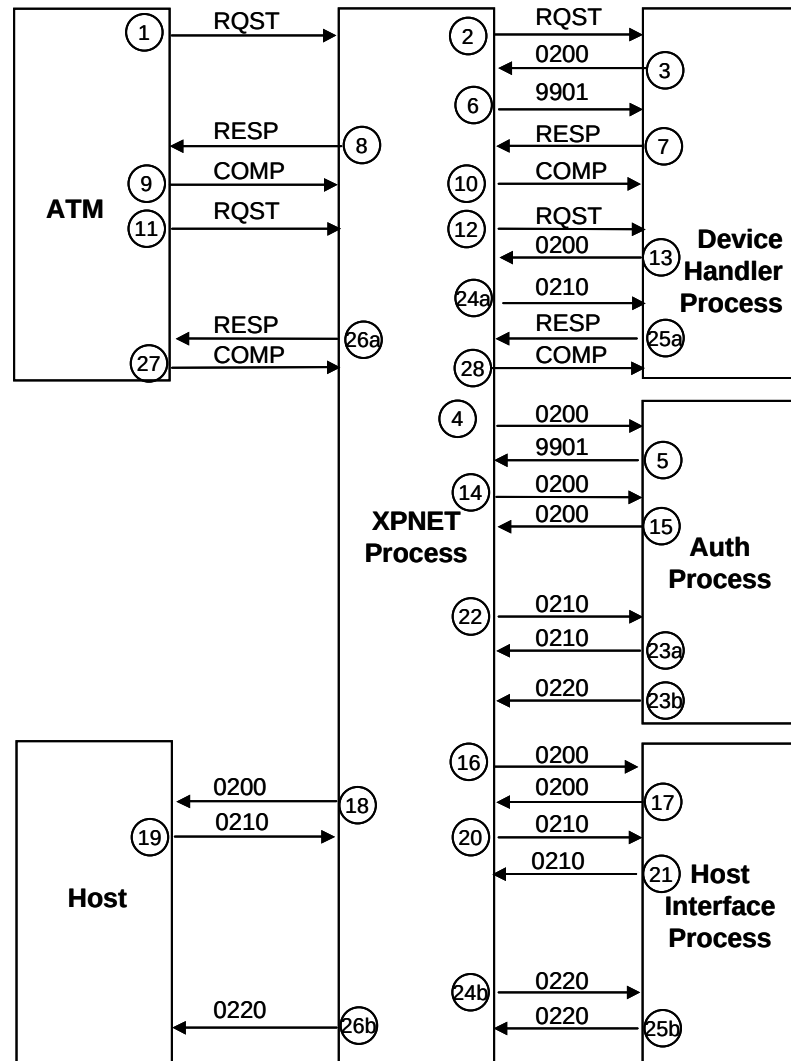
- 1, 2 The ATM sends a native-mode request message to the Device Handler process.
- 3, 4 The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Authorization process.

Steps	Processing
5, 6	From the Card Prefix File (CPF), the Authorization process locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 1. The Authorization process sends the 0200 request message to the Host Interface process.
7, 8	The Host Interface process reformats the internal message into an external message, sets an outbound request timer, and sends the 0200 request message to the host.
9, 10	The host approves the request and returns a 0210 response message to the Host Interface process.
11, 12	The Host Interface process reformats the external message into an internal message, deletes its outbound request timer, and sends the 0210 response message to the Authorization process.
13, 14	The Authorization process performs the following steps: a. Logs the response to the Transaction Log File (TLF) and sends the 0210 response to the Device Handler process. Processing continues with step 15a. b. Sends a 0220 completion message to the Host Interface process when the COMPLETION REQUIRED TO HOST field on the IPCF screen is set to 1 or 4. Processing continues with step 15b.
15a, 16a	The Device Handler process deletes its authorization timer and reformats the 0210 response message into a native-mode response that it sends to the ATM. Processing continues with step 17.
15b,16b	The Host Interface process reformats the internal message into an external message and sends the 0220 completion message to the host when the COMPLETION REQUIRED TO HOST field on the IPCF screen is set to 1 or 4. This leg of processing is now complete.
17, 18	The ATM sends a native-mode message to the Device Handler process to indicate that the transaction did not complete as authorized.

Steps	Processing
19, 20	The Device Handler process performs the following steps: a. Generates a 0420 reversal message and sends it to the Authorization process. Processing continues with step 21a. b. Sends a native-mode response to the ATM indicating to print the receipt for however the transaction was completed. Processing continues with step 21b.
21a, 22a	The Authorization process logs the reversal to the TLF and sends the 0420 reversal message to the Host Interface process. Processing continues with step 23.
21b, 22b	The ATM sends a native-mode message to the Device Handler process indicating that the transaction has been completed. This leg of processing is now complete.
23, 24	The Host Interface process reformats the internal message into an external message and sends the 0420 reversal message to the host.

Approved by Host — Surcharge Applied — Completions Required

The diagram below illustrates the flow of a transaction that originates at an ATM, has an acquirer fee (surcharge) assessed, and is approved by the host. The host requires completions. This scenario occurs with the standard surcharging application (interactive request notification processing).



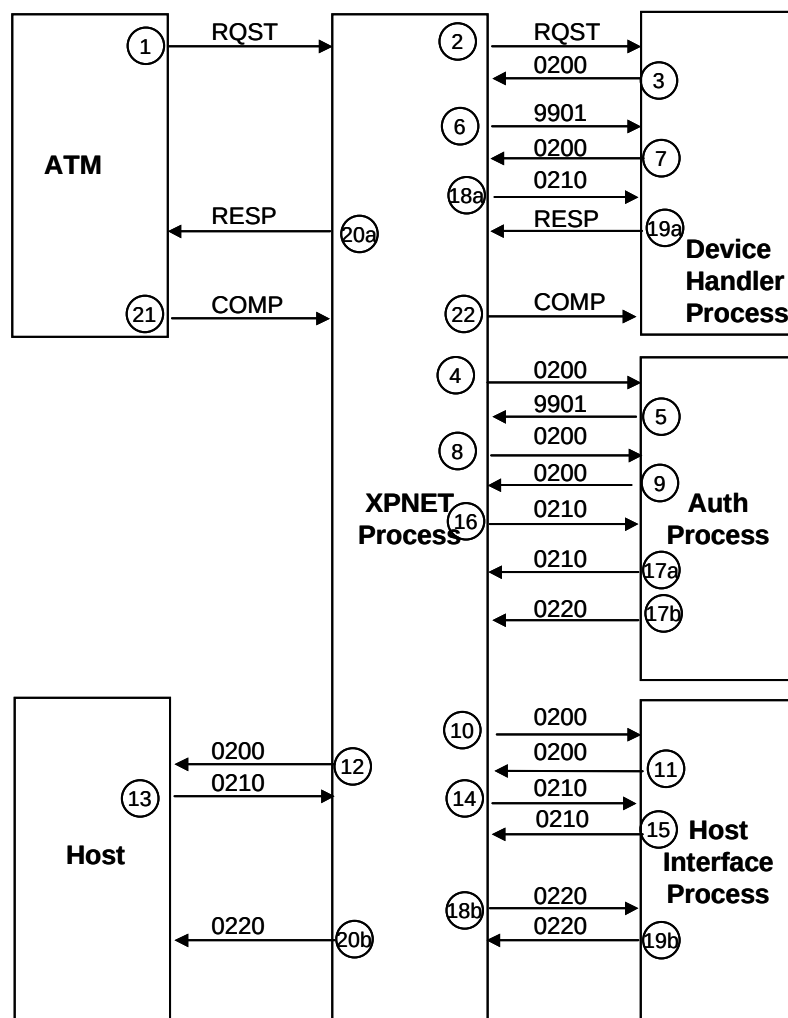
Steps	Processing
1, 2	The ATM sends a native-mode request message to the Device Handler process.
3, 4	The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Authorization process. The message contains the Surcharge Data token with the surcharge profile ID from the BASE24-atm Terminal Data files.
5, 6	The Authorization process uses the surcharge profile ID and the card profile in the Card Prefix File (CPF) to determine the surcharge to be assessed. The Authorization process then formats a 9901 surcharge notification and sends it to the Device Handler process.
7, 8	The Device Handler process formats a native-mode response indicating to the ATM that it needs to ask the cardholder for acceptance of the surcharge.
9, 10	The ATM sends a native-mode completion message to the Device Handler process.
11, 12	The ATM sends a request to the Device Handler process indicating that the customer has accepted the surcharge.
13, 14	The Device Handler process formats a 0200 message with the Surcharge Data token that includes the surcharge amount and sends it to the Authorization process.
15, 16	The Authorization process determines the authorization and routing parameters and sends the 0200 request message (which includes the Surcharge Data token and the surcharge amount) to the Host Interface process.
17, 18	The Host Interface process reformats the internal message into an external message, sets an outbound request timer, and sends the 0200 request message to the host. The message includes the Surcharge Data token with the surcharge amount. The surcharge is also reflected in the transaction amount (the AMT-1 field of the STM).
19, 20	The host approves the request and returns a 0210 response message to the Host Interface process.

Steps	Processing
21, 22	The Host Interface process reformats the external message into an internal message, deletes its outbound request timer, and sends the 0210 response message to the Authorization process. The message includes the Surcharge Data token with the surcharge amount.
23, 24	The Authorization process performs the following steps: - Formats a Transaction Log File (TLF) record, writes it to the TLF, and sends the 0210 response message to the Device Handler process. The message includes the Surcharge Data token with the surcharge amount. Processing continues with step 25a. The token information may or may not be logged to the TLF, depending on the configuration of the Token File (TKN). - Sends a 0220 completion message to the Host Interface process when the COMPLETION REQUIRED TO HOST field on the IPCF screen is set to 1 or 4. Processing continues with step 25b.
25a, 26a	The Device Handler process deletes its authorization timer and reformats the 0210 response message into a native-mode response that it sends to the ATM. The response contains the surcharge information mapped into the receipt data using the Receipt File (RCPT). Processing continues with step 27.
25b, 26b	The Host Interface process reformats the internal message into an external message and sends the 0220 completion message to the host when the COMPLETION REQUIRED TO HOST field on the IPCF screen is set to 1 or 4. This leg of processing is now complete.
27, 28	The ATM sends a native-mode message to the Device Handler process indicating that the transaction has been completed.

Approved by Host — Surcharge Applied, Noninteractive Message Sequence — Completions Required

The diagram below illustrates the flow of a transaction that originates at a dial-up ATM, has an acquirer fee (surcharge) assessed using noninteractive surcharging, and is approved by the host. The host requires completions. This scenario applies to the standard surcharging application (interactive request notification processing).

Note: The dial-up ATM feature is available with the Diebold 10XX/478X and NCR 5XXX Device Handler processes only.



Steps

Processing

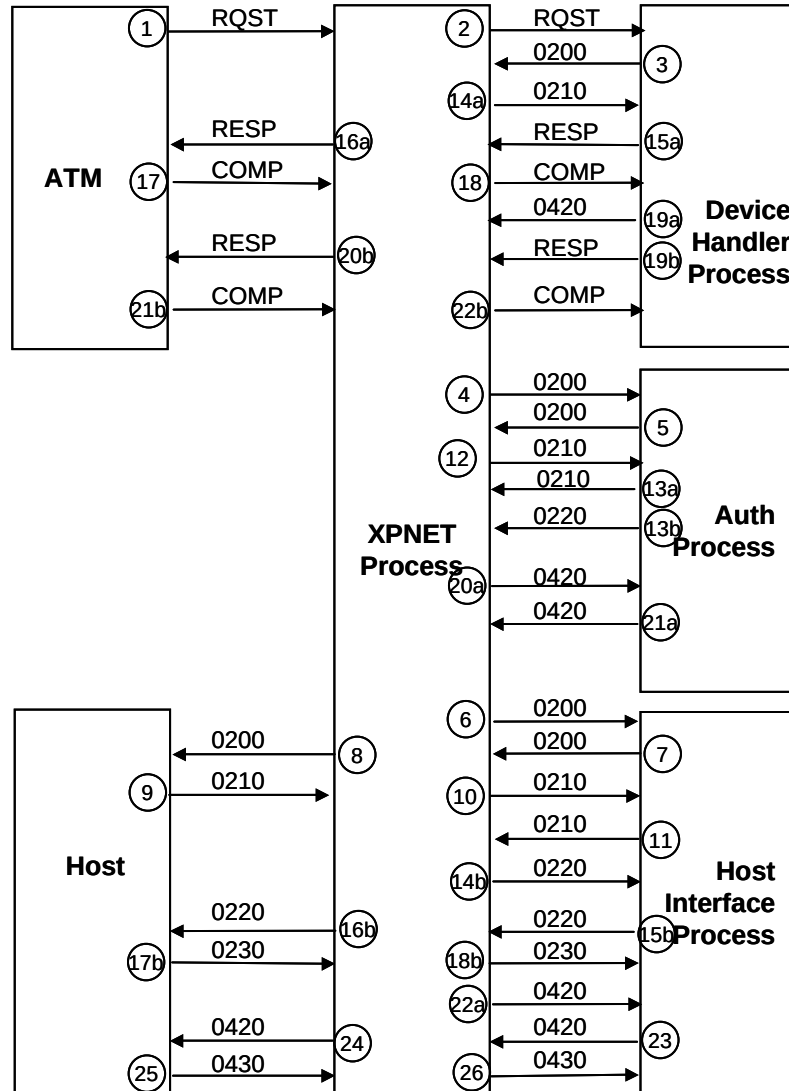
- 1, 2 The ATM sends a native-mode request message to the Device Handler process.

Steps	Processing
3, 4	The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Authorization process. The message contains the Surcharge Data token with the surcharge profile ID from the BASE24-atm Terminal Data files.
5, 6	The Authorization process uses the surcharge profile ID and the card profile in the Card Prefix File (CPF) to determine the surcharge to be assessed. The Authorization process then formats a 9901 surcharge notification and sends it to the Device Handler process.
7, 8	The Device Handler process compares the surcharge amount in the 9901 surcharge notification to the value in the DOWNLOAD SURCHARGE AMT field on ATD screen 8. If the amount in the 9901 surcharge notification is less than or equal to the value in the DOWNLOAD SURCHARGE AMT field, the Device Handler process formats a 0200 message with the Surcharge Data token that includes the surcharge amount and sends it to the Authorization process.
9, 10	The Authorization process determines the authorization and routing parameters and sends the 0200 request message (which includes the Surcharge Data token and the surcharge amount) to the Host Interface process.
11, 12	The Host Interface process reformats the internal message into an external message, sets an outbound request timer, and sends the 0200 request message to the host. The message includes the Surcharge Data token with the surcharge amount. The surcharge is also reflected in the transaction amount (the AMT-1 field of the STM).
13, 14	The host approves the request and returns a 0210 response message to the Host Interface process.
15, 16	The Host Interface process reformats the external message into an internal message, deletes its outbound request timer, and sends the 0210 response message to the Authorization process. The message includes the Surcharge Data token with the surcharge amount.

Steps	Processing
17, 18	The Authorization process performs the following steps: - Formats a Transaction Log File (TLF) record, writes it to the TLF, and sends the 0210 response message to the Device Handler process. The message includes the Surcharge Data token with the surcharge amount. Processing continues with step 19a. The token information may or may not be logged to the TLF, depending on the configuration of the Token File (TKN). - Sends a 0220 completion message to the Host Interface process when the COMPLETION REQUIRED TO HOST field on the IPCF screen is set to 1 or 4. Processing continues with step 19b.
19a, 20a	The Device Handler process deletes its authorization timer and reformats the 0210 response message into a native-mode response that it sends to the ATM. The response contains the surcharge information mapped into the receipt data using the Terminal Receipt File (TRF) and the Receipt File (RCPT). Processing continues with step 21.
19b, 20b	The Host Interface process reformats the internal message into an external message and sends the 0220 completion message to the host when the COMPLETION REQUIRED TO HOST field on the IPCF screen is set to 1 or 4. This leg of processing is now complete.
21, 22	The ATM sends a native-mode message to the Device Handler process indicating that the transaction has been completed.

Approved by Host — Transaction Reversed — Completions Required — Host Acknowledgments Required

The diagram below illustrates the flow of a transaction that originates at an ATM, is approved by a host, and is then reversed. The host requires, and must acknowledge, completions.

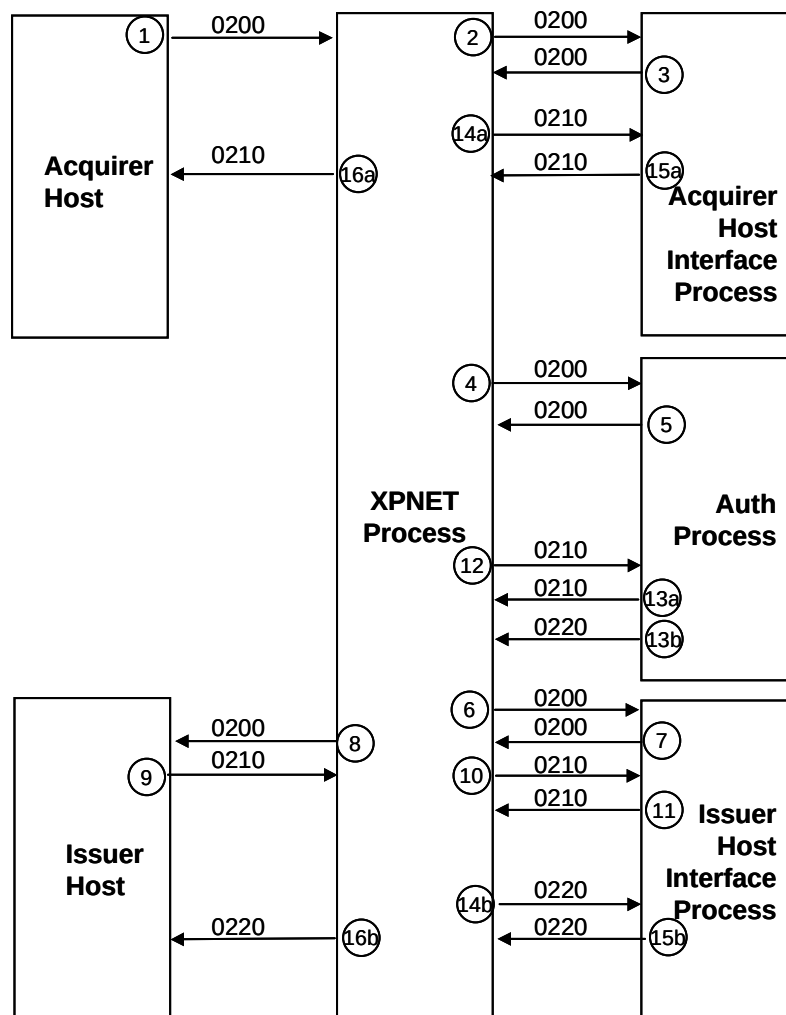


Steps	Processing
1, 2	The ATM sends a native-mode request message to the Device Handler process.
3, 4	The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Authorization process.
5, 6	From the Card Prefix File (CPF), the Authorization process locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 1. The Authorization process sends the 0200 request message to the Host Interface process.
7, 8	The Host Interface process reformats the internal message into an external message, sets an outbound request timer, and sends the 0200 request message to the host.
9, 10	The host approves the request and returns a 0210 response message to the Host Interface process.
11, 12	The Host Interface process reformats the external message into an internal message, deletes its outbound request timer, and sends the 0210 response message to the Authorization process.
13, 14	The Authorization process performs the following steps: a. Logs the response to the Transaction Log File (TLF) and sends the 0210 to the Device Handler process. Processing continues with step 15a. b. Sends a 0220 completion message to the Host Interface process when the COMPLETION REQUIRED TO HOST field on the IPCF screen is set to 1 or 4. Processing continues with step 15b.
15a, 16a	The Device Handler process deletes its authorization timer and reformats the 0210 response message into a native-mode response that it sends to the ATM. Processing continues with step 17a.

Steps	Processing
15b, 16b	The Host Interface process reformats the internal message into an external message and sends the 0220 completion message to the host when the COMPLETION REQUIRED TO HOST field on the IPCF screen is set to 1 or 4. Processing continues with step 17b.
17a, 18a	The ATM sends a native-mode message to the Device Handler process to indicate that the transaction did not complete as authorized. Processing continues with step 19.
17b, 18b	The host generates a 0230 acknowledgment message and sends it to the Host Interface process. The Host Interface process recognizes the acknowledgment and drops it. This leg of processing is now complete.
19, 20	The Device Handler process performs the following steps: a. Sends a 0420 reversal message to the Authorization process. Processing continues with step 21a. b. Sends a native-mode response to the ATM indicating to print the receipt for however the transaction was completed. Processing continues with step 21b.
21a, 22a	The Authorization process logs the reversal to the TLF and sends the 0420 message to the Host Interface process. Processing continues with step 23.
21b, 22b	The ATM sends a native-mode message to the Device Handler process indicating that the transaction has been completed. This leg of processing is now complete.
23, 24	The Host Interface process reformats the internal message into an external message and sends the 0420 message to the host.
25, 26	The host generates a 0430 reversal acknowledgment message and sends it to the Host Interface process. The Host Interface process recognizes the acknowledgment and drops it.

Approved by Host — Originates at Acquirer Host — Issuer Host Requires Completions

The diagram below illustrates the flow of a transaction that originates at an acquirer host and is approved by an issuer host. The issuer host requires completions.



Steps

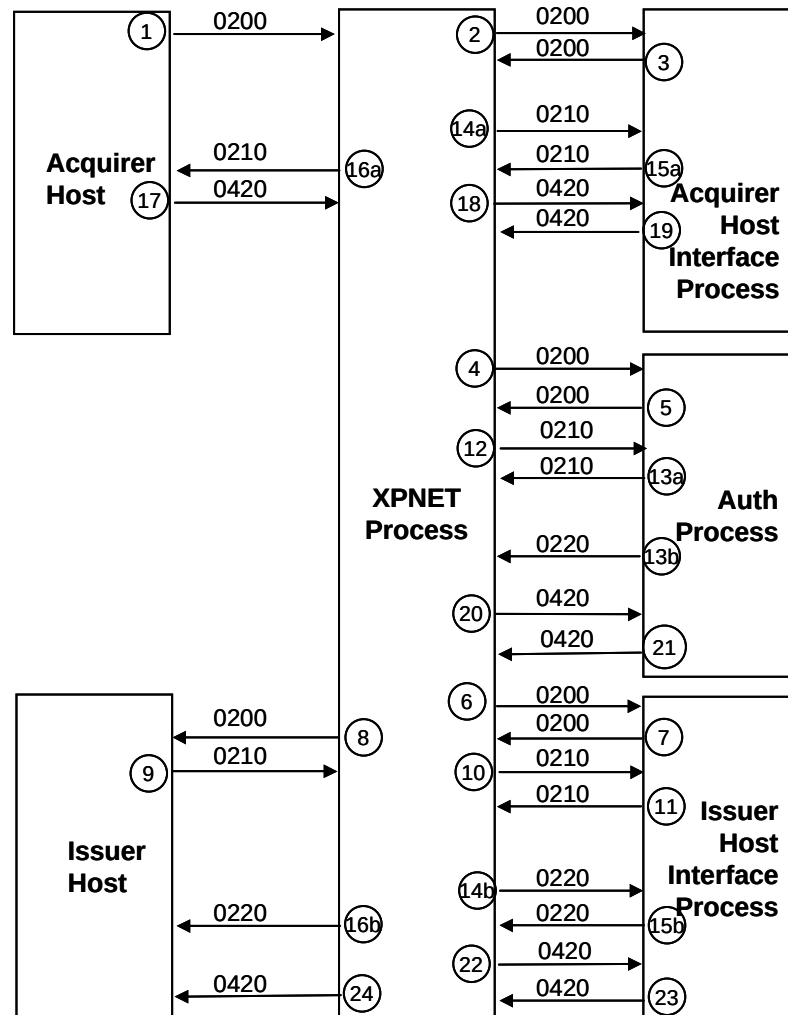
Processing

- | | |
|------|--|
| 1, 2 | The acquirer host sends a 0200 request message to the acquirer Host Interface process. |
| 3, 4 | The acquirer Host Interface process reformats the external message into an internal message, sets an inbound request timer, and sends the 0200 request message to the Authorization process. |

Steps	Processing
5, 6	From the Card Prefix File (CPF), the Authorization process locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 1. The Authorization process sends the 0200 request message to the issuer Host Interface process.
7, 8	The issuer Host Interface process reformats the internal message into an external message, sets an outbound request timer, and sends the 0200 request message to the issuer host.
9, 10	The issuer host approves the request and returns a 0210 response message to the issuer Host Interface process.
11, 12	The issuer Host Interface process reformats the external message into an internal message, deletes its outbound request timer, and sends the 0210 response message through the Authorization process.
13, 14	The Authorization process performs the following steps: - Logs the response to the Transaction Log File (TLF) and sends the 0210 response message to the acquirer Host Interface process. Processing continues with step 15a. - Sends the 0220 completion message to the issuer Host Interface process when the COMPLETION REQUIRED TO HOST field on the IPCF screen is set to 1 or 4. Processing continues with step 15b.
15a, 16a	The acquirer Host Interface process reformats the internal message into an external message, deletes its inbound request timer, and sends the 0210 response message to the acquirer host.
15b, 16b	The issuer Host Interface process reformats the internal message into an external message and sends the 0220 completion message to the issuer host when the COMPLETION REQUIRED TO HOST field on the IPCF screen is set to 1 or 4.

Approved by Host — Originates at Acquirer Host — Transaction Reversed — Issuer Host Requires Completions

The diagram below illustrates the flow of a transaction that originates at an acquirer host, is approved by an issuer host, and is then reversed. The issuer host requires completions.



Steps

Processing

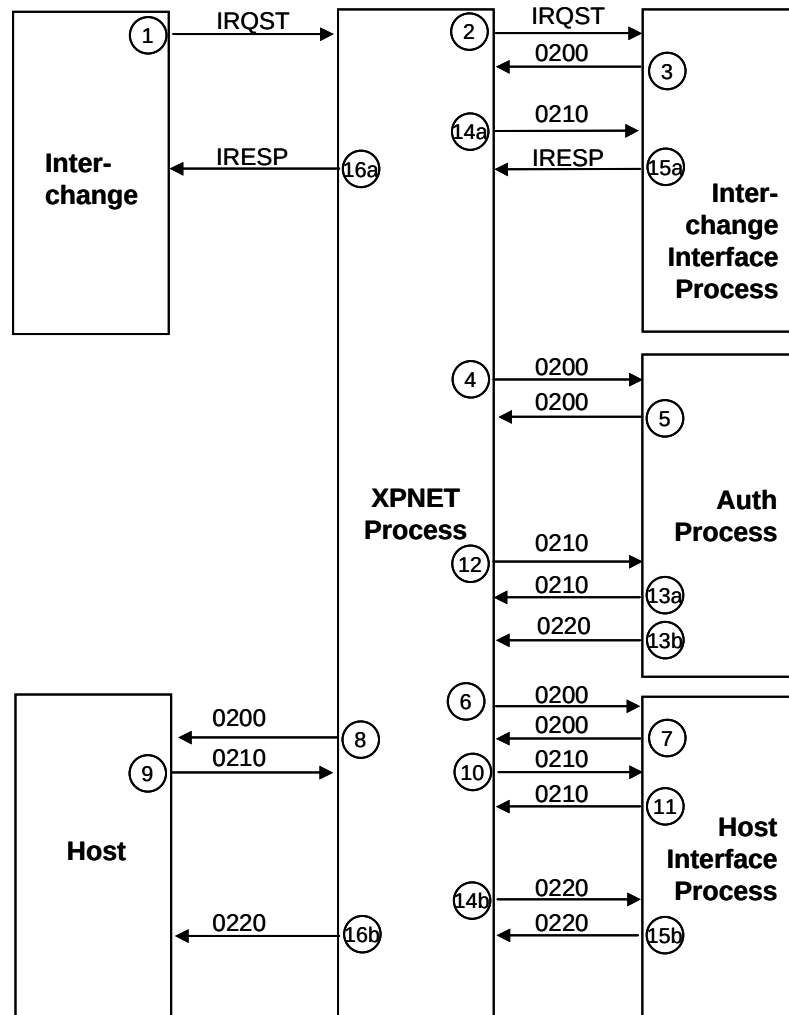
- 1, 2 The acquirer host sends a 0200 request message to the acquirer Host Interface process.
- 3, 4 The acquirer Host Interface process reformats the external message into an internal message, sets an inbound request timer, and sends the 0200 request message to the Authorization process.

Steps	Processing
5, 6	From the Card Prefix File (CPF), the Authorization process locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 1. The Authorization process sends the 0200 request message to the issuer Host Interface process.
7, 8	The issuer Host Interface process reformats the internal message into an external message, sets an outbound request timer, and sends the 0200 request message to the issuer host.
9, 10	The issuer host approves the request and returns a 0210 response message to the issuer Host Interface process.
11, 12	The issuer Host Interface process reformats the external message into an internal message, deletes its outbound request timer, and sends the 0210 response message to the Authorization process.
13, 14	The Authorization process performs the following steps: - Logs the response to the Transaction Log File (TLF) and sends the 0210 response message to the acquirer Host Interface process. Processing continues with step 15a. - Sends the 0220 completion message to the issuer Host Interface process when the COMPLETION REQUIRED TO HOST field on the IPCF screen is set to 1 or 4. Processing continues with step 15b.
15a, 16a	The acquirer Host Interface process reformats the internal message into an external message, deletes its inbound request timer, and sends the 0210 response message to the acquirer host. Processing continues with step 17.
15b, 16b	The issuer Host Interface process reformats the internal message into an external message and sends the 0220 completion message to the issuer host when the COMPLETION REQUIRED TO HOST field on the IPCF screen is set to 1 or 4. This leg of processing is now complete.
17, 18	The transaction does not complete as authorized, and the acquirer host sends a 0420 reversal message to the acquirer Host Interface process.

Steps	Processing
19, 20	The acquirer Host Interface process reformats the external message into an internal message and sends the 0420 reversal message to the Authorization process.
21, 22	The Authorization process logs the 0420 reversal message to the TLF and sends the 0420 reversal message to the issuer Host Interface process.
23, 24	The issuer Host Interface process reformats the internal message into an external message and sends the 0420 reversal message to the issuer host.

Approved by Host — Originates at an Interchange — Completions Required

The diagram below illustrates the flow of a transaction that originates at an interchange and is approved by a host. The Interchange Interface process is set up to send requests to the Authorization process. The host requires completions.



Steps

1, 2

Processing

The interchange sends a request message to the Interchange Interface process.

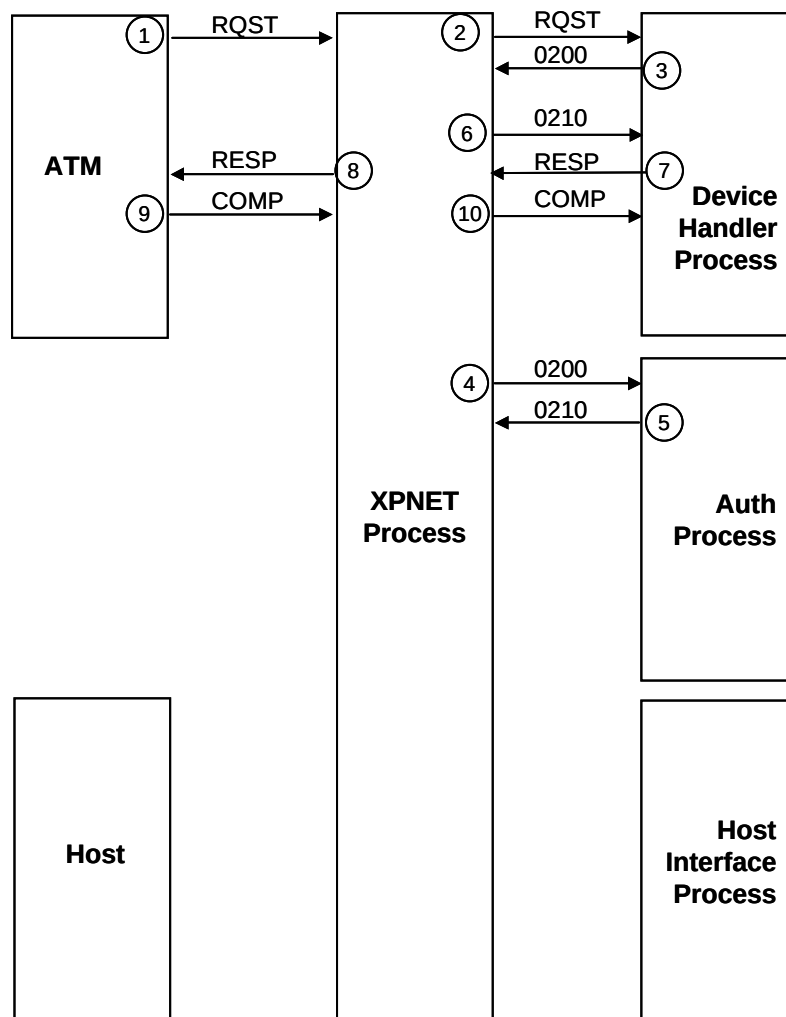
Steps	Processing
3, 4	The Interchange Interface process reformats the interchange message into an internal 0200 request message, sets an inbound request timer, and sends the 0200 request message to the Authorization process.
5, 6	From the Card Prefix File (CPF), the Authorization process locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 1. The Authorization process sends the 0200 request message to the Host Interface process.
7, 8	The Host Interface process reformats the internal message into an external message, sets an outbound request timer, and sends the 0200 request message to the host.
9, 10	The host approves the request and returns a 0210 response message to the Host Interface process.
11, 12	The Host Interface process reformats the external message into an internal message, deletes its outbound request timer, and sends the 0210 response message to the Authorization process.
13, 14	The Authorization process performs the following steps: a. Logs the response to the Transaction Log File (TLF) and sends the 0210 response message to the Interchange Interface process. Processing continues with step 15a. b. Sends the 0220 completion message to the Host Interface process when the COMPLETION REQUIRED TO HOST field on the IPCF screen is set to 1 or 4. Processing continues with step 15b.
15a, 16a	The Interchange Interface process reformats the 0210 response message into a response message understood by the interchange, logs the response in the Interchange Log File (ILF), deletes its inbound request timer, and sends the response message to the interchange.
15b, 16b	The Host Interface process reformats the 0220 internal message into an external message and sends it to the host.

BASE24-atm Transactions: Authorization Level 2

The following pages contain message flows for transactions being processed by BASE24-atm using level 2 (offline) authorization processing. This authorization level requires that BASE24-atm authorize all transactions. The host is not involved in the authorization process, although it can receive completions.

Approved by BASE24-atm — Completions Not Required

The diagram below illustrates the flow of a transaction that originates at an ATM and is approved by BASE24-atm. The host does not require completions.



Steps

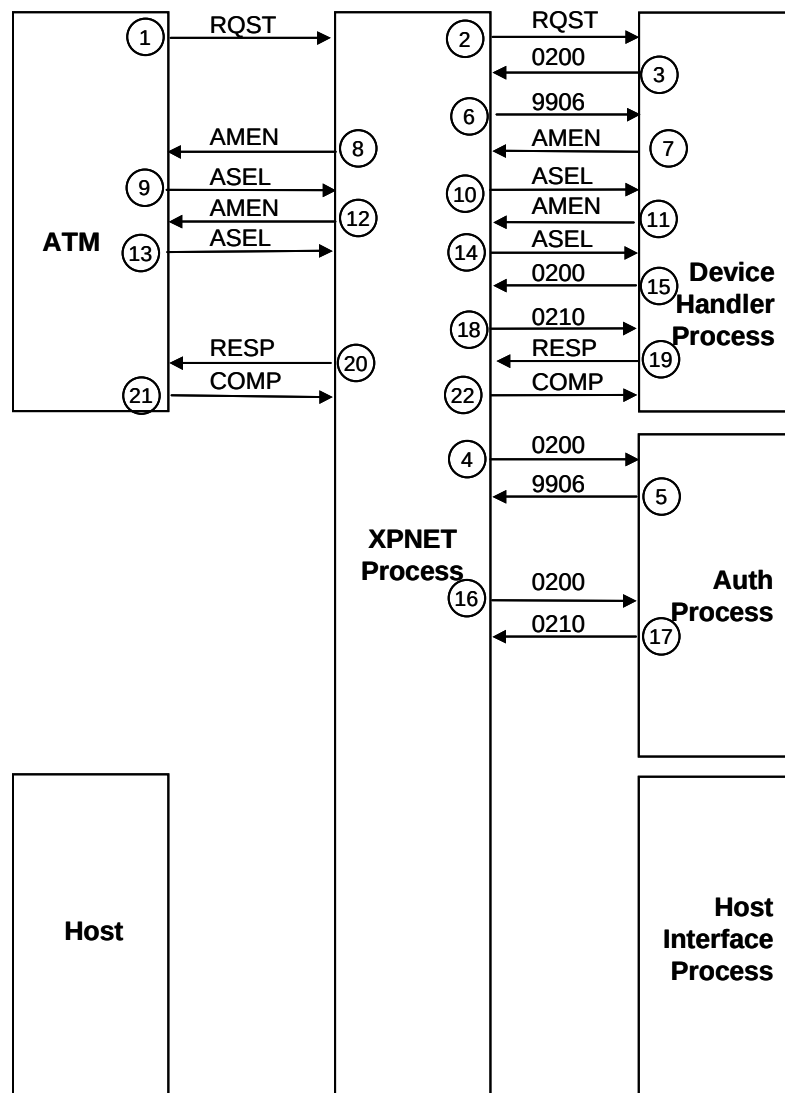
Processing

- | | |
|------|---|
| 1, 2 | The ATM sends a native-mode request message to the Device Handler process. |
| 3, 4 | The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Authorization process. |

Steps	Processing
5, 6	From the Card Prefix File (CPF), the Authorization process locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 2. The Authorization process approves the request, logs the response to the Transaction Log File (TLF), updates the database files appropriately, and returns a 0210 response message to the Device Handler process.
7, 8	The Device Handler process deletes its authorization timer and reformats the 0210 response message into a native-mode response that it sends to the ATM.
9, 10	The ATM sends a native-mode message to the Device Handler process to indicate that the transaction completed as authorized.

Approved by BASE24-atm — Transaction Requires Multiple Account Select — Completions Not Required

The diagram below illustrates the flow of a two-sided transaction that originates at an ATM, is approved by BASE24-atm, and requires a multiple account selection step. The host does not require completions.



Steps

1, 2

The ATM sends a native-mode request message to the Device Handler process.

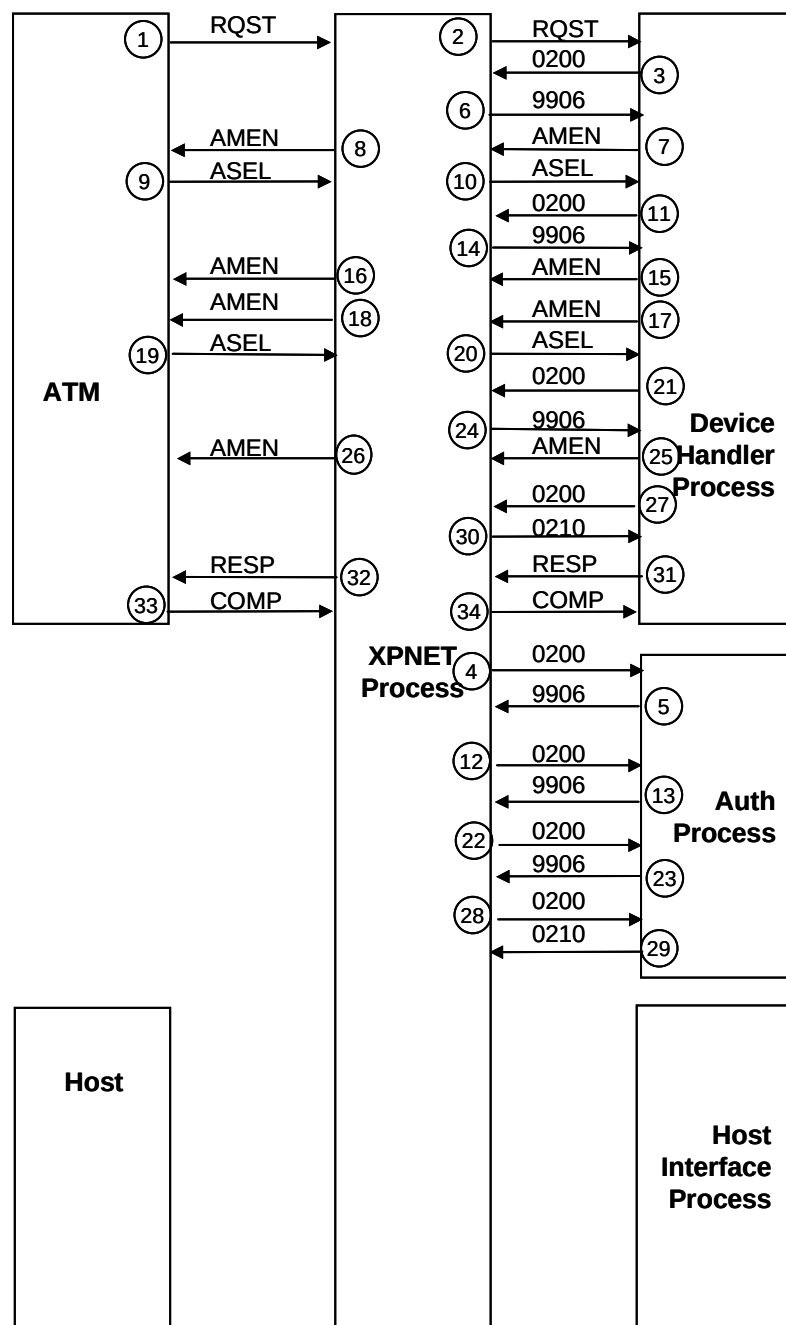
Processing

Steps	Processing
3, 4	The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Authorization process.
5, 6	The Authorization process returns an account data selection (9906) message to the Device Handler process. The 9906 message contains account data for selection by a cardholder for transactions involving a choice among multiple accounts of one type.
7, 8	The Device Handler process formats a native-mode message containing a selection menu to allow the cardholder to select a specific account for the <i>from</i> account of the transaction and sends the native-mode message to the ATM.
9, 10	The ATM returns a native-mode message containing the account number selected by the cardholder as the <i>from</i> account to the Device Handler process.
11, 12	The Device Handler process formats a second native-mode message containing a selection menu as it did in step 7 for the <i>to</i> account of the transaction and sends it to the ATM.
13, 14	The ATM returns a native-mode message to the Device Handler process containing the account number selected as the <i>to</i> account.
15, 16	The Device Handler process retrieves the 0200 message formatted in step 3, adds the account number or numbers selected by the cardholder, and sends the message to the Authorization process.
17, 18	From the Card Prefix File (CPF), the Authorization process locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 2. The Authorization process approves the request, logs the response to the Transaction Log File (TLF), updates the database files appropriately, and returns a 0210 response message to the Device Handler process.

Steps	Processing
19, 20	The Device Handler process deletes its authorization timer and reformats the 0210 response message into a native-mode response that it sends to the ATM.
21, 22	The ATM sends a native-mode message to the Device Handler process to indicate that the transaction completed as authorized.

Approved by BASE24-atm — Transaction Requires Multiple Account Select with Open Account Relationship Enhancement — Completions Not Required

The diagram on the following page illustrates the flow of a two-sided transaction that originates at an ATM, is approved by BASE24-atm, and requires a multiple account selection step with the Open Account Relationship enhancement. The host does not require completions.



Steps

Processing

- 1, 2 The ATM sends a native-mode request message to the Device Handler process.

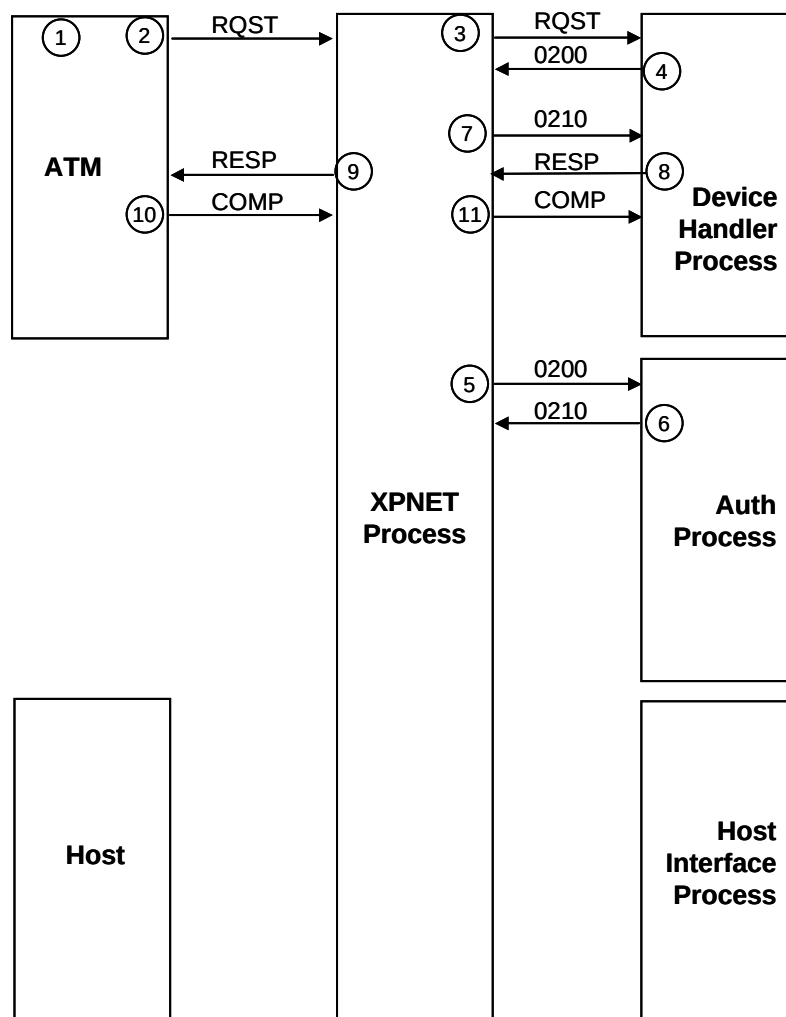
Steps	Processing
3, 4	The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Authorization process.
5, 6	The Authorization process returns an account data selection (9906) message to the Device Handler process. The 9906 message contains account data for up to five accounts for selection by a cardholder for transactions involving a choice among multiple accounts of one type. If there are five or fewer accounts of one type available, the Authorization process sends the 9906 message to the Device Handler process. If there are more than five accounts of one type available, the Authorization process adds the Multiple Account Selection token to the end of the 9906 message and sends it to the Device Handler process.
7,8	The Device Handler process checks the 9906 message for the Multiple Account Selection token. The Device Handler process formats a native-mode message containing a selection menu that enables the cardholder to select a specific account for the <i>from</i> account and sends it to the ATM. If the Multiple Account Selection token is not present (that is, there are five or fewer accounts of one type), all accounts are contained in the 9906 message. If the Multiple Account token is present (that is, there are more than five accounts of one type), the first five accounts and a selection of “More” are included in the message.
9, 10	The ATM returns a native-mode message to the Device Handler process containing the account number selected by the cardholder as the <i>from</i> account or a code indicating that the cardholder wants to see more accounts. • If the native-mode message contains the <i>from</i> account number selected by the cardholder, processing continues with step 17. • If the native-mode message contains a code indicating that the cardholder wants to see more <i>from</i> accounts, processing continues with step 11.
11, 12	The Device Handler process sets the MORE field in the Multiple Account Selection token to indicate that the customer wants to see more <i>from</i> accounts and sends the 0200 message to the Authorization process.

Steps	Processing
13, 14	The Authorization process checks the 0200 message for the Multiple Account Selection token. Since the token is present, the Authorization process retrieves the next five accounts from the CAF, updates the information in the Multiple Account Selection token, and returns a 9906 message to the Device Handler process.
15, 16	The Device Handler process formats a native-mode message containing a selection menu with the next five accounts and sends it to the ATM. Steps 9 through 16 are repeated until the cardholder selects a <i>from</i> account. When the native-mode message contains an account number, processing continues with step 17.
17, 18	The Device Handler process formats a second native-mode message containing a selection menu as it did in step 7 for the <i>to</i> account of the transaction and sends it to the ATM.
19, 20	The ATM returns a native-mode message to the Device Handler process containing the account number selected by the cardholder as the <i>to</i> account or a code indicating that the cardholder wants to see more accounts. • If the native-mode message contains the <i>to</i> account number or numbers selected by the cardholder, processing continues with step 27. • If the native-mode message contains a code indicating that the cardholder wants to see more <i>to</i> accounts, processing continues with step 21.
21, 22	The Device Handler process sets the MORE field in the Multiple Account Selection token to indicate that the customer wants to see more <i>to</i> accounts and sends the 0200 message to the Authorization process.
23, 24	The Authorization process checks the 0200 message for the Multiple Account Selection token. Since the token is present, the Authorization process retrieves the next five accounts from the CAF, updates the information in the Multiple Account token, and returns a 9906 message to the Device Handler process.

Steps	Processing
25, 26	The Device Handler process formats a native-mode message containing a selection menu with the next five accounts and sends it to the ATM. Steps 19 through 26 are repeated until the cardholder selects a <i>to</i> account. When the native-mode message contains an account number, processing continues with step 27.
27, 28	The Device Handler process retrieves the 0200 message formatted in step 3, adds the account numbers selected, drops the Multiple Account Selection token, and sends the 0200 message to the Authorization process.
29, 30	The Authorization process checks the 0200 message for the Multiple Account Selection token. Since it is not present, the Authorization process can begin authorizing the transaction. From the Card Prefix File (CPF), the Authorization process locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 2. The Authorization process approves the request, logs the response to the Transaction Log File (TLF), updates the database files appropriately, and returns a 0210 response message to the Device Handler process.
31, 32	The Device Handler process deletes its authorization timer and reformats the 0210 response message into a native-mode response that it sends to the ATM.
33, 34	The ATM sends a native-mode message to the Device Handler process to indicate that the transaction completed as authorized.

Approved by BASE24-atm — Transaction Requires Multiple Account Select by Account Qualifier — Completions Not Required

The diagram below illustrates the flow of a transaction that originates at an ATM, uses multiple account selection by qualifier, and is approved by BASE24-atm. The host does not require completions.



Steps

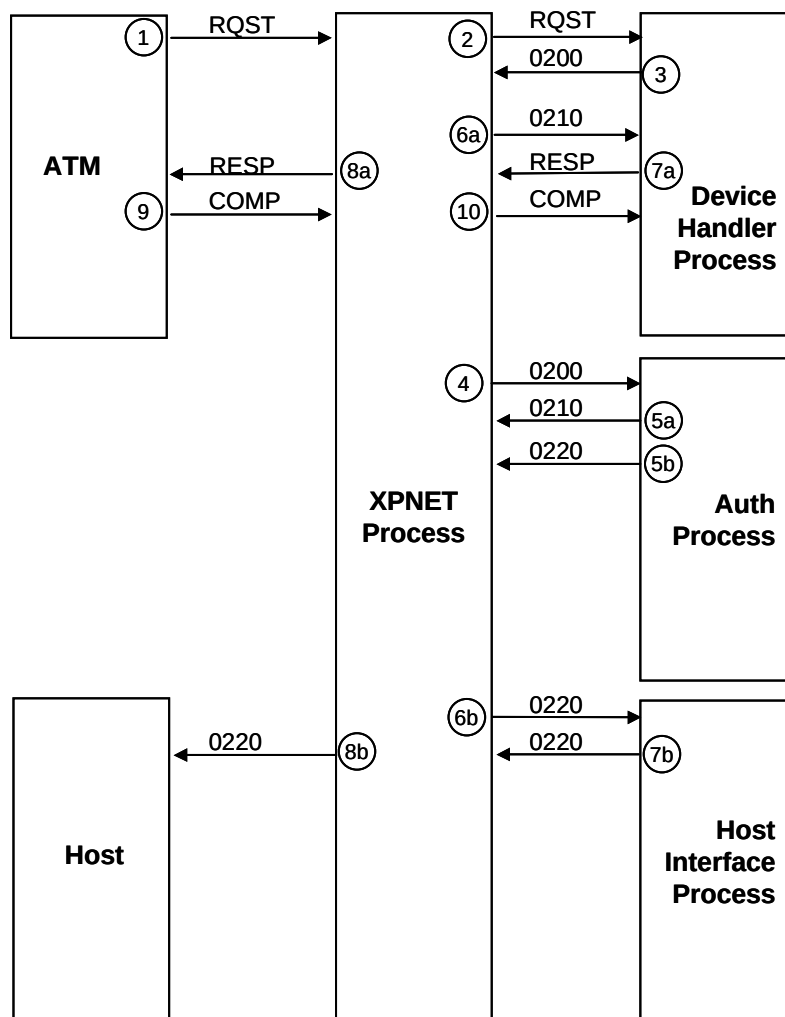
Processing

- 1 The cardholder initiates a transaction and specifies an account using the account qualifiers presented on the ATM screen.
- 2, 3 The ATM formats a native-mode request message that contains the account qualifier and sends it to the Device Handler process.

Steps	Processing
4, 5	The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, adds the Account Qualifier token to indicate the account selected, and sends the 0200 request message to the Authorization process.
6, 7	From the Card Prefix File (CPF), the Authorization process locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 2. The Authorization process uses the third digit of the account qualifier taken from the Account Qualifier token to retrieve the account from the CAF. The Authorization process approves the request, logs the response to the Transaction Log File (TLF), updates the database files appropriately, and returns a 0210 response message to the Device Handler process.
8, 9	The Device Handler process deletes its authorization timer and reformats the 0210 response message into a native-mode response that it sends to the ATM.
10, 11	The ATM sends a native-mode message to the Device Handler process to indicate that the transaction completed as authorized.

Approved by BASE24-atm — Completions Required

The diagram below illustrates the flow of a transaction that originates at an ATM and is approved by BASE24-atm. The host requires completions.



Steps

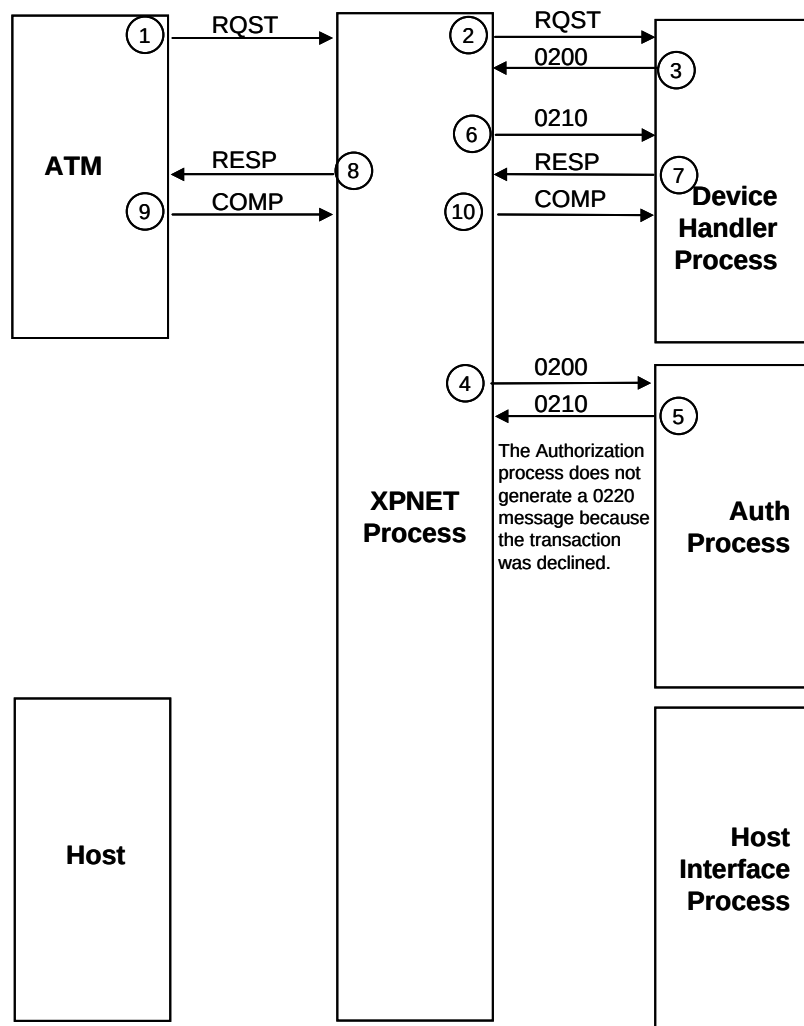
Processing

- | | |
|------|---|
| 1, 2 | The ATM sends a native-mode request message to the Device Handler process. |
| 3, 4 | The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Authorization process. |

Steps	Processing
5, 6	The Authorization process performs the following steps: - Uses the Card Prefix File (CPF) to locate an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 2. The Authorization process approves the request, logs the response to the Transaction Log File (TLF), updates the database files appropriately, and returns a 0210 response message to the Device Handler process. Processing continues with step 7a. - Sends a 0220 completion message to the Host Interface process when the COMPLETION REQUIRED TO HOST field on the IPCF screen is set to a nonzero value. Processing continues with step 7b.
7a, 8a	The Device Handler process deletes its authorization timer and reformats the 0210 response message into a native-mode response that it sends to the ATM. Processing continues with step 9.
7b, 8b	The Host Interface process reformats the internal message into an external message and sends the 0220 completion message to the host when the COMPLETION REQUIRED TO HOST field on the IPCF screen is set to a nonzero value. This leg of processing is now complete.
9, 10	The ATM sends a native-mode message to the Device Handler process to indicate that the transaction completed as authorized.

Declined by BASE24-atm — Completions Required

The diagram below illustrates the flow of a transaction that originates at an ATM and is declined by BASE24-atm. The host requires completions.



Steps

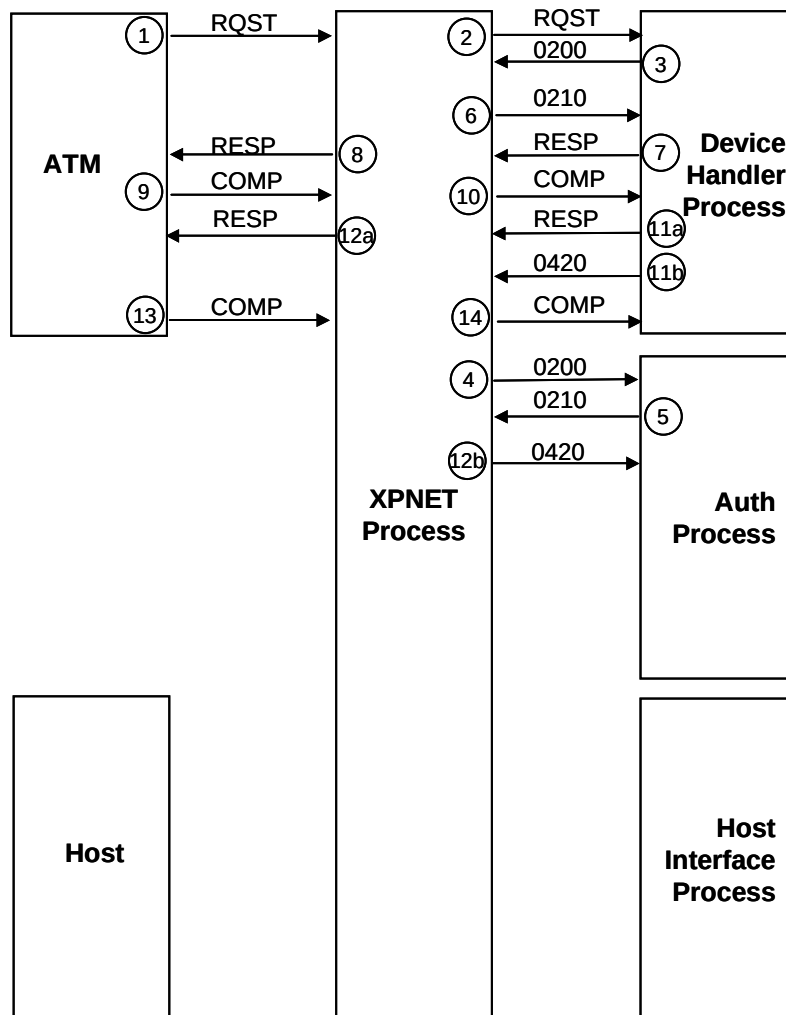
Processing

- | | |
|------|---|
| 1, 2 | The ATM sends a native-mode request message to the Device Handler process. |
| 3, 4 | The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Authorization process. |

Steps	Processing
5, 6	From the Card Prefix File (CPF), the Authorization process locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 2. The Authorization process declines the request, logs the response to the Transaction Log File (TLF), updates the database files appropriately, and returns a 0210 response message to the Device Handler process. The Authorization process generates a 0220 completion message when the COMPLETION REQUIRED TO HOST field on the IPCF screen is set to 4 or 6.
7, 8	The Device Handler process deletes its authorization timer and reformats the 0210 response message into a native-mode response that it sends to the ATM.
9, 10	The ATM sends a native-mode message to the Device Handler process to indicate that the declined transaction was completed.

Approved by BASE24-atm — Transaction Reversed — Completions Not Required

The diagram below illustrates the flow of a transaction that originates at an ATM, is approved by BASE24-atm, and is then reversed. The host does not require completions.



Steps

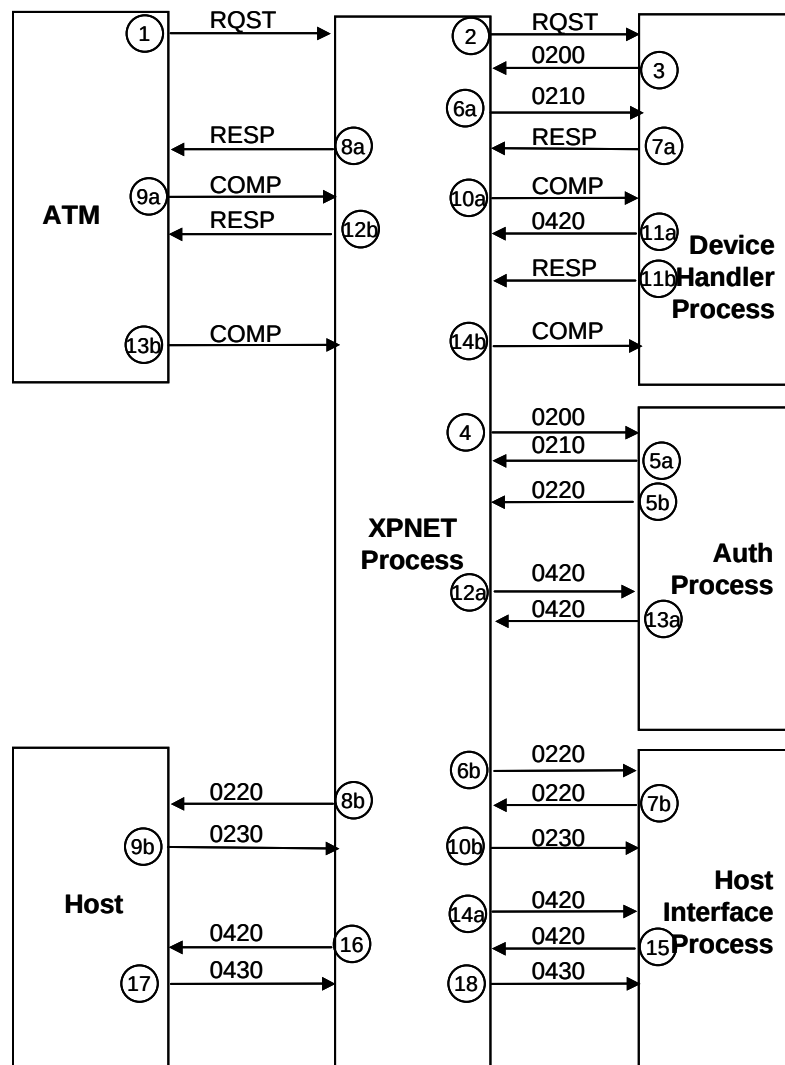
Processing

- | | |
|------|---|
| 1, 2 | The ATM sends a native-mode request message to the Device Handler process. |
| 3, 4 | The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Authorization process. |

Steps	Processing
5, 6	From the Card Prefix File (CPF), the Authorization process locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 2. The Authorization process approves the request, logs the response to the Transaction Log File (TLF), updates the database files appropriately, and returns a 0210 response message to the Device Handler process.
7, 8	The Device Handler process deletes its authorization timer and reformats the 0210 response message into a native-mode response that it sends to the ATM.
9, 10	The ATM sends a native-mode message to the Device Handler process to indicate that the transaction did not complete as authorized.
11, 12	The Device Handler process performs as follows: a. Generates a reversal (0420) message, logs it to the TLF, and sends the reversal to the Authorization process. The Authorization process then updates the BASE24 database appropriately. This leg of processing is now complete. b. Sends a native-mode response to the ATM indicating to print the receipt for however the transaction was completed. Processing continues with step 13.
13, 14	The ATM sends a native-mode message to the Device Handler process indicating that the transaction has been completed.

Approved by BASE24-atm — Transaction Reversed — Completions Required — Host Acknowledgments Required

The diagram below illustrates the flow of a transaction that originates at an ATM, is approved by BASE24-atm, and is then reversed. The host requires, and must acknowledge, completions.

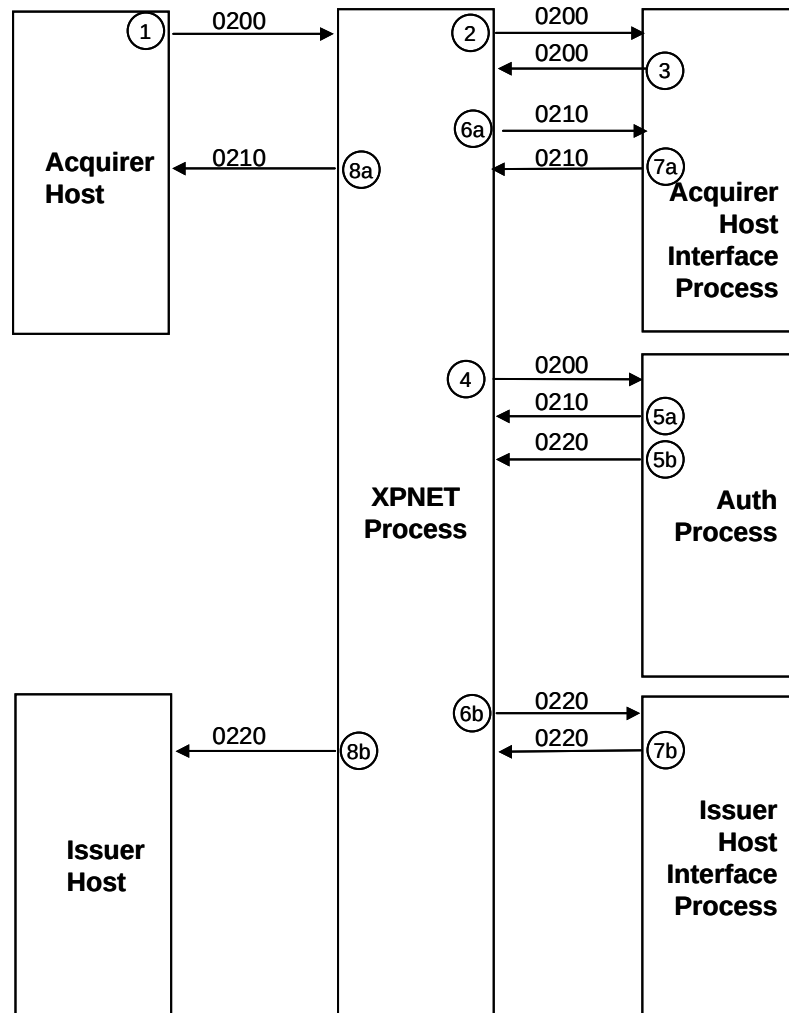


Steps	Processing
1, 2	The ATM sends a native-mode request message to the Device Handler process.
3, 4	The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Authorization process.
5, 6	The Authorization process performs the following steps: a. From the Card Prefix File (CPF), the Authorization process locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 2. The Authorization process approves the request, logs the response to the Transaction Log File (TLF), updates the database files appropriately, and returns a 0210 response message to the Device Handler process. Processing continues with step 7a. b. The Authorization process sends a 0220 completion message to the Host Interface process when the COMPLETION REQUIRED TO HOST field on the IPCF screen is set to a nonzero value.
7a, 8a	The Device Handler process deletes its authorization timer and reformats the 0210 response message into a native-mode response that it sends to the ATM. Processing continues with step 9a.
7b, 8b	The Host Interface process reformats the internal message into an external message and sends the 0220 completion message to the host when the COMPLETION REQUIRED TO HOST field on the IPCF screen is set to a nonzero value. Processing continues with step 9b.
9a, 10a	The ATM sends a native-mode message to the Device Handler process to indicate that the transaction did not complete as authorized. Processing continues with step 11.
9b, 10b	The host sends a 0230 acknowledgment message to the Host Interface process. The Host Interface process recognizes the acknowledgment and drops it. This leg of processing is now complete.

Steps	Processing
11, 12	The Device Handler process performs the following steps: - Generates a reversal and sends a 0420 message to the Authorization process. Processing continues with step 13a. - Sends a native-mode response to the ATM indicating to print the receipt for however the transaction was completed. Processing continues with step 13b.
13a, 14a	The Authorization process logs the reversal to the Transaction Log File (TLF), updates the database files, and sends the 0420 reversal message to the Host Interface process. Processing continues with step 15.
13b, 14b	The ATM sends a native-mode message to the Device Handler process indicating that the transaction has been completed. This leg of processing is now complete.
15, 16	The Host Interface process reformats the internal message into an external message and sends the 0420 reversal message to the host.
17, 18	The host sends a 0430 acknowledgment of reversal message to the Host Interface process. The Host Interface process recognizes the acknowledgment and drops it.

Approved by BASE24-atm — Originates at Acquirer Host — Issuer Host Requires Completions

The diagram below illustrates the flow of a transaction that originates at an acquirer host and is approved by BASE24-atm. The issuer host requires completions.



Steps

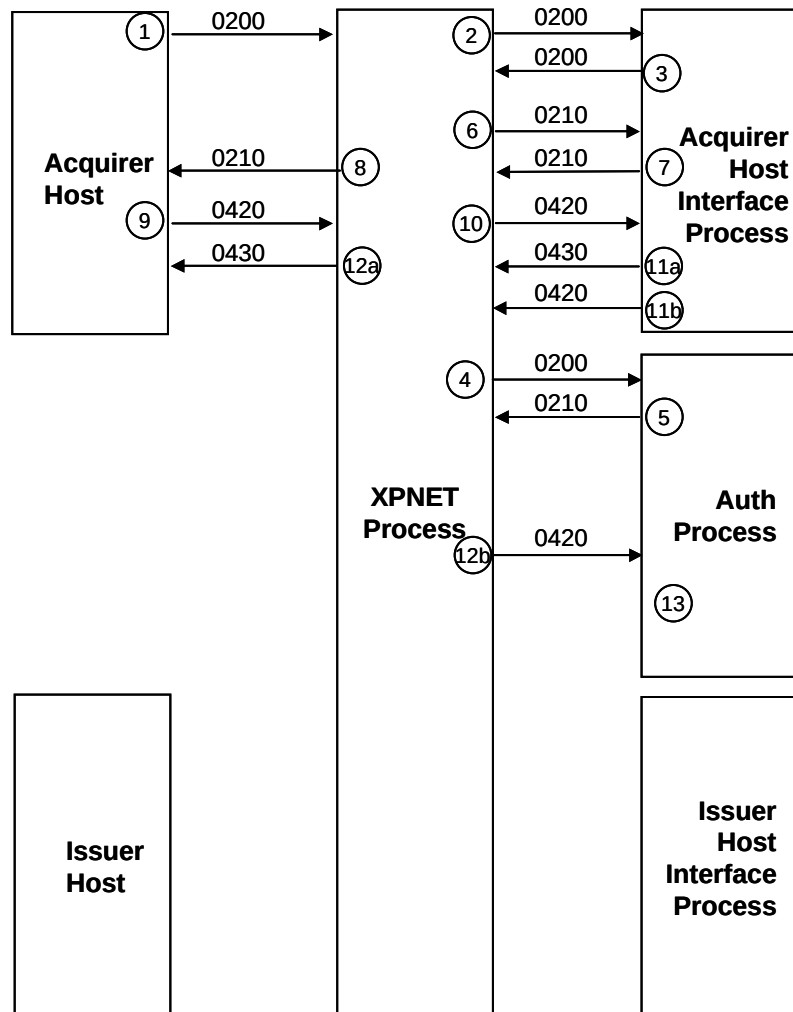
Processing

- | | |
|------|--|
| 1, 2 | The acquirer host sends a 0200 request message to the acquirer Host Interface process. |
| 3, 4 | The acquirer Host Interface process reformats the external message into an internal message, sets an inbound request timer, and sends the 0200 request message to the Authorization process. |

Steps	Processing
5, 6	The Authorization process performs as follows: - Uses the Card Prefix File (CPF) to locate an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 2. The Authorization process approves the request, logs the response to the Transaction Log File (TLF), updates the database files appropriately, and returns a 0210 response message to the acquiring Host Interface process. Processing continues with step 7a. - Sends a 0220 completion message to the issuer Host Interface process when the COMPLETION REQUIRED TO HOST field on the IPCF screen is set to a nonzero value. Processing continues with step 7b.
7a, 8a	The acquirer Host Interface process reformats the internal message into an external message, deletes its inbound request timer, and sends the 0210 response message to the acquirer host.
7b, 8b	The issuer Host Interface process reformats the internal message into an external message and sends the 0220 completion message to the issuer host when the COMPLETION REQUIRED TO HOST field on the IPCF screen is set to a nonzero value.

Approved by BASE24-atm — Originates at Acquirer Host — Transaction Reversed — Acquirer Host Requires Acknowledgments — Issuer Host Does Not Require Completions

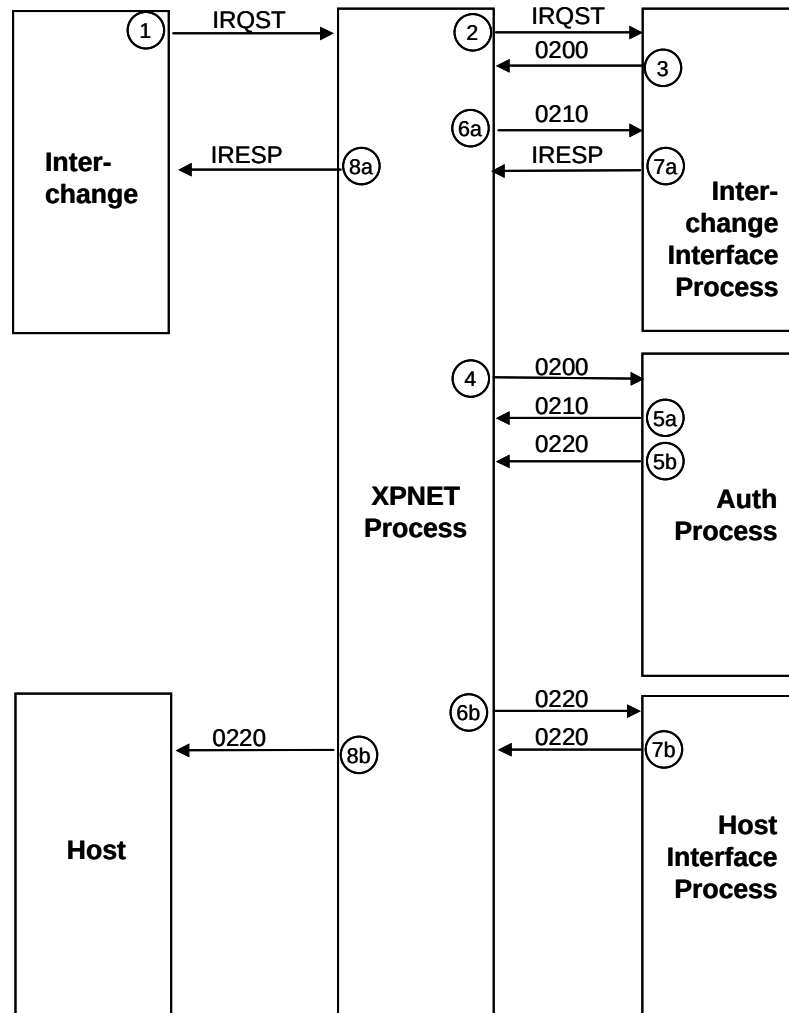
The diagram below illustrates the flow of a transaction that originates at an acquirer host, is approved by BASE24-atm, and is then reversed. The acquirer host requires acknowledgments. The issuer host does not require completions.



Steps	Processing
1, 2	The acquirer host sends a 0200 request message to the acquirer Host Interface process.
3, 4	The acquirer Host Interface process reformats the external message into an internal message, sets an inbound request timer, and sends the 0200 request message to the Authorization process.
5, 6	From the Card Prefix File (CPF), the Authorization process locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 2. The Authorization process approves the request, logs the response to the Transaction Log File (TLF), updates the database files appropriately, and returns a 0210 response message to the acquirer Host Interface process.
7, 8	The acquirer Host Interface process reformats the internal message into an external message, deletes its inbound request timer, and sends the 0210 response message to the acquirer host.
9, 10	The acquirer host formats a 0420 reversal message and sends it to the acquirer Host Interface process.
11, 12	The acquirer Host Interface process performs as follows: a. Since the acquirer host requires acknowledgments, the acquirer Host Interface process sends a 0430 reversal acknowledgment message to the acquirer host. This leg of processing is now complete. b. The acquirer Host Interface process reformats the external message into an internal message and sends the 0420 reversal message to the Authorization process.
13	The Device Handler process logs the reversal to the TLF and updates the Authorization files.

Approved by BASE24-atm — Originates at an Interchange — Completions Required

The diagram below illustrates the flow of a transaction that originates at an interchange and is approved by BASE24-atm. The Interchange Interface process is set up to send requests to an Authorization process. The host requires completions.



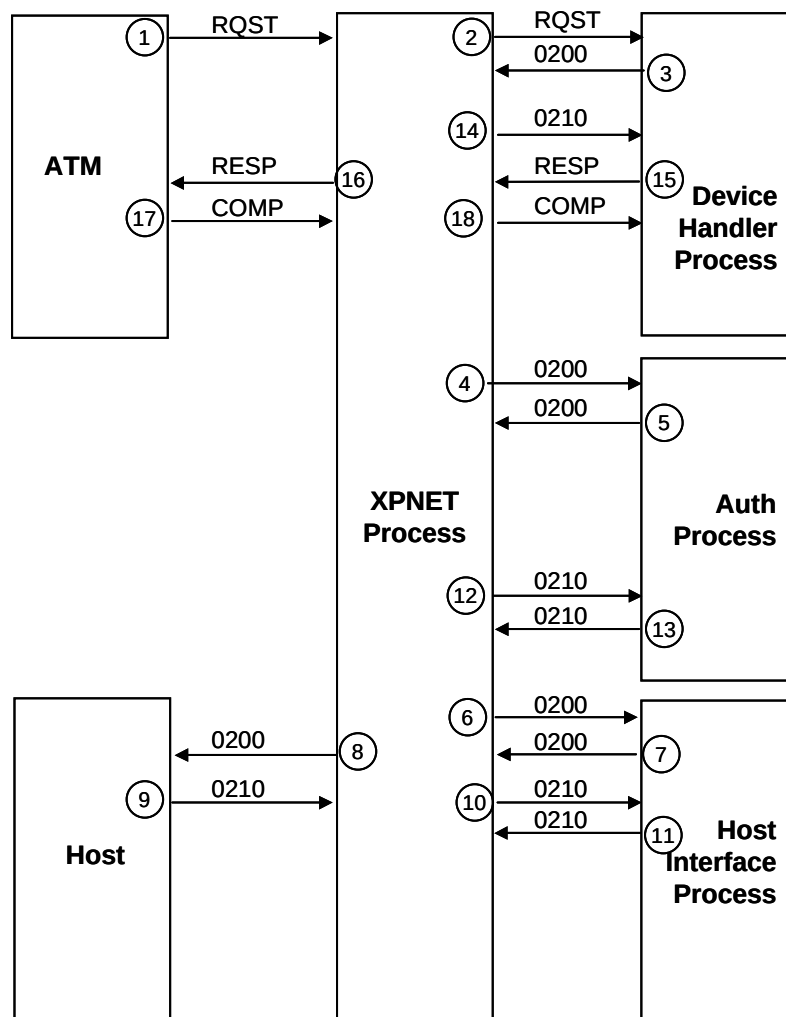
Steps	Processing
1, 2	The interchange sends a request message to the Interchange Interface process.
3, 4	The Interchange Interface process reformats the interchange message into an internal 0200 request message, sets an inbound request timer, and sends the 0200 request message to the Authorization process.
5, 6	The Authorization process performs the following steps: a. Uses the Card Prefix File (CPF) to locate an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 2. The Authorization process approves the request, logs the response to the Transaction Log File (TLF), updates the database files appropriately, and returns a 0210 response message to the Interchange Interface process. Processing continues with step 7a. b. Since the host requires completions, the Authorization process sends a 0220 completion message to the Host Interface process when the COMPLETION REQUIRED TO HOST field on the IPCF screen is set to a nonzero value. Processing continues with step 7b.
7a, 8a	The Interchange Interface process reformats the 0210 response message into a response message understood by the interchange, logs the response in the Interchange Log File (ILF), deletes its inbound request timer, and sends the response message to the interchange.
7b, 8b	The Host Interface process reformats the 0220 internal message into an external message and sends it to the host.

BASE24-atm Transactions: Authorization Level 3

The following pages contain message flows for transactions being processed by BASE24-atm using level 3 (online/offline) authorization processing. This authorization level requires that transactions be sent to a host for authorization, but allows BASE24-atm to stand in and authorize transactions when the host is not available.

Approved by Host — Completions Not Required

The diagram below illustrates the flow of a transaction that originates at an ATM and is approved by a host. The host does not require completions.



Steps

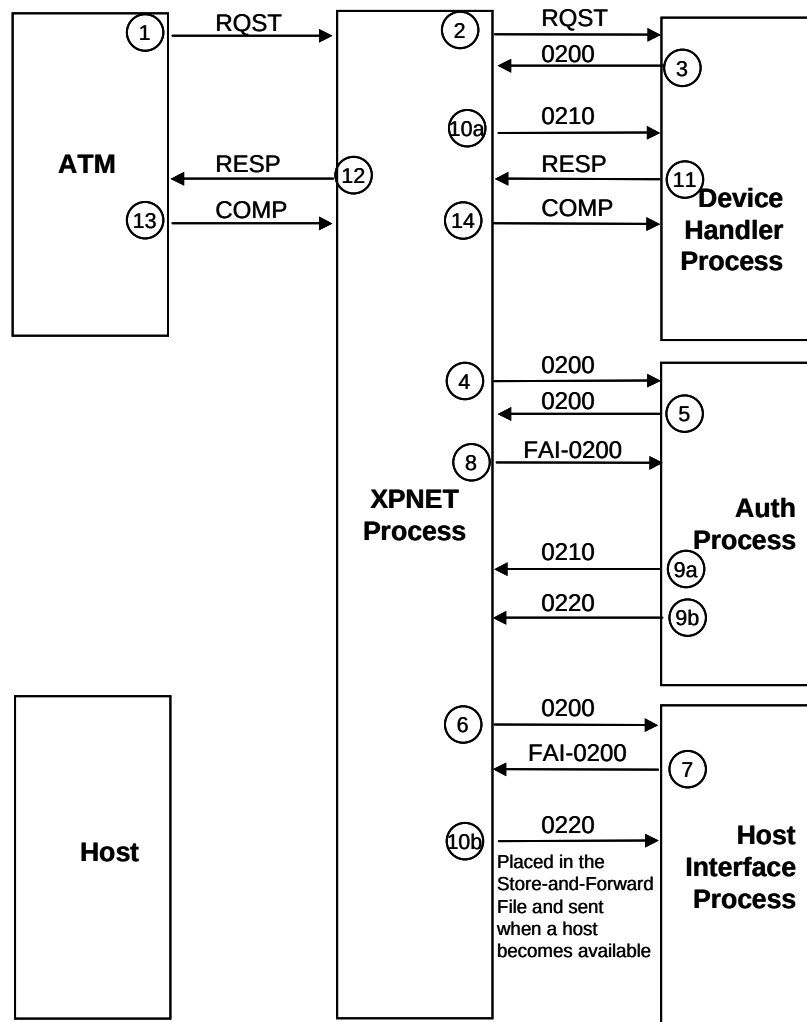
Processing

- 1, 2 The ATM sends a native-mode request message to the Device Handler process.
- 3, 4 The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Authorization process.

Steps	Processing
5, 6	From the Card Prefix File (CPF), the Authorization process locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 3. The Authorization process sends the 0200 request message to the Host Interface process.
7, 8	The Host Interface process reformats the internal message into an external message, sets an outbound request timer, and sends the 0200 request message to the host.
9, 10	The host approves the request and returns a 0210 response message to the Host Interface process.
11, 12	The Host Interface process reformats the external message into an internal message, deletes its outbound request timer, and sends the 0210 response message to the Authorization process.
13, 14	The Authorization process logs the response to the Transaction Log File (TLF), updates the database files, and sends the 0210 response message to the Device Handler process.
15, 16	The Device Handler process deletes its authorization timer and reformats the 0210 response message into a native-mode response that it sends to the ATM.
17, 18	The ATM sends a native-mode message to the Device Handler process to indicate that the transaction completed as authorized.

Approved by BASE24-atm — Completions Required — Completion Placed in Store-and-Forward File

The diagram below illustrates the flow of a transaction that originates at an ATM and is approved by BASE24-atm, which stands in for the host because communications to the host are down. The host requires completions; however, since communications are down, BASE24-atm places the completion message in the Store-and-Forward File (SAF) to be sent to the host at a later time.

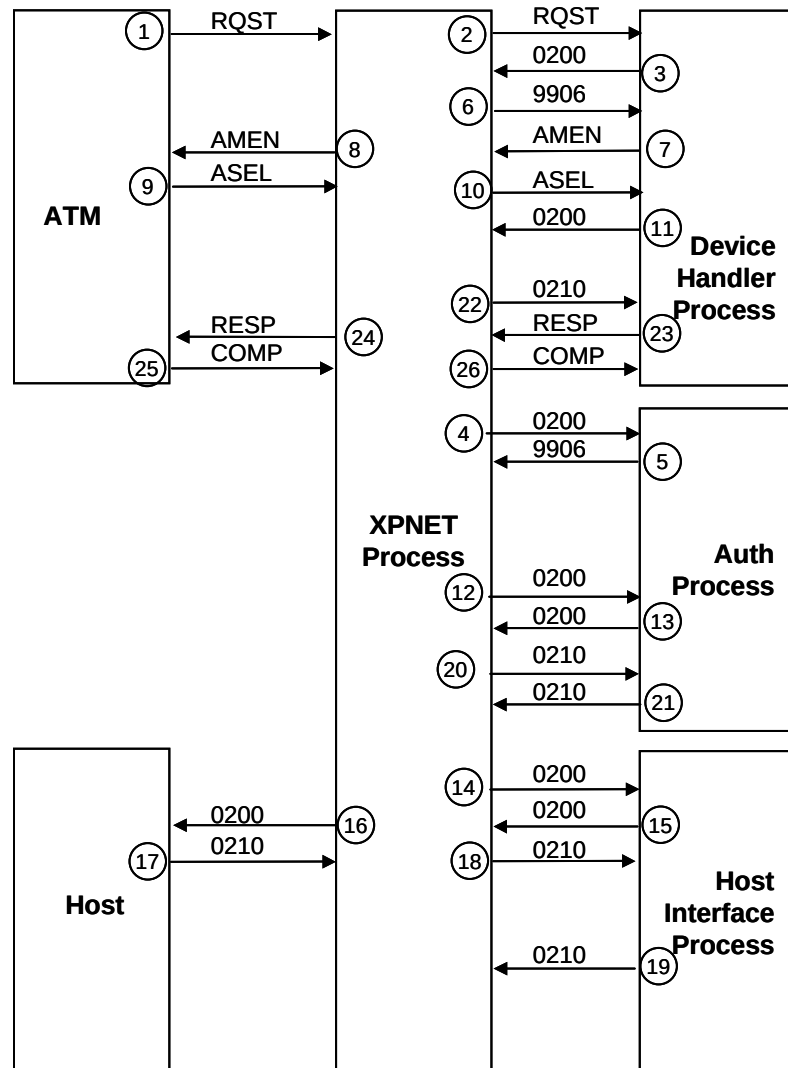


Steps	Processing
1, 2	The ATM sends a native-mode request message to the Device Handler process.
3, 4	The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Authorization process.
5, 6	From the Card Prefix File (CPF), the Authorization process locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 3. The Authorization process sends the 0200 request message to the Host Interface process.
7, 8	Because the host is unavailable, the Host Interface process changes the 0200 request message to a failed 0200 request message and sends it to the Authorization process.
9, 10	The Authorization process performs the following steps: - Stands in and approves the transaction, changes the failed 0200 request message to a 0210 response message, logs the response to the Transaction Log File (TLF), updates the database files for the cardholder, and sends the 0210 response message to the Device Handler process. Processing continues with step 11. - Sends a 0220 completion message to the Host Interface process. The Host Interface process is aware that communications to the host are down. Therefore, it does not attempt to send the advice to the host. Instead, the Host Interface process logs the message to the Store-and-Forward File (SAF) for storage. This leg of processing is now complete. Later, when communications with the host are restored, the transactions in the SAF can be transmitted to the host.

Steps	Processing
11, 12	The Device Handler process deletes its authorization timer and reformats the 0210 response message into a native-mode response that it sends to the ATM.
13, 14	The ATM sends a native-mode message to the Device Handler process to indicate that the transaction completed as authorized.

Approved by Host — Transaction Requires Multiple Account Select — Completions Not Required

The diagram below illustrates the flow of a transaction that originates at an ATM, is approved by a host, and requires a multiple account selection step. The host does not require completions.

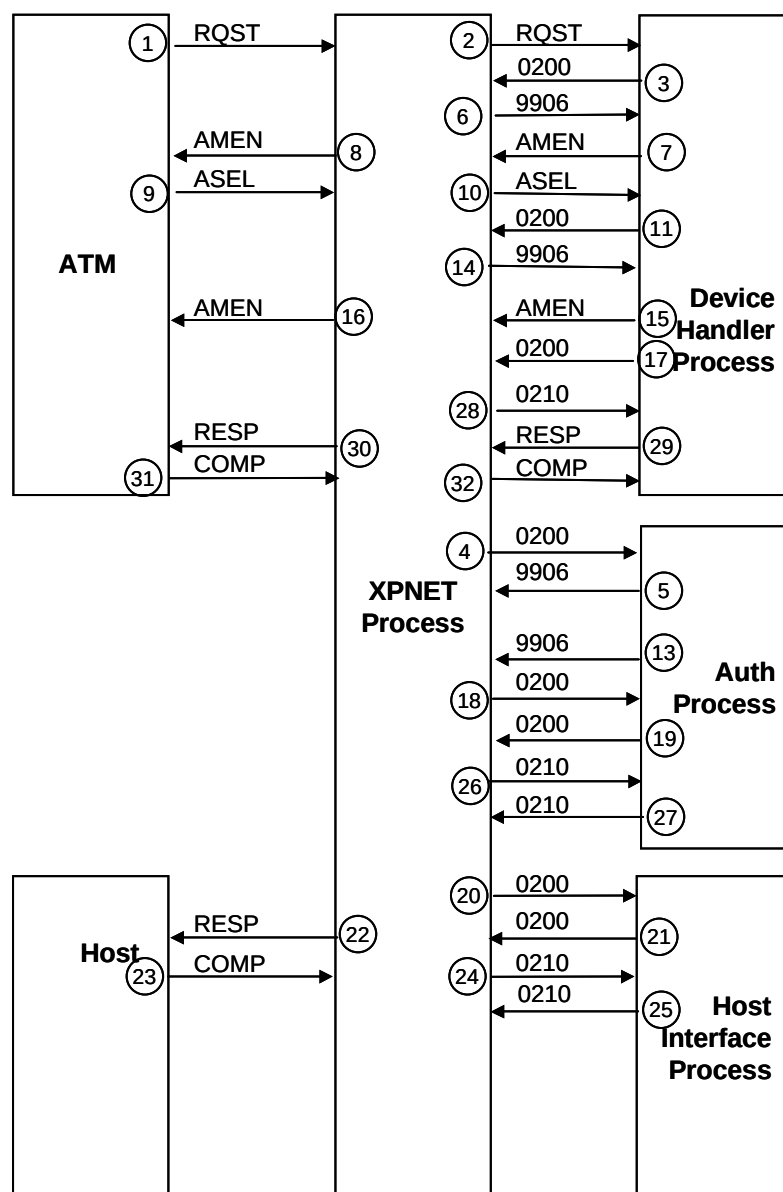


Steps	Processing
1, 2	The ATM sends a native-mode request message to the Device Handler process.
3, 4	The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Authorization process.
5, 6	The Authorization process returns an account data selection (9906) message to the Device Handler process. The 9906 message contains account data for selection by a cardholder for transactions involving multiple accounts.
7, 8	The Device Handler process formats a native-mode message containing a selection menu to allow the cardholder to select a specific account for the transaction and sends it to the ATM.
9, 10	The ATM returns a native-mode message containing the account number selected by the cardholder to the Device Handler process.
11, 12	The Device Handler process retrieves the 0200 message formatted in step 3, adds the account number or numbers selected by the cardholder, and sends the message to the Authorization process.
13, 14	From the Card Prefix File (CPF), the Authorization process locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 3. The Authorization process sends the 0200 request message to the Host Interface process.
15, 16	The Host Interface process reformats the internal message into an external message, sets an outbound request timer, and sends the 0200 request message to the host.
17, 18	The host approves the request and changes the 0200 request into a 0210 response message that it sends to the Host Interface process.

Steps	Processing
19, 20	The Host Interface process reformats the external message into an internal message, deletes its outbound request timer, and sends the 0210 response message to the Authorization process.
21, 22	The Authorization process logs the response to the Transaction Log File (TLF), updates the cardholder database files, and sends the 0210 response to the Device Handler process.
23, 24	The Device Handler process deletes its authorization timer and reformats the 0210 response message into a native-mode response that it sends to the ATM.
25, 26	The ATM sends a native-mode message to the Device Handler process to indicate that the transaction completed as authorized.

Approved by Host — Transaction Requires Multiple Account Select with the Open Account Relationship Enhancement — Completions Not Required

The diagram on the following page illustrates the flow of a transaction that originates at an ATM, is approved by the host, and requires a multiple account select step with the Open Account Relationship enhancement. The host does not require completions.



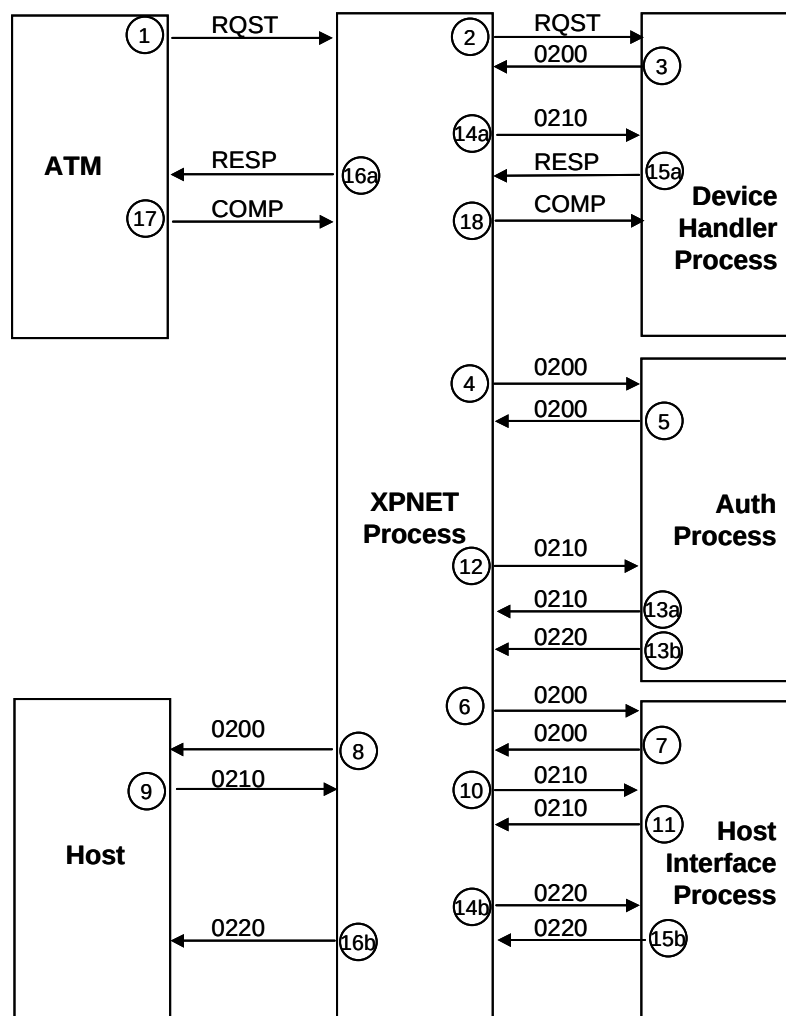
Steps	Processing
1, 2	The ATM sends a native-mode request message to the Device Handler process.
3, 4	The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Authorization process.
5, 6	The Authorization process returns an account data selection (9906) message to the Device Handler process. The 9906 message contains account data for up to five accounts for selection by a cardholder for transactions involving a choice among multiple accounts of one type. If there are five or fewer accounts of one type available, the Authorization process sends the 9906 message to the Device Handler process. If there are more than five accounts of one type available, the Authorization process adds the Multiple Account Selection token to the end of the 9906 message and sends it to the Device Handler process.
7,8	The Device Handler process checks the 9906 message for the Multiple Account Selection token. The Device Handler process formats a native-mode message containing a selection menu to allow the cardholder to select a specific account and sends it to the ATM. If the Multiple Account Selection token is not present (that is, there are five or fewer accounts of one type), all accounts are contained in the 9906 message. If the Multiple Account Selection token is present (that is, there are more than five accounts of one type), the first five accounts and a selection of “More” are included in the message.
9, 10	The ATM returns a native-mode message to the Device Handler process containing the account number selected by the cardholder or a code indicating that the cardholder wants to see more accounts. • If the native-mode message contains the account number selected by the cardholder, processing continues with step 17. • If the native-mode message contains a code indicating that the cardholder wants to see more accounts, processing continues with step 11.
11, 12	The Device Handler process sets the MORE field in the Multiple Account Selection token to indicate that the customer wants to see more accounts and sends the 0200 message to the Authorization process.

Steps	Processing
13, 14	The Authorization process checks the 0200 message for the Multiple Account Selection token. Since the token is present, the Authorization process retrieves the next five accounts from the CAF, updates the information in the Multiple Account Selection token, and returns a 9906 message to the Device Handler process.
15, 16	The Device Handler process formats a native-mode message containing a selection menu with the next five accounts and sends it to the ATM. Steps 9 through 16 are repeated until the cardholder selects an account. When the native-mode message contains an account number, processing continues with step 17.
17, 18	The Device Handler process retrieves the 0200 message formatted in step 3, adds the account numbers selected, drops the Multiple Account Selection token, and sends the 0200 message to the Authorization process.
19, 20	The Authorization process checks the 0200 message for the Multiple Account token. Since the token is not present, the Authorization process can send the transaction to the host for authorization. From the Card Prefix File (CPF), the Authorization process locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 3. The Authorization process sends the 0200 request message to the Host Interface process.
21, 22	The Host Interface process reformats the internal message into an external message, sets an outbound request timer, and sends the 0200 request message to the host.
23, 24	The host approves the request and returns a 0210 response message to the Host Interface process.
25, 26	The Host Interface process reformats the external message into an internal message, deletes its outbound request timer, and sends the 0210 response message to the Authorization process.
27, 28	The Authorization process logs the response to the Transaction Log File (TLF), updates the database files, and sends the 0210 response message to the Device Handler process.

Steps	Processing
29, 30	The Device Handler process deletes its authorization timer and reformats the 0210 response message into a native-mode response that it sends to the ATM.
31, 32	The ATM sends a native-mode message to the Device Handler process to indicate that the transaction completed as authorized.

Approved by Host — Completions Required

The diagram below illustrates the flow of a transaction that originates at an ATM and is approved by a host. The host requires completions.



Steps

Processing

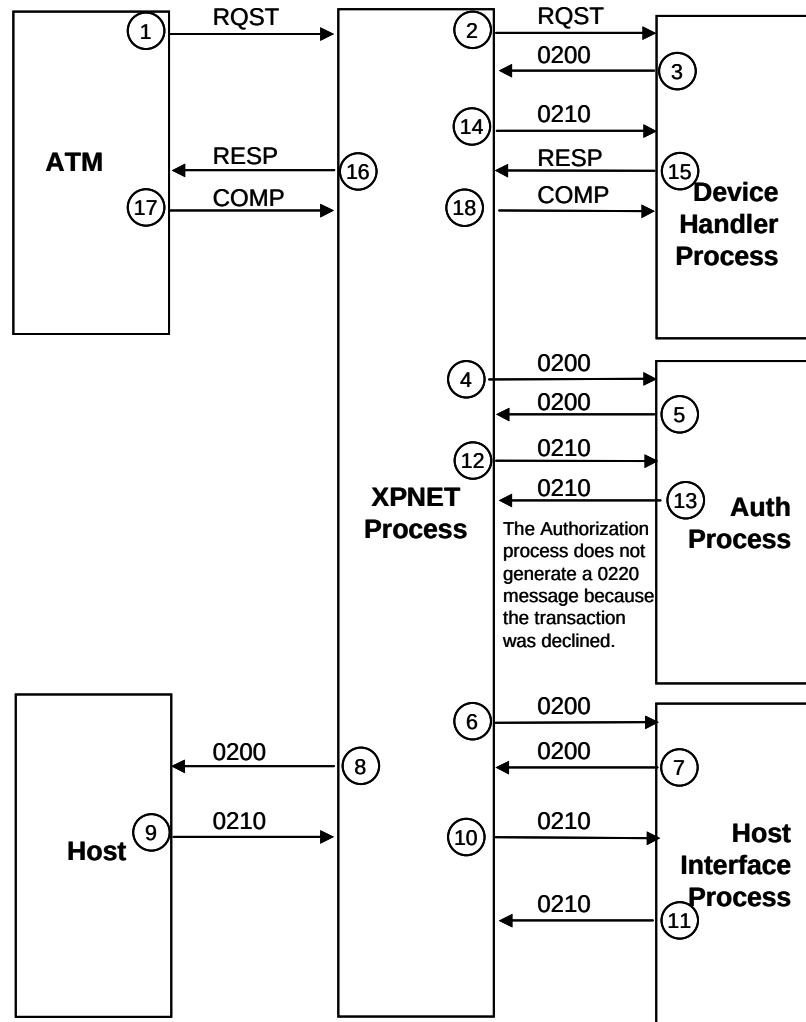
- 1, 2 The ATM sends a native-mode request message to the Device Handler process.
- 3, 4 The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Authorization process.

Steps	Processing
5, 6	From the Card Prefix File (CPF), the Authorization process locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 3. The Authorization process sends the 0200 request message to the Host Interface process.
7, 8	The Host Interface process reformats the internal message into an external message, sets an outbound request timer, and sends the 0200 request message to the host.
9, 10	The host approves the request and changes the 0200 request into a 0210 response message that it sends to the Host Interface process.
11, 12	The Host Interface process reformats the external message into an internal message, deletes its outbound request timer, and sends the 0210 response message to the Authorization process.
13, 14	The Authorization process performs the following steps: a. Logs the response to the Transaction Log File (TLF), updates the cardholder database files, and sends the 0210 response message to the Device Handler process. Processing continues with step 15a. b. Sends a 0220 completion message to the Host Interface process when the COMPLETION REQUIRED TO HOST field on the Issuer Processing Code File (IPCF) screen is set to 1 or 4. Processing continues with step 15b.
15a, 16a	The Device Handler process deletes its authorization timer and reformats the 0210 response message into a native-mode response that it sends to the ATM. Processing continues with step 17.
15b, 16b	The Host Interface process reformats the internal message into an external message and sends the 0220 completion message to the host when the COMPLETION REQUIRED TO HOST field on the Issuer Processing Code File (IPCF) screen is set to 1 or 4. This leg of processing is now complete.

Steps	Processing
17, 18	The ATM sends a native-mode message to the Device Handler process to indicate that the transaction completed as authorized.

Declined by Host — Completions Required

The diagram below illustrates the flow of a transaction that originates at an ATM and is declined by a host. The host requires completions.



Steps

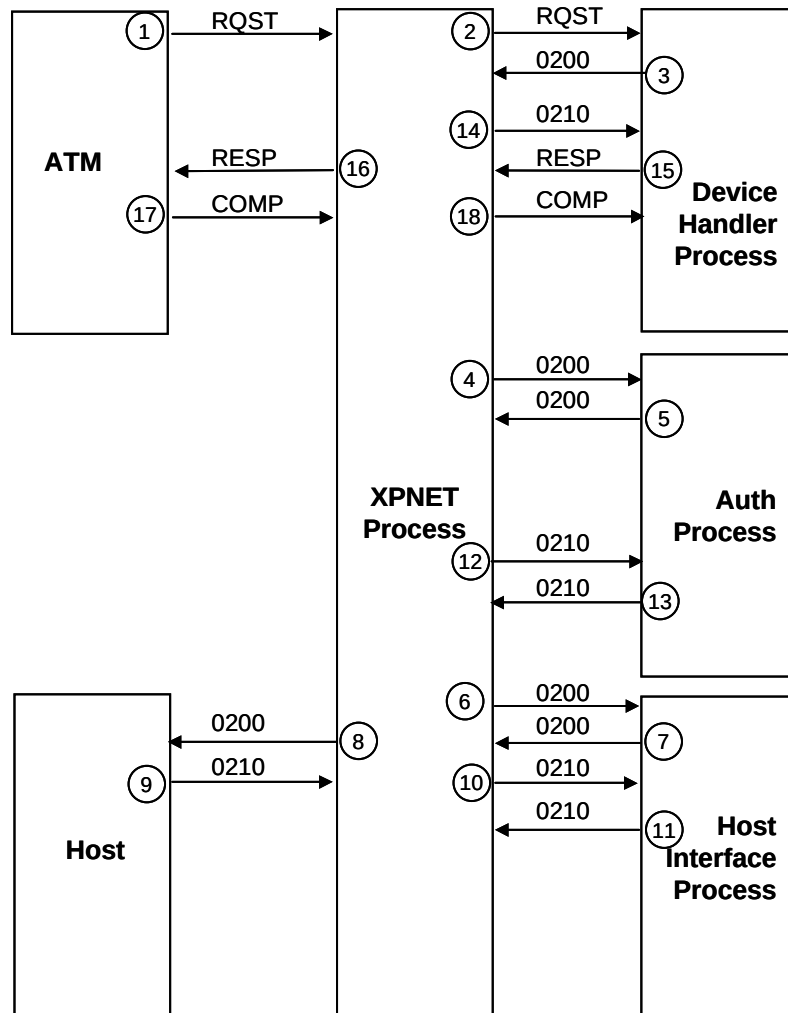
Processing

- | | |
|------|---|
| 1, 2 | The ATM sends a native-mode request message to the Device Handler process. |
| 3, 4 | The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Authorization process. |

Steps	Processing
5, 6	From the Card Prefix File (CPF), the Authorization process locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 3. The Authorization process sends the 0200 request message to the Host Interface process.
7, 8	The Host Interface process reformats the internal message into an external message, sets an outbound request timer, and sends the 0200 request message to the host.
9, 10	The host declines the request and changes the 0200 request into a 0210 response message that it sends to the Host Interface process.
11, 12	The Host Interface process reformats the external message into an internal message, deletes its outbound request timer, and sends the 0210 response message to the Authorization process.
13, 14	The Authorization process logs the response to the Transaction Log File (TLF), updates the cardholder database files if applicable, and sends the 0210 response message to the Device Handler process. The Authorization process generates a 0220 completion message when the COMPLETION REQUIRED TO HOST field on the IPCF screen is set to 4.
15, 16	The Device Handler process deletes its authorization timer and reformats the 0210 response message into a native-mode response that it sends to the ATM.
17, 18	The ATM sends a native-mode message to the Device Handler process to indicate that the declined transaction was completed.

Approved by Host — Transaction is a Balance Inquiry — Completions Required

The diagram below illustrates the flow of a balance inquiry transaction that originates at an ATM and is approved by a host. Although the host requires completions, a balance inquiry transaction requires no completion.



Steps	Processing
1, 2	The ATM sends a native-mode request message to the Device Handler process.
3, 4	The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Authorization process. Note: If a customer receipt is required for approved balance inquiry transactions, the customer receipt printer is faulted or down, and the optional receipts feature is not used, the Device Handler process denies the transaction and does not forward it to the Authorization process.
5, 6	From the Card Prefix File (CPF), the Authorization process locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 3. The Authorization process sends the 0200 request message to the Host Interface process.
7, 8	The Host Interface process reformats the internal message into an external message, sets an outbound request timer, and sends the 0200 request message to the host.
9, 10	The host approves the request and changes the 0200 request into a 0210 response message that it sends to the Host Interface process.
11, 12	The Host Interface process reformats the external message into an internal message, deletes its outbound request timer, and sends the 0210 response message to the Authorization process.
13, 14	The Authorization process logs the response to the Transaction Log File (TLF) and sends the 0210 response message to the Device Handler process. The Authorization process generates a 0220 completion message when the COMPLETION REQUIRED TO HOST field on the IPCF screen is set to 1 or 4.
15, 16	The Device Handler process deletes its authorization timer and reformats the 0210 response message into a native-mode response that it sends to the ATM.

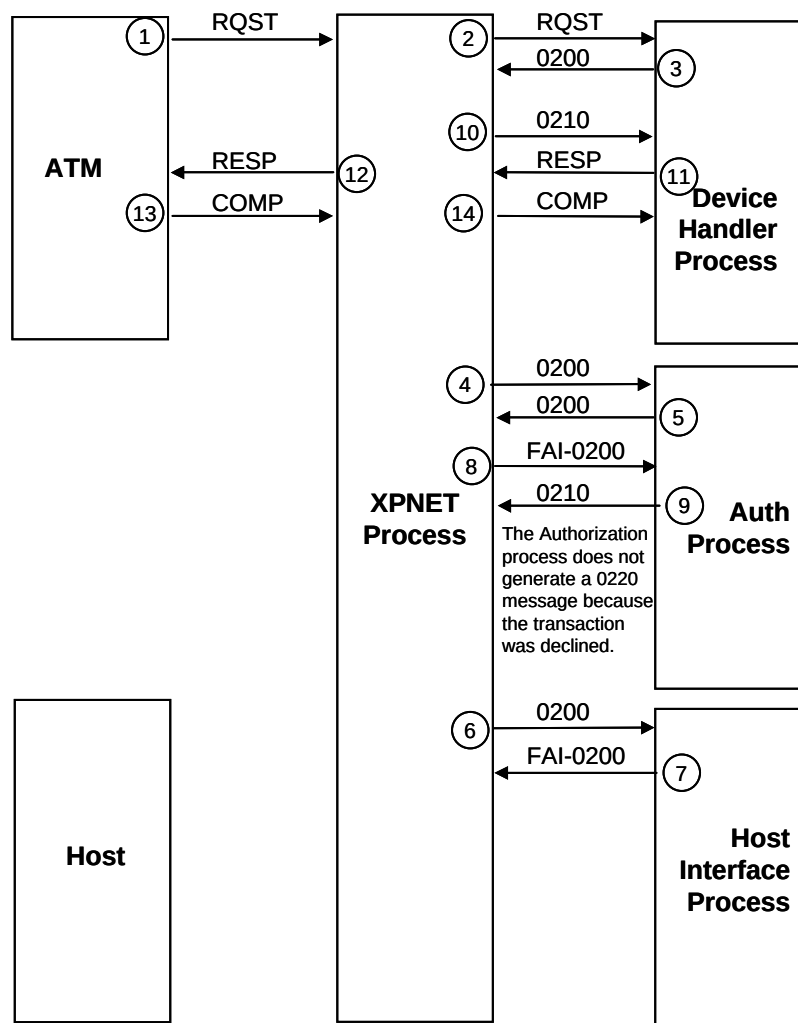
Steps**Processing**

17, 18

The ATM sends a native-mode message to the Device Handler process to indicate that the transaction completed as authorized.

Declined by BASE24-atm — Completions Required

The diagram below illustrates the flow of a transaction that originates at an ATM and is declined by BASE24-atm, which stands in for the host because communications to the host are down. The host requires completions.



Steps

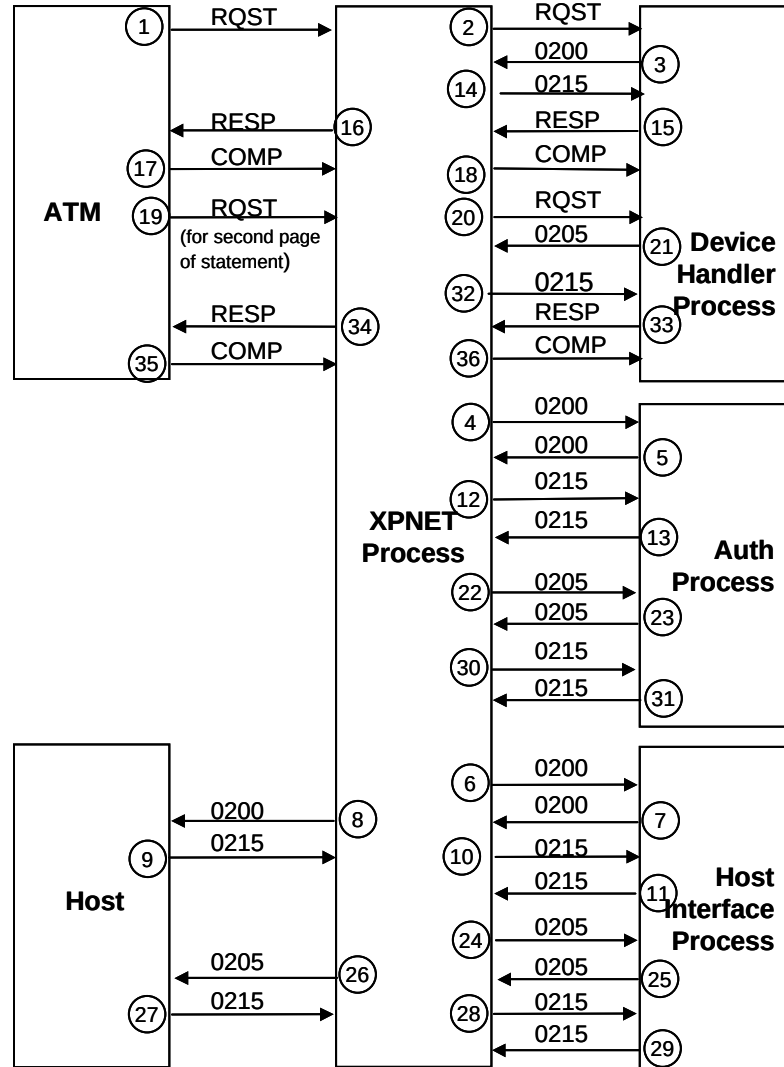
Processing

- | | |
|------|---|
| 1, 2 | The ATM sends a native-mode request message to the Device Handler process. |
| 3, 4 | The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Authorization process. |

Steps	Processing
5, 6	From the Card Prefix File (CPF), the Authorization process locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 3. The Authorization process sends the 0200 request message to the Host Interface process.
7, 8	Because the host is unavailable, the Host Interface process changes the 0200 request message to a failed 0200 request message and sends it to the Authorization process.
9, 10	The Authorization process stands in and declines the request, changes the failed 0200 request message to a 0210 response message, logs the response to the Transaction Log File (TLF), updates the cardholder database files if applicable, and sends the 0210 response to the Device Handler process. The Authorization process generates a 0220 completion message when the COMPLETION REQUIRED TO HOST field on the IPCF screen is set to 4 or 6.
11, 12	The Device Handler process deletes its authorization timer and reformats the 0210 response message into a native-mode response that it sends to the ATM.
13, 14	The ATM sends a native-mode message to the Device Handler process to indicate that the declined transaction was completed.

Approved by Host — Transaction is a Statement Print Transaction — Completions Not Required

The diagram below illustrates the flow of a statement print transaction that originates at an ATM and is approved by a host. The host does not require completions. Because of the amount of information carried in the statement print transaction message, several request and response messages may be sent between BASE24-atm and the host.

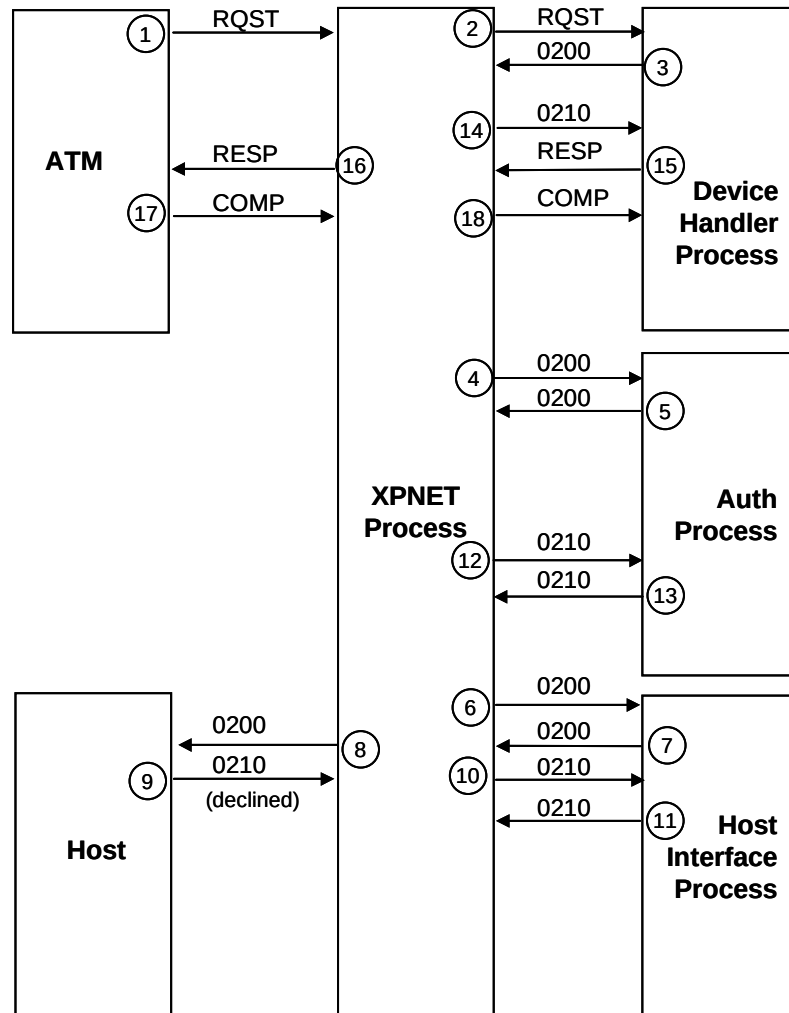


Steps	Processing
1, 2	The ATM sends a native-mode request message to the Device Handler process.
3, 4	The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Authorization process.
5, 6	From the Card Prefix File (CPF), the Authorization process locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 3. The Authorization process sends the 0200 request message to the Host Interface process.
7, 8	The Host Interface process reformats the internal message into an external message, sets an outbound request timer, and sends the 0200 request message to the host.
9, 10	The host processes the 0200 request message and returns a 0215 response message to the Host Interface process. The 0215 message carries the statement information (which may or may not be all of the information needed).
11, 12	The Host Interface process reformats the external message into an internal message, deletes its outbound request timer, and sends the 0215 response message to the Authorization process.
13, 14	The Authorization process logs the response to the Transaction Log File (TLF) and sends the 0215 message to the Device Handler process.
15, 16	The Device Handler process reformats the internal message into a native-mode response message that it sends to the ATM.
17, 18	The ATM sends a native-mode message to the Device Handler process to indicate it has received the statement information.
19, 20	The ATM sends a native-mode request message to the Device Handler process requesting additional statement information.

Steps	Processing
21, 22	The Device Handler process reformats the native-mode message into a 0205 message, requesting additional statement information for the transaction in progress, sets an authorization timer, and sends the 0205 message to the Authorization process.
23, 24	The Authorization process sends the 0205 message to the Host Interface process.
25, 26	The Host Interface process reformats the internal message into an external message, sets an outbound request timer, and sends the 0205 message to the host.
27, 28	The host processes the 0205 message and returns a 0215 response message to the Host Interface process. The 0215 message carries additional statement information.
29, 30	The Host Interface process reformats the external message into an internal message, deletes its outbound request timer, and sends the 0215 message to the Authorization process.
31, 32	The Authorization process sends the 0215 message to the Device Handler process.
33, 34	The Device Handler process deletes its authorization timer and reformats the 0215 message into a native-mode response that it sends to the ATM.
35, 36	The ATM sends a native-mode message to the Device Handler process to indicate that the transaction completed as authorized. If additional information is needed, steps 19–34 are repeated until all information has been sent.

Declined by Host — Transaction is a Statement Print Transaction — Completions Not Required

The diagram below illustrates the flow of a statement print transaction that originates at an ATM and is declined by a host. The host does not require completions.



Steps

Processing

- | | |
|------|---|
| 1, 2 | The ATM sends a native-mode request message to the Device Handler process. |
| 3, 4 | The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Authorization process. |

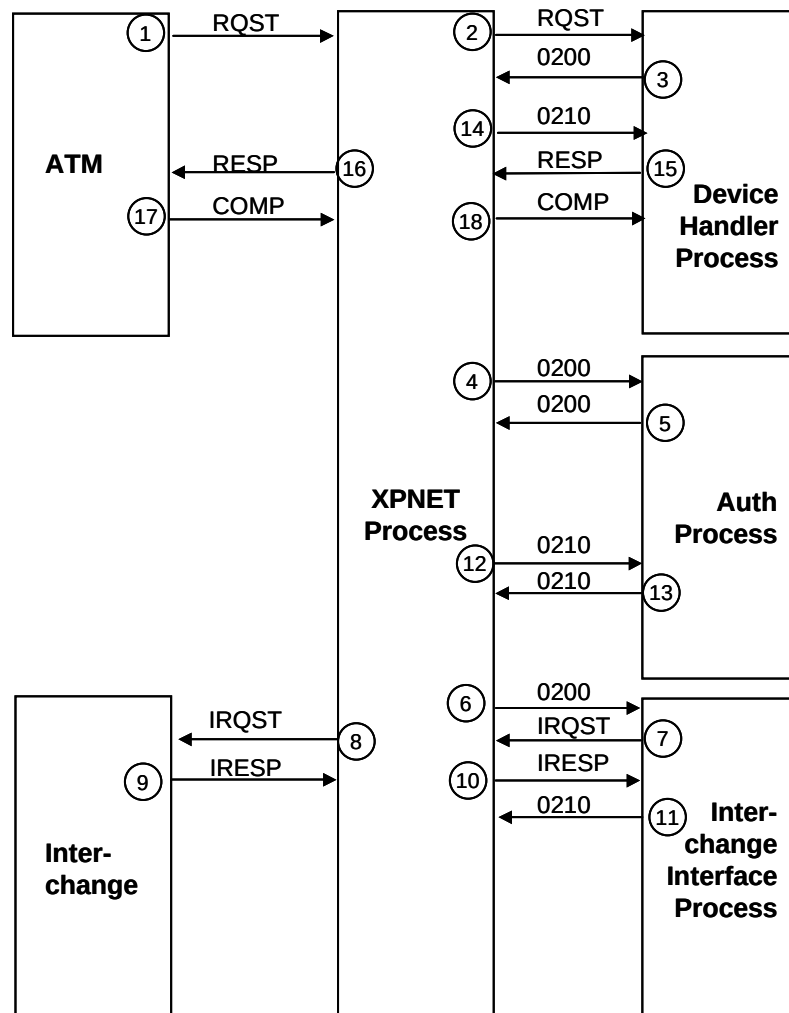
Steps	Processing
5, 6	From the Card Prefix File (CPF), the Authorization process locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 3. The Authorization process sends the 0200 request message to the Host Interface process.
7, 8	The Host Interface process reformats the internal message into an external message, sets an outbound request timer, and sends the 0200 request message to the host.
9, 10	The host declines the transaction and returns a 0210 response message to the Host Interface process.
11, 12	The Host Interface process reformats the external message into an internal message, deletes its outbound request timer, and sends the 0210 response message to the Authorization process.
13, 14	The Authorization process logs the response to the Transaction Log File (TLF) and sends the 0210 response message to the Device Handler process.
15, 16	The Device Handler process deletes its authorization timer and reformats the 0210 response message into a native-mode response that it sends to the ATM.
17, 18	The ATM sends a native-mode message to the Device Handler process to indicate that the declined transaction was completed.

Interchange Transactions: Requests Sent to an Interchange

The following pages contain transaction diagrams for situations where transactions are sent to an interchange for authorization.

Approved by an Interchange

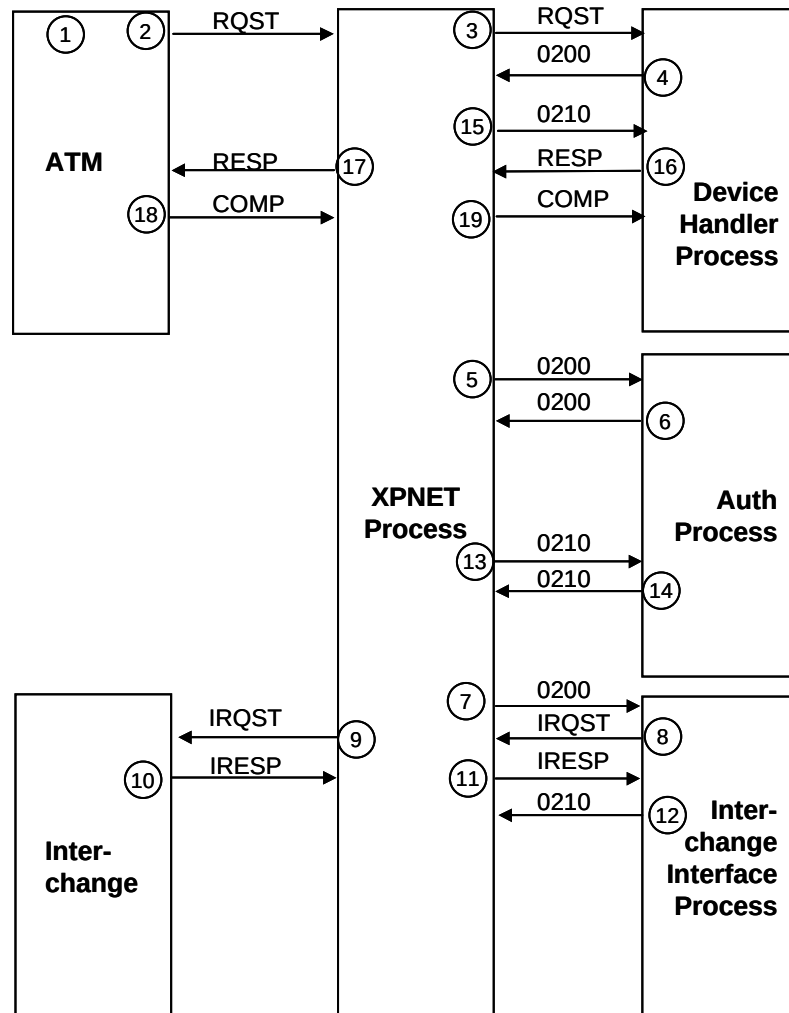
The diagram below illustrates the flow of a transaction that originates at an ATM and is approved by an interchange.



Steps	Processing
1, 2	The ATM sends a native-mode request message to the Device Handler process.
3, 4	The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Authorization process.
5, 6	The Authorization process does not recognize the card prefix and uses the SPROUTE File (SPROUTE) to determine the authorizer. The Authorization process sends the 0200 request message to the Interchange Interface process.
7, 8	The Interchange Interface process reformats the internal 0200 request into an external message format understood by the interchange, sets an outbound request timer, and sends the request message to the interchange.
9, 10	The interchange approves the transaction and sends the response to the Interchange Interface process.
11, 12	The Interchange Interface process reformats the response into a 0210 internal response message, logs the response in the Interchange Log File (ILF), deletes its outbound request timer, and sends it to the Authorization process.
13, 14	The Authorization process logs the response to the Transaction Log File (TLF) and sends the 0210 response message to the Device Handler process.
15, 16	The Device Handler process deletes its authorization timer and reformats the 0210 response message into a native-mode response that it sends to the ATM.
17, 18	The ATM sends a native-mode message to the Device Handler process to indicate that the transaction completed as authorized.

Approved by an Interchange — Multiple Account Selection by Account Qualifier

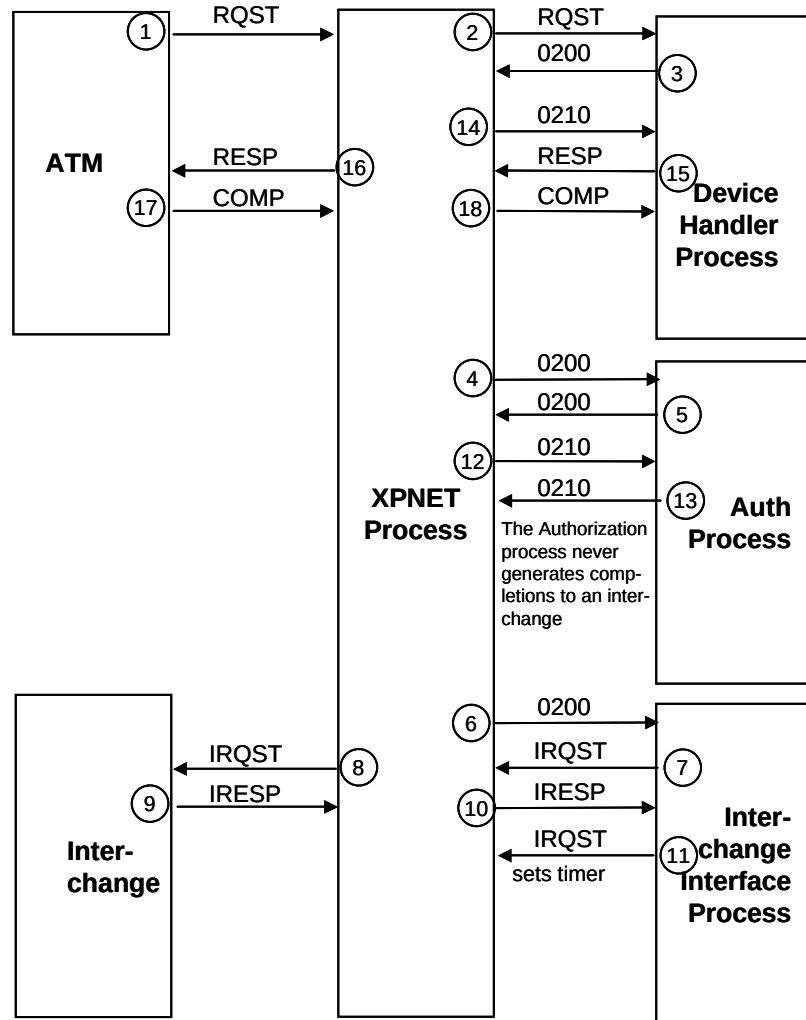
The diagram below illustrates the flow of a transaction that originates at an ATM and is approved by an interchange. The cardholder selects an account using multiple account selection by account qualifier.



Steps	Processing
1	The cardholder initiates a transaction and specifies an account using the account qualifiers presented on the ATM screen.
2, 3	The ATM sends a native-mode request message to the Device Handler process.
4, 5	The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, adds the Account Qualifier token to specify the account selected, and sends the 0200 request message to the Authorization process.
6, 7	The Authorization process does not recognize the card prefix and uses the SPROUTE File (SPROUTE) to determine the authorizer. The Authorization process sends the 0200 request message, including the Account Qualifier token to the Interchange Interface process.
8, 9	The Interchange Interface process reformats the internal 0200 request into an external message format understood by the interchange, sets an outbound request timer, and sends the request message to the interchange.
10, 11	The interchange approves the transaction and sends the response to the Interchange Interface process.
12, 13	The Interchange Interface process reformats the response into a 0210 internal response message, logs the response in the Interchange Log File (ILF), deletes its outbound request timer, and sends it to the Authorization process.
14, 15	The Authorization process logs the response to the Transaction Log File (TLF) and sends the 0210 response message to the Device Handler process.
16, 17	The Device Handler process deletes its authorization timer and reformats the 0210 response message into a native-mode response that it sends to the ATM.
18, 19	The ATM sends a native-mode message to the Device Handler process to indicate that the transaction completed as authorized.

Approved by an Interchange — Advices Required

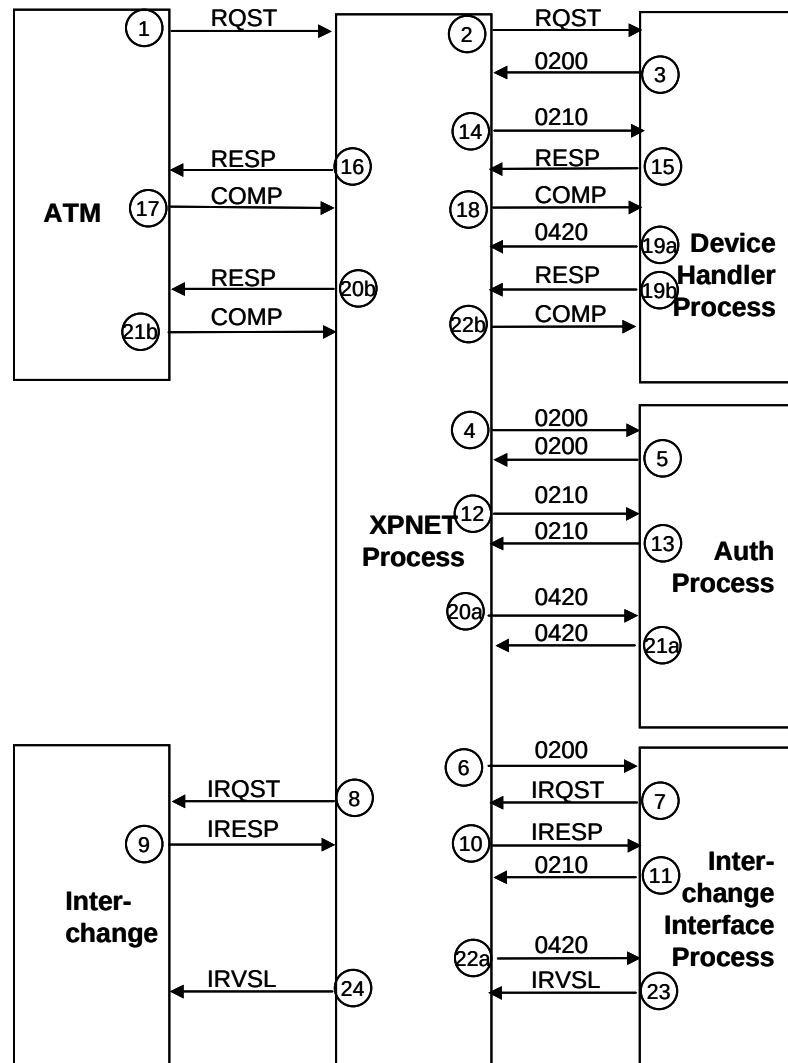
The diagram below illustrates the flow of a transaction that originates at an ATM and is approved by an interchange. Advices are required by the interchange. If the Interchange Interface process has not received a reversal from the terminal by the time an advice timer expires, the Interchange Interface process generates and sends an advice to the interchange to notify the interchange that the transaction completed as authorized.



Steps	Processing
1, 2	The ATM sends a native-mode request message to the Device Handler process.
3, 4	The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Authorization process.
5, 6	The Authorization process does not recognize the card prefix and uses SPROUTE to determine the authorizer. The Authorization process sends the 0200 request message to the Interchange Interface process.
7, 8	The Interchange Interface process reformats the 0200 request into an external message format understood by the interchange, sets an outbound request timer, and sends the request to the interchange.
9, 10	The interchange approves the request and sends the response to the Interchange Interface process.
11, 12	The Interchange Interface process deletes its outbound request timer, sets an advice timer, reformats the response into a 0210 internal response message, logs the response in the Interchange Log File (ILF), and sends the message to the Authorization process.
13, 14	The Authorization process sends the 0210 response message to the Device Handler process.
15, 16	The Device Handler process deletes its authorization timer and reformats the 0210 response message into a native-mode response that it sends to the ATM.
17, 18	The ATM sends a native-mode message to the Device Handler process to indicate that the transaction completed as authorized.

Approved by an Interchange — Transaction Reversed — Advices Required — Advice Not Sent

The diagram below illustrates the flow of a transaction that originates at an ATM, is approved by an interchange, and is then reversed. The interchange requires advices, and the reversal reaches the Interchange Interface process prior to its sending an advice to the interchange.

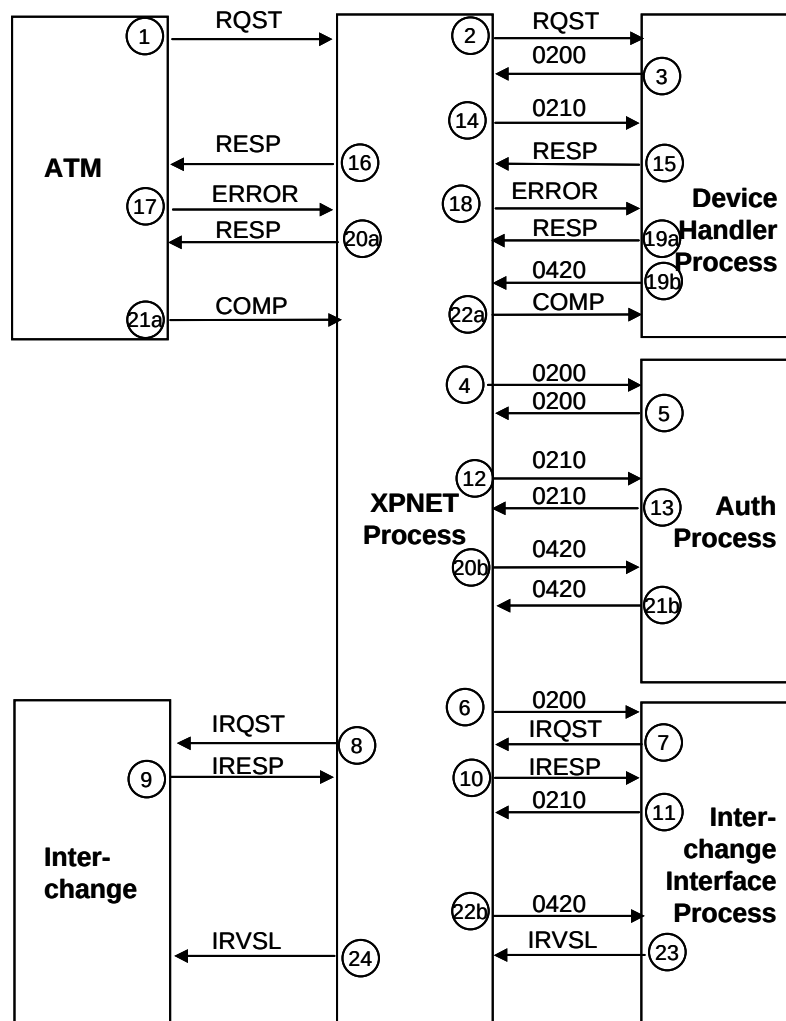


Steps	Processing
1, 2	The ATM sends a native-mode request message to the Device Handler process.
3, 4	The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Authorization process.
5, 6	The Authorization process does not recognize the card prefix and uses SPROUTE to determine the authorizer. The Authorization process sends the 0200 request message to the Interchange Interface process.
7, 8	The Interchange Interface process reformats the 0200 request into an external message format understood by the interchange, sets an outbound request timer, and sends the request to the interchange.
9, 10	The interchange approves the request and sends the response to the Interchange Interface process.
11, 12	The Interchange Interface process deletes its outbound request timer, sets an advice timer, reformats the response into a 0210 internal response message, logs the response in the Interchange Log File (ILF), and sends the message to the Authorization process.
13, 14	The Authorization process logs the response to the Transaction Log File (TLF) and sends the 0210 response message to the Device Handler process.
15, 16	The Device Handler process deletes its authorization timer and reformats the 0210 response message into a native-mode response that it sends to the ATM.
17, 18	The ATM sends a native-mode message to the Device Handler process to indicate that the transaction did not complete as authorized.

Steps	Processing
19, 20	The Device Handler process performs the following steps: a. The Device Handler process generates a 0420 reversal message that it sends to the Authorization process. Processing continues with step 21a. b. Sends a native-mode response to the ATM indicating to print the receipt for however the transaction was completed. Processing continues with step 21b.
21a, 22a	The Authorization process logs the reversal to the TLF and sends the 0420 message to the Interchange Interface process. Processing continues with step 23.
21b, 22b	The ATM sends a native-mode message to the Device Handler process indicating that the transaction has been completed. This leg of processing is now complete.
23, 24	The Interchange Interface process advice timer has not expired, so the Interchange Interface process reformats the reversal message into an external message format understood by the interchange and sends the reversal message to the interchange.

Approved by an Interchange — Transaction Reversed — Advices Required — Advice Sent

The diagram below illustrates the flow of a transaction that originates at an ATM, is approved by an interchange, and is then reversed. The interchange requires advices, and the reversal reaches the Interchange Interface process after it has sent an advice to the interchange.



Steps

Processing

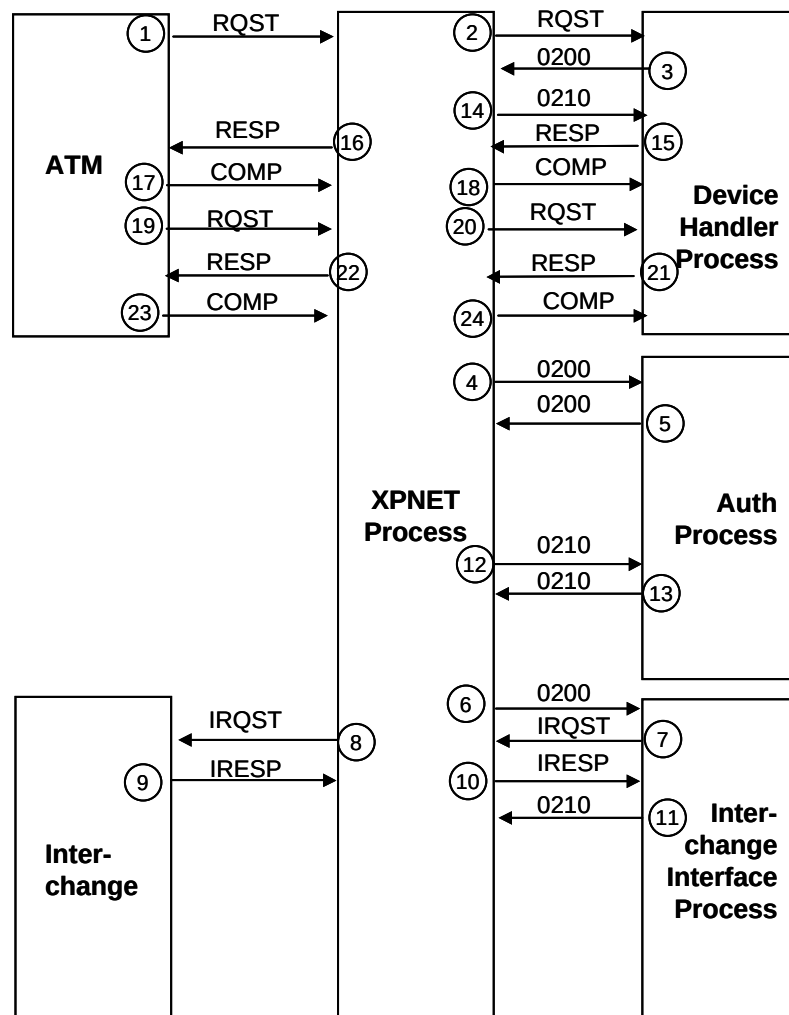
- | | |
|------|---|
| 1, 2 | The ATM sends a native-mode request message to the Device Handler process. |
| 3, 4 | The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Authorization process. |

Steps	Processing
5, 6	The Authorization process does not recognize the card prefix and uses SPROUTE to determine the authorizer. The Authorization process sends the 0200 request message to the Interchange Interface process.
7, 8	The Interchange Interface process reformats the 0200 request into an external message format understood by the interchange, sets an outbound request timer, and sends the request to the interchange.
9, 10	The interchange approves the request and sends the response to the Interchange Interface process.
11, 12	The Interchange Interface process deletes its outbound request timer, sets an advice timer, reformats the response into a 0210 internal response message, logs the response in the Interchange Log File (ILF), and sends the message to the Authorization process.
13, 14	The Authorization process logs the response to the Transaction Log File (TLF) and sends the 0210 response message to the Device Handler process.
15, 16	The Device Handler process deletes its authorization timer and reformats the 0210 response message into a native-mode response that it sends to the ATM.
17, 18	The ATM sends a native-mode message to the Device Handler process to indicate that the transaction did not complete as authorized.
19, 20	The Device Handler process performs the following steps: a. Generates a 0420 reversal message that it sends to the Authorization process. Processing continues with 21a. b. Sends a native-mode response to the ATM indicating to print the receipt for however the transaction was completed. Processing continues with step 21b.
21a, 22a	The Authorization process logs the reversal to the TLF and sends the 0420 message to the Interchange Interface process. Processing continues with step 23.

Steps	Processing
21b, 22b	The ATM sends a native-mode message to the Device Handler process indicating that the transaction has been completed. This leg of processing is now complete.
23, 24	The Interchange Interface process reformats the reversal message into an external message format understood by the interchange and sends the reversal message to the interchange.

Approved by an Interchange — Surcharge and Issuer Fee Applied — Interactive Response Notification Processing — Cardholder Accepts Transaction

The diagram below illustrates the flow of a transaction that originates at an ATM and is approved at an interchange. The institution uses interactive response notification processing; that is the transaction is routed to the issuer for authorization before the cardholder is notified of any fees. In this scenario, a surcharge and an issuer fee (or rebate) are applied, and the cardholder elects to proceed with the transaction.

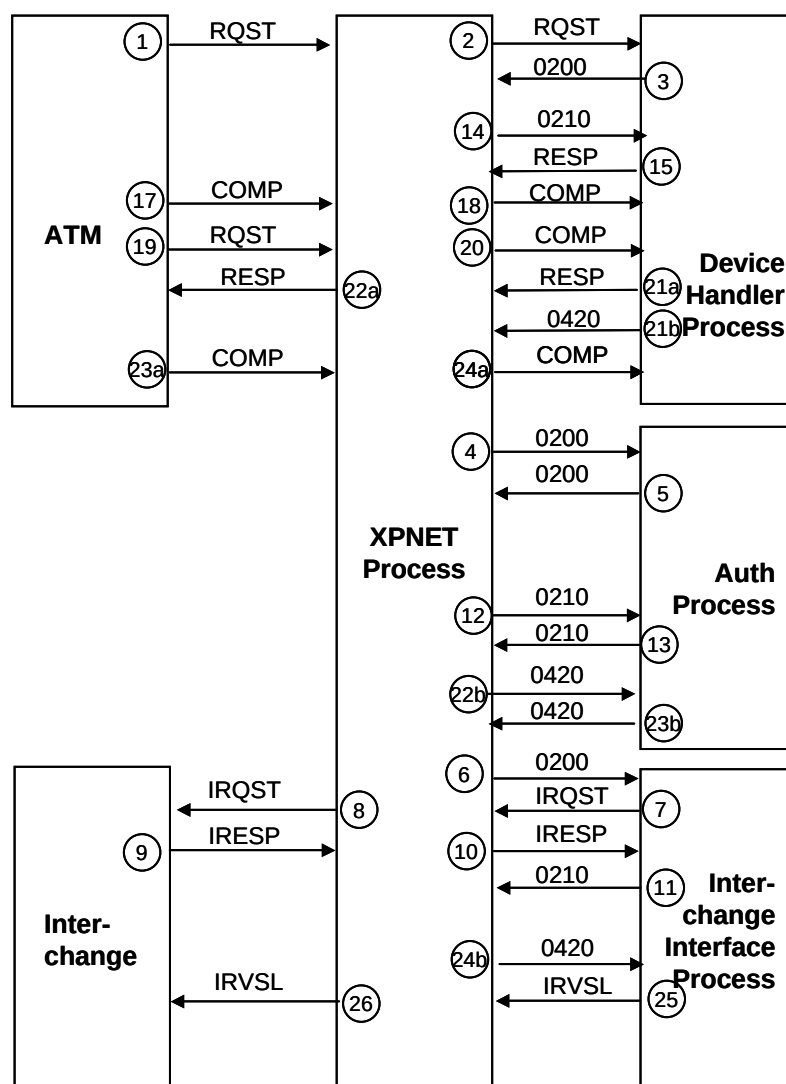


Steps	Processing
1, 2	The ATM sends a native-mode request message to the Device Handler process.
3, 4	The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Authorization process. The message contains the Surcharge Data token with the terminal surcharge profile ID specified in the SURCHARGE PROFILE field on ATD screen 1. The AUTH-IND field of the token is set to a value of 1, indicating that the Authorization process routes the transaction with the surcharge without notifying the cardholder first.
5, 6	The Authorization process uses the terminal surcharge profile ID and the card profile in the Card Prefix File (CPF) to determine the surcharge to be assessed. The Authorization process then determines the authorization and routing parameters and sends the 0200 request message (which includes the Surcharge data token and the surcharge amount) to the Interchange Interface process.
7, 8	The Interchange Interface process reformats the internal message into an external message format understood by the interchange, sets an outbound request timer, and sends the 0200 request message to the interchange. The message includes the Surcharge Data token with the surcharge amount. The surcharge is also reflected in the transaction amount (the AMT-1 field of the STM).
9, 10	The interchange approves the request, adds an issuer fee, and sends the response to the Interchange Interface process.
11, 12	The Interchange Interface process reformats the external message into an internal message, deletes its outbound request timer, and sends the 0210 response message to the Authorization process. The message includes the Surcharge Data token with the surcharge amount and the Issuer Fee Rebate token with the issuer fee or rebate amount.

Steps	Processing
13, 14	The Authorization process formats a Transaction Log File (TLF) record, writes it to the TLF, and sends the 0210 response message to the Device Handler process. The message includes the Surcharge Data token with the surcharge amount and the Issuer Fee Rebate token with the issuer fee or rebate amount. The token information may or may not be logged to the TLF, depending on the configuration of the Token File (TKN).
15, 16	The Device Handler process deletes its authorization timer and reformats the 0210 response message to include a surcharge or issuer fee or rebate notify screen and sends it to the ATM.
17, 18	The ATM sends a native-mode completion message to the Device Handler process and displays the corresponding surcharge or issuer fee or rebate screen to the cardholder.
19, 20	The cardholder elects to proceed with the transaction, and the ATM sends a native-mode request message to the Device Handler process.
21, 22	The Device Handler process sends a native-mode response message to the ATM. The response contains the surcharge and issuer fee or rebate information mapped into the receipt data using the Receipt File (RCPT).
23, 24	The ATM sends a completion message to the Device Handler process.

Approved by an Interchange — Surcharge and Issuer Fee Applied — Interactive Response Notification Processing — Cardholder Cancels Transaction

The diagram below illustrates the flow of a transaction that originates at an ATM and is approved at an interchange. The institution uses interactive response notification processing; that is the transaction is routed to the issuer for authorization before the cardholder is notified of any fees. In this scenario, a surcharge and an issuer fee (or rebate) are applied, and the cardholder elects to cancel the transaction.



Steps	Processing
1, 2	The ATM sends a native-mode request message to the Device Handler process.
3, 4	The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Authorization process. The message contains the Surcharge Data token with the terminal surcharge profile ID specified in the SURCHARGE PROFILE field on ATD screen 1. The AUTH-IND field of the token is set to a value of 1, indicating that the Authorization process routes the transaction with the surcharge without notifying the cardholder first.
5, 6	The Authorization process uses the terminal surcharge profile ID and the card profile in the Card Prefix File (CPF) to determine the surcharge to be assessed. The Authorization process then determines the authorization and routing parameters and sends the 0200 request message (which includes the Surcharge data token and the surcharge amount) to the Interchange Interface process.
7, 8	The Interchange Interface process reformats the internal message into an external message format understood by the interchange, sets an outbound request timer, and sends the 0200 request message to the interchange. The message includes the Surcharge Data token with the surcharge amount. The surcharge is also reflected in the transaction amount (the AMT-1 field of the STM).
9, 10	The interchange approves the request, adds an issuer fee, and sends the response to the Interchange Interface process.
11, 12	The Interchange Interface process reformats the external message into an internal message, deletes its outbound request timer, and sends the 0210 response message to the Authorization process. The message includes the Surcharge Data token with the surcharge amount and the Issuer Fee Rebate token with the issuer fee amount.

Steps	Processing
13, 14	The Authorization process formats a Transaction Log File (TLF) record, writes it to the TLF, and sends the 0210 response message to the Device Handler process. The message includes the Surcharge Data token with the surcharge amount and the Issuer Fee Rebate token with the issuer fee or rebate amount. The token information may or may not be logged to the TLF, depending on the configuration of the Token File (TKN).
15, 16	The Device Handler process deletes its authorization timer and reformats the 0210 response message to include a surcharge or issuer fee notify screen and sends it to the ATM.
17, 18	The ATM sends a native-mode completion message to the Device Handler process and displays the corresponding surcharge or issuer fee screen to the cardholder.
19, 20	The cardholder chooses to cancel the transaction, and the ATM sends a native-mode request message to the Device Handler process indicating the transaction is to be reversed.
21, 22	The Device Handler performs the following steps: a. Sends a native-mode response message to the ATM indicating the customer cancelled the transaction. b. Formats the completion into a 0420 reversal message and sends it to the Authorization process.
23a, 24a	The ATM sends a completion message to the Device Handler process. This leg of processing is now complete.
23b, 24b	The Authorization process logs the reversal to the TLF and sends the 0420 reversal message to the Interchange Interface process. Processing continues with step 25.
25, 26	The Interchange Interface process reformats the 0420 reversal message into an external message and sends it to the interchange.

Interchange Transactions: Adjustments from an Interchange

The following pages contain transaction diagrams for adjustment transactions (5400 messages) sent to BASE24-atm by an interchange.

The processing of adjustments is controlled by the HOST ADJ. PROCESSING flag on screen 3 of the Institution Definition File (IDF) and the authorization level at which an institution is being processed.

The first digit of the HOST ADJ. PROCESSING flag in the IDF can contain the following values:

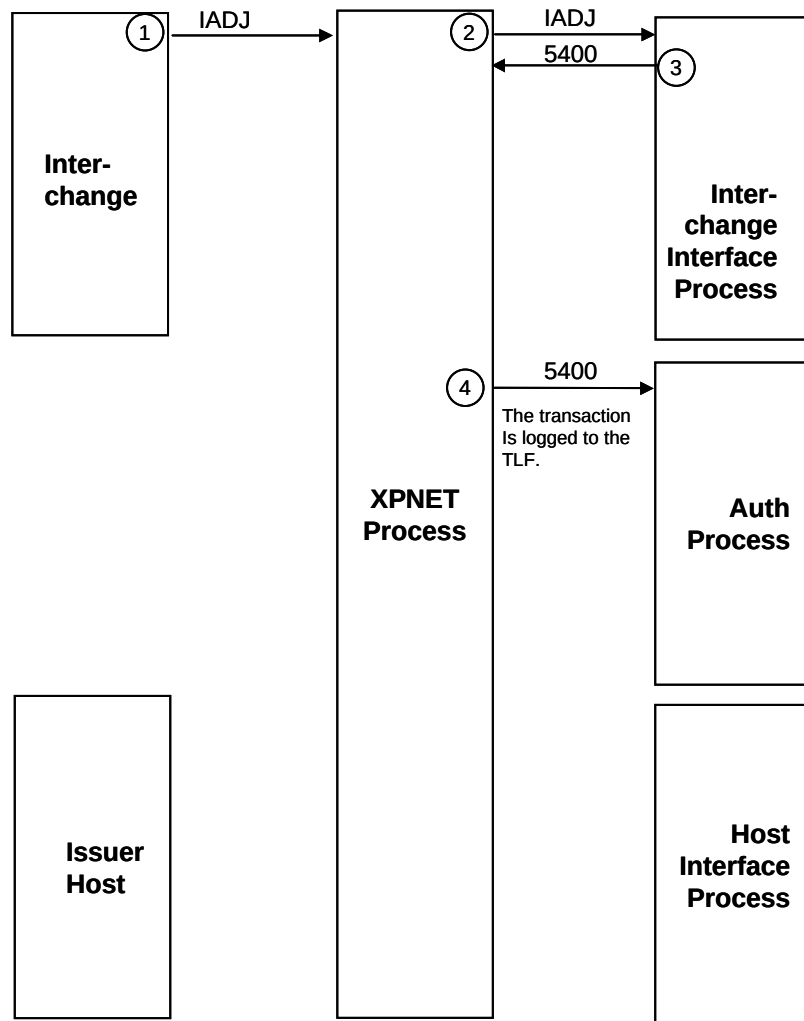
- 0 = Process adjustments manually.
- 1 = Process adjustments on the HP NonStop system only.
- 2 = Process adjustments on the HP NonStop system and at the host.

The interaction of the IDF flag settings and the different authorization levels is illustrated in the following diagrams.

Manual Adjustment from an Interchange

The diagram below illustrates the flow of an adjustment coming to BASE24-atm from an interchange. The authorization level used by the institution has no effect on how the adjustment is processed.

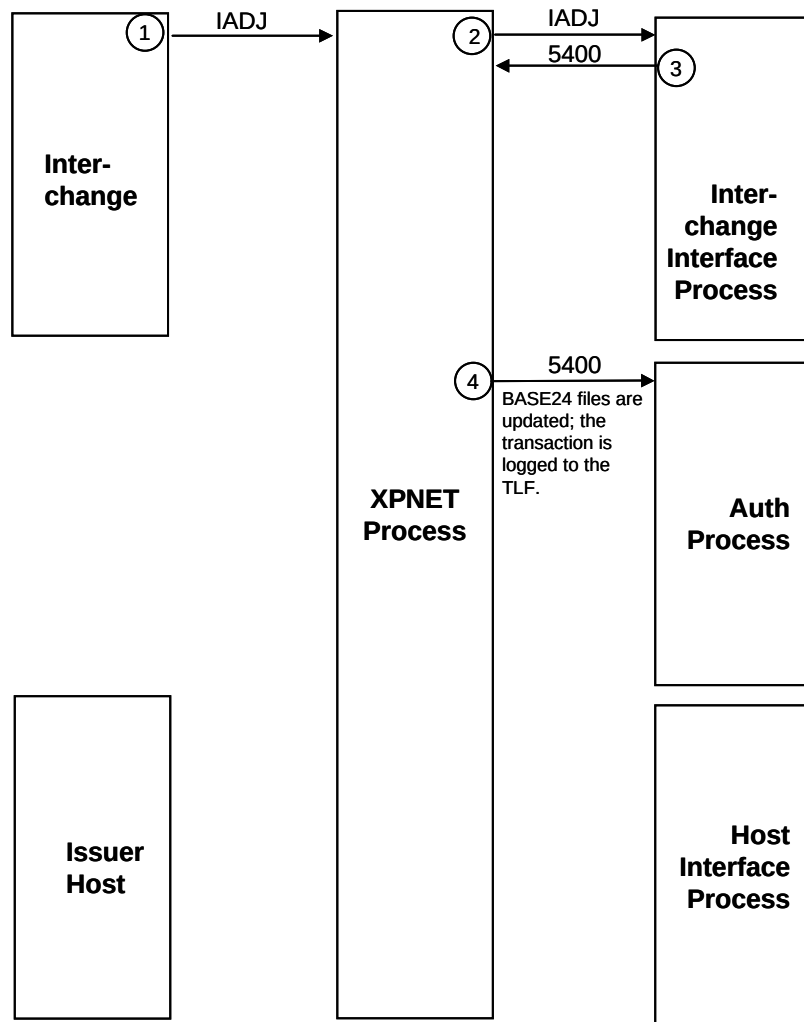
The institution specifies that adjustments must be processed manually by setting the first digit of the HOST ADJ. PROCESSING field on IDF screen 3 to 0.



Steps	Processing
1, 2	An interchange sends a transaction adjustment to its Interchange Interface process.
3, 4	The Interchange Interface process reformats the adjustment message into an internal 5400 adjustment message, logs the adjustment in the Interchange Log File (ILF), and sends the 5400 message to the Authorization process. The Authorization process logs the adjustment to the Transaction Log File (TLF).

Adjustment from an Interchange — Processed by BASE24-atm Only

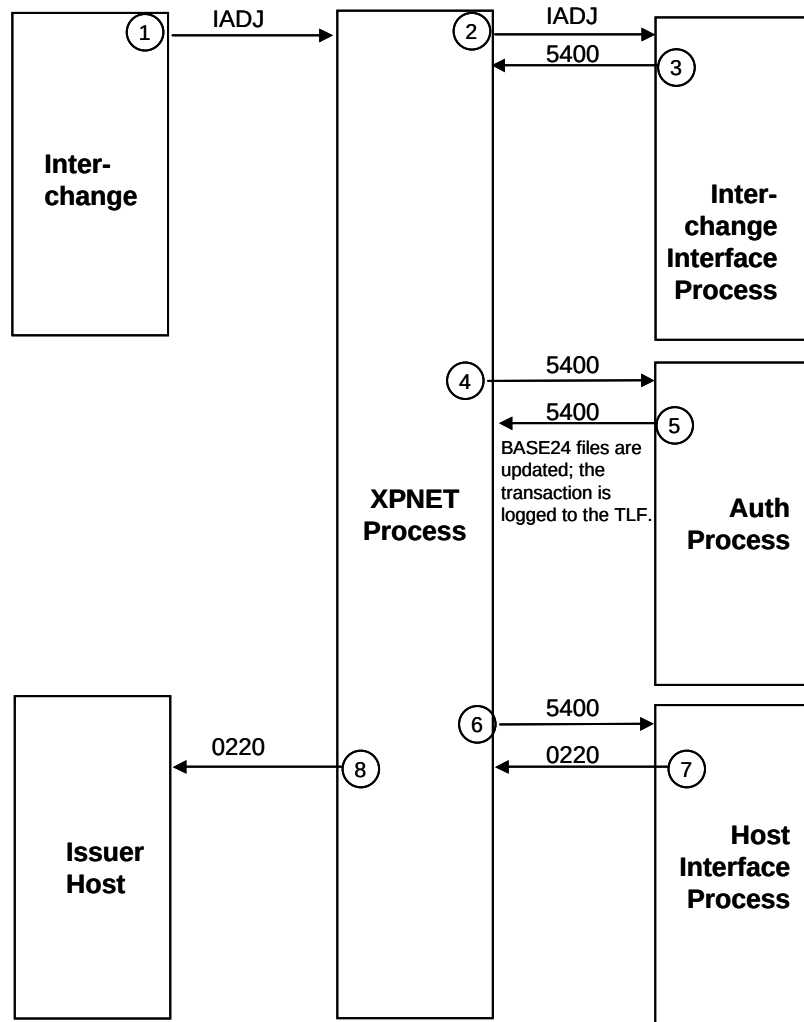
The diagram below illustrates the flow of an adjustment coming to BASE24-atm from an interchange. The first digit of the HOST ADJ. PROCESSING field on IDF screen 3 is set to 1, indicating that adjustments are processed on the BASE24 transaction processing system only. The authorization level used by the institution has no effect on how the adjustment is processed.



Steps	Processing
1, 2	An interchange sends a transaction adjustment to its Interchange Interface process.
3, 4	The Interchange Interface process reformats the adjustment message into an internal 5400 adjustment message, logs the adjustment in the Interchange Log File (ILF), and sends the 5400 message to the Authorization process. The Authorization process logs the adjustment to the Transaction Log File (TLF) and updates the BASE24 files with the information in the 5400 adjustment message.

Adjustment from an Interchange — Processed by BASE24-atm and Host

The diagram below illustrates the flow of an adjustment coming to BASE24-atm from an interchange. The first digit of the HOST ADJ. PROCESSING field on IDF screen 3 is set to 2, indicating that adjustments are processed on the BASE24 transaction processing system and also at the host. The authorization level used is 1 (online) or 3 (online/offline).



Steps	Processing
1, 2	An interchange sends a transaction adjustment to its Interchange Interface process.
3, 4	The Interchange Interface process reformats the adjustment message into an internal 5400 adjustment message, logs the adjustment in the Interchange Log File (ILF), and sends the 5400 message to the Authorization process. The Authorization process logs the adjustment to the Transaction Log File (TLF) and updates the BASE24-atm database appropriately.
5, 6	The Authorization process sends the 5400 adjustment message to the Host Interface process.
7, 8	The Host Interface process reformats the 5400 message to a 0220 external message and sends the 0220 message to the issuer host.

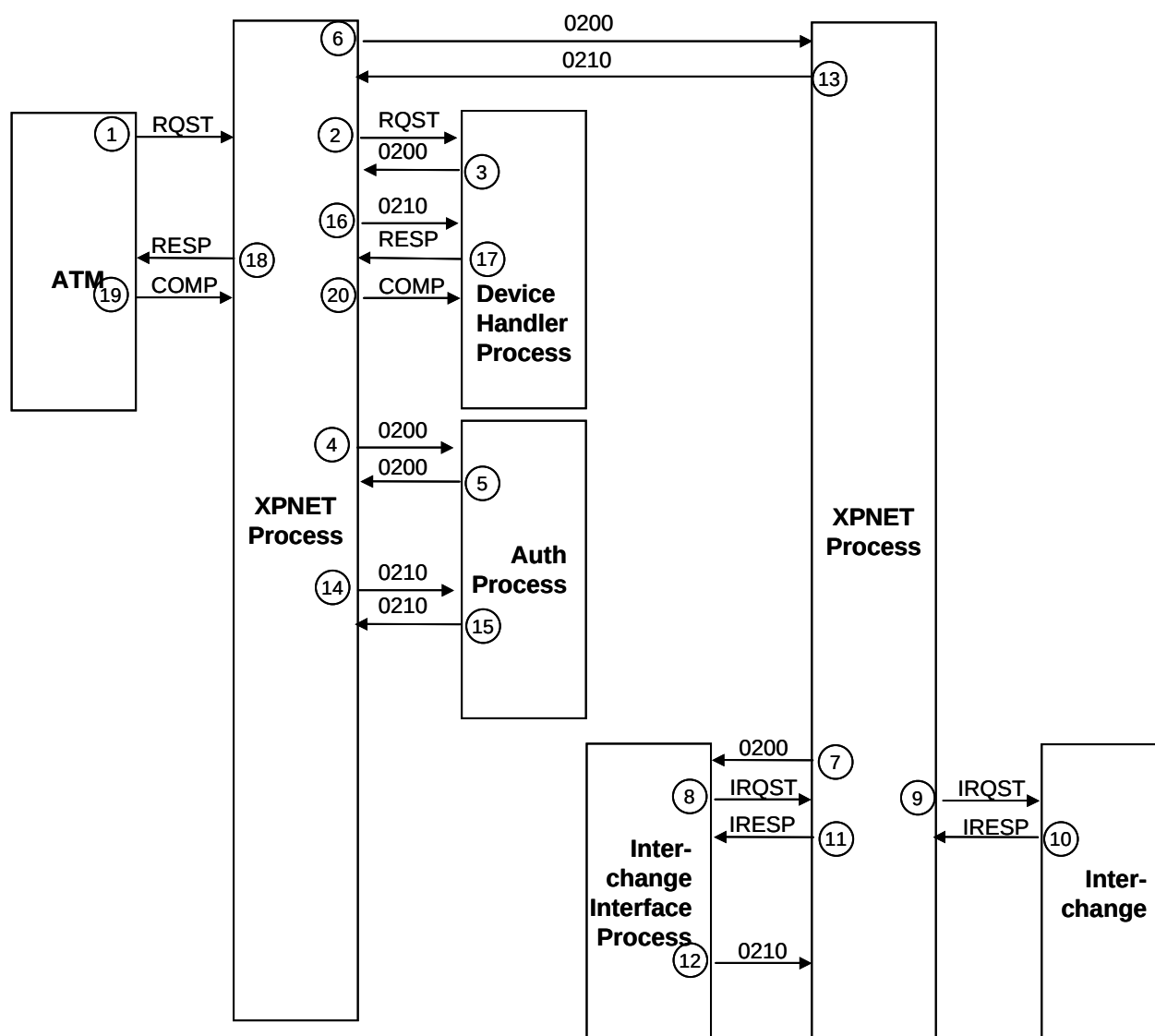
Multinodal Transactions

The following pages contain transaction diagrams for transactions that require traffic between two XPNET nodes.

Approved on Another XPNET Node

The diagram on the following page illustrates the flow of a transaction that originates at an ATM on one XPNET node and is approved by an interchange connected to an Interchange Interface process on a different XPNET node.

In this situation, the Authorization process on the first XPNET node does not have access to the cardholder record, and the transaction must be routed elsewhere for processing. The Authorization process determines the process to which to route the transaction for processing—in this case, an Interchange Interface—using SPROUTE.

**Steps****Processing**

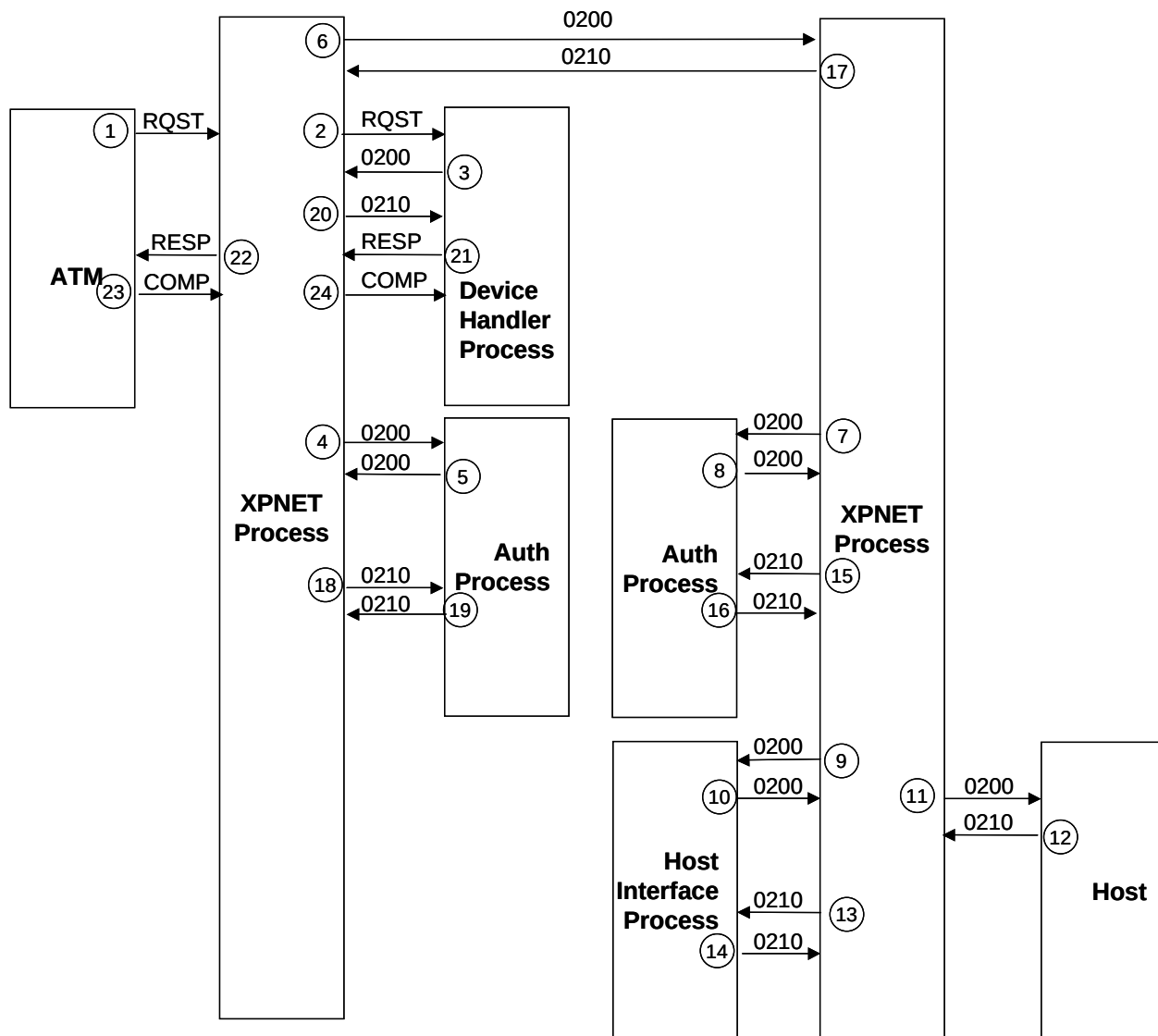
- 1, 2 The ATM sends a native-mode request message to the Device Handler process.
- 3, 4 The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Authorization process.

Steps	Processing
5	The Authorization process does not recognize the card prefix and uses SPROUTE to determine the authorizer. In this situation, the authorizer is an interchange connected to an Interchange Interface process in another XPNET node. The Authorization process places the name of the Interchange Interface process in the 0200 message and sends the message to the XPNET process.
6	The XPNET process does not recognize the Interchange Interface process as a process in its own XPNET node. Using its system configuration information, the XPNET process identifies the XPNET node with which the Interchange Interface process is associated and sends the transaction to the XPNET process corresponding to the XPNET node. This communication occurs through the link functionality of each XPNET process, which allows all of the XPNET processes configured in a system to communicate with one another.
7	The XPNET process of the second XPNET node recognizes the Interchange Interface process as one of the processes in its XPNET node and forwards the 0200 request message to the Interchange Interface process.
8, 9	The Interchange Interface process reformats the 0200 request message into an external message format understood by the interchange, sets an outbound request timer, and sends the request to the interchange.
10, 11	The interchange approves the transaction and returns a response to the Interchange Interface process.
12	The Interchange Interface process reformats the response into an internal 0210 response message, logs the response in the Interchange Log File (ILF), deletes its outbound request timer, and sends the 0210 response message to the Authorization process that originated the transaction.

Steps	Processing
13	The XPNET process does not recognize the Authorization process as a process in its own XPNET node. Using its system configuration information, the XPNET process identifies the XPNET node with which the Authorization process is associated and sends the transaction to the XPNET process corresponding to the XPNET node. This communication occurs through the link functionality of each XPNET process, which allows all of the XPNET processes configured in a system to communicate with one another.
14	The XPNET process of the first XPNET node recognizes the Authorization process as one of the processes in its XPNET node and forwards the 0210 response message to the Authorization process.
15, 16	The Authorization process logs the transaction to the Transaction Log File (TLF) and sends the 0210 response message to the Device Handler process.
17, 18	The Device Handler process deletes its authorization timer and reformats the 0210 response message into a native-mode response that it sends to the ATM.
19, 20	The ATM sends a native-mode message to the Device Handler process to indicate that the transaction completed as authorized.

Approved on Another Logical Network

The diagram below illustrates the flow of a transaction that originates at an ATM in one logical network and must be sent to another logical network for processing. In this situation, the Authorization process on the first XPNET node does not have access to the cardholder record, and the transaction must be routed elsewhere for processing. The Authorization process determines the process to which to route the transaction for processing—in this case, an Authorization process in a different logical network—using SPROUTE.



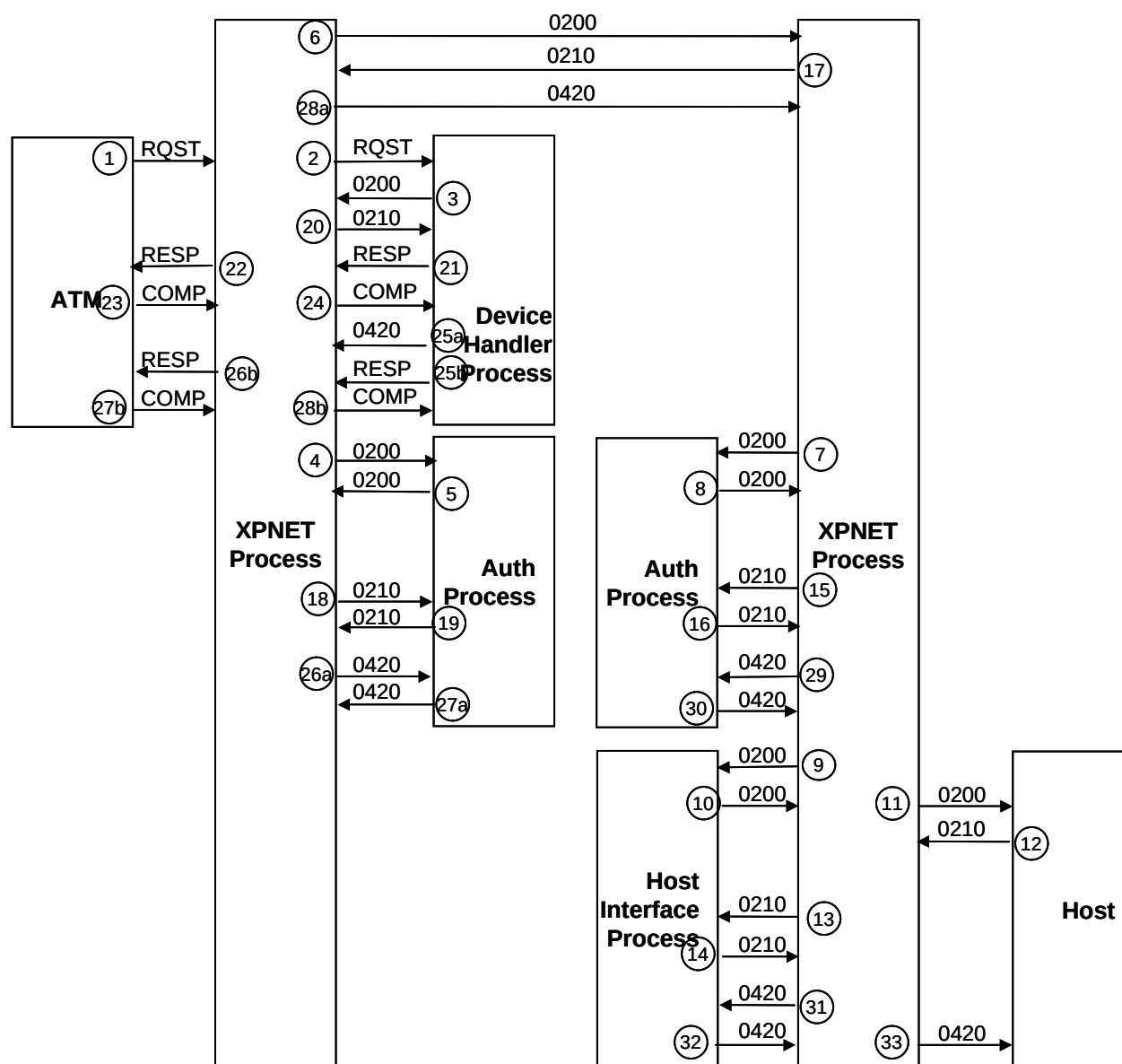
Steps	Processing
1, 2	The ATM sends a native-mode request message to the Device Handler process.
3, 4	The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Authorization process.
5	The Authorization process does not recognize the card prefix and uses SPROUTE to determine the authorizer. From SPROUTE, the Authorization process determines that it must send the request to an Authorization process in another logical network. The Authorization process then sends the 0200 request message to the Authorization process specified in SPROUTE.
6	The XPNET process does not recognize the Authorization process as a process in its own XPNET node. Using its system configuration information, the XPNET process identifies the XPNET node with which the Authorization process is associated and sends the transaction to the XPNET process corresponding to the XPNET node. This communication occurs through the link functionality of each XPNET process, which allows all of the XPNET processes configured in a system to communicate with one another.
7	The XPNET process of the second XPNET node recognizes the Authorization process as one of the processes in its XPNET node and forwards the 0200 request message to the Authorization process.
8, 9	From the Card Prefix File (CPF), the Authorization process locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed by a host. The Authorization process sends the 0200 request message to the Host Interface process.
10, 11	The Host Interface process reformats the 0200 request message into an external message format, sets an outbound request timer, and sends the 0200 request message to the host.
12,13	The host approves the transaction and sends the message to the Host Interface process.

Steps	Processing
14,15	The Host Interface process reformats the response into a 0210 internal response message, deletes its outbound request timer, and sends the message to the Authorization process.
16	The Authorization process logs the transaction to the Transaction Log File (TLF) and sends the 0210 response message to the Authorization process that sent the request.
17	The XPNET process does not recognize the Authorization process as a process in its own XPNET node. Using its system configuration information, the XPNET process identifies the XPNET node with which the Authorization process is associated and sends the transaction to the XPNET process corresponding to the XPNET node. This communication occurs through the link functionality of each XPNET process, which allows all of the XPNET processes configured in a system to communicate with one another.
18	The XPNET process of the first XPNET node recognizes the Authorization process as one of the processes in its XPNET node and forwards the 0210 response message to the Authorization process.
19, 20	The Authorization process logs the transaction to the Transaction Log File (TLF) and sends the 0210 response message to the Device Handler process.
21, 22	The Device Handler process deletes its authorization timer and reformats the 0210 response message into a native-mode response that it sends to the ATM.
23, 24	The ATM sends a native-mode message to the Device Handler process to indicate that the transaction completed as authorized.

Approved on Another Logical Network — Transaction Reversed

The diagram on the following page illustrates the flow of a transaction that originates at an ATM in one logical network and must be sent to another logical network for processing. The transaction does not complete and is subsequently reversed.

In this situation, the Authorization process on the first XPNET node does not have access to the cardholder record, and the transaction must be routed elsewhere for processing. The Authorization process determines the process to which to route the transaction for processing—in this case, an Authorization process in a different logical network—using SPROUTE.

**Steps****Processing**

- 1, 2 The ATM sends a native-mode request message to the Device Handler process.
- 3, 4 The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Authorization process.

Steps	Processing
5	The Authorization process does not recognize the card prefix and uses SPROUTE to determine the authorizer. From SPROUTE, the Authorization process determines that it must send the request to an Authorization process in another logical network. The Authorization process then sends the 0200 request message to the Authorization process specified in SPROUTE.
6	The XPNET process does not recognize the Authorization process as a process in its own XPNET node. Using its system configuration information, the XPNET process identifies the XPNET node with which the Authorization process is associated and sends the transaction to the XPNET process corresponding to the XPNET node. This communication occurs through the link functionality of each XPNET process, which allows all of the XPNET processes configured in a system to communicate with one another.
7	The XPNET process of the second XPNET node recognizes the Authorization process as one of the processes in its XPNET node and forwards the 0200 request message to the Authorization process.
8, 9	From the Card Prefix File (CPF), the Authorization process locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed by a host. The Authorization process sends the 0200 request message to the Host Interface process.
10, 11	The Host Interface process reformats the 0200 request message into an external message format, sets an outbound request timer, and sends the 0200 request message to the host.
12, 13	The host approves the transaction and sends the message to the Host Interface process.
14, 15	The Host Interface process reformats the response into a 0210 internal response message, deletes its outbound request timer, and sends the message to the Authorization process.

Steps	Processing
16	The Authorization process logs the transaction to the Transaction Log File (TLF) and sends the 0210 response message to the Authorization process that sent the request.
17	The XPNET process does not recognize the Authorization process as a process in its own XPNET node. Using its system configuration information, the XPNET process identifies the XPNET node with which the Authorization process is associated and sends the transaction to the XPNET process corresponding to the XPNET node. This communication occurs through the link functionality of each XPNET process, which allows all of the XPNET processes configured in a system to communicate with one another.
18	The XPNET process of the first XPNET node recognizes the Authorization process as one of the processes in its XPNET node and forwards the 0210 response message to the Authorization process.
19, 20	The Authorization process logs the transaction to the Transaction Log File (TLF) and sends the 0210 response message to the Device Handler process.
21, 22	The Device Handler process deletes its authorization timer and reformats the 0210 response message into a native-mode response that it sends to the ATM.
23, 24	The ATM sends a native-mode message to the Device Handler process to indicate that the transaction did not complete as authorized.
25, 26	The Device Handler process performs the following steps: - Based on the completion message returned from the ATM, the Device Handler process determines that the transaction did not complete as authorized and generates a 0420 reversal message that it sends to the Authorization process. Processing continues with step 27a. - Sends a native-mode response to the ATM indicating how to finish the transaction. Processing continues with step 27b.

Steps	Processing
27a	Once again, the Authorization uses SPROUTE to determine where to route the reversal message. The Authorization process sends the 0420 message to the Authorization process for the logical network specified in SPROUTE. Processing continues with step 28a.
28a	The XPNET process does not recognize the Authorization process as a process in its own XPNET node. Using its system configuration information, the XPNET process identifies the XPNET node with which the Authorization process is associated and sends the transaction to the XPNET process corresponding to the XPNET node. This communication occurs through the link functionality of each XPNET process, which allows all of the XPNET processes configured in a system to communicate with one another. Processing continues with step 29.
27b, 28b	The ATM sends a native-mode completion message to the Device Handler process. This leg of processing is now complete.
29	The XPNET process of the second XPNET node recognizes the Authorization process as one of the processes in its XPNET node and forwards the 0420 message to the Authorization process.
30, 31	From the Card Prefix File (CPF), the Authorization process locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed by a host. The Authorization process sends the 0420 message to the Host Interface process.
32, 33	The Host Interface process reformats the 0420 message into an external message format and sends the 0420 message to the host.

Administrative Transaction

The following pages contain a transaction diagram for an administrative transaction performed at the ATM using an administrative card. The administrative transactions available through BASE24-atm include the following transactions:

- Balance with current cash
- Balance with standard replenishment
- Terminal subtotals
- Increase cash
- Decrease cash

The balance with current cash and balance with standard replenishment transactions allow terminals to be balanced with either the standard replenishment balancing method or the current cash balancing method.

The terminal subtotals transaction allows activity totals since the last balancing transaction up to the time the transaction is initiated to be printed on the audit or receipt printers. Terminal totals are not cleared with this transaction.

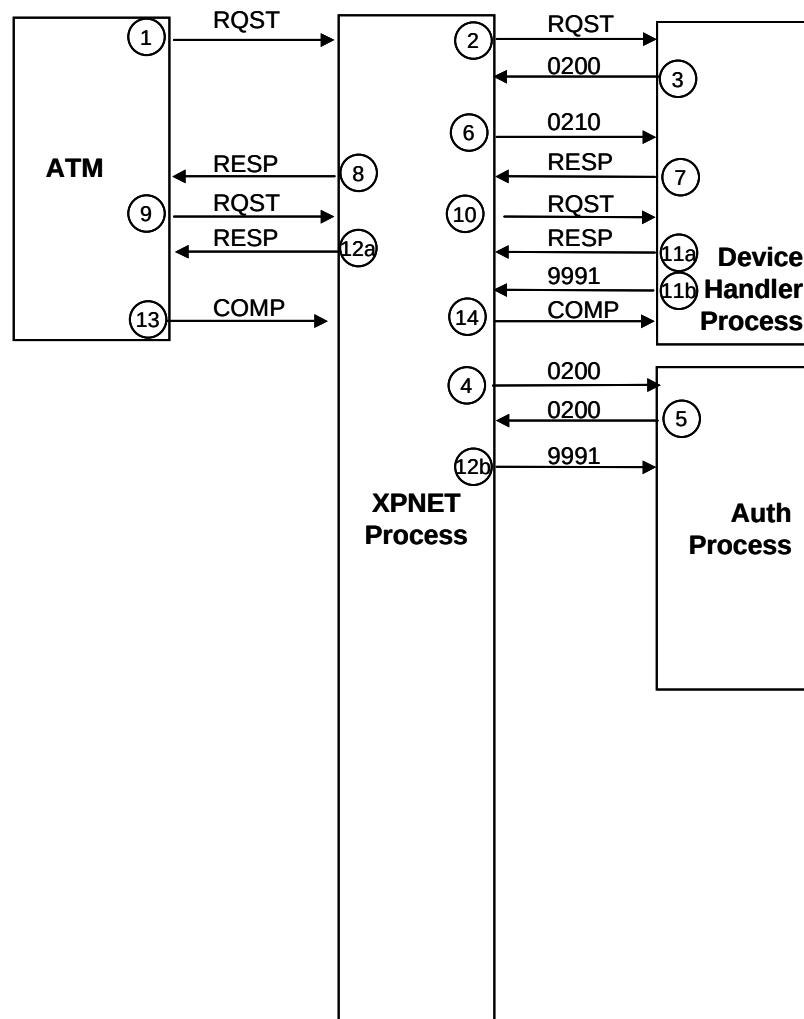
The increase cash and decrease cash transactions allow terminals to be replenished by increasing or decreasing the high or low bill denominations without balancing the terminal.

For more information on administrative transactions, refer to the ***BASE24-atm Settlement and Reporting Manual***.

Note: Not all terminals allow all of the above transactions. Refer to the appropriate device support manual for a list of supported administrative transactions.

Administrative Transaction

The diagram below illustrates the flow of an administrative transaction. These procedures are not applicable to all device types.



Steps

Processing

- | | |
|------|---|
| 1, 2 | The ATM sends a native-mode request message to the Device Handler process. This native-mode request is for a standard customer transaction such as a withdrawal. |
| 3, 4 | The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Authorization process. |

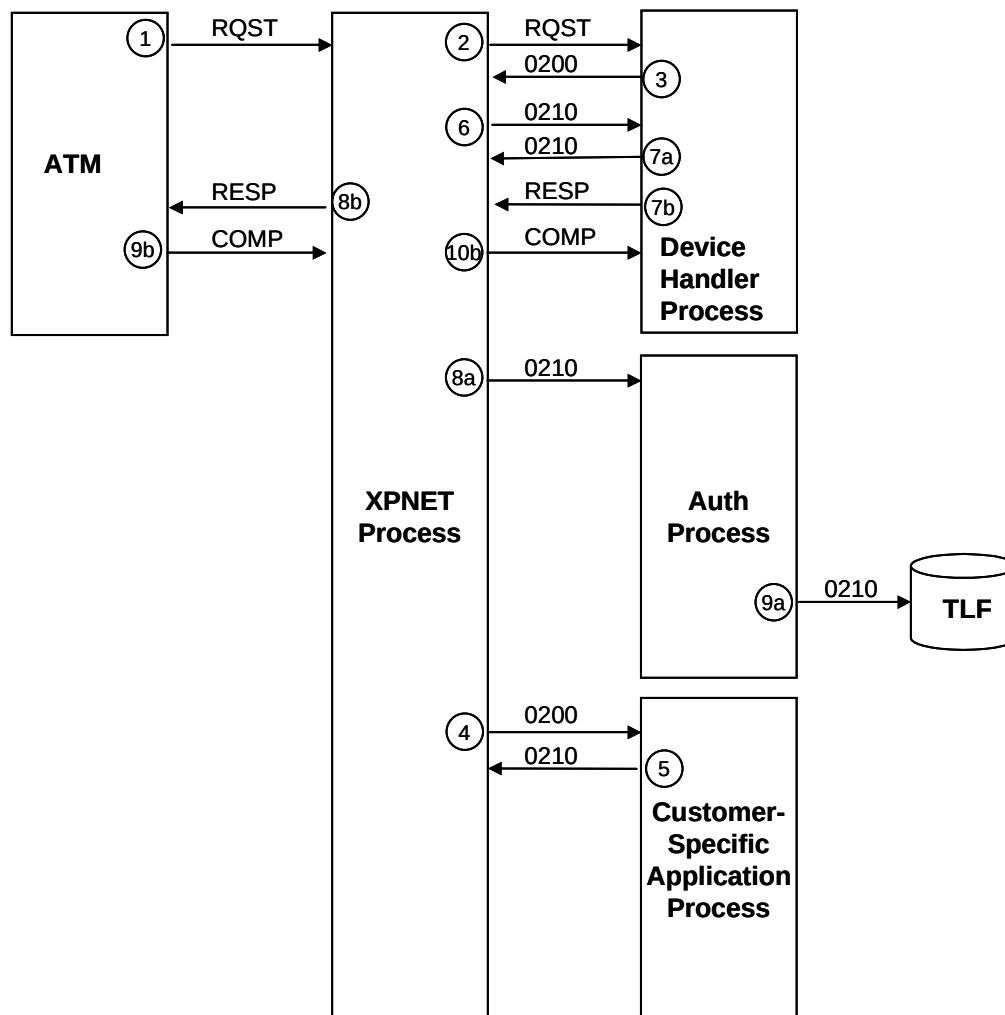
Steps	Processing
5, 6	The Authorization process recognizes that the card initiating the transaction is an administrative card. The process checks the balancing parameters in the Institution Definition File (IDF) to determine whether an administrative balancing transaction can be approved within the balancing window (response code 90), outside the balancing window (response code 91), or at any time (response code 92). The Authorization process sends the appropriate response code in a 0210 response message to the Device Handler process.
7, 8	The Device Handler process deletes its authorization timer and determines that the response code sent from the Authorization process is for an administrative transaction. The Device Handler process saves the response code and sends a native-mode response to the ATM instructing the device to display the administrative menu to the service person at the ATM.
9, 10	The service person selects an administrative transaction and the ATM sends a native-mode request to the Device Handler process.
11, 12	The Device Handler process performs the following steps: - If the transaction was a balancing or an adjustment transaction, the Device Handler process formats the correct Transaction Log File (TLF) records and sends them to the Authorization process in log only transaction (9991). The Authorization process receives the message and logs the TLF records to the TLF. This leg of processing is now complete. - Determines from the saved response code whether the transaction is allowed. The process generates a native-mode response that it sends to the ATM. Processing continues with step 13. Note: More than one message may be sent to the device because the data to be printed on an administrative receipt could not fit in a single message buffer.
13, 14	The ATM sends a native-mode message to the Device Handler process to indicate that the transaction completed as authorized.

APCF-Routed Transactions

The following pages contain message flows for transactions being processed by the authorizer specified at the processing code level. This authorizer is specified in the AUTHORIZATION DESTINATION field on APCF screen 2. In the following transactions the AUTHORIZATION DESTINATION field specifies an customer-specific process outside the BASE24-atm system for authorization.

Acquired at the ATM—Approved by a Customer-Specific Process—Transaction Logged to TLF

The diagram below illustrates a transaction acquired at the ATM and approved by a customer-specific process. Completions are required, and the transaction is configured to be logged to the TLF.



Steps

1, 2

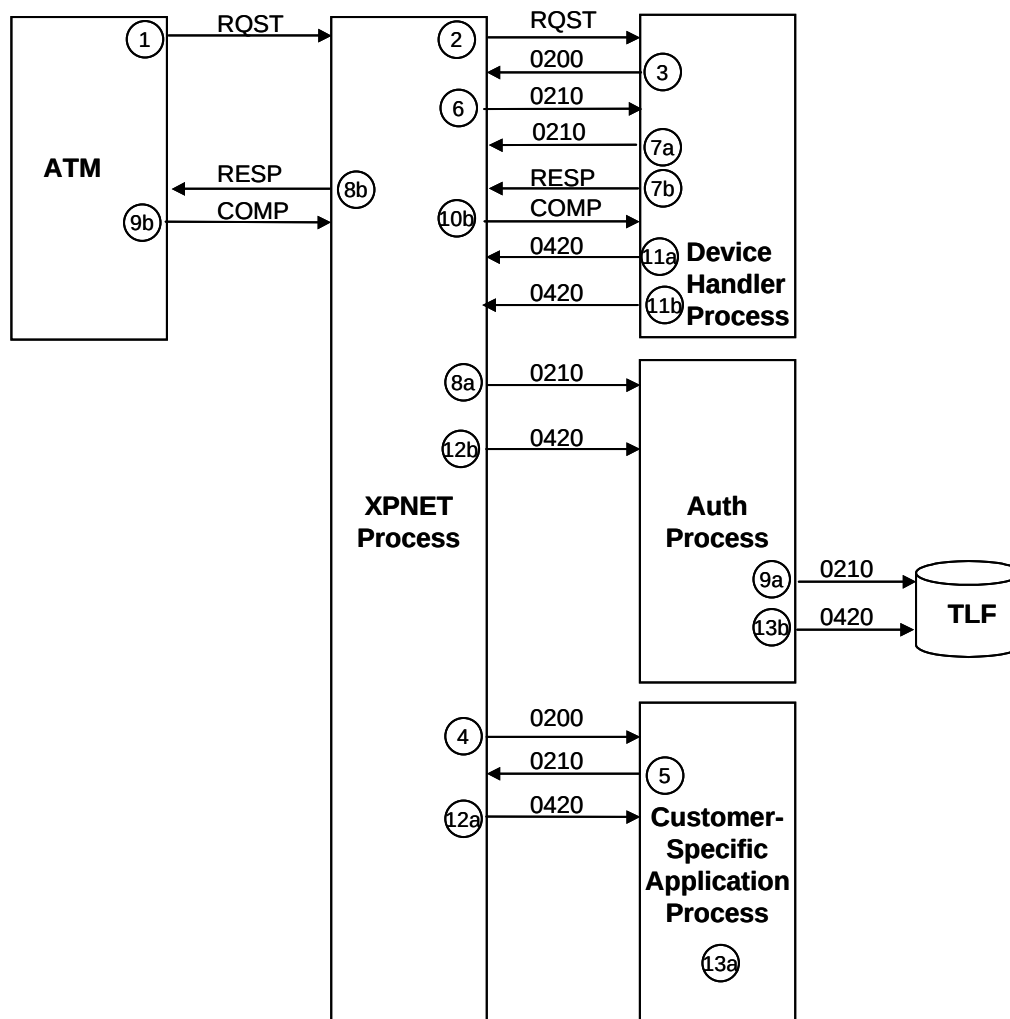
Processing

The ATM sends a native-mode request message to the Device Handler process.

Steps	Processing
3, 4	The Device Handler process reformats the native-mode message into a 0200 request message, determines the message should be sent to a customer-specific application process, sets an authorization timer, and sends the message to the application process.
5, 6	The application process approves the transaction and sends a 0210 response message to the Device Handler process.
7, 8	The Device Handler process deletes its authorization timer and performs the following steps: a. Determines the response must be logged by a BASE24-atm Authorization process to the TLF. The Device Handler process sends a copy of the transaction to the Authorization process. b. Reformats the 0210 message into a native-mode response message and sends it to the ATM.
9a	The Authorization process logs the transaction to the TLF.
9b, 10b	The ATM sends a native-mode completion message to the Device Handler process.

Acquired at the ATM—Approved by a Customer-Specific Process—Transaction Logged to TLF—Transaction is Reversed

The diagram below illustrates a transaction acquired at the ATM and approved by a customer-specific process. Completions are required, and the transaction is configured to be logged to the TLF. The transaction is subsequently reversed.



Steps

1, 2

The ATM sends a native-mode request message to the Device Handler process.

Processing

Steps	Processing
3, 4	The Device Handler process reformats the native-mode message into a 0200 request message, determines the message should be sent to a customer-specific application process, sets an authorization timer, and sends the message to the application process.
5, 6	The application process approves the transaction and sends a 0210 response message to the Device Handler process.
7, 8	The Device Handler process deletes its authorization timer and performs the following steps: a. Determines the response must be logged by a BASE24-atm Authorization process to the TLF. The Device Handler process sends a copy of the transaction to the Authorization process. b. Sends a native-mode completion message to the ATM.
9a	The Authorization process logs the transaction to the TLF. This leg of processing is complete. Processing continues with step 11.
9b, 10b	The ATM sends a native-mode message to the Device Handler process indicating that the transaction did not complete as authorized. Processing continues with step 11.
11, 12	The Device Handler process formats a 0420 reversal message and performs the following steps: a. Sends the 0420 reversal message to the application process. b. Sends a copy of the 0420 reversal message in a log-only message to the Authorization process.
13a	The application process receives the reversal and processes the transaction.
13b	The Authorization process logs the reversal to the TLF.

Mobile Top-Up Transactions

The following pages contain message flows for standard mobile top-up transactions, not an enhanced mobile top-up routing environment, using either the definitions- or message-based models. The definitions-based model uses locally defined values for mobile telephone operators (i.e., the actual service providers), the amount of time that can be purchased, and other information needed to complete the purchase of mobile telephone time. The message-based model of mobile top-up processing uses a series of messages between BASE24 and a telecommunication service provider to gather the information needed to complete the purchase.

For standard mobile top-up processing, the transaction is processed first by the Device Handler process and then the Transaction Context Manager process, which routes the transaction, based on settings in the Split Transaction Routing File (STRF), to the BASE24-atm Authorization process and the mobile operator or telecommunication service provider.

In an enhanced mobile top-up routing environment, the Device Handler process sends the transaction to the BASE24-atm Authorization process. The Authorization process sends the transaction to the host process configured in the Institution Definition File (IDF) based on the MTU PREFIX ROUTING field on CPF screen 4. This process, configured in the IDF, will be sent to a network interface. This network interface will send the transaction to the network interchange. The network interchange is then responsible for splitting the transaction into the funds side and the top-up side, and finally, sending it to the appropriate entities for authorizing. The funds side will be sent to a funds authorizer; in this case the ATM auth process for authorization.

At this time, the transaction will no longer use the MTU PREFIX ROUTING field, but will use either the EMV PREFIX ROUTING field on CPF screen 11 or the default PREFIX ROUTING field on CPF screen 1 for authorization. The transaction will be authorized and returned to the interchange network. The interchange network will coordinate the merging of the funds response and the top-up response. It will return a single response to BASE24-atm.

BASE24-atm customers utilizing the enhanced mobile top-up routing environment must be aware of any specific network interchange requirements for routing and configuration that are outside the scope of BASE24.

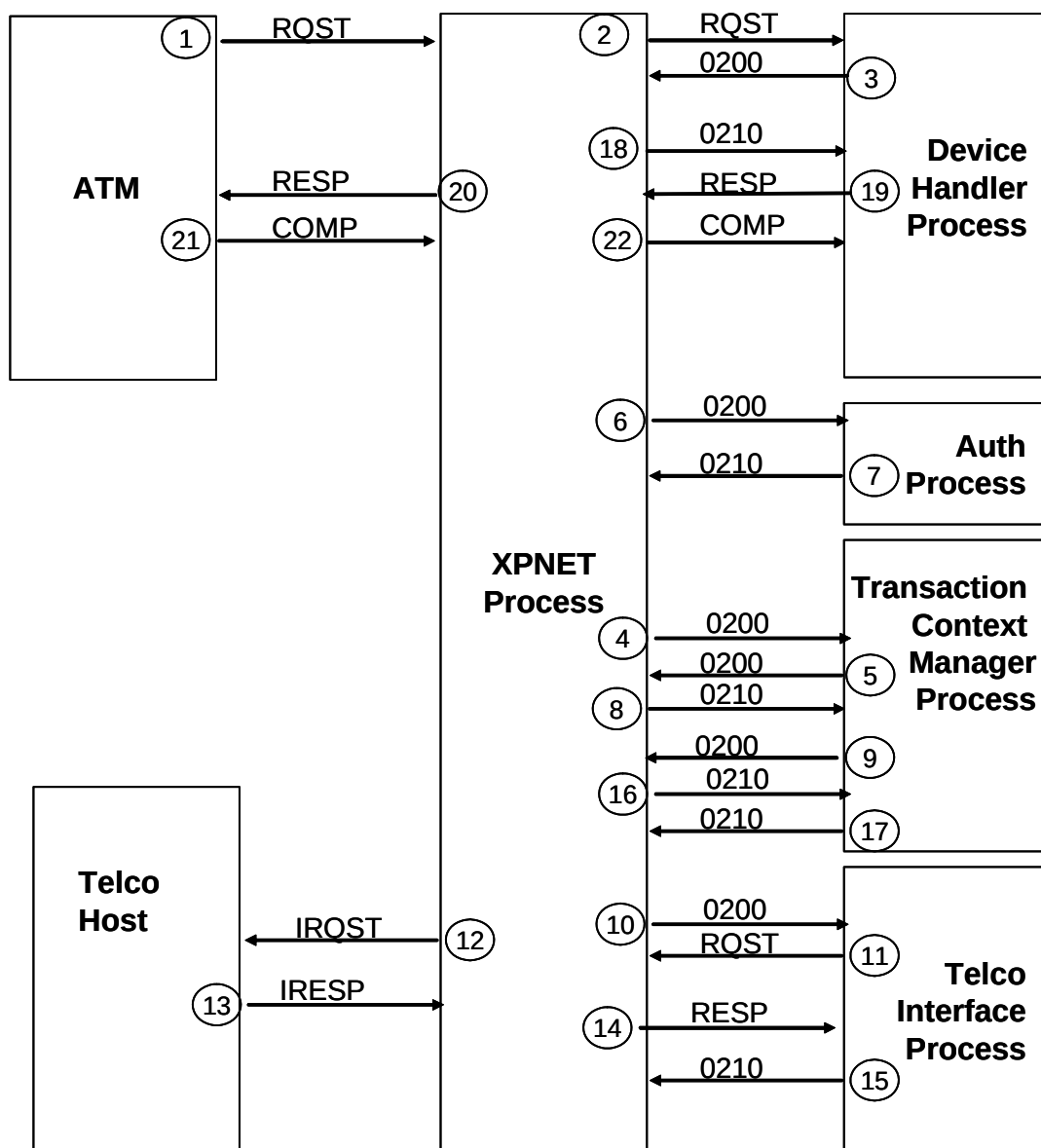
For the following transaction flows, the Split Transaction Routing File (STRF) is configured to route to funds authorization and then to the secondary service (Telco Interface process) for the definitions-based model.

The STRF is configured to route to the secondary service (Telco Interface process), then to funds authorization, and finally, back to the secondary service (Telco Interface process) for the message-based model.

Refer to the device-specific device support manuals for more information about the BASE24-atm Mobile Top-Up Device Handler extension.

Definitions-based Model (Approved)

The diagram below illustrates a mobile top-up transaction that is approved by BASE24-atm and the telephone company (telco) host.



Steps

1, 2

The ATM sends a native-mode request message to the Device Handler process.

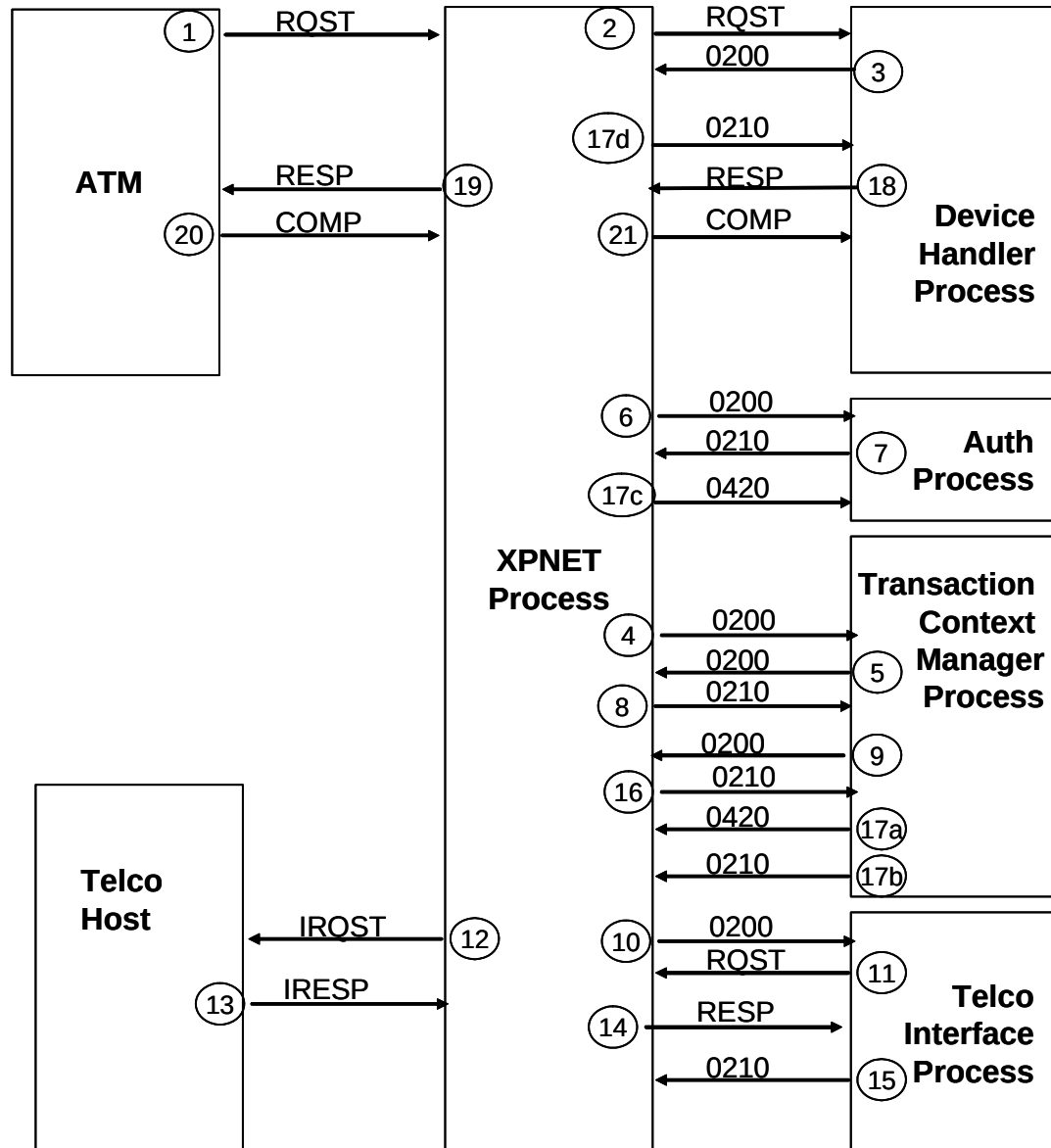
Processing

Steps	Processing
3, 4	The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Transaction Context Manager process.
5, 6	The Transaction Context Manager process determines from the Split Transaction Routing File (STRF) that the transaction must be sent to the Authorization process. The Transaction Context Manager process sends a 0200 request to the Authorization process.
7, 8	The Authorization process determines how to process the transaction request. The Authorization process approves the request, logs the response to the Transaction Log File (TLF), updates the database files appropriately, and returns a 0210 response message to the Transaction Context Manager process.
9, 10	The Transaction Context Manager process determines from the Split Transaction Routing token that a request must be sent to the Telco Interface. The Transaction Context Manager process sends a 0200 request to the Telco Interface process configured in the Split Transaction Routing File (STRF).
11, 12	The Telco Interface process reformats the 0200 request into an external message format that is understood by the telco host, sets an outbound request timer, and sends the request to the telco host.
13, 14	The telco host approves the transaction request, formats a response, and sends the response to the Telco Interface process.
15, 16	The Telco Interface process deletes the outbound request timer, reformats the response into a 0210 internal response message, logs the 0210 response to the Interchange Log File (ILF), and sends the response to the Transaction Context Manager process.
17, 18	The Transaction Context Manager process logs the 0210 response to the Transaction Log File (TLF) and sends the 0210 response to the Device Handler process.
19, 20	The Device Handler process deletes its authorization timer and reformats the 0210 response message into a native-mode response that it sends to the ATM.

Steps	Processing
21, 22	The ATM sends a native-mode message to the Device Handler process to indicate that the transaction completed as authorized.

Definitions-based Model (Declined by Telco)

The diagram below illustrates a mobile top-up transaction that is declined by the telephone company (telco) host and the generation of a reversal response.



Steps

Processing

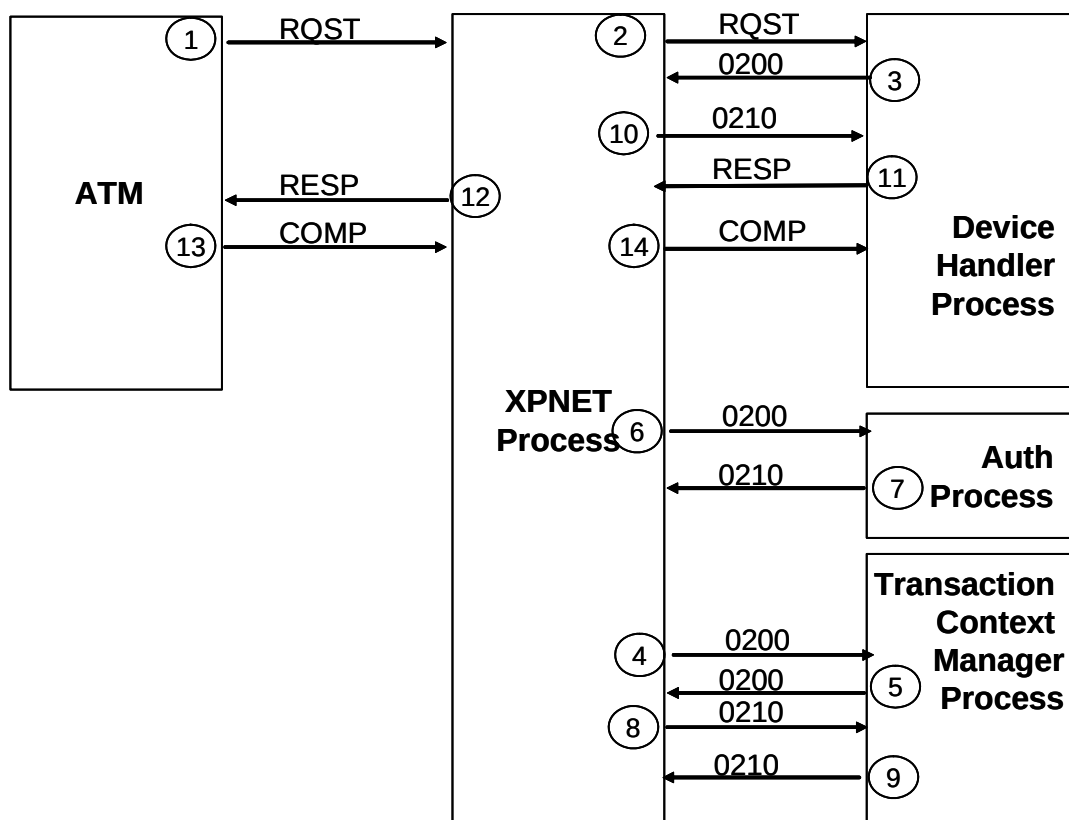
- 1, 2 The ATM sends a native-mode request message to the Device Handler process.

Steps	Processing
3, 4	The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Transaction Context Manager process.
5, 6	The Transaction Context Manager process determines from the Split Transaction Routing File (STRF) that the transaction must be sent to the Authorization process. The Transaction Context Manager process sends a 0200 request to the Authorization process.
7, 8	The Authorization process determines how to process the transaction request. The Authorization process approves the request, logs the response to the Transaction Log File (TLF), updates the database files appropriately, and returns a 0210 response message to the Transaction Context Manager process.
9, 10	The Transaction Context Manager process determines from the Split Transaction Routing token that a request must be sent to the Telco Interface. The Transaction Context Manager process sends a 0200 request to the Telco Interface process configured in the Split Transaction Routing File (STRF).
11, 12	The Telco Interface process reformats the 0200 request into an external message format that is understood by the telco host, sets an outbound request timer, and sends the request to the telco host.
13, 14	The telco host declines the transaction request, formats a response, and sends the response to the Telco Interface process.
15, 16	The Telco Interface process deletes the outbound request timer, reformats the response into a 0210 internal response message, logs the 0210 response to the Interchange Log File (ILF), and sends the response to the Transaction Context Manager process.
17a-17d	The Transaction Context Manager process generates a 0420 reversal to the Authorization process and logs the 0210 response to the Transaction Log File (TLF). The Transaction Context Manager process sends the 0210 response to the Device Handler process. Processing continues with step 18.

Steps	Processing
18, 19	The Device Handler process deletes its authorization timer and reformats the 0210 response message into a native-mode response that it sends to the ATM.
20, 21	The ATM sends a native-mode message to the Device Handler process to indicate that the transaction completed as authorized.

Definitions-based Model (Declined by BASE24-atm)

The diagram below illustrates a mobile top-up transaction that is declined by BASE24-atm.



Steps

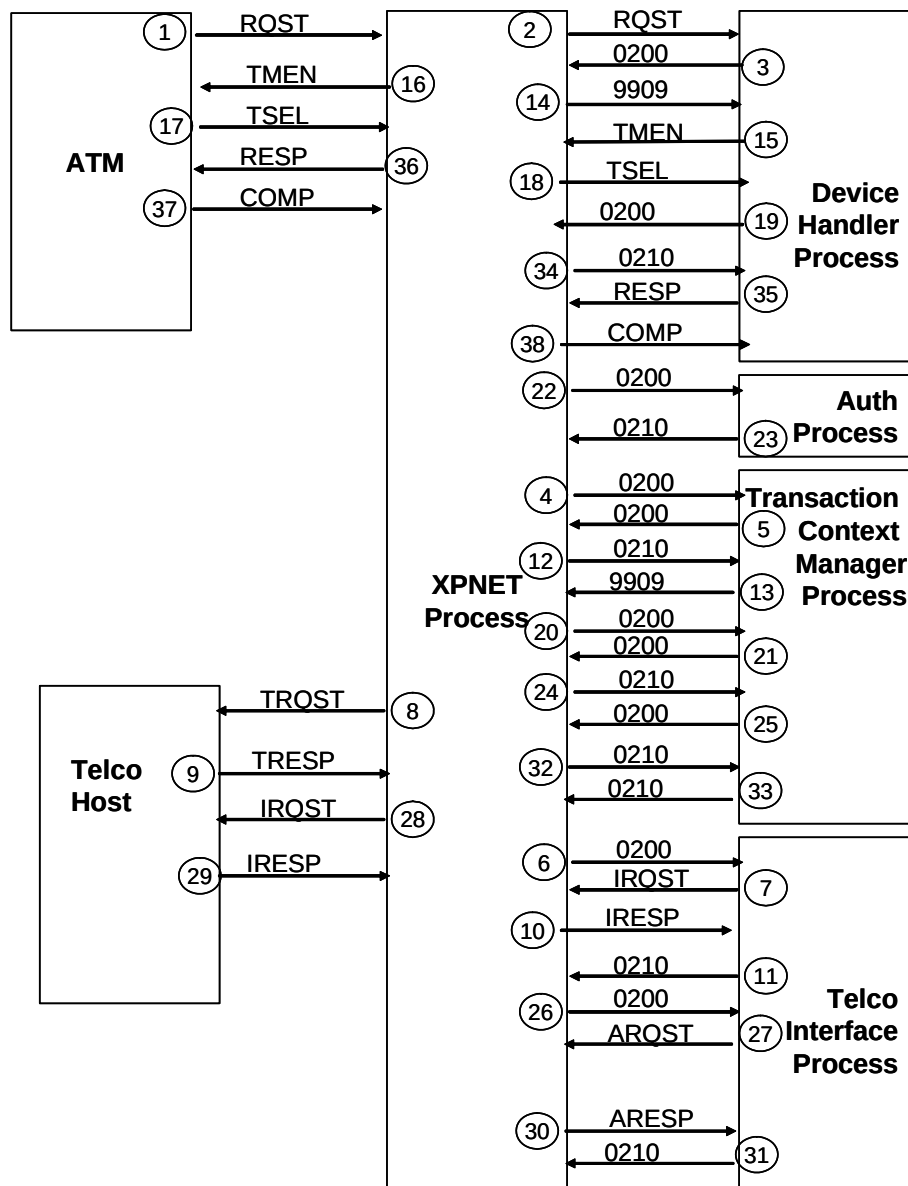
Processing

- 1, 2 The ATM sends a native-mode request message to the Device Handler process.
- 3, 4 The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Transaction Context Manager process.
- 5, 6 The Transaction Context Manager process determines from the Split Transaction Routing File (STRF) that the transaction must be sent to the Authorization process. The Transaction Context Manager process sends a 0200 request to the Authorization process.

Steps	Processing
7, 8	The Authorization process determines how to process the transaction request. The Authorization process denies the request, logs the response to the Transaction Log File (TLF), and returns a 0210 response message to the Transaction Context Manager process.
9, 10	The Transaction Context Manager process sends the 0210 response message to the Device Handler process.
11, 12	The Device Handler process deletes its authorization timer and reformats the 0210 response message into a native-format response that it sends to the ATM.
13, 14	The ATM sends a native-mode message to the Device Handler process to indicate that the transaction completed as authorized.

Message-based Model (Approved with Additional Cardholder Information)

The diagram below illustrates a mobile top-up transaction that is approved by BASE24-atm and the telephone company (telco) host once additional information about the cardholder is provided.



Steps

1, 2

The ATM sends a native-mode request message to the Device Handler process.

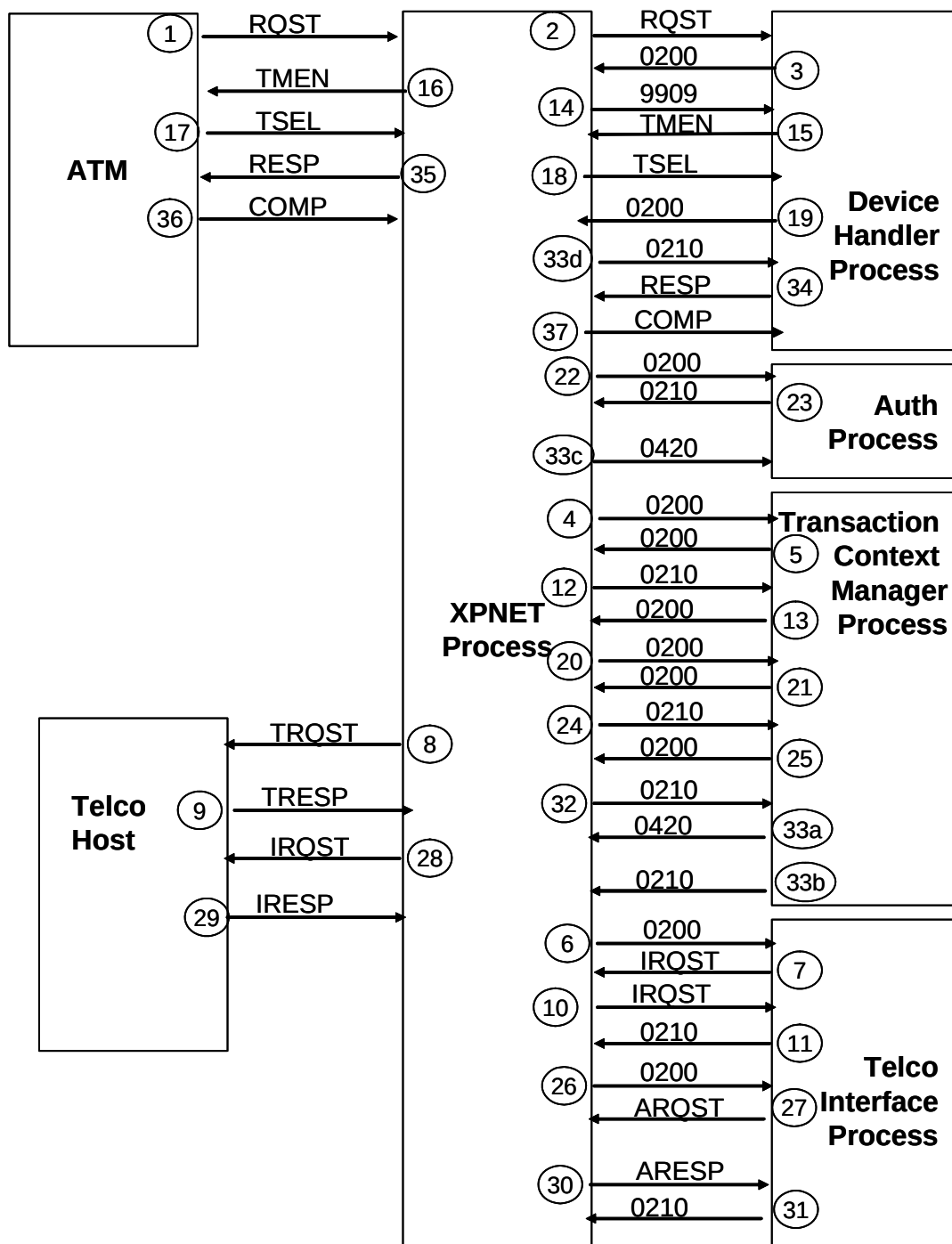
Processing

Steps	Processing
3, 4	The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Transaction Context Manager process.
5, 6	The Transaction Context Manager process sends a 0200 request to the Telco Interface process, configured in the Split Transaction Routing File (STRF).
7, 8	The Telco Interface process reformats the 0200 request into an external message format understood by the telco host, sets an outbound request timer, and sends the information-only request to the telco host.
9, 10	The telco host approves the information-only transaction request, formats a response, and sends the response to the Telco Interface process.
11, 12	The Telco Interface process deletes the outbound request timer, reformats the response into a 0210 internal response message, and sends the response to the Transaction Context Manager process.
13, 14	The Transaction Context Manager process returns a mobile top-up notification/selection (9909) message to the Device Handler process. The 9909 message contains mobile top-up data for acknowledgement/selection by a cardholder for transactions involving a choice among carriers, amounts, or taxes.
15, 16	The Device Handler process checks the 9909 message for the Pre-Pay Response token. The Device Handler process formats a native-mode message containing a selection menu that enables the cardholder to make a selection and sends it to the ATM.
17, 18	The ATM returns a native-mode message to the Device Handler process containing the information selected by the cardholder. Steps 3-18 continue until the cardholder has selected a valid carrier and amount, and accepted the tax amount.
19, 20	The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Transaction Context Manager process.

Steps	Processing
21, 22	The Transaction Context Manager process sends a 0200 request to the Authorization process.
23, 24	The Authorization process determines how to process the transaction request. The Authorization process approves the request, logs the response to the Transaction Log File (TLF), updates the database files appropriately, and returns a 0210 response message to the Transaction Context Manager process.
25, 26	The Transaction Context Manager process sends a 0200 request to the Telco Interface process configured in the Split Transaction Routing File (STRF).
27, 28	The Telco Interface process reformats the 0200 request into an external message format that is understood by the telco host, sets an outbound request timer, and sends the request to the telco host.
29, 30	The telco host approves the transaction request, formats a response, and sends the response to the Telco Interface process.
31, 32	The Telco Interface process deletes the outbound request timer, reformats the response into a 0210 internal response message, logs the 0210 response to the Interchange Log File (ILF), and sends the response to the Transaction Context Manager process.
33, 34	The Transaction Context Manager process logs the 0210 response to the Transaction Log File (TLF) and sends the 0210 response to the Device Handler process.
35, 36	The Device Handler process deletes its authorization timer and reformats the 0210 response message into a native-mode response that it sends to the ATM.
37, 38	The ATM sends a native-mode message to the Device Handler process to indicate that the transaction completed as authorized.

Message-based Model (Declined by Telco)

The diagram on the following page illustrates a mobile top-up transaction that is declined by the telephone company (telco) host and the generation of a reversal response.

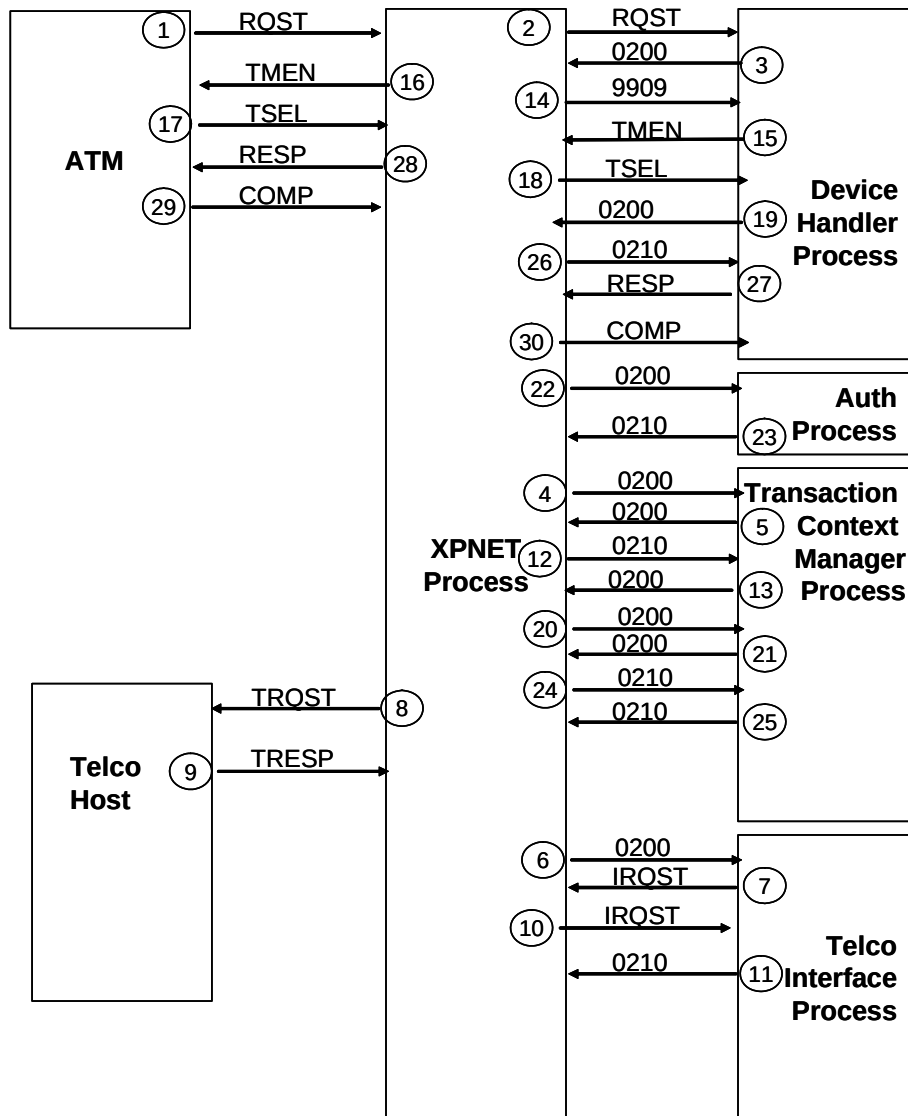


Steps	Processing
1, 2	The ATM sends a native-mode request message to the Device Handler process.
3, 4	The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Transaction Context Manager process.
5, 6	The Transaction Context Manager process sends a 0200 request to the Telco Interface process, configured in the Split Transaction Routing File (STRF).
7, 8	The Telco Interface process reformats the 0200 request into an external message format understood by the telco host, sets an outbound request timer, and sends the information-only request to the telco host.
9, 10	The telco host approves the information-only transaction request, formats a response, and sends the response to the Telco Interface process.
11, 12	The Telco Interface process deletes the outbound request timer, reformats the response into a 0210 internal response message, and sends the response to the Transaction Context Manager process.
13, 14	The Transaction Context Manager process returns a mobile top-up notification/selection (9909) message to the Device Handler process. The 9909 message contains mobile top-up data for acknowledgement/selection by a cardholder for transactions involving a choice among carriers, amounts, or taxes.
15, 16	The Device Handler process checks the 9909 message for the Pre-Pay Response token. The Device Handler process formats a native-mode message containing a selection menu that enables the cardholder to make a selection and sends it to the ATM.
17, 18	The ATM returns a native-mode message to the Device Handler process containing the information selected by the cardholder. Steps 3-18 continue until the cardholder has selected a valid carrier and amount, and accepted the tax amount.

Steps	Processing
19, 20	The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Transaction Context Manager process.
21, 22	The Transaction Context Manager process sends a 0200 request to the Authorization process.
23, 24	The Authorization process determines how to process the transaction request. The Authorization process approves the request, logs the response to the Transaction Log File (TLF), updates the database files appropriately, and returns a 0210 response message to the Transaction Context Manager process.
25, 26	The Transaction Context Manager process sends a 0200 request to the Telco Interface process configured in the Split Transaction Routing File (STRF).
27, 28	The Telco Interface process reformats the 0200 request into an external message format that is understood by the telco host, sets an outbound request timer, and sends the request to the telco host.
29, 30	The telco host declines the transaction request, formats a response, and sends the response to the Telco Interface process.
31, 32	The Telco Interface process deletes the outbound request timer, reformats the response into a 0210 internal response message, logs the 0210 response to the Interchange Log File (ILF), and sends the response to the Transaction Context Manager process.
33a-33d	The Transaction Context Manager process generates a 0420 reversal and logs the 0210 response to the Transaction Log File (TLF). The Transaction Context Manager process sends the 0210 response to the Device Handler process. Processing continues with step 34.
34, 35	The Device Handler process deletes its authorization timer and reformats the 0210 response message into a native-mode response that it sends to the ATM.
36, 37	The ATM sends a native-mode message to the Device Handler process to indicate that the transaction completed as authorized.

Message-based Model (Declined by BASE24-atm)

The diagram below illustrates a mobile top-up transaction that is declined by BASE24-atm.



Steps

1, 2

Processing

The ATM sends a native-mode request message to the Device Handler process.

Steps	Processing
3, 4	The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Transaction Context Manager process.
5, 6	The Transaction Context Manager process sends a 0200 request to the Telco Interface process, configured in the Split Transaction Routing File (STRF).
7, 8	The Telco Interface process reformats the 0200 request into an external message format understood by the telco host, sets an outbound request timer, and sends the information-only request to the telco host.
9, 10	The telco host approves the information-only transaction request, formats a response, and sends the response to the Telco Interface process.
11, 12	The Telco Interface process deletes the outbound request timer, reformats the response into a 0210 internal response message, and sends the response to the Transaction Context Manager process.
13, 14	The Transaction Context Manager process returns a mobile top-up notification/selection (9909) message to the Device Handler process. The 9909 message contains mobile top-up data for acknowledgement/selection by a cardholder for transactions involving a choice among carriers, amounts, or taxes.
15, 16	The Device Handler process checks the 9909 message for the Pre-Pay Response token. The Device Handler process formats a native-mode message containing a selection menu that enables the cardholder to make a selection and sends it to the ATM.
17, 18	The ATM returns a native-mode message to the Device Handler process containing the information selected by the cardholder. Steps 3-18 continue until the cardholder has selected a valid carrier and amount, and accepted the tax amount.
19, 20	The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Transaction Context Manager process.

Steps	Processing
21, 22	The Transaction Context Manager process sends a 0200 request to the Authorization process.
23, 24	The Authorization process determines how to process the transaction request. The Authorization process denies the request, logs the response to the Transaction Log File (TLF), and returns a 0210 response message to the Transaction Context Manager process.
25, 26	The Transaction Context Manager process sends the 0210 response message to the Device Handler process.
27, 28	The Device Handler process deletes its authorization timer and reformats the 0210 response message into a native-format response that it sends to the ATM.
29, 30	The ATM sends a native-mode message to the Device Handler process to indicate that the transaction completed as authorized.

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6: BASE24-atm Transaction Impacts on the Database

This section describes the impact of BASE24-atm transactions on the BASE24-atm database. It describes how transactions affect the balance and accumulation totals in the Positive Balance File (PBF), Cardholder Authorization File (CAF), and Usage Accumulation File (UAF). It also describes how transactions affect totals in the BASE24-atm Terminal Data files.

PBF and CAF/UAF Impacts

The table later in this subsection shows the impacts of approved transactions on PBF balances and CAF/UAF accumulation totals. The balance and accumulation totals included in the impact table are described below.

PBF Totals

The following is a list of balance totals whose transaction impacts are included in the impact table. The captions used in the impact table match the field names of the corresponding totals on PBF screen 1. Several PBF totals have different meanings depending on whether the PBF record represents a credit or non-credit account.

PBF Totals	
Noncredit	Credit
AVAILABLE BALANCE	AVAILABLE CREDIT
LEDGER BALANCE	CREDIT LIMIT
AMOUNT ON HOLD	CREDIT BALANCE
CASH OUT	

CAF/UAF Totals

The following summarizes those CAF/UAF accumulation totals whose transaction impacts are included in the impact table. Not every CAF/UAF totals field is listed in the table. Rather, the impact table includes representative CAF/UAF totals, which can be tied to corresponding fields on the CAF and UAF screens.

CAF and UAF accumulation totals are functionally identical. They are simply in different files used with different authorization methods. Throughout this section, they are treated identically.

Also, CAF and UAF accumulation fields include both total and offline accumulators. Here again, for the purpose of impacts, these are identical. Total versus offline accumulation is based on the authorization level being used and

whether BASE24 or a host approved the transaction. When the total and offline accumulators are impacted by a transaction, the method and amount by which they are impacted are the same.

Cash Withdrawal Column

The Cash Withdrawal column in the impact table describes how transactions impact the following CAF and UAF fields:

CAF screen 1 — TOTAL CASH WDL
CAF screen 1 — OFFLINE CASH WDL

CAF screen 8 — TOTAL CASH WDL
CAF screen 8 — OFFLINE CASH WDL

UAF screen 1 — TOTAL CASH WDL
UAF screen 1 — OFFLINE CASH WDL

UAF screen 4 — TOTAL CASH WDL
UAF screen 4 — OFFLINE CASH WDL

Cash Advance Column

The Cash Advance column in the impact table describes how transactions impact the following CAF and UAF fields:

CAF screen 1 — TOTAL CASH ADV
CAF screen 1 — OFFLINE CASH ADV

CAF screen 8 — TOTAL CASH ADV
CAF screen 8 — OFFLINE CASH ADV

UAF screen 1 — TOTAL CASH ADV
UAF screen 1 — OFFLINE CASH ADV

UAF screen 4 — TOTAL CASH ADV
UAF screen 4 — OFFLINE CASH ADV

Times Used Column

The Times Used column in the impact table describes how transactions impact the following CAF and UAF fields:

CAF screen 8 — TIMES USED THIS PERIOD

UAF screen 4 — TIMES USED THIS PERIOD

Transaction Impact Table

The table beginning on the following page describes the impacts of approved transactions on the PBF and CAF/UAF balance and accumulation fields.

The impacts are shown using the amount fields from the ATM Standard Internal Message (STM), which carry the pertinent amounts for each ATM transaction. The STM is described in appendix A.

The values used in the impact table are described below.

Value	Description
A1	Represents amount 1 (RQST.AMT-1) in the STM. A plus sign (+) or minus sign (–) indicates whether the amount is added to or subtracted from the total.
A2	Represents amount 2 (RQST.AMT-2) in the STM. A plus sign (+) or minus sign (–) indicates whether the amount is added to or subtracted from the total.
A3	Represents amount 3 (RQST.AMT-3) in the STM. A plus sign (+) or minus sign (–) indicates whether the amount is added to or subtracted from the total.
DC1	Represents the deposit credit amount (RQST.DEP-BAL-CR) in the STM. For single-account deposits, this is the deposit credit amount calculated for the account to which the deposit is being made. For split deposits, this is the deposit credit amount calculated for the first of the two accounts to which the deposit is being made. If an institution does not grant deposit credit, this amount will be 0. A plus sign (+) or minus sign (–) indicates whether the amount is added to or subtracted from the total.
DC2	Represents a second deposit credit amount in the STM (DEP-BAL-CR in the Self-Service Banking Base token) used for split deposits only. For split deposits, this is the deposit credit amount calculated for the second of the two accounts to which the deposit is being made. If an institution does not grant deposit credit, this amount will be 0. A plus sign (+) or minus sign (–) indicates whether the amount is added to or subtracted from the total.

Tran Code	Transaction Description		PBF Totals				CAF/UAF Totals		
			Cash Out	Amount On Hold / Credit Balance	Ledger Balance / Credit Limit	Available Balance / Available Credit	Times Used	Cash Advance	Cash Withdrawal
03 01 00	Check guarantee			+A1	−A1	−A1			
10 00 00	Fast cash withdrawal		+A1		−A1	−A1	+1		+A1
10 01 00	Withdrawal from checking		+A1		−A1	−A1	+1		+A1
10 11 00	Withdrawal from savings		+A1		−A1	−A1	+1		+A1
10 31 00	Withdrawal from credit account		+A1	+A1		−A1	+1	+A1	
11 00 00	Cash check						+1		+A1
20 00 01	Deposit to checking			+A1 −DC1	+A1	+DC1			
20 00 11	Deposit to savings			+A1 −DC1	+A1	+DC1			
20 01 01	Split deposit	to checking 1		+A1 −DC1	+A1	+DC1			
		to checking 2		+A2 −DC2	+A2	+DC2			

Tran Code	Transaction Description		PBF Totals				CAF/UAF Totals		
			Cash Out	Amount On Hold / Credit Balance	Ledger Balance / Credit Limit	Available Balance / Available Credit	Times Used	Cash Advance	Cash Withdrawal
20 01 11	Split deposit	to checking		+A1 -DC1	+A1	+DC1			
		to savings		+A2 -DC2	+A2	+DC2			
20 11 11	Split deposit	to savings 1		+A1 -DC1	+A1	+DC1			
		to savings 2		+A2 -DC2	+A2	+DC2			
24 00 01	Deposit to checking with cash back		+A2	+A1 -DC1	+A1 -A2	+DC1 -A2	+1		+A2
24 00 11	Deposit to savings with cash back		+A2	+A1 -DC1	+A1 -A2	+DC1 -A2	+1		+A2
40 01 01	Transfer	from checking 1			-A1	-A1			
		to checking 2			+A1	+A1			
40 01 11	Transfer	from checking			-A1	-A1			
		to savings			+A1	+A1			

Tran Code	Transaction Description		PBF Totals				CAF/UAF Totals		
			Cash Out	Amount On Hold / Credit Balance	Ledger Balance / Credit Limit	Available Balance / Available Credit	Times Used	Cash Advance	Cash Withdrawal
40 11 01	Transfer	from savings			−A1	−A1			
		to checking			+A1	+A1			
40 11 11	Transfer	from savings 1			−A1	−A1			
		to savings 2			+A1	+A1			
40 31 01	Transfer	from credit account		+A1		−A1			
		to checking			+A1	+A1			
50 01 31	Payment	from checking			−A1	−A1			
		to credit account		−A1		+A1			
50 11 31	Payment	from savings			−A1	−A1			
		to credit account		−A1		+A1			

BASE24-atm Terminal Data Files Impacts

Screen 6 of the BASE24-atm Terminal Data files (ATD) contains terminal activity information. The table on the following page describes how the transaction amounts carried in the BASE24-atm internal message for each type of financial transaction impact the accumulation totals in the BASE24-atm Terminal Data files. Refer to the ***BASE24-atm Files Maintenance Manual*** for more information about ATD screen 6. Hopper information is carried on ATD screens 4 and 5. ATD screen 10 carries hopper information for the NCR 56XX terminals with coin dispensing capabilities.

The impacts are shown using the amount fields from the ATM Standard Internal Message (STM), which carry the pertinent amounts for each ATM transaction. The STM is described in appendix A.

The values used in the table are described below.

Value	Description
A1	Represents amount 1 (RQST.AMT-1) in the STM. A plus sign (+) or minus sign (–) indicates whether the amount is added to or subtracted from the total. If the surcharging feature is used, this value represents only the amount requested by the cardholder (AMT-1 – fee). The BASE24-atm Terminal Data files totals do not reflect any fee assessment.
A2	Represents amount 2 (RQST.AMT-2) in the STM. A plus sign (+) or minus sign (–) indicates whether the amount is added to or subtracted from the total.
CO	Represents the cash out amount—the amount actually dispensed from the hopper—for each hopper affected by the transaction. A plus sign (+) or minus sign (–) indicates whether the cash out amount is added to or subtracted from the total.
*	Indicates an on-us transaction.

On-U's Credits Total																				
On-U's Debits Total	+A1*																			
Terminal Credits Total	+A1																			
Terminal Debits Total		+A1																		
Checks Amount																				
Checks Count																				
Payments Total																				
Payments Count																				
Commercial Deposits Total																				
Commercial Deposits Count																				
Deposits Amount																				
Deposits Count																				
Message Count																				
Log Only Count																				
Hopper (X) Cash Out	+CO																			
Hopper (X) End Cash	−CO																			
Transaction																				
Withdrawal																				
Deposit																				
Commercial Deposit																				
Deposit with Cash Back																				
Commercial Deposit with Cash Back																				
Transfer																				
Payment																				
Payment Enclosed																				
Check Guarantee																				
Cash Check																				
Check Deposit with Cash Back																				
Split Deposit (envelope)																				

On-Us Credits Total	+A1* +A2*																			
On-Us Debits Total																				
Terminal Credits Total																				
Terminal Debits Total	+A1 +A2	+A1	+A1 +A2																	
Checks Amount	+A1 +A2	+A1																		
Checks Count	+1	+1																		
Payments Total																				
Payments Count																				
Commercial Deposits Total																				
Commercial Deposits Count			+1																	
Deposits Amount																				
Deposits Count																				
Message Count				+1																
Log Only Count					+1															
Hopper (X) Cash Out																				
Hopper (X) End Cash																				
Transaction	Split Deposit (check)	Check Deposit	Commercial Split Deposit (envelope)	Message to Institution	Log-Only	Mobile Top-Up														

ACI Worldwide, Inc.

A: BASE24-atm Internal Message

BASE24-atm internal messages consist of a series of core fields—known as the Standard Internal Message (STM)—followed by any additional message tokens that are required for processing.

This appendix describes the fields in the STM. Tokens are mentioned briefly; however, in-depth descriptions of the various tokens supported by BASE24-atm are contained in the ***BASE24 Tokens Manual***.

Tokens and the BASE24-atm Internal Message

As transactions are processed and additional information is required that is not carried in the STM, BASE24-atm processes add tokens to the internal message. A token is a collection of related data required to perform a specific function. Tokens can be made up of one field or of a series of related fields.

Tokens are only added to the BASE24-atm internal message if they are required to process a specific transaction. Therefore, it is possible for an internal message to have several tokens associated with it—or none at all. When the first token is added to the internal message, BASE24 adds two tokens: the first is a Header token and the second is the specific token needed for processing. Once the Header token is in place, subsequent tokens can be added to the message individually as required. The general structure of an internal message with message tokens is illustrated below.

BASE24-atm Internal Message with Tokens

STM	Header Token	Token	Token	Token	...
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The following is a partial list of tokens that can be carried in the BASE24-atm internal message, depending on the features and add-on products being used. All tokens that can be carried in BASE24-atm internal messages are described in the ***BASE24 Tokens Manual***.

- Statement Print token (token 02)
- BASE24-atm Release 5.0 token (token 03)
- PIN Change token (token 06)
- Self-Service Banking Base token (token 07)
- Self-Service Banking Check token (token 14)
- Self-Service Banking Check Terminal Settlement token (token 15)
- Surcharge token (token 25)

Standard Internal Message (STM)

The following pages describe the Standard Internal Message (STM) used by BASE24-atm. It is 872 bytes in length.

Position	Level	Field Name and Description	Data Type
1–4	02	TYP The type of message described by the STM. Valid values are as follows: 0200 = Financial Transaction Request 0205 = Statement Print Transaction Request 0210 = Financial Transaction Request Response 0215 = Statement Print Transaction Response 0220 = Financial Transaction Advice 0420 = Reversal 5400 = Financial Transaction Correction 9901 = Surcharge Notification 9906 = Account Data Selection 9907 = Card Review Account Selection 9909 = Mobile Top-Up Notification/Selection 9980 = Informational (Money in Drawer) 9991 = Log-Only Transaction This field is set by the Device Handler, Authorization, Interchange Interface, or Host Interface process.	PIC 9(4)
5–6	02	PROD-ID The BASE24 product originating this transaction. For BASE24-atm transactions, this field is set to 01. This field is set by the Device Handler, Interchange Interface, or Host Interface process.	PIC X(2)
7–8	02	DPC-NUM The Data Processing Center (DPC) number identifying the DPC to which the transaction is to be sent.	TYPE BINARY 16

Position	Level	Field Name and Description	Data Type
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This field is set by the Authorization process using the RT-TBL.PRI-DPC field in the BASE24-atm segment of the IDF. The Host Interface process uses this field to identify the DPC to receive the message.

9-10	02	TRAN-CDE	PIC X(2)
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The type of transaction. The following codes are supported by BASE24-atm. Not all of the codes listed are supported by all ATMs.

Note 1: Transaction codes 25 and 26 are only supported with the definitions-based model of BASE24-atm Mobile Top-Up. The BASE24-atm Transaction Context Manager and certain Mobile Operator interfaces process these transaction codes. BASE24-atm Authorization does not recognize these codes.

Note 2: Transaction code 70 also supports passbook update transactions with subtype APU0.

Valid values are as follows:

03	=	Check Guarantee
04	=	Check Verify
10	=	Withdrawal
11	=	Cash Check
20	=	Deposit
24	=	Deposit with Cash Back
25	=	Mobile Top-Up (Note 1)
26	=	Mobile Top-Up with Funds (Note 1)
30	=	Balance Inquiry
40	=	Transfer
41	=	Load Value
50	=	Payment from/to
51	=	Payment Enclosed
60	=	Message Enclosed to Financial Institution
61	=	Log Only
62	=	Card Review
70	=	Statement print (Note 2)
81	=	PIN change
82	=	EMV PIN unblock
U1	=	SVC Balance Inquiry

Position	Level	Field Name and Description	Data Type
11–12	02	FROM-ACCT-TYP The type of <i>from</i> account. Valid values are as follows: 00 = No account needed 01 = Checking 11 = Savings 31 = Credit 70 = Electronic purse Note: For log-only transactions, this field is used to determine the type of log-only transaction being sent. BASE24 supports up to four different types of log-only transactions, which are defined by the institution. Values for log-only transactions are 01, 02, 03, and 04. For split-deposit transactions, this field specifies the type for a second <i>to</i> account.	PIC X(2)
13–14	02	TO-ACCT-TYP The type of <i>to</i> account. Valid values are as follows: 00 = No account needed 01 = Checking 11 = Savings 31 = Credit 70 = Electronic purse Note: For split-deposit transactions, this field specifies a type for a first <i>to</i> account.	PIC X(2)
15–52	02	RTE The following fields are used to route messages to the appropriate BASE24 processes.	
15–16	03	STAT The status of a message at the system level. Valid values are as follows: 00 = No error 01 = Processing error	PIC 99

Position	Level	Field Name and Description	Data Type
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02 = Security device failure
11 = Time out after exit from BASE24
12 = Line down

If this field is set to 00, the STM response code (RQST.RESP field or the STMT-INFO.RESP field) is used to determine the disposition of the message. Nonzero values indicate a failed message.

This field is set by the Host Interface process and used by the Authorization process.

17	03	NOT-ON-US-FLG	PIC 9
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A code indicating whether the transaction is on-us or not-on-us. Valid values are as follows:

0 = Terminal-owner and card-issuer are the same (on-us).
1 = Terminal-owner and card-issuer are not the same (not-on-us).

On-us transactions are initiated by the institution's cardholder at one of the institution's terminals.

Not-on-us transactions are initiated by the institution's cardholder from a different institution's terminal.

To set this field, the Authorization process compares the value in the TERM-OWNER-FIID field to the value in the CRD-FIID field. If they are equal, this field is set to 0; otherwise, this field is set to 1.

For transactions originating through an interchange, the Interchange Interface process sets this field to 1.

The Authorization process uses this field to establish sharing parameters. The Device Handler process uses this field to determine if the transaction amount should be written to the ON-US-CR or ON-US-DB fields in the BASE24-atm Terminal Data files.

Position	Level	Field Name and Description	Data Type
18–33	03	ORIG-PRO-NAME The symbolic name of the BASE24-atm process that received the transaction. The Authorization processes use this name for return routing. This field is set by the Device Handler, Host Interface, or Interchange Interface process. The symbolic name of the Device Handler process is placed in this field for transactions that originate from terminals connected to BASE24-atm. The symbolic name of the Host Interface process is placed in this field for transactions from a host. The symbolic name of the Interchange Interface process is placed in this field for transactions from an interchange. If the Authorization process cannot handle a request and must route the request elsewhere for authorization, the value of this field is moved to the RTE.HLD-ORIG-PRO-NAME field and the symbolic name of the Authorization process routing the transaction is placed in this field, except for statement print (0205) transactions. For statement print requests, the Authorization process leaves this field unaltered to avoid unneeded processing time in the return transaction path.	PIC X(16)
34–49	03	HLD-ORIG-PRO-NAME The symbolic name of the first BASE24-atm process to receive the transaction. The Authorization process uses this name for return routing. If the Authorization process must route a transaction, it moves the value of the RTE.ORIG-PRO-NAME field to this field and places its own symbolic name in the RTE.ORIG-PRO-NAME field, except for statement print (0205) transactions. For statement print requests, the Authorization process leaves the RTE.ORIG-PRO-NAME field unaltered and sets this field to all zeros. This avoids unneeded processing time incurred by the transaction path described next.	PIC X(16)

Position	Level	Field Name and Description	Data Type
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The RTE.ORIG-PRO-NAME field is then used for return routing to the Authorization process, and the Authorization process uses the RTE.HLD-ORIG-PRO-NAME field for return routing to the originating process.

50	03	ORIGINATOR	PIC X
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Indicates where the transaction originated.

This applies to request (0200) messages, completion (0220) messages, and reversal (0420) messages. Valid values are as follows:

- 1 = Device controlled by BASE24
- 2 = Device Handler process
- 3 = Authorization process
- 4 = Host Interface process
- 5 = Host
- 6 = Interchange Interface process
- 7 = Interchange

This field is set by the Authorization, Device Handler, Interchange Interface, or Host Interface process. The BASE24 reporting programs use this field to identify interchange transactions.

51	03	RESPONDER	PIC X
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Indicates where a response message (0210) was generated. Valid values are as follows:

- 0 = Default (response not generated yet)
- 1 = Device controlled by BASE24
- 2 = Device Handler process
- 3 = Authorization process
- 4 = Host Interface process
- 5 = Host
- 6 = Interchange Interface process
- 7 = Interchange
- S = Secondary Authorizer

This field is set by the Authorization, Device Handler, Interchange Interface, Transaction Context Manager, or Host Interface process.

Position	Level	Field Name and Description	Data Type
52	03	EXCPT-FLG A flag indicating the transaction could not be processed due to invalid data (for example, invalid business date or Track 2 data). This field is set by the Authorization process. Valid values are as follows: 0 = Normal 1 or 2 = Exception	PIC X
53–56	02	CRD-LN The logical network of the card issuer. This field is set by the Authorization or Interchange Interface process. The Authorization process sets this field using the LOGICAL-NET param in the LCONF; the Interchange Interface process sets the field using the LN field in the Base segment of the Interchange Configuration File (ICF) or Enhanced Interchange Configuration File (ICFE).	PIC X(4)
57–67	02	RCV-INST-ID-NUM The routing/identification number of the card issuer. For BASE24-atm card issuers, this field is set by the Authorization process using the INST-ID-NUM field in the Base segment of the IDF.	PIC 9(11)
68–71	02	CRD-FIID The identifier of the card-issuing institution. If the Authorization process identifies the prefix of the card as being one that it can process, this field is set using the FIID field in the Base segment of the CPF. The Interchange Interface process sets this field by using the ISSUER-ROUTING-NUM field in the ANSI-based Standard External Message.	PIC X(4)

Position	Level	Field Name and Description	Data Type
72–95	02	SHRG-GRP	PIC X OCCURS 24 TIMES
These fields identify the sharing groups of which the originating terminal is a part. They are set by the Device Handler, Interchange Interface, or Host Interface process. For transactions originating at BASE24-atm terminals, these fields are set by the Device Handler process using the SHRG-GRP fields in the BASE24-atm Terminal Data files. For transactions originating from interchanges, these fields are set by the Interchange Interface process using the SHRG-GRP fields in the BASE24-atm segment of the ICF or ICFE. For transactions originating from an ISO host, these fields are set by the ISO Host Interface process using the element P-48 of the ISO-based external message. For transactions originating from an ANSI host, these fields are set by the Host Interface process using the SHARING.SHARING-GRP fields in the ANSI-based Standard External Message. If the transaction is not-on-us, the Authorization process looks for a match between one of the identifiers in this field and those found in the IDF. When the first match is found, the Authorization process writes the matching identifier in the first byte of this field.			
96–101	02	TRAN-DAT	PIC X(6)
The date (YYMMDD) the transaction was initiated. This field is set by the transaction originator. The originator could be the Device Handler, Interchange Interface, or Host Interface process.			
102–109	02	TRAN-TIM	PIC X(8)
The time (hhmmssstt) the transaction was initiated. This field is set by the transaction originator. The originator could be the Device Handler, Interchange Interface, or Host Interface process.			

Position	Level	Field Name and Description	Data Type
		For transactions originating at a device, the Device Handler process sets this field using the current time and the device time offset (TIM-OFST field, below) from the BASE24-atm Terminal Data files.	
110	02	USER-FLD1	PIC X
111–112	02	TIM-OFST	TYPE BINARY 16
		The time difference between the location of the originating terminal and the location of the HP NonStop processor for BASE24.	
		This number is added to the BASE24-atm system local time to obtain the terminal local time. This field is set by the Device Handler process.	
113–118	02	POST-DAT	PIC X(6)
		The date (YYMMDD) of the TLF to which the transaction will be posted. This field is set by the Device Handler, Interchange Interface, or Host Interface process.	
119–124	02	ACQ-ICHG-SETL-DAT	PIC X(6)
		The date (YYMMDD) the transaction will be settled by the interchange if an interchange originated the transaction. Otherwise, this field is zero filled.	
		This field is set by the Interchange Interface process using the POST-DAT field in the Base segment of the ICF or ICFE.	
125–130	02	ISS-ICHG-SETL-DAT	PIC X(6)
		The date (YYMMDD) the transaction will be settled by the interchange if an interchange authorized the transaction. Otherwise, this field is zero filled.	
		This field is set by the Interchange Interface process using the POST-DAT field in the Base segment of the ICF or ICFE.	

Position	Level	Field Name and Description	Data Type
131–134	02	TERM-OWNER-FIID Identifies the owner of the originating terminal. This field is set by the Device Handler, Host Interface, or Interchange Interface process. For transactions that originate at BASE24-atm terminals, this field is set by the Device Handler process using the TERM-OWNER.FIID field in the BASE24-atm Terminal Data files. For transactions that originate from an ISO host, this field is set by the Host Interface process using element P-60 of the ISO-based external message. For transactions that originate from an ANSI host, this field is set by the Host Interface process using the TERM.OWNER field in the ANSI-based Standard External Message. For transactions that originate from interchanges, this field is set by the Interchange Interface process, using the PRI-KEY.FIID field in the ICF or ICFE.	PIC X(4)
135–150	02	TERM-ID Identifies the originating terminal. This field is set by the Device Handler, Host Interface, or Interchange Interface process. For transactions that originate at BASE24-atm terminals, this field is set by the Device Handler process from the TERM-ID field in the BASE24-atm Terminal Data files. For transactions that originate from ISO hosts, this field is set by the Host Interface process, using element P-41 of the ISO-based external message. For transactions that originate from ANSI hosts, this field is set by the Host Interface process, using the TERM.ATM-ID field in the ANSI-based Standard External Message. For transactions that originate from an interchange, this field is set by the Interchange Interface process using the DFLT-TERM-NUM field in Base segment of the ICF or ICFE if the interchange does not provide a terminal ID.	PIC X(16)

Position	Level	Field Name and Description	Data Type
151–154	02	TERM-LN Identifies the logical network of the originating terminal. This field is set by either the Device Handler or Interchange Interface process. For transactions that originate at BASE24-atm terminals, this field is set by the Device Handler process using the TERM-OWNER.LOGICAL-NET field in the BASE24-atm Terminal Data files. For transactions that originate from interchanges, this field is set by the Interchange Interface process using the LOGICAL-NET field in the ICF or ICFE.	PIC X(4)
155–158	02	BRCH-ID A code identifying the branch of the originating terminal. This field is set for transactions originating at BASE24-atm terminals. This field is set by the Device Handler process using the TERM-OWNER.BRCH-ID field in the BASE24-atm Terminal Data files.	PIC X(4)
159–162	02	REGN-ID A code identifying the region of the originating terminal. This field is set for transactions originating at BASE24-atm terminals. This field is set by the Device Handler process using the TERM-OWNER.REGION-ID field in the BASE24-atm Terminal Data files.	PIC X(4)
163–174	02	SEQ-NUM The sequence number associated with this transaction. If the ATM does not generate a sequence number, the associated process generates one. This field remains unchanged through the life of the transaction. This field is set by the Device Handler, Interchange Interface, or Host Interface process.	PIC X(12)

Position	Level	Field Name and Description	Data Type
175	02	DEP-SETL-IMP-FLG A code indicating whether deposits are to impact settlement. This field is used in the creation of settlement reports; it has no effect on whether a transaction is logged to the TLF. Valid values are as follows: 0 = Deposits do not have settlement impact. 1 = Envelope deposits have settlement impact. 2 = Commercial deposits have settlement impact (for example: Securomatic deposits). 3 = Both envelope and commercial deposits have settlement impact. The Authorization process uses the DEP-SETL-IMP-FLG field in the BASE24-atm segment of the IDF to set this field for BASE24-atm card issuers.	PIC X
176	02	ADJ-SETL-IMP-FLG A code indicating whether adjustments are to impact settlement. For adjustment (5400) messages sent to BASE24-atm card issuers, this field is set by the Authorization process from the second position of the HOST-ADJ-PROCESSING field in the BASE24 segment of the IDF. Valid values are as follows: 0 = Adjustments do not have settlement impact. 1 = Adjustments have settlement impact.	PIC X
177	02	PROC-ACCT-IND A code indicating how the <i>from</i> and <i>to</i> accounts are processed by the DPC on a two-sided transaction. Valid values are as follows: 0 = Process both the <i>from</i> and <i>to</i> accounts. 1 = Process only the <i>from</i> account. 2 = Process only the <i>to</i> account. This field is set to 1 or 2 by the Authorization process when different hosts are involved for each side of a two-sided transaction.	PIC X(1)

Position	Level	Field Name and Description	Data Type
178	02	TERM-CAPTURE-FLG A code indicating whether the terminal has the capability to capture the card. For example, a terminal that accepts cards by way of a card swipe mechanism cannot capture cards. Valid values are as follows: 0 = Terminal can capture card. 1 = Terminal cannot capture card. The default value is zero. This field is set by the Device Handler process and the Interchange Interface process.	PIC X(1)
178	02	USER-FLD2 REDEFINES TERM-CAPTURE-FLG This field was redefined for programs that reference USER-FLD2.	PIC X(1)
179–658	02	RQST The following fields are used for financial transaction request (0200), reversal advice (0420), and adjustment (5400) messages.	
179–218	04	TRACK2 The Track 2 data from the card initiating the transaction. The Authorization processes and report programs use this field. This field is set by the Device Handler, Interchange Interface, or Host Interface process. If the information was read only from Track 2 of the magnetic stripe on the card, the data is the information encoded on Track 2 of the magnetic stripe on the back of the card originating the transaction, including start and end sentinel and longitudinal redundancy check (LRC) characters.	PIC X(40)

Position	Level	Field Name and Description	Data Type
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The content of Track 2 data (between the start sentinel and end sentinel) is specified in the ISO 7813 standard, **Identification Cards—Financial Transaction Cards**.

1. Primary account number (PAN), left justified (up to 19 digits)
2. Field separator (=)
3. Country code (if present; 3 digits)
4. Expiration date (YYMM)
5. Service code (if present; 3 digits)
6. Discretionary data (balance of available space)

If the information was read only from Track 1 of the magnetic stripe on the card, the Track 1 data is moved to the Track 1 token, and this field is constructed from other information in the message. The first byte contains an M, followed by the card number, a field separator, the expiration date (YYMM), and an end sentinel.

If full track data is not available (e.g., in an advice or reversal message), or if discretionary data suppression is implemented, this field is constructed from either the suppressed track or other information in the incoming message. The first byte contains an M, followed by the card number, a field separator, the expiration date (YYMM), and an end sentinel. If the expiration date is not available, it is replaced with the value 0000.

219–234	04	PIN-OFST	PIC X(16)
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The PIN offset value. This value is calculated and filled in by the Authorization process when the institution allows its cardholders to select or change their PINs.

235–237	04	MBR-NUM	PIC X(3)
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The member number or card sequence number for the card used to initiate the transaction.

This field is used to access the cardholder files. For BASE24-atm card issuers, this field is set and used by the Authorization process.

Position	Level	Field Name and Description	Data Type
238–239	04	TERM-TYP A code indicating the type of terminal from which the transaction originated. For transactions originating at a BASE24 terminal, the Device Handler process sets this field using the TERM-TYP field in the BASE24-atm Terminal Data files. This field is used by the Super Extract process.	PIC 9(2)
240–242	04	ORIG-CRNCY-CDE The type of currency used by the originating terminal. It is set by the Device Handler, Interchange Interface, or Host Interface process. For transactions originating at BASE24-atm terminals, this field is set by the Device Handler process using either the HOPPER _x .CRNCY-CDE field (where <i>x</i> is the hopper number) or the AMT-CRNCY-CDE in the BASE24-atm Terminal Data files. The HOPPER _x .CRNCY-CDE field is used when the TRAN-CDE field is set to 10 (withdrawal), 11 (cash check), or 24 (deposit with cash back). The AMT-CRNCY-CDE field is used when the TRAN-CDE field contains any other value. For transactions that originate from an interchange, this field is set by the Interchange Interface process using the CRNCY-CDE field in Base segment of the ICF or ICFE. For transactions that originate from an ISO host, this field is set by the Host Interface process using element P-49 of the ISO-based external message. For transactions that originate from an ANSI host, this field is set by the Host Interface process using the CURRENCY-CDE in the ANSI-based standard external message.	PIC 9(3)
243–272	04	MULT-CRNCY The following group of fields apply only in multicurrency systems. Multicurrency systems are not currently available with BASE24-atm. These fields are redefined as a user-field for single currency systems.	

Position	Level	Field Name and Description	Data Type
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Note: These fields are not used for the BASE24-atm Multiple Currency add-on product.

243–245	06	AUTH-CRNCY-CDE	PIC 9(3)
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A code indicating the type of currency used in the response from the authorizing entity.

For on-us and not-on-us transactions, the Authorization process retrieves the value for this field from the CRNCY-CDE field in the Base segment of the IDF.

For transactions coming from an interchange, the Interchange Interface process retrieves the value for this field from the CRNCY-CDE field in the Base segment of the ICF or ICFE.

246–248	06	SETL-CRNCY-CDE	PIC 9(3)
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The currency code of the settlement entity.

249–256	06	AUTH-CONV-RATE	PIC 9(8)
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The exchange rate of the authorizing entity used to compute the amount that will affect the customer's balance. The first digit contains the offset of the decimal point from the right-hand side. Unless multiple currencies are involved, the default value is 61000000.

257–264	06	SETL-CONV-RATE	PIC 9(8)
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The exchange rate of the settlement entity used to compute the final settlement amount. The first digit contains the offset of the decimal point from the right-hand side. Unless multiple currencies are involved, the default value is 61000000.

265–272	06	CONV-DAT-TIM	TYPE BINARY 64
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A timestamp indicating when the actual conversion between the currency of the transaction and the currency of the database took place. This field is displayed in Greenwich Mean Time (GMT).

Position	Level	Field Name and Description	Data Type
243–272	04	USER-FLD3 (NOT USED)	REDEFINES MULT-CRNCY PIC X(30)
273–283	04	ACQ-INST-ID-NUM The routing/identification number of the terminal owner. For transactions originating at BASE24-atm terminals, this field is set by the Device Handler process using the ID-NUM field in the BASE24-atm Terminal Data files.	PIC 9(11)
284	04	MULT-ACCT A code indicating to the Authorization process whether a terminal allows multiple account selection. Valid values are as follows: 0 = The terminal does not allow multiple account selection; default to the primary account. 1 = The terminal allows multiple account selection. 2 = A fast cash transaction was selected; default to the primary account. 3 = Prescreen the transaction even though accounts are already present in the request message. The Olivetti Device Handler process sets this field to a value of 3 when a card review transaction was just performed at an Olivetti device. The ISO Host Interface process sets this field to a value of 3 for incoming request messages when the HOST-SELECTED-ACCT param in the LCONF is set to true. This field is set by either the Device Handler process or the Interchange Interface process. For transactions that originate at BASE24-atm terminals, this field is set by the Device Handler process using the MULT-ACCT field in the BASE24-atm Terminal Data files.	PIC 9

Position	Level	Field Name and Description	Data Type
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For transactions originating from the host, the ISO Host Interface process sets this field according to the value of the HOST-SELECTED-ACCT param.

For transactions originating from interchanges, the Interchange Interface process sets this field to 0.

285–286	04	RVSL-CDE	PIC 99
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A code indicating the reason for the reversal or adjustment. This field is used only for reversal (0420) and adjustment (5400) transactions.

Values for reversals are as follows:

- 01 = Timeout
- 02 = Command reject
- 03 = Destination not available
- 08 = Customer canceled transaction
- 10 = Hardware error
- 20 = Suspect transaction
- 21 = Message Authentication Code (MAC) failure
- 22 = MAC key synchronization error
- 23 = Message replay error
- 24 = Invalid MAC
- 30 = Funds not approved for Load Value
- 31 = Load Value not successful
- 40 = Split Transaction Routing enabled; secondary service not approved

Values for adjustment codes are as follows:

- 12 = Original amount incorrect
- 13 = ATM malfunction
- 14 = Suspicious reversal override
- 15 = Misdispense reversal override
- 16 = Duplicate transaction
- 17 = Reconciliation error
- 18 = PLUS add cash withdrawal or advance

Position	Level	Field Name and Description	Data Type
287–305	04	FROM-ACCT The account number of the <i>from</i> account of a transaction. If the <i>from</i> account is not needed or not known, this field contains zeros.	PIC X(19)
306	04	USER-FLD4 (NOT USED)	PIC X
307–325	04	TO-ACCT The account number of the <i>to</i> account of a transaction. If the <i>to</i> account is not needed or not known, this field contains zeros.	PIC X(19)
326–341	04	PIN The PIN entered by the cardholder initiating the transaction. If the PIN has already been verified, this field contains zeros.	PIC X(16)
342–343	04	PIN-SIZE The length of the PIN (for example, 01 equals one digit in length; 04 equals four digits in length).	PIC 9(2)
344	04	PIN-TRIES A count of the number of bad PIN attempts at the ATM. The value in this field is increased by the Authorization process. The Authorization process sets this to 1, 2, 3, and so on, as invalid PIN tries are detected during a transaction. Once a PIN has been verified as correct, this field is set to Z.	PIC X
345–360	04	PIN-KEY The encrypted version of the actual key used to encrypt PINs before they are transmitted from an ATM. This field is used when keys are managed by a security device.	PIC X(16)

Position	Level	Field Name and Description	Data Type
		For transactions originating at BASE24-atm terminals, this field is set by the Device Handler process using the ENCR-PIN.ENCR-KEYS.P-KEY field in the BASE24-atm Terminal Data files.	
361	04	PIN-FRMT Identifies the format of the PIN field. Valid values are as follows: 0 = Clear PIN 1 = Encrypted ANSI PIN block 3 = Encrypted PIN/PAD PIN block	PIC 9
362	04	PINPAD-CHAR The fill character used to pad the PIN block if the PIN was not long enough to fill the PIN block. This field is used by the system to determine the length of the PIN.	PIC X
363–364	04	ANSI-OFST A value indicating the starting position of the twelve PAN digits. It is used to encrypt the PIN if the system is using the ANSI PIN block. The first digit in the PAN has a positional value of 0.	TYPE BINARY 16
365–380	04	USER-KEY The previous PIN key used by the interchange to encrypt PINs. This field is set by the Interchange Interface process after a key exchange. If the PIN verification with the new key fails with a sanity check error, the Authorization process attempts to verify the PIN with the key in this field.	PIC X(16)
381–382	04	USER-FLD4A Bytes 17 and 18 of the IBM PIN block format. This is required when the value in the PIN-FRMT field is 2 (the PIN is encrypted by IBM 3624 PIN block) and the PIN is encrypted.	PIC X(2)

Position	Level	Field Name and Description	Data Type
383	04	CUST-BAL-INFO A code indicating the type of customer balance information given at an ATM. This field is set by the Authorization process using the CUST-BAL-INFO field in the BASE24-atm segment of the IDF. Valid values are as follows: 0 = No information given. 1 = RQST.AMT-2 field given only. 2 = RQST.AMT-3 field given only. 3 = RQST.AMT-2 and RQST.AMT-3 fields given; AMT-2 preferred. 4 = RQST.AMT-2 and RQST.AMT-3 fields given; AMT-3 preferred. For all transactions except balance inquiry transactions, this field determines what balances are displayed. However, if the transaction is a balance inquiry transaction and this field is set to 0 (no information given), the Device Handler process determines what balances to display using the CUST-BAL-INFO field in the BASE24-atm Terminal Data files.	PIC 9
384	04	CUST-BAL-DSPY A code indicating whether the customer balance information should be displayed, printed, both, or neither. This field is set by the Authorization process using the CUST-BAL-DSPY field in the BASE24-atm segment of the IDF. Valid values are as follows: 0 = Do not print or display balance information. 1 = Display balance information on screen only. 2 = Print balance information on receipt only. 3 = Print and display balance information. For all transactions except balance inquiry transactions, this field determines where balances are displayed. However, if the transaction is a balance inquiry transaction and this field is set to 0 (do not print or display), the Device Handler process determines how to display the balances using the CUST-BAL-DSPY field in the BASE24-atm Terminal Data files.	PIC 9

Position	Level	Field Name and Description	Data Type
385–392	04	AMT-1 The transaction amount requested. In the case of force post (0220) or reversal (0420) messages, this field contains the amount originally requested. For adjustment (5400) messages, this field indicates the original completion amount.	TYPE BINARY 64
393–400	04	AMT-2 For reversal (0420) messages this field contains the actual completion amount of the transaction. For response (0210) messages, this field contains a balance amount. This is the credit balance for credit accounts; and it is the ledger balance for non-credit accounts. For most request (0200) messages, this field contains zeros. However, for deposits with cash-back messages, this field contains the cash-back amount. For split deposits, this field contains the amount going to the second account. For adjustment (5400) messages, this field indicates the updated completion amount.	TYPE BINARY 64
401–408	04	AMT-3 For response (0210) messages, this field contains a balance amount. This is the available balance for non-credit accounts; and it is the available credit for credit accounts. For a reversal (0420) message that is the result of a deposit with cash back transaction, this field contains the deposit amount if the deposit part of the transaction was not completed or the cash back amount if the deposit was completed but the cash back part was not. The reversal of a deposit with cash back is the only type of reversal that uses this field.	TYPE BINARY 64
409–412	04	DEP-BAL-CR The amount of credit given on a deposit. This field is used for deposit reversal processing.	TYPE BINARY 32

Position	Level	Field Name and Description	Data Type
413	04	AVAIL-DEP-TYP The type of depository attached to the originating terminal. For transactions originating at BASE24-atm terminals, the Device Handler process sets this field using the AVAIL-DEP-TYP field in the BASE24-atm Terminal Data files, as well as the deposit method the ATM indicates is being used. Valid values are as follows: 0 = Normal envelope depository 1 = Commercial depository (for example: Securomatic) 2 = Envelope and commercial depository 3 = Check depository 4 = Currency acceptor depository 5 = Bulk check depository	PIC 9
414	04	DEP-TYP The type of depository to be used at the terminal. This field is set by the Authorization process based on the card type that initiated this transaction. Valid values are as follows: 0 = Normal envelope depository 1 = Commercial depository (for example: Securomatic) 2 = Envelope and commercial depository 3 = Check depository 4 = Currency acceptor depository 5 = Bulk check depository	PIC 9
415	04	COMPL-REQ A code indicating whether the host requires a completion message. Valid values are as follows: 0 = No completion required. This value is set from the Issuer Processing Code File (IPCF) by the Authorization process. 1 = Completion required for approved transactions. This value is set from the IPCF by the Authorization process.	PIC 9

Position	Level	Field Name and Description	Data Type
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|--|--|---|--|
| | | 2 = Host not available. The Authorization process generates a completion for approved transactions regardless of the IPCF settings. | |
| | | 3 = Completion acknowledgment required. This value is set by an Interchange Interface process on deposit transactions so that the Authorization process returns a 0230 message containing the deposit credit amount. This amount is used if a subsequent reversal is generated. | |
| | | 4 = Completion required for all transactions. This value is set from the IPCF by the Authorization process. | |
| | | 5 = Completion required for BASE24 authorized transactions that have been approved. This value is set from the IPCF by the Authorization process. | |
| | | 6 = Completion required for BASE24 authorized transactions that have either been approved or declined, regardless of the response code. This value is set from the IPCF by the Authorization process. | |

416–418	04	RESP	PIC X(3)
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A three-digit response code. The first digit indicates whether the card was returned or retained. Values for the first digit are as follows:

- 0 = Return the card.
- 1 = Retain the card.

The next two digits indicate the reason why the transaction was approved or denied. For a complete list of the BASE24 transaction response codes, refer to appendix B.

419–443	04	TERM-NAME-LOC	PIC X(25)
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The name and location of the originating terminal.

For transactions originating at BASE24-atm terminals, this field is set by the Device Handler process using the TERM-NAME-LOC field in the BASE24-atm Terminal Data files.

Position	Level	Field Name and Description	Data Type
444–465	04	TERM-OWNER-NAME The name of the financial institution that owns the ATM. For transactions originating at BASE24-atm terminals, this field is set by the Device Handler process using the TERM-OWNER-NAME field in the BASE24-atm Terminal Data files.	PIC X(22)
466–478	04	TERM-CITY The city of the originating terminal. For transactions originating at BASE24-atm terminals, this field is set by the Device Handler process using the TERM-CITY field in the BASE24-atm Terminal Data files.	PIC X(13)
479–481	04	TERM-ST-X Identifies the state, province, or political boundary of the originating terminal. In the United States, this is the two-digit state code, left-justified, and blank-filled. For transactions originating at BASE24-atm terminals, this field is set by the Device Handler process using the TERM-ST-X field in the BASE24-atm Terminal Data files.	PIC X(3)
482–483	04	TERM-CNTRY-X Identifies the country of the originating terminal. For transactions originating at BASE24-atm terminals, this field is set by the Device Handler process using the TERM-CNTRY-X field in the BASE24-atm Terminal Data files.	PIC X(2)
484–511	04	ORIG The original data elements apply to transaction reversals (0420) and adjustments (5400) only. If this transaction is a reversal or adjustment, these fields contain information about the original transaction.	

Position	Level	Field Name and Description	Data Type
484–495	06	SEQ-NUM The sequence number of the original transaction.	PIC X(12)
496–499	06	TRAN-DAT The date (MMDD) the original transaction was processed.	PIC X(4)
500–507	06	TRAN-TIM The time (hhmmssstt) from the original transaction timestamp.	PIC X(8)
508–511	06	B24-POST-DAT The BASE24 posting date of the original transaction.	PIC X(4)
512	04	TERM-TRAN-ALLOWED A code indicating at what level a given transaction is allowed for not-on-us cardholders at this terminal. Levels are as follows: 0 = Not allowed 1 = Allowed within the county 2 = Allowed within the state 3 = Allowed nationally 4 = Allowed internationally For transactions originating at BASE24-atm terminals, the Device Handler process sets this field using the NOT-ON-US-CRD field in the BASE24-atm Terminal Data files or the TXN-ALWD field in the Acquirer Processing Code File (APCF). For transactions originating through interchanges, the Interchange Interface process sets this field using the NOT-ON-US field in the ICF or the TXN-ALWD field in the APCF.	PIC 9

Position	Level	Field Name and Description	Data Type
513–518	04	TRACE-NUM The trace number assigned to the transaction. This is set and used by the Host Interface process to match transactions to their correct response timers when responses are received.	PIC 9(6)
519–520	04	TERM-ST The state in which the originating terminal is located. In the United States, this field contains the ANSI state code. The Authorization process compares this field to the FI-ST field in the Base segment of the IDF when determining if a given transaction is allowed at the state level. For transactions originating at BASE24-atm terminals, the Device Handler process sets this field using the TERM-STATE field in the BASE24-atm Terminal Data files.	PIC 9(2)
521–523	04	TERM-CNTY The county in which the originating terminal is located. In the United States, this field contains the ANSI county code for the county. The Authorization process compares this field to the FI-CNTY field in the Base segment of the IDF when determining if a given transaction is allowed at the county level. For transactions originating at BASE24-atm terminals, the Device Handler process sets this field using the TERM-CNTY field in the BASE24-atm Terminal Data files.	PIC 9(3)
524–526	04	TERM-CNTRY The ISO country code, indicating the country in which the originating terminal is located. The Authorization process compares this field to the FI-CNTRY field in the Base segment of the IDF when determining if a given transaction is allowed at the country level.	PIC 9(3)

Position	Level	Field Name and Description	Data Type
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For transactions originating at BASE24-atm terminals, the Device Handler process sets this field using the TERM-CNTRY field in the BASE24-atm Terminal Data files.

527–542	04	USER-FLD5 (NOT USED)	PIC X(16)
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543–558	04	PRI-DEST	PIC X(16)
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The primary destination for alternate routing regarding foreign transactions.

This field is initially zero-filled by the Device Handler process. The Authorization process places the symbolic name of the primary destination in this field.

559–574	04	ALT-DEST	PIC X(16)
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The alternate destination for routing foreign transactions.

This field is initially zero-filled by the Device Handler process. The Authorization process places the symbolic name of the alternate destination in this field.

575–590	04	AUTH-DEST	PIC X(16)
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The symbolic name of the entity that performed the authorization. This field is used for reversal (0420) messages.

This field is initially zero-filled by the Device Handler process.

591	04	DEST-ORDER	PIC X
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The routing order, so that the receiving entity can determine whether to forward the request.

Once a response (0210) message is received the field can be reset to reflect who authorized the transaction. Valid values are as follows:

A = Alternate destination

P = Primary destination

Position	Level	Field Name and Description	Data Type
592–602	04	RTE-GRP The routing group to which the terminal belongs for routing and processing of foreign transactions. The default value is zeros. For transactions originating at BASE24-atm terminals, the Device Handler process sets this field using the RTE-GRP field of the BASE24-atm Terminal Data files. For transactions originating at an interchange, the Interchange Interface process sets this field using the DFLT-RTG-GRP field in the BASE24-atm segment of the ICF or ICFE.	PIC X(11)
603–606	04	SAVE-ACCT The actual <i>to</i> and <i>from</i> account types. The following fields are set by the Authorization process.	
603–604	06	FROM-ACCT-TYP The type of <i>from</i> account. For a split-deposit transaction, this field specifies the type for a second <i>to</i> account. Valid values are as follows: 01 = Checking 11 = Savings 31 = Credit	PIC X(2)
605–606	06	TO-ACCT-TYP The type of <i>to</i> account. Valid values are as follows: 01 = Checking 11 = Savings 31 = Credit	PIC X(2)
607–616	04	POSTAL-CDE The postal code of the terminal. This field is set by the Device Handler or Interchange Interface process using the POSTAL-CDE field in the BASE24-atm Terminal Data files. For	PIC X(10)

Position	Level	Field Name and Description	Data Type
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transactions originating from an ISO host, this field is set by the ISO Host Interface process using the element P-62 of the ISO-based external message.

617–654	04	USER-FLD6 (NOT USED)	PIC X(38)
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655	04	FRMT	PIC X
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A code indicating whether the Device Handler process supports the BASE24-atm Statement Print Data Compression format on statement print transactions. Currently, the NCR 5XXX Device Handler process and the Diebold 10XX/478X Device Handler process support data compression. Valid values are as follows:

- 0 = Columnar format (data compression is not supported).
- 1 = BASE24-atm data compression format (data compression is supported).

The NCR 5XXX Device Handler process sets this value from the N5080.STMT-PRNTR-CNTL.FRMT field in the BASE24-atm Terminal Data files. The Diebold 10XX/478X Device Handler process sets this value from the DIEBOLD.FRMT field in the BASE24-atm Terminal Data files.

656–658	04	PRNT-SIZ	PIC X(3)
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The maximum number of characters per line supported by the Device Handler process when BASE24-atm Statement Print Data Compression is implemented for statement print transactions. Currently, the NCR 5XXX Device Handler process and the Diebold 10XX/478X Device Handler process support data compression; however the Diebold 10XX/478X Device Handler process does not use this field for data compression. If data compression is not supported, this field is zero filled.

The NCR 5XXX Device Handler process sets this value by subtracting the N5080.STMT-PRNTR-CNTL-LEFT.MARGIN field from the N5080.STMT-PRNTR-CNTL-RIGHT.MARGIN field in the BASE24-atm Terminal Data files.

Position	Level	Field Name and Description	Data Type
179–658	02	ACCT	REDEFINES RQST
		These fields are set by the Authorization process. They contain cardholder accounts for multiple account selection. The accounts to be placed in these fields are taken from the Cardholder Authorization File (CAF).	
179–486	04	MULT	OCCURS 2 TIMES
		The first occurrence of this field contains the cardholder <i>from</i> accounts for multiple account selection; the second occurrence of this field contains the cardholder <i>to</i> accounts for multiple account selection.	
		179–332 = <i>From</i> account fields (154 positions in length)	
		333–486 = <i>To</i> account fields (154 positions in length)	
	06	ACCT-TYP	PIC 9(2)
		The type of accounts included in the ACCT.MULT.INFO fields.	
	06	CNT	PIC X
		Actual number of occurrences of the ACCT.MULT.INFO fields.	
	06	USER-FLD7 (NOT USED)	PIC X
	06	INFO	OCCURS 5 TIMES
		The following fields contain account information used for account selection messages. Each occurrence is 30 positions in length.	
	08	NUM	PIC X(19)
		The account number.	

Position	Level	Field Name and Description	Data Type
	08	USER-FLD8 (NOT USED)	PIC X
	08	DESCR The account description.	PIC X(10)
487	04	PIN-VRFY-FLG A code indicating whether the PIN has been verified in the 9906 message. This field is set by the Authorization process and used by the Device Handler process.	PIC 9
488–658	04	SAVE-AREA (NOT USED)	PIC X(171)
179–658	02	CRD-REVIEW The following fields are used when the Olivetti device is being used, multi-account cards are configured at the ATM, BASE24 is the authorizer of the transaction, and a positive authorization method is being used. The BASE24 Authorization process initiates these fields using the respective CAF field.	REDEFINES RQST
179	04	CRD-STAT The card status. This field is set by the Authorization process using the CRD-STAT field in the Base segment of the CAF. It is used by the Authorization and Device Handler processes.	PIC X
180	04	USER-FLD8A (NOT USED)	PIC X
181–182	04	PIN-TRIES The number of card PIN tries. This field is used by the Authorization and Device Handler processes and is set from the BAD-PIN-TRIES field in the Base segment of the CAF.	TYPE BINARY 16

Position	Level	Field Name and Description	Data Type
183–190	04	LIMIT The card limit. This field is used by the Authorization and Device Handler processes and is set from the GRP-LMT. AGGR-LMT field in the Base segment of the CAF.	TYPE BINARY 64
191–192	04	CNT The number of accounts associated with the current card. This field is used by the Authorization and Device Handler processes and is set from the ACCT-CNT field in the Accounts segment of the CAF.	TYPE BINARY 16
193–544	04	ACCTS The following fields contain account information used by the Authorization and Device Handler processes. These fields are set from the Accounts segment of the CAF. Each occurrence is 22 positions in length.	OCCURS 16 TIMES
	06	TYP The type of account included in the CRD-REVIEW.ACCTS field.	PIC 9(2)
	06	NUM The account number.	TYPE X(19)
	06	STAT The status of the account.	PIC X
545–658	04	SAVE-AREA	PIC X(114)
179–658	02	STMT-INFO The following fields are used for statement print transaction request (0205) and statement print response (0215) messages.	REDEFINES RQST

Position	Level	Field Name and Description	Data Type
179–194	04	HI-PRO-NAME The symbolic process name of the Host Interface process used by the Authorization process to send transactions to a host. The Authorization process sets this field using the RT-TBL. SYM-NAME fields in BASE24-atm segment of the IDF.	PIC X(16)
195–234	04	TRACK2 The Track 2 data, as sent by the ATM to the Device Handler process.	PIC X(40)
235–236	04	SAVE-FROM-ACCT-TYP The actual <i>from</i> account type.	PIC X(2)
237–239	04	MBR-NUM The member number or card sequence number of the card used to initiate the transaction.	PIC X(3)
240–241	04	TERM-TYP A code indicating the type of terminal from which the transaction originated. For transactions originating at a BASE24 terminal, the Device Handler process sets this field using the TERM-TYP field in the BASE24-atm Terminal Data files.	PIC 9(2)
242–252	04	ACQ-INST-ID-NUM The routing or issuer identification number of the terminal owner. For transactions originating at a BASE24 terminal, the Device Handler process sets this field using the ID-NUM field in the BASE24-atm Terminal Data files.	PIC 9(11)

Position	Level	Field Name and Description	Data Type
253–271	04	FROM-ACCT The <i>from</i> account number of the transaction. If no <i>from</i> account is needed or known, this field is set to zeros by the Device Handler process. In the case of primary accounts, this field is set by the Authorization process.	PIC X(19)
272–274	04	RESP A three-digit response code. The first digit indicates whether the card should be returned or retained. The next two digits indicate the reason why the transaction was approved or denied. For a complete list of the BASE24 transaction response codes, refer to appendix B.	PIC X(3)
275–658	04	STMT The statement data coming from the host.	
275–276	06	PAGE-IND If numeric, this is the current page number of the item request. Otherwise, valid values are as follows: LP = Last page of the statements SP = Summary page 1P = Account statements fill only one page Note: Because the IBM 4730 is limited to one message, this field contains either 1P or LP if the IBM 4730 is the ATM involved.	PIC X(2)
277–282	06	LAST-STMT-DAT The earliest date (YYMMDD) that the information in the statement starts, sent by the host. Note: Due to length restrictions of the IBM 4730 response message, the LAST-STMT-DAT field is not used by the IBM 4730 Device Handler process.	PIC X(6)

Position	Level	Field Name and Description	Data Type
283–284	06	HEADER-LINES The number of lines in the header, as defined and sent by the host.	PIC X(2)
285–286	06	COLUMN-LINES The number of characters per line, as defined and sent by the host.	PIC X(2)
287–646	06	STMT-FLD The statement lines and header information, as sent by the host.	PIC X(360)
647–654	06	USER-FLD9 (NOT USED)	PIC X(8)
655	04	FRMT A code indicating whether the Device Handler process supports the BASE24-atm Statement Print Data Compression format on statement print transactions. Currently, the NCR 5XXX Device Handler process and the Diebold 10XX/478X Device Handler process support data compression. Valid values are as follows: 0 = Columnar format (data compression is not supported). 1 = BASE24-atm data compression format (data compression is supported). The NCR 5XXX Device Handler process sets this value from the N5080.STMT-PRNTR-CNTL.FRMT field in the BASE24-atm Terminal Data files. The Diebold 10XX/478X Device Handler process sets this value from the DIEBOLD.FRMT field in the BASE24-atm Terminal Data files.	PIC X
656–658	04	PRNT-SIZ The maximum number of characters per line the Device Handler process supports when BASE24-atm Statement Print Data Compression is implemented for statement print	PIC X(3)

Position	Level	Field Name and Description	Data Type
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transactions. Currently, the NCR 5XXX Device Handler process and the Diebold 10XX/478X Device Handler process support data compression; however the Diebold 10XX/478X Device Handler process does not use this field for data compression. If data compression is not supported, this field is zero filled.

The NCR 5XXX Device Handler process sets this value by subtracting the N5080.STMT-PRNTR-CNTL-LEFT.MARGIN field from the N5080.STMT-PRNTR-CNTL-RIGHT.MARGIN field in the BASE24-atm Terminal Data files.

659–666	02	ENTRY-TIM	TYPE BINARY 64
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The timestamp indicating when the transaction entered the BASE24 system for the first time. This field is set by the Device Handler, Interchange Interface, or Host Interface process.

The Device Handler process loads the timestamp when the STM is initialized. The Interchange Interface or Host Interface process sets this field when the transaction originates from the DPC or interchange.

667–674	02	EXIT-TIM	TYPE BINARY 64
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The timestamp, expressed in Greenwich Mean Time (GMT), indicating when the transaction exited the BASE24 system for authorization, either through the Host Interface or Interchange Interface process.

675–682	02	RE-ENTRY-TIM	TYPE BINARY 64
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The timestamp, expressed in Greenwich Mean Time (GMT), indicating when the response to a transaction request to an interchange or a DPC reentered the BASE24 system.

Position	Level	Field Name and Description	Data Type
683–693	02	CRD-ACCPT-ID-NUM The institution identification number of the card acceptor on a request (0200) message originating from an acquirer host (in the event that the acquirer is not the actual card acceptor). This field is for logging purposes only. Information in this field is right-justified and blank-filled.	PIC 9(11)
694–704	02	FRWD-INST-ID-NUM The value of the ATM-FORWARD-INST-ID param in the LCONF. This field is for logging purposes only. It equates to a routing transit number representing BASE24. This field is set by the BASE24-atm Authorization process.	PIC 9(11)
705–715	02	CRD-ISS-ID-NUM The institution identification number of the card issuer on a response (0210) message from the authorizing host, in case the receiving institution is not the actual card issuer. The information in this field is right-justified and blank-filled.	PIC 9(11)
716–721	02	AUTH-ID-RESP The host transaction sequence number. This field is for logging and extract purposes only.	PIC X(6)
722–828	02	TRACK3 An exact image of Track 3 as sent from the ATM. If the Track 3 information was not sent, this field contains zeros.	PIC X(107)
829–871	02	MICR-DATA (NOT USED)	PIC X(43)
872	02	USER-FLD10 (NOT USED)	PIC X

B: BASE24-atm Card Disposition and Authorization Response Codes

This appendix contains descriptions of the card disposition codes and authorization response codes supported by BASE24-atm. These codes are used by transaction authorizers to indicate how they want transactions to be handled. When a transaction is authorized, the authorizer returns these codes to inform the party requesting the authorization of the actions to take. BASE24-atm carries both of these codes in the RQST.RESP field of the Standard Internal Message (STM).

This appendix includes a description of each code and how it can be generated by the BASE24-atm Authorization process. The following abbreviations have been used throughout this appendix for the specified file names and authorization methods:

File Names

ARF	=	Account Routing File
ATD	=	BASE24-atm Terminal Data files
CAF	=	Cardholder Authorization File
CCF	=	Corporate Check File
CNRF	=	Chip Number Range File
CPF	=	Card Prefix File
CSF	=	Check Status File
IDF	=	Institution Definition File
KEYA	=	Key Authorization File
KEYI	=	ICC KEY File
NEG	=	Negative Card File
PBF	=	Positive Balance File
SPF	=	Stop Payment File
UAF	=	Usage Accumulation File

Authorization Methods

NEG-only = Negative Authorization without Usage Accumulation method
NEG/UAF = Negative Authorization with Usage Accumulation method
CAF-only = Positive Authorization method
CAF/PBF = Positive Balance Authorization method

Card Disposition Codes

Card disposition codes are one-position codes indicating whether a transaction authorizer wants the card used in a transaction to be returned to the cardholder or retained. These codes enable institutions to retrieve cards that have been lost or stolen or are to be taken out of circulation on behalf of the card issuer.

Card disposition codes are carried in the first position of the RQST.RESP field of the STM. The RQST.RESP field is a three-position field, with the second and third positions carrying the response code. Thus, in BASE24-atm, the card disposition functions separately from the response code.

Card disposition code values are as follows:

- 0 = Return card
- 1 = Retain card

Under most circumstances in BASE24-atm processing, even when a transaction is denied, the card is returned. However, the BASE24-atm Authorization process does specify that cards be retained in the following instances:

- A NEG record exists for the card and the NEG capture code is not equal to 00 (return card).
- The status code in the CAF is equal to 2 (lost) or 3 (stolen).
- The cardholder has exceeded his or her PIN-retry limit and the BAD PIN ACTION field in the card issuer IDF or CPF indicates that the card is to be retained.
- The card has expired and the EXP CHECK DISP field in the card issuer IDF record indicates that the card is to be retained.

Authorization Response Codes

Authorization response codes are two-position codes, indicating whether a transaction authorizer wants a transaction to be approved or denied and why.

Authorization response codes are carried in the second and third positions of the RQST.RESP field of the STM. The RQST.RESP field is a three-position field, with the first position carrying the card disposition code.

The following table provides a description of each response code supported by BASE24-atm, including the circumstances under which the BASE24-atm Authorization process generates the code.

Code	Description
00	APPROVED WITH BALANCES The transaction was approved and account balances were returned with the transaction response. The BASE24-atm Authorization process returns a 00 response code when it approves the following transactions under CAF/PBF processing: - Credit withdrawals - Checking withdrawals - Savings withdrawals - Deposits - Inquiries
01	APPROVED—NO BALANCES FOR DISPLAY The transaction was approved and account balances were not returned with the transaction response. The BASE24-atm Authorization process returns a 01 response code when it approves transactions as follows: a) Approves transactions under NEG-only, NEG/UAF, and CAF-only processing. b) Approves the following transactions under CAF/PBF processing: - Deposit with cash back - Split deposit - Payment from checking to credit account <i>Continued on the following page</i>

Code	Description
01 <i>continued</i>	Payment from savings to credit account Transfer Check verification Check guarantee Cash check Payment enclosed Message to the financial institution c) Approves a log-only transaction.
50	UNAUTHORIZED USAGE The transaction was denied due to a prohibitive status in the cardholder's file. The BASE24-atm Authorization process returns a response code of 50 under one of the following circumstances: a) The capture code in the NEG is nonzero. b) The reason code in the NEG is 01 (lost), 02 (stolen), 05 (denial), or 11 (closed). c) The card status in the CAF is 2 (lost), 3 (stolen), 6 (BASE24-pos use), or 9 (closed). d) The card status in the CAF is 4 (restricted) and the transaction is not one of the following: Deposit Deposit with cash back Balance inquiry Payment enclosed Message to financial institution Log-only transaction e) The card status in the CAF is 0 (issued, but not active) and the institution or card prefix is not set up to allow cardholders to select their own PINs. f) The card status in the CAF is 0 (issued, but not active) and the institution or card prefix is set up to allow cardholder-selected PINs, but there is already a PIN offset in the CAF.
51	EXPIRED CARD The transaction was denied because the card used in the transaction is expired.

Code	Description
52	INVALID CARD The transaction was denied because, under CAF-only or CAF/PBF processing, a CAF record could not be retrieved for the card number and member number involved in the transaction.
53	INVALID PIN The transaction was denied because of one of the following conditions: a) The cardholder's PIN was entered incorrectly or did not match the PIN associated with the card used in the transaction. b) The two PINs entered on a PIN change transaction were not the same.
54	DATABASE PROBLEM The transaction was denied due to a problem with the BASE24 database. The BASE24-atm Authorization process returns a response code of 54 under one of the following circumstances: a) The TLF is full. b) Appropriate authorization and routing parameters cannot be derived from the IDF for the transaction. c) The currency code used in the transaction does not match either the currency code of the card issuer or the currency code of the mobile operator in the Mobile Operator File (MOF) for mobile top-up transactions. d) Under CAF/PBF processing, PBF records are not available for the accounts involved in the transaction. e) Under CAF/PBF processing, the status in the PBF does not allow for the transaction because the account is inactive or the account is closed. f) Under NEG/UAF processing, a UAF record cannot be created or accessed. g) No KEYA record can be found for PIN verification. h) No KEYA record can be found for PIN offset generation. <i>Continued on the following page</i>

Code	Description
54 <i>continued</i>	i) The Authorization process cannot generate a PIN offset for a PIN change transaction. j) Under NEG or NEG/UAF processing, a PIN change transaction is attempted. PIN change transactions are not supported under NEG or NEG/UAF processing. k) Under CAF or CAF/PBF processing, the PIN offset is not located in the CAF. For PIN change transactions, the PIN offset must be stored in the CAF. l) A PIN change transaction is attempted and the PIN offset is located on Track 2 or Track 3 of the card (local PIN verification). PIN change transactions are not supported with local PIN verification. m) A PIN change transaction is attempted and the institution is set up for Identikey PIN verification. PIN change transactions are not supported under this PIN verification method because a PIN offset cannot be generated. n) Invalid KEYA or card data is found during a card verification transaction. o) The BASE24-atm Authorization process is configured to access a Thales e-Security (Racal) security module and PIN verification parameters are set up to perform PIN verification using the Diebold algorithm. p) The CPF EMV segment does not exist, or the ISSUER APPL DATA FORMAT field in the CPF EMV contains an invalid value. q) The top-up transaction amount does not match the configured amounts for the mobile operator in the Mobile Operator File (MOF).

Code	Description
55	INELIGIBLE TRANSACTION The transaction was denied due to processing restrictions imposed on the type of transaction being performed. The BASE24-atm Authorization process returns a response code of 55 under one of the following circumstances: a) An administrative card is being used, but the card issuer does not own the terminal from which the transaction originated. b) The attempted transaction is not supported by BASE24-atm. c) A deposit-only card is being used, but the transaction is other than a deposit or message to financial institution. d) A transfer or split deposit is being attempted between or to accounts of the same type. This is not allowed under Level 1 (online) processing; level 2 (offline) processing if BASE24 is set up for NEG-only or NEG/UAF processing; or level 3 (online/offline) processing if the host is not available and BASE24 is set up for NEG-only or NEG/UAF processing. Under CAF-only or CAF/PBF processing, a transfer is being attempted where the <i>from</i> and <i>to</i> accounts are the same account. e) A statement print transaction is being attempted under level 2 (offline) processing. f) A balance inquiry is being attempted under level 2 (offline) processing and BASE24 is set up for NEG-only, NEG/UAF, or CAF-only processing. In this case, account balances are not available from BASE24. g) A balance inquiry is being attempted using a Super Teller card. h) The Super Teller terminal ID in the transaction does not match the terminal ID in the CAF or NEG record for the Super Teller card. i) The transaction has neither a <i>from</i> account nor <i>to</i> account and is not one of the following transactions: fast cash withdrawal, cash check, payment enclosed, message to financial institution, card review, PIN change, log only, balance inquiry from default, or EMV PIN unblock. j) The authorization level retrieved from the Institution Definition File (IDF) routing table is not 1, 2, or 3.

Code	Description
56	INELIGIBLE ACCOUNT The transaction was denied because of restrictions on, or difficulties with, the application accounts specified in the cardholder CAF record. The BASE24-atm Authorization process returns a response code of 56 under one of the following circumstances: a) The cardholder CAF record does not contain an appropriate account for the transaction (for example, a savings account for a savings account withdrawal). b) The cardholder CAF record contains duplicate account numbers (two identical account numbers of the same type of account). c) The application account selected for the transaction has a status in the CAF of 0 (no relationship), 9 (closed), 2 (restricted), or 4 (restricted primary). d) A fast cash withdrawal and balance inquiry from default require that the Authorization process provide the account number to use, but no primary account is available in the CAF. e) Under CAF-only and CAF/PBF processing, the account numbers provided for the transaction are not in the CAF. f) Under CAF-only and CAF/PBF processing, a cash check transaction is being attempted, but there are no accounts listed in the CAF.
57	TRANSACTION NOT SUPPORTED The transaction was denied because the terminal owner or card issuer does not allow the transaction. The BASE24-atm Authorization process returns a response code of 57 under one of the following circumstances: a) The terminal owner does not allow the attempted transaction. b) The card issuer does not allow the attempted transaction. c) A card review transaction is requested and the authorization level is 1 or the authorization method is NEG or NEG/UAF <i>Continued on the following page</i>

Code	Description
57 <i>continued</i>	d) A card review transaction is requested and the Authorization process is unable to retrieve account information from the CAF under the CAF or CAF/PBF authorization method. e) The transaction is a PIN Unblock transaction (transaction code 82), and the OFFLINE PIN MANAGEMENT ACTION field on CPF screen 12 is set to 0, which disallows PIN management transactions. f) The transaction is a PIN Unblock transaction (transaction code 82), and there is no EMV segment present in the CAF record. g) The Authorization Multiple Currency enhancement is installed and the transaction is a deposit with cash back, a split deposit, or a generic inquiry. h) The Bulk Check Infrastructure is installed and the transaction is a split deposit that is performed as a bulk check transaction (i.e., several checks in a single transaction with a portion of the deposit to go to one account and the rest to another).
58	INSUFFICIENT FUNDS NO AMOUNT 3 The transaction was denied because it requested funds from an account in which the available funds were less than or equal to zero. The BASE24-atm Authorization process returns a response code of 58 under CAF/PBF processing only. In this case, the Authorization process returns zeros in the RQST.AMT-3 field of the STM.
59	INSUFFICIENT FUNDS WITH AMOUNT 3 The transaction was denied because it requested funds from an account in which the available funds were greater than zero, but less than the requested amount. The BASE24-atm Authorization process returns a response code of 59 under CAF/PBF processing only. In this case, the Authorization process places the amount of available funds in the account in RQST.AMT-3 field of the STM.
60	USES LIMIT EXCEEDED The transaction was denied because the cardholder had already reached the maximum number of withdrawals allowed during the current usage accumulation period.

Code	Description
61	WITHDRAWAL LIMIT WOULD BE EXCEEDED The transaction was denied because it would have caused the cardholder's withdrawal limit to be exceeded for the current usage accumulation period.
62	PIN TRIES EXCEEDED The transaction was denied because the cardholder had already reached the maximum number of PIN retries allowed for the current usage accumulation period.
63	WITHDRAWAL LIMIT ALREADY REACHED The transaction was denied because the cardholder had already reached his or her withdrawal limit for the current usage accumulation period.
64	INVALID CREDIT CARD CASH ADVANCE AMOUNT The transaction was denied because it did not meet the criteria for credit set by the card issuer for card transactions. The BASE24-atm Authorization process returns a response code of 64 under one of the following circumstances: a) The amount specified for a credit withdrawal or transfer from credit account is below the minimum specified in the CPF. b) The difference between the withdrawal amount and the minimum specified is not divisible by the standard credit account increment specified in the CPF.
65	NO STATEMENT INFORMATION FOR THE ACCOUNT The transaction was denied as a result of host processing. The BASE24-atm Authorization process does not generate this response code. It is sent by a host.
66	STATEMENT INFORMATION NOT AVAILABLE The transaction was denied as a result of host processing. The BASE24-atm Authorization process does not generate this response code. It is sent by a host.

Code	Description
67	INVALID CASH BACK AMOUNT The transaction was denied because the amount requested back in a deposit with cash back transaction exceeded the amount of the deposit.
68	EXTERNAL DECLINE The transaction was denied as a result of processing by an external system (host or interchange), which returned a response code that could not be translated to a more specific response code by BASE24. The BASE24-atm Authorization process does not generate this response code.
69	NO SHARING BETWEEN THE CARD ISSUER AND TERMINAL OWNER The transaction was denied because the terminal owner does not have a sharing relationship with the card issuer. The BASE24-atm Authorization process returns a response code of 69 when the card issuer is not the terminal owner and none of the sharing groups in the BASE24-atm Terminal Data files, ICF or ICFE match the sharing groups in the card issuer IDF.
70	SYSTEM ERROR The transaction was denied because the BASE24-atm Authorization process could not locate an IDF record for the card issuer, the card issuer IDF record was set with a fatal error, or there was a hardware problem with a security module. In addition, the Transaction Context Manager process uses this response code if it denies a transaction for any reason.
71	CONTACT CARD ISSUER The transaction was denied because the PAN was less than five digits long or for one of the following reasons related to BASE24-atm self-service banking (SSB): a) A withdrawal, deposit with cash back, or cash check transaction was attempted and the amount of cash to be dispensed was less than the minimum allowable cash withdrawal specified in the CPF. <i>Continued on the following page</i>

Code	Description
71 <i>continued</i>	b) The MICR data contained an amount sign (\$) indicating that the check has already been authorized. c) An ARF record was located for the given check, but the insertion of data into the account data in the MICR token would make the MICR data longer than the defined length of the MICR token. d) An SPF record was located for the given check indicating a stop payment. e) A CSF record was located for the given check and either the record status indicated that the check should be rejected or the check amount was greater than the check limit in the CSF record. f) A check-cashing card was used but no CCF record could be located for the given check. g) A CCF record was located for the given check and either the record status indicated that the check should be rejected or the check amount was greater than the check limit in the CCF record.
72	DESTINATION NOT AVAILABLE The transaction was denied because the destination to which the Authorization process routed the transaction was unavailable. The BASE24-atm Authorization process returns a response code of 72 under one of the following circumstances: a) A transaction request is forwarded by the Authorization process but cannot be delivered to its destination. b) The transaction request cannot be delivered to a back-end host, and BASE24 is not set up to stand-in for the host. c) A balance inquiry cannot get through to a host and cannot be approved by BASE24 because the authorization method is not CAF/PBF. d) A statement print transaction cannot get through to a host. e) A log-only transaction requires processing by a host but the host is unavailable. <i>Continued on the following page</i>

Code	Description
72 <i>continued</i>	f) The authorization level is offline but the requested transaction is not supported offline. g) A PIN Change or PIN Unblock cannot get through to a host and cannot be approved by BASE24 because the authorization method is not CAF or CAF/PBF.
73	ROUTING LOOK UP PROBLEM The transaction was denied because of a configuration problem. The BASE24-atm Authorization process returns a response code of 73 under one of the following circumstances: a) The Authorization process is the alternate destination but it does not recognize the prefix. b) The Authorization process does not recognize the prefix associated with the transaction and cannot determine another destination to which to send the transaction.
74	MESSAGE EDIT ERROR The transaction was denied because invalid data was encountered in the request message. The BASE24-atm Authorization process returns a response code of 74 under one of the following circumstances: a) Invalid data is encountered in editing the request message. b) The Track 2 data cannot be obtained from the request message. This could be the result of inappropriate values in the CPF or a bad Track 2 stripe on the card. c) The posting date in the request message is not equal to today's or tomorrow's date. d) An invalid (nonnumeric) member number, PAN, expiration date, or PIN offset is derived from Track 2. e) Authorization cannot retrieve and validate the BASE24-atm Release 5.0 token, and the token is required to process the transaction. f) An invalid expiration date is found for a card verification transaction. <i>Continued on the following page</i>

Code	Description
74 <i>continued</i>	g) Invalid or missing CVV data is detected during the request message processing. h) Refer to the CRD-VRFY-FLG field in the BASE24-atm Release 5.0 token for additional information about card verification results.
75	RESERVED BASE24 CODE The BASE24-atm Authorization process does not generate this response code.
80	SIGNATURE CHECK FAILED The Authorization process returns a response code of 80 when the signature in the load value transaction cannot be verified because a CNRF record for the stored value card (SVC) cannot be found or the signature is invalid.
81	SECURITY MODULE PARAMETER ERROR The BASE24-atm Authorization process returned the transaction because of an error message received from the security module when the BASE24-atm Authorization process used it to verify a transaction.
82	SECURITY MODULE FAILURE The BASE24-atm Authorization process returned the transaction because the security module did not respond when the BASE24-atm Authorization process used it to verify a transaction.
83	KEYI RECORD NOT FOUND The BASE24-atm Authorization process returned the transaction because it could not locate a valid KEYI record for this card prefix.
84	ATC CHECK FAILURE The BASE24-atm Authorization process returned the transaction because it failed the application transaction counter (ATC) test. One ATC is maintained by the microchip on the card and a corresponding value is held in the CAF. A comparison of these values is designed to detect fraudulent cards.

Code	Description
85	CVR DECLINE The BASE24-atm Authorization process returned the transaction because it failed the card verification results (CVR) test based on preset rules defined for a card type.
86	TVR DECLINE The BASE24-atm Authorization process returned the transaction because it failed the terminal verification results (TVR) test based on preset rules defined for a card type.
87	REQUEST CRYPTOGRAM FAILURE The BASE24-atm Authorization process returned the transaction because the authorization request cryptogram included in the transaction data did not match the authorization request cryptogram generated by BASE24.
88	FALLBACK DECLINE The BASE24-atm Authorization process returned the transaction because the terminal could not read an integrated chip card, and generated a “fallback” transaction. This can occur as a result of a faulty chip reader or a faulty chip, but could also be an indication of a fraudulent card.
90	APPROVED TRANSACTION—INSIDE WINDOW The Authorization process returns a response code of 90 when the terminal is set up to allow administrative transactions within the balancing window specified in the IDF and an administrative transaction is entered during the terminal balancing window.
91	APPROVED TRANSACTION—OUTSIDE WINDOW The Authorization process returns a response code of 91 when the terminal is set up to allow administrative transactions within the balancing window specified in the IDF and an administrative transaction is entered outside the terminal balancing window.
92	APPROVED TRANSACTION—BALANCE ANYTIME The Authorization process returns a response code of 92 when the terminal is set up to allow administrative transactions anytime.

C: BASE24-atm Device Handler Processing

This appendix describes Device Handler transaction processing concepts. This discussion includes the basic transaction sequence and how BASE24-atm Device Handler processes enforce that transaction sequence.

Note: Different Device Handlers process transactions differently depending on the requirements of the ATMs to which they correspond. However, most Device Handlers process transactions according to the general concepts.

Device Handler Transaction Processing Concepts

At a basic level, most BASE24-atm transactions—whether they are customer or administrative transactions—are processed by Device Handler processes in a similar fashion.

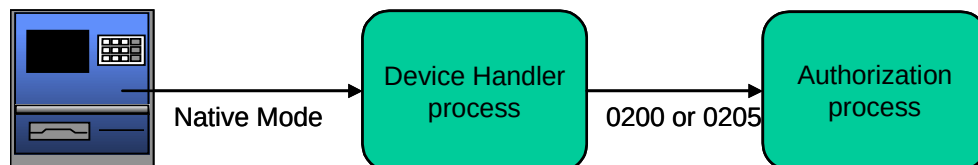
The Device Handler process receives the terminal request and translates the request into BASE24-atm Standard Internal Message (STM) format. The Device Handler process sends the STM to the Authorization process, which returns an authorization response message in the STM format. The Device Handler process translates the authorization response into the terminal native message format and sends it to the terminal. The Device Handler process then waits for a completion message from the terminal. If the transaction does not complete as authorized, the Device Handler process sends a reversal message in the Standard Internal Message (STM) format to the Authorization process.

Note: An interactive step (for example, a PIN change or multiple account select message) also can be included for requests and responses.

Transaction Sequence

The following steps illustrate the basic message flows through the Device Handler process for customer and administrative transactions. The messages that are, or can be, involved in the processing sequence are described below:

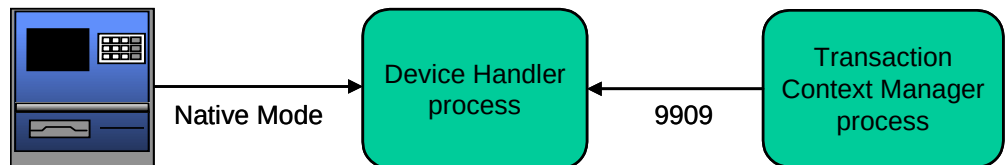
1. **Terminal Requests** are the result of transactions being initiated at the ATM. The Device Handler process translates a terminal request into an STM as a financial request (0200) or statement print (0205) message and forwards the request to the Authorization process as shown below. Initially, all request messages are sent as 0200 messages. When an ATM requires more information for a statement print transaction, a 0205 message is sent after the initial 0200 message.



2. **Account Data Selection** messages are returned to the Device Handler process by the Authorization process when a choice of accounts is required from the customer. Account selection (9906) messages, as shown below, are returned only when the customer has multiple accounts and the ATM supports multiple account selection.



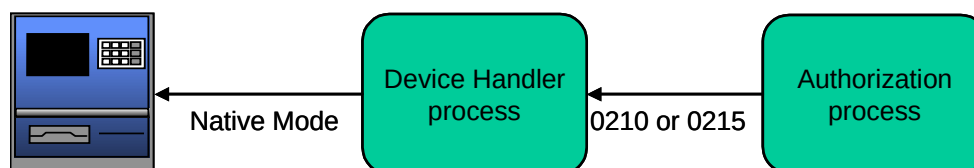
3. **Mobile Top-Up Notification/Selection** messages are returned to the Device Handler process by the Transaction Context Manager process when selections are required from the customer. Mobile Top-Up Notification/Selection (9909) messages, as shown below, are returned only when mobile top-up has been licensed.



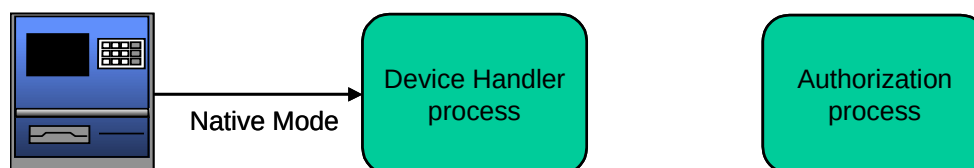
4. **Interactive Terminal Responses** are the result of an account selection performed at the ATM by a customer. The Device Handler process translates the terminal responses into internal messages as financial request (0200) messages and forwards them to the Authorization process as shown below.



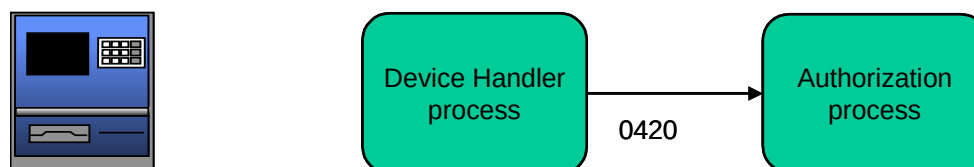
5. **Authorization Responses** are returned to the Device Handler process by the Authorization process when the transaction has been approved or declined. Authorization responses are returned in an internal message as Financial Transaction Request Response (0210) or Statement Print Response (0215) messages. The Device Handler process translates these response messages into the native-mode format of the ATM and sends them to the ATM as shown below.



6. **Completions** are returned from the ATM to indicate the outcome of a transaction as shown below. Device Handler processes evaluate completions to determine whether transactions completed as authorized.



7. **Reversals** are generated by Device Handler processes when transactions fail to complete as authorized. Reversal (0420) messages are sent by the Device Handler process to the Authorization process as shown below.



Enforcing the Transaction Sequence

As the primary BASE24 interface to ATMs, one of the principal functions of the BASE24-atm Device Handler process is to enforce the basic transaction sequence.

To enforce the proper transaction sequence, the Device Handler process uses several flags and fields that reside in extended memory or in the BASE24-atm Terminal Data files but are not accessible from BASE24 file maintenance screens. These flags and fields are discussed below.

STATUS Flag

The Diebold 10XX/478X Device Handler process and the NCR 5XXX Device Handler process set a STATUS flag in the dynamic portion of the BASE24-atm Terminal Data files each time they process a transaction message. (Other Device Handler processes set the STATUS flag in the DEV-DAT portion of the Terminal Data File.) This flag indicates the next message expected in the transaction sequence. If the anticipated message is not received, the Device Handler process takes the necessary steps to correct the situation.

The STATUS flag values used in processing transactions are shown below. Not all Device Handler processes use the same STATUS flag; however, all Device Handler processes have some type of flag which performs the function of controlling the transaction sequence.

STATUS Flag Values	Description
Expect Terminal Request	Indicates that there is no transaction currently in progress and that the Device Handler process can accept a new transaction from this ATM.
Expect Authorization Response	Indicates that the Device Handler process is expecting a reply from the Authorization process for this ATM.
Expect Interactive Terminal Response	Indicates that the Device Handler process is expecting an interactive terminal response from the ATM. In this case, the Device Handler process has sent multiple accounts to the ATM for selection.

STATUS Flag Values	Description
Expect Completion	Indicates that the Device Handler process is expecting a completion message from the ATM. This status is used for approved transactions. If a Device Handler process is expecting a completion and none is received, it reverses the transaction.
Expect Void	Indicates that the Device Handler process is expecting a completion message from the ATM. This status is used for declined and cancelled transactions. It differs from the expect completion status in that if the Device Handler process is expecting a void and does not receive a completion, the Device Handler process does not reverse the transaction.

TRANSACTION-IN-PROGRESS Flag

The Device Handler process sets a TRANSACTION-IN-PROGRESS flag in the BASE24-atm Terminal Data files each time it begins processing a transaction for a terminal and then resets it once the transaction is complete. This flag resides in the DEV-DAT portion of the BASE24-atm Terminal Data files and indicates that a transaction is in progress for the terminal. The Settlement Initiator process will not attempt to cut over a terminal with a transaction in progress. Rather, it will set the value in the CUTOVER-PEND field to indicate that the terminal needs to be cut over.

TRAN-SEQ-NUM Field

The Device Handler process uses the TRAN-SEQ-NUM field in the BASE24-atm Terminal Data files to store the sequence number of the last transaction processed from the terminal.

Last Message Information

The Diebold 10XX/478X and NCR 5XXX Device Handler processes store the last transaction message sent to or received from the terminal in extended memory. If the transaction is reversed and the Device Handler process loses the STM data necessary to reverse the transaction, it can obtain that information from the

ATDD1. Token data for the transaction can be retrieved from the ATDD2, the TLF, or not at all, depending on the configuration at the terminal or institution level.

Device Handler processes other than the Diebold 10XX/478X and NCR 5XXX use the LAST-MSG field in the BASE24-atm Terminal Data files to store the last transaction message sent to, or received from, the terminal. This field is used to store messages so that if a transaction times out or needs to be reversed, the transaction information is available to the Device Handler process.

CUTOVER-PEND Field

The Device Handler process checks the CUTOVER-PEND field in the BASE24-atm Terminal Data files each time it completes a transaction to determine whether the Settlement Initiator process has attempted to cut over the terminal while a transaction was in progress. If this flag is set, the Device Handler process cuts the terminal over before accepting another transaction.

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D: Pregon Utility

This appendix describes the Pregon utility and procedures used to build a new or revised SPROUTE File (SPROUTE). It describes the commands used by the utility, the procedures required to run the utility, and the possible error messages that the utility can generate. It also describes the procedures for warmbooting SPROUTE after it is built.

The Pregen Utility

The PREMAP File (PREMAP) and SPROUTE File (SPROUTE) contain the transaction routing logic used by the BASE24-atm Authorization process to route transactions. The Pregen utility builds the routing files from information contained in the Pregen Command File (PRECMD).

The Pregen utility should be run anytime after you add or delete card or interchange prefixes or you modify the routing logic. No system down-time is required to run the utility. It can be executed while the XPNET system is running.

The Pregen utility is a command-driven utility. The utility reads the commands from PRECMD. The commands provide the Pregen utility with the information it needs to create the routing files. There are six Pregen commands:

BUILDFILESIZE	Specifies the size of the primary and secondary extents for PREMAP and SPREFIXN.
BUILDSPROUTE	Specifies the location for PREMAP and SPROUTE.
PRADD	Specifies the transaction routing logic.
PREND	Indicates the end of the PRADD commands.
REMOTESPROUTE	Specifies the location for PREMAP and SPROUTE on remote HP NonStop systems.
SPROUTEND	Indicates the end of the Pregen command file.

These commands are described below in the order that they appear in PRECMD.

BUILDFILESIZE Command

The BUILDFILESIZE command alters the number of pages used for the primary and secondary extents of PREMAP or SPREFIXN. Each page can contain 2048 bytes of data. This command is optional. If this command is used, it must be the first command in PRECMD. If this command is not included in the command file, the Pregen utility uses a default of 5 pages for both the primary and secondary extents for PREMAP and 1 page for both the primary and secondary extents for SPREFIXN. You may need to change the default settings for large configurations.

Syntax

BUILDFILESIZE [PREMAP | SPREFIXN] , *primary-extents* , *secondary-extents* ;

primary-extents — The number of pages for the first extent of the file. Valid values are from 0 to 65535 pages.

secondary-extents — The number of pages for the secondary extents of the file. Valid values are from 0 to 65535 pages.

Example

BUILDFILESIZE PREMAP, 10, 10;

BUILDFILESIZE SPREFIXN, 4, 4;

BUILDSPROUTE Command

The BUILDSPROUTE command directs the Pregon utility to build SPROUTE. This command is required when the Pregon utility is run.

Syntax

BUILDSPROUTE *location*;

location — Specifies the HP NonStop system name (optional), volume, and subvolume where the Pregon utility should create SPROUTE. Typically, SPROUTE is placed on the XPNET system data subvolume. The LCONF assign SPANCTL must be set to the same SPROUTE location.

Example

BUILDSPROUTE \$DATA1.PRODDATA;

PRADD Command

The PRADD command provides the physical description of how transactions are to be routed from source processes within the named logical network to interchanges and destination processes throughout the system. It identifies a source logical network or interchange, source process, and routing group (that is, the key fields) to the system as a transaction source. For each transaction source, the command also identifies valid symbolic destinations to which the source process can route transactions, along with the method of routing to each

destination. Also, for transactions to be routed using the LGNT method, the prefix file and file type (that is, Card Prefix File or Interchange Prefix File) is indicated for each destination. There must be one PRADD command for each logical network/source process/routing group combination in the XPNET system.

For more information on PREMAP routing entries, see the topic “SPROUTE Routing” provided in section 4 of this manual.

Syntax

*PRADD source-ln , source-process , routing-group ,
method , [card-type] , dest , [file-name] , [type] , [alt-dest] ,
[method , [card-type] , dest , [file-name] , [type] , [alt-dest]] ;*

source-ln — The name of a logical network (or interchange) from which a transaction can originate. Logical network (or interchange) names can be up to 4 characters in length. Valid source logical network names include all logical networks configured in the XPNET system and interchange logical networks. Together with the source process, this element uniquely identifies the source of a transaction.

source-process — The name of a process that invokes its own routing utilities to determine the destination for a transaction, and then routes the transaction accordingly. Typically, source process names are one of the following:

- The symbolic name of a BASE24-atm Authorization process. These processes invoke their routing utilities (SKEL) when a CPF record for the card cannot be found in the local logical network.
- An asterisk (*). An asterisk indicates that routing for a specific process is not required.

Symbolic process names are up to 16 characters long.

routing-group — A routing group associated with an ATM terminal or an interchange.

An ATM is assigned to a routing group in its BASE24-atm Terminal Data files record. Each routing group defined in the BASE24-atm Terminal Data files can have a corresponding entry in the PREMAP table, where the routing methodology for all devices in the routing group is defined.

An asterisk can be entered in this field if routing based on the routing group is not required. The asterisk is used with the last two entries in the selection hierarchy when the BASE24-atm Authorization process is searching

SPROUTE for a PREMAP entry. For more information on the selection hierarchy, see the “Selection Hierarchy” topic provided in section 4 of this manual.

Valid routing groups for ATM terminals are 1 to 11 alphanumeric characters in length, while valid routing groups for interchanges are 1 to 11 numeric characters in length.

method — A routing method, or algorithm, to use in determining whether to route the transaction using the current set of action fields. Valid routing methods are as follows:

AMEX	= American Express algorithm
BNET or INAS	= MasterCard algorithm
DSCV	= Discover algorithm
LGNT	= Logical network routing method
VISA or BASE	= Visa algorithm
DFLT	= Default routing method

For more information on routing methods and algorithms, see the “Routing Methods” topic provided in section 4 of this manual.

card-type — The card type to be used for the transaction. Valid card types are 1 to 2 characters in length and always begin with a letter. For more information on reserved card types and for a description of valid values, see the **BASE24 Base Files Maintenance Manual**.

If a card type is not included in the entry, the comma is still required as a placeholder.

dest — The symbolic name of the process to which the transaction is to be forwarded by the BASE24-atm Authorization process if the routing method tests are passed. The primary destination must be a symbolic process name defined in the XPNET system. For example, when an interchange is the primary destination, the symbolic name of the Interchange Interface process is specified in this action field.

file-name — The fully-qualified name of the prefix file to be checked for the LGNT routing method. The routing utilities check this file for the card prefix associated with the transaction to determine whether to use this set of action fields.

Fully-qualified prefix file names can be from 1 to 36 characters in length. It is recommended that prefix file name entries be compatible with the Expand product to accommodate communication between HP NonStop systems.

If the prefix file name is included in the entry, the file type must also be included in the entry. If a prefix file name is not included in the entry, the comma is still required as a placeholder.

type — The type of file named in the *file-name* variable. Valid values are as follows:

CP = Card Prefix File (CPF)
IP = Interchange Prefix File (IPF)

If the prefix file name is included in the entry, a value must be included for this variable. If this variable is not included in the entry, the comma is still required as a placeholder.

alt-dest — The symbolic name of the process to which a transaction is to be forwarded by the BASE24-atm Authorization process if the primary destination is not available to authorize the transaction.

Note: While an Authorization process will place an alternate destination in the internal message if one is found in SPROUTE, it does not route to an alternate destination if a message is found to be undeliverable to the primary destination.

Valid combinations for the action variables (method, card type, destination, file name, type, and alternate destination) are shown in the table below:

<i>method</i>	<i>card-type</i>	<i>dest</i>	<i>file-name</i>	<i>type</i>	<i>alt-dest</i>
AMEX	blanks	any symbolic destination	blanks	blanks	any symbolic destination or blanks
BNET or INAS	blanks	any symbolic destination	blanks	blanks	any symbolic destination or blanks
DSCV	blanks	any symbolic destination	blanks	blanks	any symbolic destination or blanks
VISA or BASE	blanks	any symbolic destination	blanks	blanks	any symbolic destination or blanks

<i>method</i>	<i>card-type</i>	<i>dest</i>	<i>file-name</i>	<i>type</i>	<i>alt-dest</i>
LGNT	blanks	any symbolic destination	fully qualified CPF name	CP	any symbolic destination or blanks
LGNT	blanks	any symbolic destination	fully qualified IPF name	IP	any symbolic destination or blanks
DFLT	any	any symbolic destination	blanks	blanks	any symbolic destination or blanks

Examples

```
PRADD  PRO1,    *    , 000000000000,
      LGNT,    , P2A^AUTH1, \SYS.$DATA.PRO2DATA.CPF, CP, P2A^AUTH2,
      DFLT,    , P1A^CIRR ,           , ,      ;

PRADD  PRO2,    *    , 000000000000,
      LGNT,    , P1A^AUTH1, \SYS.$DATA.PRO1DATA.CPF, CP, P1A^AUTH2,
      DFLT,    , P1A^CIRR ,           , ,      ;
```

PREND Command

The PREND command marks the end of the PREMAP routing commands. This command is required every time the Pregon utility is run.

Syntax

```
PREND;
```

REMOTESPROUTE Command

The REMOTESPROUTE command specifies the locations on remote HP NonStop systems where the Pregon utility should place a copy of PREMAP and SPROUTE. There must be one REMOTESPROUTE command for each remote HP NonStop system in the XPNET system. The command can be repeated for each remote HP NonStop system. These commands are optional and must follow the PREND command.

After the Pregen utility successfully creates SPROUTE on the local system, the Pregen utility creates and obeys a FUP obey file called PREFUPIN to copy SPROUTE to each of the specified remote HP NonStop systems.

Syntax

REMOTESPROUTE *location* ;

location — Specifies the HP NonStop system name, volume, and subvolume on which PREMAP and SPROUTE should be placed. Typically, this command specifies the XPNET system data subvolume on the remote HP NonStop system.

Example

REMOTESPROUTE \SYS2.\$DATA1.PRODDATA;

SPROUTEND Command

The SPROUTEND command marks the end of SPROUTE commands. This command is required every time the Pregen utility is run and must be the last command in the Pregen Command File.

Syntax

SPROUTEND;

Pregen Command File

The Pregen Command File (PRECMD) is composed of a series of Pregen commands used to create SPROUTE. PRECMD is used as the input file when the Pregen utility is executed. It resides on the XPNET system control subvolume (e.g., \$DATA.PRODCNTL).

The following is the PRECMD command order required to create SPROUTE:

```
[BUILDFILESIZE]
BUILDSPROUTE
PRADD
PREND
[REMOTESPROUTE]
SPROUTEND
```

An example of a PRECMD is shown below. System managers are strongly urged to use a comment box like the one at the top of this example. The comment box helps you identify the files and processes impacted by the Pregon utility.

```

!*****
!*                               PREGEN COMMAND FILE (PRECMD)                               *
!*****
!*                               S O U R C E                                           *
!* LN          PROCESS          ROUTING GROUP                                           *
!*                               *                                                       *
!* ORDERED    CARD    PRIMARY          ALT                                           *
!* METHOD      TYPE    DEST              DEST              PREFIX FILE NAME           *
!*=====
!* PRO1      P1A^AUTH1  00000000001                                           *
!*                               *                                                       *
!* LGNT      (none)  P1A^CIRR  (none)  \SYS1.$DATA.PRO1DATA.IPFCIRR                *
!* LGNT      (none)  P1A^PLUS  (none)  \SYS1.$DATA.PRO1DATA.IPFPLUS                *
!* LGNT      (none)  P1A^BICI  (none)  \SYS1.$DATA.PRO1DATA.IPFBICI                *
!*                               *                                                       *
!* PRO1      *          00000000002                                           *
!*                               *                                                       *
!* LGNT      (none)  P1A^VNET  (none)  \SYS1.$DATA.PRO1DATA.IPFVISA                *
!* DFLT      (none)  P1A^PLUS  (none)  \SYS1.$DATA.PRO1DATA.IPFPLUS                *
!*****
BUILDSPROUTE $DATA.PRODDATA;
PRADD        PRO1,  P1A^AUTH1  , 00000000001,
  LGNT, ,  P1A^CIRR  , \SYS1.$DATA.PRO1DATA.IPFCIRR,IP, ,
  LGNT, ,  P1A^PLUS  , \SYS1.$DATA.PRO1DATA.IPFPLUS,IP, ,
  LGNT, ,  P1A^BICI  , \SYS1.$DATA.PRO1DATA.IPFBICI,IP, ,;
PRADD        PRO1,      *      , 00000000002,
  LGNT, ,  P1A^VNET  , \SYS1.$DATA.PRO1DATA.IPFVISA,IP, ,
  DFLT, ,  P1A^PLUS  , \SYS1.$DATA.PRO1DATA.IPFPLUS,IP, ,;
PREND;
REMOTESPROUTE \SYS2.$DATA.PRODDATA;
REMOTESPROUTE \SYS3.$DATA.PRODDATA;
SPROUTEND;

```

Pregon FUP Obey File

Each time you run the Pregon utility, it creates and executes a FUP obey file to perform the following:

- Rename the old SPROUTE to SPROUTEO
- Rename the new SPROUTE created during the run from SPROUTEN to SPROUTE

- Purge the temporary SPREFIX File called SPREFIXN
- Copy the new SPROUTE to any remote HP NonStop system specified in a REMOTESPROUTE command

An example of the PREFUPIN FUP obey file created and executed by the Pregen utility is shown below. This example assumes that one REMOTESPROUTE command was found in PRECMD. The VOLUME, PURGE, RENAME, and DUP commands are repeated for each remote HP NonStop system to distribute the new SPROUTE. The number of errors and warnings allowed is increased by four for each remote HP NonStop system. The Pregen utility creates and executes this file after it successfully creates a new SPROUTE. The file is placed on the XPNET system control subvolume.

```
ALLOW 012 ERRORS
ALLOW 012 WARNINGS
CONFIGURE ECHO OBEY
VOLUME \SYS1.$DATA.PRODDATA
PURGE SPROUTEO!
RENAME SPROUTE, SPROUTEO
RENAME SPROUTEN, SPROUTE
PURGE SPREFIXN!
VOLUME \SYS2.$DATA.PRODDATA
PURGE SPROUTEO!
RENAME SPROUTE, SPROUTEO
DUP \SYS1.$DATA.PRODDATA.SPROUTE, *, SOURCEDATE
```

Running the Pregen Utility

You can execute the Pregen utility directly from a TACL prompt or by obeying the Pregen Go File (GOPRE) located on the XPNET system control subvolume. GOPRE identifies PRECMD as the input file.

GOPRE should contain the following TACL commands:

```
RUN $SYSTEM.XPNET.PREGEN/OUT $S.#PREG, IN PRECMD,NAME/
PERUSE
```

When execution of GOPRE completes, an HP NonStop PERUSE utility prompt is displayed. You can view or print the Pregen report from the PERUSE prompt. An example of a Pregen report is shown below.

Pregon Report

Each time the Pregon utility is run, a report is generated. The report basically reflects what the Pregon utility did during its run. It does this by producing a copy of PRECMD and then embedding messages after appropriate commands. These messages indicate how files or processes were affected and where the files are located. Each message generated by the Pregon utility on the report is preceded by two asterisks (**).

The Pregon report also indicates any errors that the Pregon utility encountered. The Pregon error messages are listed and described later in this appendix.

The structure of the report reflects the steps executed by the Pregon utility during its run. The steps are as follows:

1. Open PRECMD.
2. If an error occurs, print the error to the home terminal and abend.
3. Open the report location.
4. If an error occurs, print the error on the home terminal and abend.
5. Print the VERSION line on the report.
6. Process the Pregon build commands. Echo the comments and commands on the report.
7. If an error occurs, print the error on the report and stop.
8. Process the remainder of PRECMD, echoing commands on the report. Print the action taken where relevant.
9. Create the PREFUPIN command file and start a FUP process to obey the PREFUPIN commands.
10. Print an informational message on the report for each FUP command executed and operation performed, as well as any messages for warnings or errors encountered by FUP.
11. Print the END line on the report.
12. Stop.

An example of the report produced by the Pregon utility is shown on the following pages.

```

** PREGEN VERSION 3.00   June 24, 1997

!*****
!*          PREGEN COMMAND FILE (PRECMD)
!*****
!*          S O U R C E
!* LN  PROCESS  ROUTING GROUP
!*
!* ORDERED CARD PRIMARY  ALT
!* METHOD TYPE DEST      PREFIX FILE NAME
!*-----
!* PRO1  P1A^AUTH1  000000000001
!*
!* LGNT  (none)  P1A^CIRR  (none)  \SYS1.$DATA.PRO1DATA.IPFCIRR
!* LGNT  (none)  P1A^PLUS  (none)  \SYS1.$DATA.PRO1DATA.IPFFPLUS
!* LGNT  (none)  P1A^BICI  (none)  \SYS1.$DATA.PRO1DATA.IPFBICI
!*
!* PRO1  *  000000000002
!*
!* LGNT  (none)  P1A^VNET  (none)  \SYS1.$DATA.PRO1DATA.IPFVISA
!* DFLT  (none)  P1A^PLUS  (none)  \SYS1.$DATA.PRO1DATA.IPFFPLUS
!*
!*****
!***** BUILDSPROUTE \SYS1.$DATA.PRODATA;
!*****

** PREGEN NEXT VERSION REPORT:  Tuesday, October 3, 2000 - 10:16:05 AM USED COMMAND FILE:  $DATA.PRODCNTL.PRECMD

** PREMAP FILE:  \SYS1.$DATA.PRODDATA.PREMAP CREATED
PRADD  LGNT, , P1A^CIRR , \SYS1.$DATA.PRO1DATA.IPFCIRR,IP, ,
        LGNT, , P1A^PLUS , \SYS1.$DATA.PRO1DATA.IPFFPLUS,IP, ,
        LGNT, , P1A^BICI , \SYS1.$DATA.PRO1DATA.IPFBICI,IP, ,
PRADD  PRO1, * , 000000000002,
        LGNT, , P1A^VNET , \SYS1.$DATA.PRO1DATA.IPFVISA,IP, ,
        DFLT, , P1A^PLUS , \SYS1.$DATA.PRO1DATA.IPFFPLUS,IP, ,
PREND;

** PREMAP CONTENTS:

PRO1  P1A^AUTH1  000000000001
        0 LGNT  P1A^CIRR
        1 LGNT  P1A^PLUS
        2 LGNT  P1A^BICI
PRO1  *  000000000002
        0 LGNT  P1A^VNET
        1 DFLT  P1A^PLUS
        IP IP000000 \SYS1.$DATA.PRO1DATA.IPFCIRR
        IP IP000001 \SYS1.$DATA.PRO1DATA.IPFFPLUS
        IP IP000002 \SYS1.$DATA.PRO1DATA.IPFBICI
        IP IP000000 \SYS1.$DATA.PRO1DATA.IPFVISA
        IP IP000001 \SYS1.$DATA.PRO1DATA.IPFFPLUS

```

```

** BUILDING SPREFIX USING PREFIX FILE(S) :

\sys1.$DATA.PRO1DATA.IPFCLRR
\sys1.$DATA.PRO1DATA.IPFPLUS
\sys1.$DATA.PRO1DATA.IPFBI
\sys1.$DATA.PRO1DATA.IPFVISA
\sys1.$DATA.PRO1DATA.IPFPLUS

** SPROUTE FILE: \sys1.PRODDATA.SPROUTEN CREATED
SPROUTEND;
09:30:33 (I) - 3801 FUP command/response : OBEY \sys1.$PRODCNTL.PREFUPIN
09:30:33 (I) - 3801 FUP command/response : VOLUME \sys1.$DATA.PRODDATA
09:30:33 (I) - 3801 FUP command/response : PURGE SPROUTEO!
09:30:33 (E) - 3680 Fup warning message is : \sys1.$PRODDATA.SPROUTEO: ERR 11
09:30:33 (E) - 3690 PREGEN will continue to run
09:30:33 (E) - 3700 Please evaluate fup message
09:30:33 (E) - 3710 PREGEN may have to be run again
09:30:33 (I) - 3801 FUP command/response : 0 FILES PURGED
09:30:33 (I) - 3801 FUP command/response : RENAME SPROUTE, SPROUTEO
09:30:33 (I) - 3801 FUP command/response : RENAME SPROUTEN, SPROUTE
09:30:33 (I) - 3801 FUP command/response : PURGE SPREFIXN!
09:30:33 (I) - 3801 FUP command/response : 1 FILE PURGED

** END OF PREGEN BUILD NEXT VERSION RUN Tuesday, October 3, 2000 - 10:16:18 AM

```

Using a New SPROUTE File

Whenever you add or delete card or interchange prefixes, or change routing logic, a new SPROUTE must be created and warmbooted as follows:

1. Add or delete prefixes from the CPF or IPF as required.
2. Modify the routing logic in PRECMD as required using the commands explained earlier in this appendix.
3. Execute the GOPRE obey file to run the Pregen utility. Once run, verify the output.
4. Issue the following command from a network control facility to all XPNET nodes to ensure that the new SPROUTE file is being used:

CONTROL NODE *xpobj*, WARMBOOTSPROUTE

xpobj — Identifies the specific XPNET node or nodes affected by the command.

For more information on the CONTROL NODE and DELIVER PROCESS commands, see the ***NET24-XPNET Network Control Command Reference Manual***.

Error and Status Messages

All error and status conditions that affect normal program processing result in messages being interspersed in the report file by the Pregon utility. The system manager can then take corrective action as necessary.

The following pages provide a description of each error or status message that can be generated by the Pregon utility. The operator action required in response to each message is also provided.

Message Identification

Each error and status message from the Pregon utility is grouped into one of four categories:

Critical (C)	Indicates that a processing or hardware error has occurred. The error must be corrected and the utility rerun.
Error (E)	Indicates some type of processing or hardware error has occurred that is noncritical to normal processing. The error must be corrected and the utility rerun.
Warning (W)	Indicates a potential problem that may or may not require intervention.
Information (I)	Conveys general information and usually does not require any action.

All messages are identified by a unique numeric code.

The following error messages can be generated by the Pregen utility.

(C) - 1360 Error occurred during obey of PREFUPIN - error *number*

Where: *number* = The operating system error number

Cause: An operating system error occurred while executing the PREFUPIN obey file.

Remedy: See the appropriate HP NonStop documentation for information about the operating system error. Correct the error condition and manually rerun the utility.

(C) - 1370 Correct error and manually re-run PREFUPIN

Cause: An operating system error occurred while executing the PREFUPIN obey file.

Remedy: See the appropriate HP NonStop documentation for information about the operating system error. Correct the error condition and manually rerun the utility.

(C) - 1540 ** Write of FUP OBEY FILE (SPTFUPIN) failed - error *number*

Where: *number* = The operating system error number

Cause: An operating system error occurred while attempting to create the SPTFUPIN obey file.

Remedy: See the appropriate HP NonStop documentation for information about the operating system error. Correct the error condition and rerun the utility.

Note: This message only applies when running a previous release of BASE24-atm under NET24-XPNET release 3.0. This message does not apply for BASE24-atm release 5.3 or later.

(C) - 1550 ** Close of FUP OBEY FILE (SPTFUPIN) failed - error
number

Where: *number* = The operating system error number

Cause: An operating system error occurred while attempting to close the SPTFUPIN obey file.

Remedy: See the appropriate HP NonStop documentation for information about the operating system error. Correct the error condition and rerun the utility.

Note: This message only applies when running a previous release of BASE24-atm under NET24-XPNET release 3.0. This message does not apply for BASE24-atm release 5.3 or later.

(C) - 3010 BUILD command - no Buildsproute found *string*

Where: *string* = Data from the command file

Cause: An unexpected command was found in the command file where the required BUILDSPROUTE command was expected.

Remedy: Correct the command file and rerun the Pregen utility.

(E) - 3175 An error occurred while updating *filename*'s filecode

Where: *filename* = The name of a file

Cause: The Pregen utility was unable to mark a file as valid.

Remedy: Correct the error condition and rerun the Pregen utility.

(C) - 3220 PREMAPCREATE - Premap *filename* create error *number*

Where: *filename* = The name of a file
number = The operating system error number

Cause: An error occurred while attempting to create a file due to an operating system error.

Remedy: See the appropriate HP NonStop documentation for information about the operating system error. Correct the error condition and rerun the Pregen utility.

(C) - 3370 BUILDSPREFIX - Sprefix *filename* create error *number*

Where: *filename* = The name of the new SPREFIX File (SPREFIXN)

number = The operating system error number

Cause: The Pregen utility was unable to create SPREFIXN due to an operating system error.

Remedy: See the appropriate HP NonStop documentation for information about the operating system error. Correct the error condition and rerun the Pregen utility.

(C) - 3390 BUILDSPROUTE - no End, found *string*

Where: *string* = Data from the command file

Cause: A required semicolon (;) was missing from the end of the BUILDSPROUTE command or a REMOTESPROUTE command.

Remedy: Correct the command file and rerun the Pregen utility.

(C) - 3400 PREMAP routing - invalid command, found *string*

Where: *string* = Data from the command file

Cause: An unexpected command was found in the command file.

Remedy: Correct the command file and rerun the Pregen utility.

(C) - 3410 PREMAP routing - no empty records in PREMAPTABLE

Cause: The number of routing methods specified in the command file has exceeded the maximum (8800) allowed by the Pregen utility.

Remedy: Correct the command file and rerun the Pregen utility.

(C) - 3421 PREMAP routing - invalid METHOD *method*

Where: *method* = The routing method specified in the PRADD command

Cause: An invalid routing method was specified in the PRADD statement. Valid routing methods are LGNT, DFLT, AMEX, BASE, BNET, VISA, DSCV, or INAS.

Remedy: Correct the PRADD command and rerun the Pegen utility.

(C) - 3422 PREMAP routing - required FILENAME missing

Cause: The LGNT routing method requires an associated prefix file name.

Remedy: Correct the PRADD command and rerun the Pegen utility.

(C) - 3423 PREMAP routing - FILENAME not allowed

Cause: A file name is only allowed for the LGNT routing method.

Remedy: Correct the PRADD command and rerun the Pegen utility.

(C) - 3424 PREMAP routing - required FILE TYPE missing

Cause: The LGNT routing method requires a prefix file type of CP or IP.

Remedy: Correct the PRADD command and rerun the Pegen utility.

(C) - 3425 PREMAP routing - invalid FILE TYPE *type*

Where: *type* = The invalid file type

Cause: An invalid prefix file type was specified in a PRADD command. Valid prefix file types are CP or IP.

Remedy: Correct the PRADD command and rerun the Pegen utility.

(C) - 3426 PREMAP routing - FILE TYPE not allowed

Cause: A prefix file type was specified with a routing method other than LGNT. Prefix file types can only be specified with the LGNT routing method.

Remedy: Correct the PRADD command and rerun the Pregen utility.

(C) - 3427 PREMAP routing - logical network name too long *name*

Where: *name* = An invalid logical network name

Cause: The source logical network name is longer than four characters.

Remedy: Correct the PRADD command and rerun the Pregen utility.

(C) - 3428 PREMAP routing - symbolic source too long *name*

Where: *name* = An invalid symbolic name

Cause: The symbolic source process name is longer than 16 characters.

Remedy: Correct the command and rerun the Pregen utility.

(C) - 3429 PREMAP routing - routing group too long *group*

Where: *group* = A routing group found in a PRADD command

Cause: The routing group entered in the PRADD command is invalid because it is too long.

Remedy: Correct the command file and rerun the Pregen utility.

(C) - 3431 PREMAP routing - alternate destination too long
destination

Where: *destination* = An alternate destination found in a PRADD command

Cause: An alternate destination specified in a PRADD command is invalid because it is too long.

Remedy: Correct the command file and rerun the Pregen utility.

(C) - 3432 PREMAP routing - card type too long *card-type*

Where: *card-type* = A card type found in a PRADD command

Cause: An alternate destination specified in a PRADD command is invalid because it is too long.

Remedy: Correct the command file and rerun the Pregen utility.

(C) - 3480 GENERATE keys - invalid file format, found *type*

Where: *type* = The file type found in the command file

Cause: An invalid file format specification was found in the command file.

Remedy: Correct the command file and rerun the Pregen utility.

(C) - 3490 OPEN and BUILD - file open error, using *filename*

Where: *filename* = The name of a file

Cause: An error occurred while opening the file specified.

Remedy: Correct the error condition and rerun the Pregen utility.

(C) - 3500 OPEN and BUILD - fname expand error, using *filename*

Where: *filename* = The name of a file

Cause: The Pregen utility was unable to expand a file name in the command file.

Remedy: If the file name used is not the one desired, correct the command file and rerun the Pregen utility.

(C) - 3540 OPEN and BUILD - invalid format, found *format*

Where: *format* = The file format

Cause: An internal program error has occurred.

Remedy: Contact HELP24.

(C) - 3560 FILESIZE - primary extents convert error, used *number*

Where: *number* = The number used in place of the number in the command file

Cause: The primary extent size is specified incorrectly in the command file.

Remedy: Correct the command file and rerun the Pregen utility.

(C) - 3570 FILESIZE - secondary extents convert error, used *number*

Where: *number* = The number used in place of the number in the command file

Cause: The secondary extent size is specified incorrectly in the BUILDFILESIZE command.

Remedy: Correct the command file and rerun the Pregen utility.

(C) - 3580 FILESIZE - invalid file name, used *filename*

Where: *filename* = The name of a file

Cause: An invalid file name was specified in the BUILDFILESIZE command.

Remedy: Correct the command and rerun the Pregen utility.

(C) - 3590 FILE CODE FILLER - invalid filetype filecode

Cause: An internal program error has occurred.

Remedy: Contact HELP24.

(C) - 3600 FILE CODE FILLER - filetype filecode conversion error

Cause: An internal program error has occurred.

Remedy: Contact HELP24.

(E) - 3610 Error occurred during creation of FUP, error was *number*

Where: *number* = The error returned by the call to the NEWPROCESS procedure

Cause: The Pregon utility was unable to start FUP due to an error.

Remedy: Rerun the Pregon utility. If error persists, contact HELP24.

(E) - 3620 Error occurred during open of FUP

Cause: The Pregon utility was unable to open FUP.

Remedy: Contact HELP24.

(E) - 3630 Error occurred during write of startup message to FUP

Cause: The Pregon utility was unable to send a startup message to FUP.

Remedy: Contact HELP24.

(E) - 3640 Error *number* occurred during open of \$RECEIVE

Where: *number* = The operating system error number

Cause: The Pregon utility was unable to open \$RECEIVE due to an operating system error.

Remedy: See the appropriate HP NonStop documentation for information about the operating system error. Contact HELP24.

(E) - 3650 This is not a named process. Be sure to use the NAME field in the run command. Example: RUN PREGON /IN <command file>, OUT <out file>, NAME/

Cause: The Pregon utility was not run as a named process.

Remedy: Rerun the Pregon utility as a named process.

(E) - 3660 Error *number* occurred during readupdate of \$RECEIVE

Where: *number* = The operating system error number

Cause: A read on \$RECEIVE failed due to an operating system error.

Remedy: See the appropriate HP NonStop documentation for information about the operating system error. Contact HELP24.

(E) - 3670 Fup error message is : *text*

Where: *text* = The error message from FUP

Cause: An error message was received from the FUP process started by the Pregen utility.

Remedy: Evaluate the error message, take appropriate action, and rerun the Pregen utility.

(E) - 3680 Fup warning message is : *text*

(E) - 3690 PREGEN will continue to run

(E) - 3700 Please evaluate fup message

(E) - 3710 PREGEN may have to be run again

Where: *text* = The error message from FUP

Cause: A warning message was received from the FUP process started by the Pregen utility.

Remedy: Evaluate the warning message. If necessary, take appropriate action, and rerun the Pregen utility.

(E) - 3720 Unknown FUP message : *text*

Where: *text* = The message from FUP

Cause: A message was received from the FUP process started by the Pregen utility.

Remedy: Evaluate the message. If necessary, take appropriate action, and rerun the Pregen utility.

(E) - 3800 Fup response with no text

Cause: An internal program error has occurred.

Remedy: Contact HELP24.

(I) - 3801 FUP command/response : *text*

Where: *text* = The command or response message from FUP

Cause: An informational message indicating the current command FUP is executing or an operation FUP has performed.

Remedy: No action is required.

(E) - 3803 System STOP message received

Cause: The FUP process being used by the Pregen utility has stopped.

Remedy: Rerun the Pregen utility.

(E) - 3806 System ABEND message received

Cause: The FUP process being used by the Pregen utility has stopped.

Remedy: Rerun the Pregen utility.

(E) - 3809 Unknown system message received

Cause: An unknown system message was received. It is ignored by the Pregen utility.

Remedy: No action is required.

(E) - 3810 Problem has occurred while initializing FUP

Cause: The Pregen utility encountered a problem while initializing FUP.

Remedy: Rerun the Pregen utility. If the error persists, contact HELP24.

(E) - 3820 An error occurred during EXIT from FUP

Cause: The Pregen utility encountered an error while exiting FUP.

Remedy: Rerun the Pregen utility. If the error persists, contact HELP24.

(C) - 3909 Looking for SPROUTEND - found *string*, ignoring

Where: *string* = Data from the command file

Cause: An unexpected command was found in the command file where the required SPROUTEND command was expected.

Remedy: Correct the command file and rerun the Pregen utility.

(C) - 3920 ALLOCATESEGMENT error *number*, (size *size*, file *filename*)

Where: *number* = The error returned from ALLOCATESEGMENT
size = The size (in bytes) of the extended memory segment
filename = The name of a file

Cause: The Pregen utility was unable to allocate an extended memory segment.

Remedy: Analyze the ALLOCATESEGMENT error and take appropriate action.

(C) - 3921 Open failed for *type* file *filename*, error *number*

Where: *type* = Specifies the type of file, and indicates PREMAP or SPREFIX

filename = The name of a file

number = The OPEN error number

Cause: The Pregen utility was unable to open the specified file.

Remedy: Analyze the OPEN error and take appropriate action.

(C) - 3960 Fatal error *number* while reading PREMAP file *filename*

Where: *number* = The READ error number
 filename = The name of a file

Cause: The Pregon utility was unable to read PREMAP.

Remedy: Analyze the error and take appropriate action.

(C) - 3980 Fatal error *number* while reading SPREFIX file *filename*

Where: *number* = The READ error number
 filename = The name of a file

Cause: The Pregon utility was unable to read the temporary SPREFIX.

Remedy: Analyze the error and take appropriate action.

(C) - 3981 Fatal error, Sprefix record does not match a premap
sprefix key

Cause: The Pregon utility encountered a record in the temporary SPREFIX
 with an invalid key.

Remedy: Verify that the information in the Pregon Command File
 (PRECMD), Card Prefix File (CPF), and Interchange Prefix File
 (IPF) is correct and rerun the Pregon utility.

(C) - 3982 Fatal error, more sprefix keys found than table space
allocated

Cause: The Pregon utility read more records from the temporary SPREFIX
 than it allocated space for in memory.

Remedy: Contact HELP24.

(C) - 3983 Invalid PREFIX length *length* for SPREFIX key *key*

Where: *length* = The length of a record in SPREFIX
 key = The key to the SPREFIX record

Cause: The Pregen utility encountered a SPREFIX record with an invalid prefix length.

Remedy: Correct the prefix length and rerun the Pregen utility.

(C) - 3984 Invalid PAN length *length* for SPREFIX key *key*

Where: *length* = The length of the primary account number (PAN)
 key = The key to the record in SPREFIX

Cause: The Pregen utility encountered a SPREFIX record with an invalid PAN length.

Remedy: Correct the PAN length and rerun the Pregen utility.

(W) - 3990 NEXT^PARAM - end of statement, possible missing field

Cause: The Pregen utility expected to find the next parameter for a command in the command file, but instead found the end of the command.

Remedy: Correct the Pregen Command File (PRECMD) and rerun the Pregen utility.

(C) - 9999 BUILDSPROUTE - fname expand error, using *filename*

Where: *filename* = The name of a file

Cause: The Pregen utility was unable to expand the file name specified for the BUILDSPROUTE command or a REMOTESPROUTE command.

Remedy: If the file name used is not the one desired, correct the command file and rerun the Pregen utility.

E: Base Extended Memory Table Build Utility

This section provides detailed instructions for using the Base Extended Memory Table Build utility. This utility is a stand-alone program used to build or rebuild shared extended memory tables from disk files. The Base Extended Memory Table Build utility is executed from a TACL prompt. This section discusses the following topics:

- An overview of Base Extended Memory Table Build utility processing, including procedures to make processes reallocate (warmboot) a newly built or rebuilt extended memory table
- A description of the source files from which extended memory tables are created for the BASE24-atm product
- The configuration requirements for the Base Extended Memory Table Build utility, including a list of all Logical Network Configuration File (LCONF) assigns and params
- Procedures for starting the Base Extended Memory Table Build utility from a TACL prompt
- A description of the Extended Memory Table Build Report produced when the Base Extended Memory Table Build utility is run

Note: This appendix addresses extended memory tables used by procedural BASE24-atm processes. For information on extended memory tables used by object-oriented applications, such as BASE24 Customer Service, refer to the *BASE24 ITS Kernel Configuration Manual*.

Overview

The Base Extended Memory Table Build utility builds shared extended memory tables for use by BASE24 processes. The program can build hash tables, radix tables, or custom-designed tables.

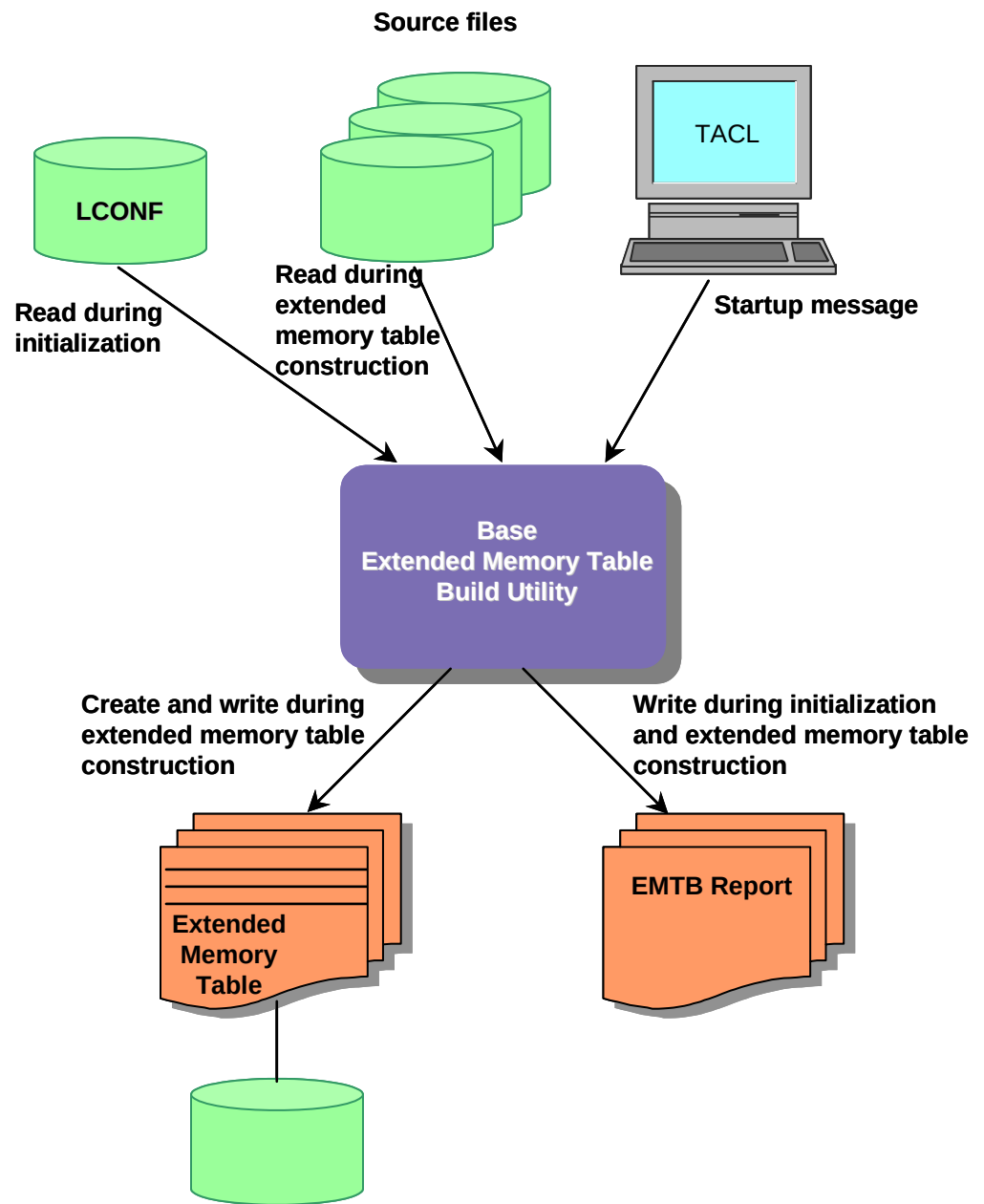
Shared extended memory tables are built in the background while the BASE24 system is processing.

The Base Extended Memory Table Build utility currently supports the following shared extended memory tables used by BASE24-atm processes.

- Acquirer Processing Code File (APCF) extended memory table
- Dynamic Currency Conversion Data (DCCD) extended memory table
- Exchange Rate File (ERF) extended memory table
- Institution Definition File (IDF) extended memory table
- Issuer Processing Code File (IPCF) extended memory table
- Note ID File (NIDF) extended memory table
- Receipt File (RCPT) extended memory table
- Split Transaction Routing File (STRF) extended memory table (used by the Transaction Context Manager process)
- Terminal Receipt File (TRF) extended memory table

The Base Extended Memory Table Build utility creates an Extended Memory Table Build report summarizing the processing performed each time it is executed. The Extended Memory Table Build report is described later in this section.

The processing performed by the Base Extended Memory Table Build utility is depicted in the following illustration. A key to the abbreviations follows the illustration.



Key

EMTB = Extended Memory Table Build utility
LCONF = Logical Network Configuration File

Allocating a Newly Built or Rebuilt Extended Memory Table

Once an extended memory table is built or rebuilt, it is not used until the application that uses the extended memory table has been informed of the change and reallocates (warmboots) the table. You can warmboot extended memory tables using the EMT Warmboot Control screen available from the Device Control Terminal (DCT) or by issuing ALTER ATTRIBUTE text commands from a network control facility. Both methods of warmbooting extended memory tables are described below.

Note: You must have network control security access to enter ALTER ATTRIBUTE commands before you can use either of these methods.

EMT Warmboot Control Screen

Extended memory tables are reread by BASE24 processes when you issue an ALTER ATTRIBUTE text command on an extended memory table from the EMT Warmboot Control screen. For example, when you enter the APCF acronym in the ALTER ATTRIBUTE EMT field on this screen, all processes configured to use the APCF extended memory table in an APCFEMT-WARMIdentifier LCONF param reallocate the APCF extended memory table. An EMT-WARMIdentifier LCONF param exists for each extended memory table that can be warmbooted from the EMT Warmboot Control screen as shown in the following table.

Extended Memory Table	LCONF Warmboot Param
APCFEMT	APCFEMT-WARMIdentifier
DCCDEMT	DCCDEMT-WARMIdentifier
ERFEMT	ERFEMT-WARMIdentifier
IDFEMT	IDFEMT-WARMIdentifier
IPCFEMT	IPCFEMT-WARMIdentifier
NIDFEMT	NIDFEMT-WARMIdentifier
RCPTENT	RCPTENT-WARMIdentifier
STRFEMT	STRFEMT-WARMIdentifier
TRFEMT	ATM-TRFEMT-WARMIdentifier

After each process successfully reallocates the specified extended memory table, a *warmboot completed* message is displayed on the network log for the process. For detailed information on the EMT Warmboot Control screen, refer to the ***BASE24 Device Control Manual***.

Network Control Facility

You can also allocate extended memory tables by entering DELIVER PROCESS commands directly from a network control facility. The text sent in these commands is “ALTER ATTRIBUTE *emt-name*,” where *emt-name* is APCFEMT, IPCFEMT, or TRFEMT. The command text is sent only to those processes identified as the object of the DELIVER PROCESS command. After each process to which the command is sent has successfully reallocated the specified extended memory table, a warmboot completed message is displayed on the network log for the process. For detailed information on the ALTER ATTRIBUTE text command, refer to the ***BASE24 Text Command Reference Manual***.

Source Files

Extended memory tables must be rebuilt anytime a change is made to the source files for the extended memory table. The source files for extended memory tables used by BASE24-atm are identified below.

Acquirer Processing Code File Extended Memory Table

The Acquirer Processing Code File (APCF) extended memory table must be rebuilt anytime a change is made to the APCF. The APCF contains one record for each combination of acquirer transaction profile, message type, and transaction code.

The following BASE24-atm processes use the APCF extended memory table:

BASE24-atm Diebold 10XX/478X Device Handler process
BASE24-atm NCR 5XXX Device Handler process

Banknet Interchange Interface process
BIC ISO Interchange Interface process
MDS/MDSM Interchange Interface process
PLUS ISO Interchange Interface process
VisaNet Interchange Interface process

Dynamic Currency Conversion Data Extended Memory Table

The Dynamic Currency Conversion Data (DCCD) extended memory table must be rebuilt anytime a change is made to the DCCD. The DCCD contains the following two records:

1. BIN currency record – maps each BIN to a currency code and issuer designator.
2. Currency data record – maps terminal DCC profile, currency code, and issuer designator to other data used in DCC processing.

Exchange Rate File Extended Memory Table

The Exchange Rate File (ERF) extended memory table must be rebuilt anytime a change is made to the ERF. The ERF contains one record for each currency, except the Base currency, that can be used by BASE24.

The BASE24-atm Authorization process uses the ERF extended memory table.

In addition, this file is also used by both the BASE24-atm Diebold 10XX/478X and NCR 5XXX Device Handler processes, if multi-currency cash deposits are allowed.

Institution Definition File Extended Memory Table

The Institution Definition File (IDF) extended memory table must be rebuilt anytime a change is made to the IDF. The IDF contains one record for each institution participating in the logical network. The following BASE24-atm processes use the IDF extended memory table:

BASE24-atm Diebold 10XX/478X Device Handler process
BASE24-atm NCR 5XXX Device Handler process

Issuer Processing Code File Extended Memory Table

The Issuer Processing Code File (IPCF) extended memory table must be rebuilt anytime a change is made to the IPCF. The IPCF contains one record for each combination of issuer transaction profile, message type, and transaction code.

The following BASE24-atm processes use the IPCF extended memory table:

BASE24-atm Authorization process

Banknet Interchange Interface process

BIC ISO Interchange Interface process

MDS/MDSM Interchange Interface process

PLUS ISO Interchange Interface process

VisaNet Interchange Interface process

Note ID File Extended Memory Table

The Note ID File (NIDF) extended memory table must be rebuilt anytime a change is made to the NIDF. The NIDF contains one record for each set of Note ID's that can be supported at a terminal.

The BASE24-atm NCR 5XXX Device Handler process uses the NIDF extended memory table.

Receipt File Extended Memory Table

The Receipt File (RCPT) extended memory table must be rebuilt anytime a change is made to the RCPT. The RCPT contains one record for each receipt template.

The following BASE24-atm processes use the RCPT extended memory table:

BASE24-atm Diebold 10XX/478X Device Handler process

BASE24-atm NCR 5XXX Device Handler process

Split Transaction Routing File Extended Memory Table

The Split Transaction Routing File (STRF) extended memory table must be rebuilt anytime a change is made to the STRF. The STRF contains one record for each transaction subtype supported in the network that requires unique routing.

The BASE24-atm Transaction Context Manager process uses the STRF extended memory table.

Terminal Receipt File Extended Memory Table

The Terminal Receipt File (TRF) extended memory table must be rebuilt anytime a change is made to the TRF. The TRF contains one record for each approved or denied transaction receipt option.

The following BASE24-atm processes use the TRF extended memory table:

BASE24-atm Diebold 10XX/478X Device Handler process

BASE24-atm NCR 5XXX Device Handler process

LCONF Assigns and Params

The following LCONF assigns and params are used by the Base Extended Memory Table Build utility on behalf of BASE24 modules or processes.

Each of these assigns and params is described in detail in the ***BASE24 Logical Network Configuration File Manual***.

Assigns

ALT-COLL
APCF
APCFEMT
ATM-NIDF
ATM-NIDFEMT
ATM-TRF
ATM-TRFEMT
DCCD
DCCDEMT
EMT-SWAPVOL
ERF
ERFEMT
IDF
IDFEMT
IPCF
IPCFEMT
PRI-COLL
RCPT
RCPTEMT
STRF
STRFEMT

Params

ALT-TIM-LMT
APCFEMT-MAX-TBL-SIZE
ATM-NIDFEMT-MAX-TBL-SIZE
ATM-TRFEMT-MAX-TBL-SIZE
CONSOLE-PRT
DATE-DISPLAY-FORMAT
DCCDEMT-MAX-TBL-SIZE
ERFEMT-MAX-TBL-SIZE
IDFEMT-MAX-TBL-SIZE
IPCFEMT-MAX-TBL-SIZE
PRI-TIM-LMT
RCPTEMT-MAX-TBL-SIZE
STRFEMT-MAX-TBL-SIZE
STRFEMT-WARM*identifier*

Startup

This subsection describes the procedures for starting the Extended Memory Table Build utility. The utility is started from a TACL prompt and can be run using a high or low process identification number (PIN). For convenience, ACI provides an interactive TACL macro named EMTBRUN for starting the utility.

Note: If NET24-XPNET security is bound into the Extended Memory Table Build utility executable, you must have security access to the STATUS NODE network control command before using the Extended Memory Table Build utility.

EMTBRUN TACL Macro

ACI provides a TACL macro called EMTBRUN to operate the Extended Memory Table Build utility. This macro is installed on the BAxxEMTB subvolume, where xx is the number of the current release. If desired, you can add the EMTBRUN macro to the PMSEARCHLIST for the system or your TACL logon. When started from a TACL prompt, the macro prompts you for the individual file acronym of the extended memory table to be built (e.g., APCF, IPCF, TRF, etc.) or ALL (if you want to build all available extended memory tables). The macro also prompts you to enter the fully qualified name of the Logical Network Configuration File (LCONF) containing the Extended Memory Table Build utility assigns and params.

The Extended Memory Table Build Reports generated by the utility are sent to a spooler location of \$\$S.#emt.emt-name, where *emt-name* is the acronym of the file from which the extended memory table is built.

Preparing to Use the Macro

Before using the EMTBRUN macro, you may need to add a command line specifying an alternate subvolume if the macro is not located on the same subvolume as the object files it executes (i.e., BEMTBLO and TRFEMTTO). The BEMTBLO file contains the object code for building the base extended memory tables. The TRFEMTTO file contains the object code for building the TRF extended memory table. The following lines from the macro explain the command line text that may need to be modified.

```
=====
== The BA60OBJ.BEMTBLO object and the AT60EMTB.TRFEMTTO for the      =
== BASE24-atm are assumed to be on the same volume where this macro =
== resides. If the macro is on a different volume, specify the       =
```



```

== alternate volume by uncommenting the following line, and      =
== replacing '$volume' with the different volume name.           =
==                                                                =
== #set :_emtvol $volume                                         =
==                                                                =
=====

```

Executing the Macro

When you run the Extended Memory Table Build utility using the EMTBRUN macro, the utility prompts you for all needed information. You provide the information required at each prompt by typing it and pressing the **Enter** key. You can exit the utility at any time by typing EXIT and pressing the **Enter** key or by pressing the **Break** key.

1. At the TACL prompt, enter the following command to start the Extended Memory Table Build utility, where [*\$vol.subvol*] identifies the volume and subvolume location of the TACL macro if it is not on the current volume and subvolume or defined in the PMSEARCHLIST.

```
[$vol.subvol.] EMTBRUN
```

After entering the command, the following text is displayed on the screen:

```
$vol.subvol.EMTBRUN
```

```

-----
=                                                                =
= Welcome to the EMT Build Run TACL macro.                      =
=                                                                =
= This TACL macro is used to build Extended Memory Table       =
= files (EMTs).                                                 =
=                                                                =
= An EMT will be rebuilt using the EMT Build utility.  It      =
= reads data from the appropriate database files on the        =
= logical network specified in the LCONF.  This EMT Build     =
= utility creates extended memory table files created         =
= specifically for use by BASE24 applications.                 =
=                                                                =
-----

```

Available EMT choices:

```
APCF DCCD ERF IDF IPCF RCPT STRF TRF NIDF
ALL or EXIT
```

```
Enter EMT choice ->
```

2. Enter one of the extended memory table input file (source file) acronyms or ALL for the extended memory table or tables to be built.

Prompt: Enter EMT choice ->

Entry: Source file acronym, ALL, or EXIT

After entering a valid choice, the following text is displayed on the screen:

```
Enter LCONF filename  
or EXIT
```

```
LCONF filename ->
```

3. Enter the fully qualified name of the Logical Network Configuration File (LCONF).

Prompt: LCONF filename ->

Entry: The fully qualified name of the LCONF or EXIT. If the LCONF resides on the same volume or subvolume as your current volume and subvolume, the volume or subvolume name is not required.

4. After entering the LCONF location, the utility starts running. The following is an example of the text displayed on the screen:

Running EMT Build Program

```
The LCONF file: $volume.PRODCNTL.L1CONF  
EMTS          : APCFEMT  
EMT Build Obj : $volume.BA60OBJ.BEMTBLO
```

```
$volume.BA60OBJ.BEMTBLO/NOWAIT,OUT $S.#EMT.APCF/ LCONF $volume.PRODCNTL.  
L1CONF,APCFEMT
```

```
Check Spooler to verify completion status
```

```
Spooler Name is: $s.#emt.apcf
```

```
TACL >
```

5. Check the spooler reports to make sure the requested extended memory tables were successfully built.
6. Warmboot the extended memory tables built in steps 1–4 using the EMT Warmboot screen or the ALTER ATTRIBUTE text commands.

Extended Memory Table Build Report

The Extended Memory Table Build Report provides a summary of all processing performed by the Base Extended Memory Table Build utility. A sample of the Extended Memory Table Build Report is shown below and on the following pages. Each sample page is followed by an explanation of each field appearing on the report page.

The title of the report and the current page number are printed at the top of each report page.

In addition to the fields documented here, various self-explanatory exception messages may be written to the report at any time indicating exceptions that occur during processing. If an exception causes the build utility to abend, the phrase “*emt-name* BUILD HALTED” will be written to the report, where *emt-name* is the name of the extended memory table being built when the exception occurs.

Cover Page

The cover page is printed at the beginning of the Extended Memory Table Build report. It summarizes the start date and time and lists the extended memory tables that were requested to be built.

BASE24	EXTENDED MEMORY TABLE BUILD REPORT	PAGE: 1
Start Date:	YY/MM/DD	
Start Time:	HH:MM:SS	
EMT Swap Files Requested:		
	APCFEMT	
	IPCFEMT	
	TRFEMT	

Start Date — The date, in the format specified by the DATE-DISPLAY-FORMAT LCONF param, the Base Extended Memory Table Build utility began processing.

Start Time — The time of day (HH:MM:SS) the Base Extended Memory Table Build utility began processing.

EMT Swap Files Requested — The names of the extended memory tables requested to be built.

Hash Table Summary Page

A Hash Table Summary report page is printed for each extended memory table built with the exception of some core extended memory tables that have their own report page format. For descriptions of any core-specific report pages, refer to the product-specific transaction processing manual.

The Hash Table Summary page details the hashing algorithms and bucket factors used to build the optimum hash table for a file or table.

BASE24	EXTENDED MEMORY TABLE BUILD REPORT	PAGE: n
Table Type:	APCFEMT	
Source File:	\SYS.\$DATA.PRO1DATA.APCF	
EMT Swap File:	\SYS.\$DATA.PRO1DATA.APCFEMT	
Table Size:	135470 bytes	
Total Data Records:	2356	
Maximum Record Length:	112 bytes	
Key Type:	ALL STRING	
Key Length:	24 bytes	
Key Algorithm:	DOUBLE BYTE SUMMATION	
Hash Algorithm:	MODULO DIVIDE	
Bucket Factor:	2	
Total Buckets:	29	
Total Buckets Used:	13	
Percent Buckets Used:	44%	
Total Barrels:	1	
Total Collisions:	0	
Percent Collisions:	0%	
Total Collision Chains:	0	
Longest Collision Chain:	0	
Average Collision Chain:	0	
Build Attempts:	24	

Table Type — The acronym for the extended memory table being built.

Source File — The fully qualified location of the source file or table on disk from which the extended memory table is to be built.

EMT Swap File — The fully qualified location of the swap file used to build this extended memory table. This is the name of the actual extended memory table file on disk. This file is opened by the object when it allocates the extended memory segment. The swap file is used to swap out records from disk as they are added to the table in memory.

Table Size — The total size, in bytes, required for the extended memory table.

Total Data Records — The total number of data records or rows included in the extended memory table.

Maximum Record Length — The maximum length, in bytes, of each record included in the extended memory table.

Key Type — The type of record or row key used in this extended memory table. Valid types are as follows: ALL ASCII (alphanumeric) and ALL BINARY (the binary data cannot exceed four bytes).

Key Length — The total length, in bytes, of the primary key to this extended memory table.

Key Algorithm — The algorithm used to hash the key from this file or table in building the optimum hash table. The Base Extended Memory Table Build utility tries all combinations of allowed hash algorithms, key algorithms, and bucket factors until a table with the fewest collisions is built. Valid key algorithms include the following: SINGLE BYTE XOR, DOUBLE BYTE XOR, TRIPLE BYTE XOR, QAUDRUPLE BYTE XOR, SINGLE BYTE SUMMATION, DOUBLE BYTE SUMMATION, and SINGLE BYTE POLYNOMIAL.

Hash Algorithm — The algorithm used to hash key values from this file or table in building the optimum hash table. The Base Extended Memory Table Build utility tries all combinations of allowed hash algorithms, key algorithms, and bucket factors until a table with the fewest collisions is built. Valid hash algorithms include the following: MULTIPLICATIVE, MID-SQUARE, and MODULO DIVIDE.

Bucket Factor — The bucket factor used in building the hash table. The bucket factor is the ratio of the number of buckets that will exist as opposed to the actual number of data records in the hash table. The Base Extended Memory Table Build utility tries all combinations of allowed hash algorithms, key algorithms, and bucket factors until a table with the fewest collisions is built.

Total Buckets — The total number of buckets used for all combinations of allowed hash algorithms, key algorithms, and bucket factors attempted.

Total Buckets Used — The total number of buckets used in the optimum hash table. The Base Extended Memory Table Build utility tries all combinations of allowed hash algorithms, key algorithms, and bucket factors until a table with the fewest collisions is built.

Percent Buckets Used — The percentage of buckets containing data items in the optimum hash table.

Total Barrels — The total number of barrels used in the optimum hash table. A barrel is equal to a group of buckets.

Total Collisions — The total number of collisions that occurred on the optimum hash table. A collision occurs when two or more data records or rows hash to the same bucket.

Percent Collisions — The percentage of collisions that occurred on the optimum hash table. A collision occurs when two or more data records or rows hash to the same bucket.

Total Collision Chains — The total number of collision chains that occurred on the optimum hash table. A collision chain occurs when two or more data records or rows hash to the same bucket.

Longest Collision Chain — The longest collision chain that occurred on the optimum hash table. A collision chain occurs when two or more data records or rows hash to the same bucket.

Average Collision Chain — The average length of all collision chains that occurred on the optimum hash table. A collision chain occurs when two or more data records or rows hash to the same bucket.

Build Attempts — The total number of combinations attempted to obtain the optimum hash table.

Trailer Page

The trailer page is printed at the end of the Extended Memory Table Build report. It summarizes the total number of swap files successfully built, the total number of swap files that failed, as well as the date and time processing completed.

```
BASE24                      EXTENDED MEMORY TABLE BUILD REPORT                      PAGE:  n

      EMT Swap Files Built:

                          APCFEMT
                          IPCFEMT
                          TRFEMT

      EMT Swap Files Failed:

      End Date:           YY/MM/DD
      End Time:           HH:MM:SS
```

EMT Swap Files Built — The extended memory tables successfully built in this run.

EMT Swap Files Failed — The extended memory tables that failed to be built in this run.

End Date — The date, in the format specified by the DATE-DISPLAY-FORMAT LCONF param, this run of the Base Extended Memory Table Build utility ended.

End Time — The time of day (HH:MM:SS) this run of the Base Extended Memory Table Build utility ended.

ACI Worldwide, Inc.

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